ENT ATLAS

IMAGING

Learn to read ENT imaging

Eighteen lessons, from the normal scan to the patterns of disease

18 lessons · November 2026

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IMAGING COURSE · LESSON 1

CT for the ENT surgeon

Most of the scans an ENT surgeon requests are CT, and most mistakes in reading them come from not knowing what the picture is made of. A CT image is a map of density, displayed through a window you choose, reconstructed with a filter that favours either bone or soft tissue, and sliced in planes you can rebuild at will. Once those four ideas are clear, every temporal bone, sinus and neck scan becomes easier to read.

AT A GLANCE

- CT measures density in **Hounsfield units**: air −1000, fat about −100, water 0, soft tissue 30–60, bone 400–1000+.
- The **window** decides what you see: always read bone and soft-tissue windows separately.
- Thin-slice **bone algorithm** for temporal bone and sinus; **contrast-enhanced** soft-tissue CT for neck masses, abscess and cancer.
- Learn the artefacts: dental amalgam streaks, motion, partial volume and beam hardening.
- Cone-beam CT gives sinus and ear bone detail at lower dose but no useful soft tissue.

1 CHAPTER ONE What a CT image is

A map of density.

1.1 Hounsfield units

CT measures how much each small volume of tissue (a *voxel*) absorbs X-rays, and expresses it on the **Hounsfield scale**, where water is 0 and air is −1000. Fat is negative (about −100), soft tissues sit a little above water (muscle 40 to 50), fresh blood is denser (60 to 80), iodinated contrast in vessels reaches 100 to 300, and bone ranges from a few hundred to over 1000. Placing a region of interest on a lesion and reading its number is often the quickest way to tell fat from fluid, or fluid from blood.

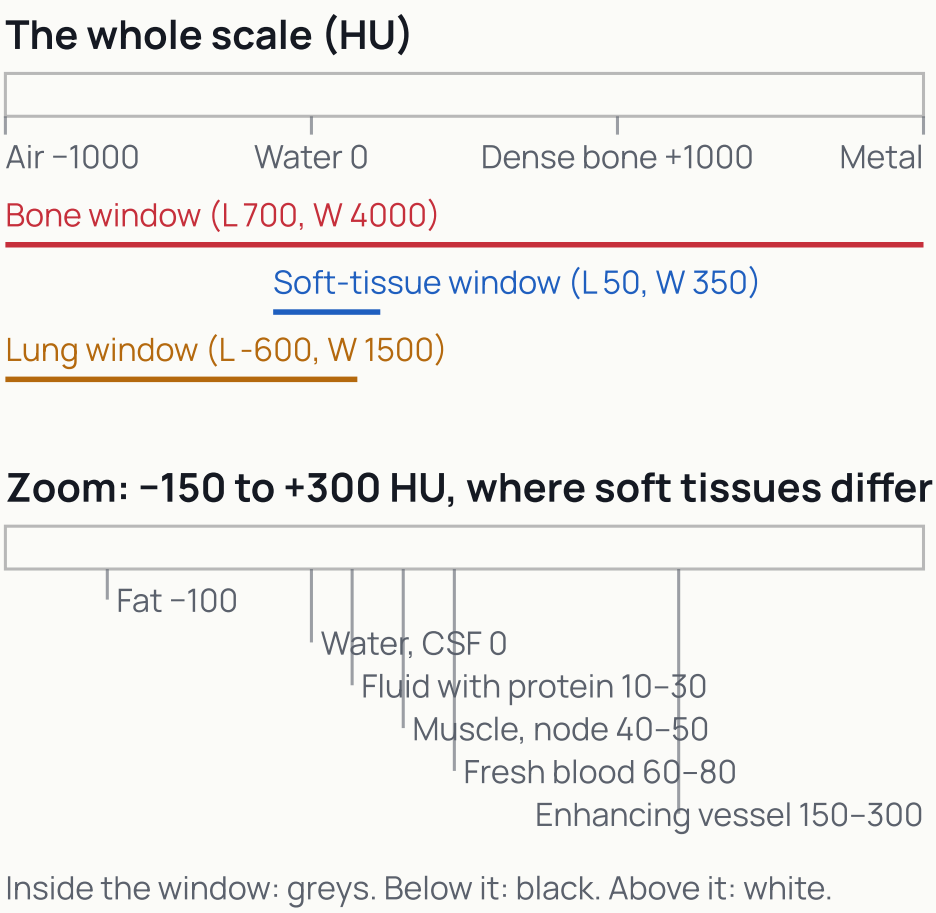


FIGURE 1 The Hounsfield scale and three common windows. The window decides which densities are shown as grey: everything below it is black and everything above it white.

Drawn for ENT Atlas

1.2 Windows

A screen can show only about 256 shades of grey, and the eye distinguishes far fewer, so the radiologist chooses a **window**: a *level* (the centre value) and a *width* (the range displayed). A **bone window** (level about 500 to 700, width 2000 to 4000) shows ossicles, the otic capsule, the skull base foramina and thin sinus walls, but makes all soft tissue look the same grey. A **soft-tissue window** (level about 40 to 60, width 350 to 400) separates muscle, fat, nodes, tumour and abscess, but turns bone uniformly white. A wide **lung window** is used for the airway and the chest.

Every CT must be read in both bone and soft-tissue windows, because each hides what the other shows: an orbital abscess is invisible on bone windows, and a dehiscent lamina papyracea is invisible on soft-tissue windows.

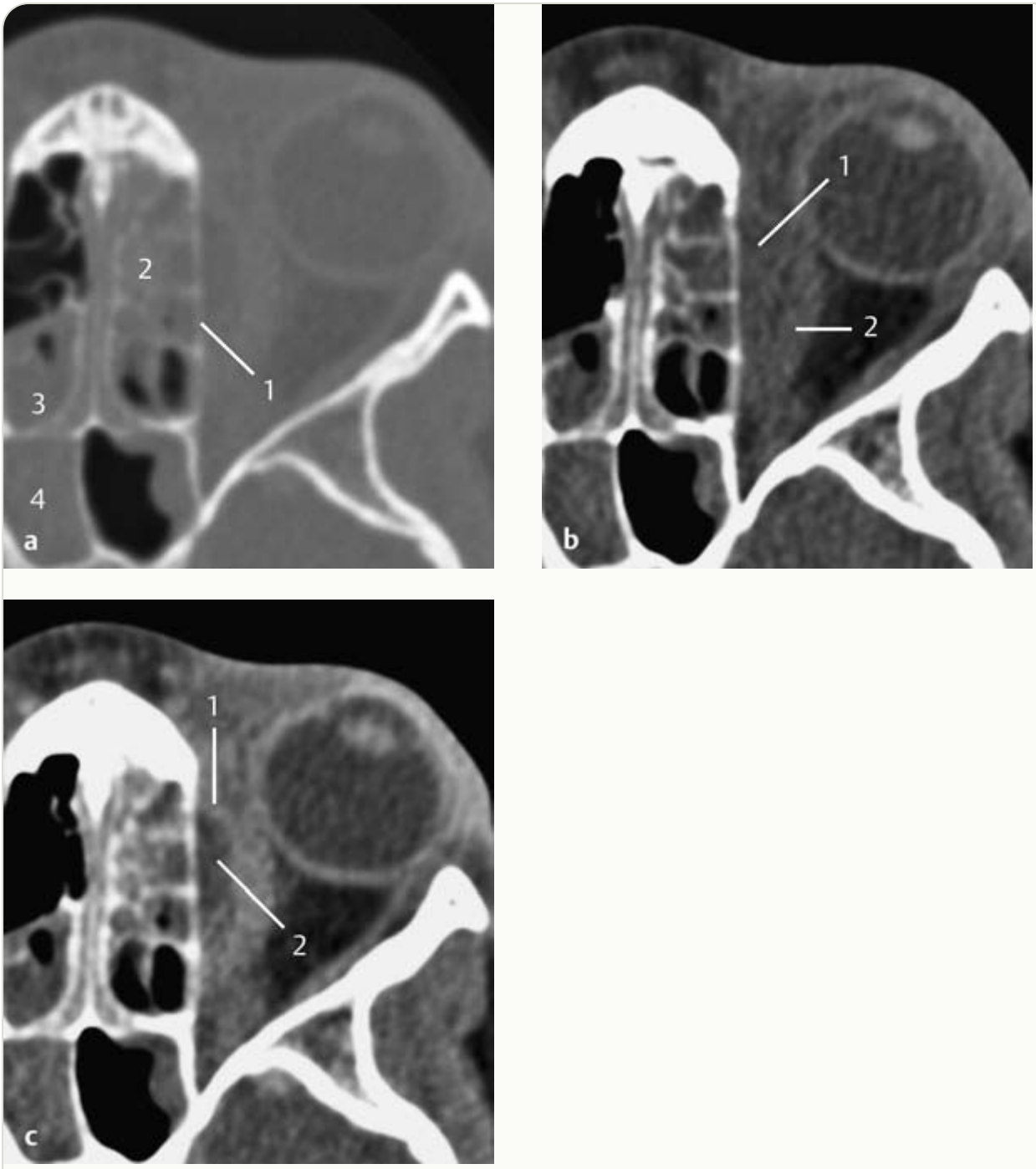


FIGURE 2 The same patient in three settings. (a) Bone window: intact lamina papyracea (1) with ethmoid and sphenoid opacification. (b) Soft-tissue window: an orbital mass (1) compressing the medial rectus (2). © After contrast: the capsule of a subperiosteal abscess enhances (1) around fluid (2).

Dunnebier, Imaging for Otolaryngologists

READ THE WINDOW BEFORE THE SCAN

Before interpreting any image, look at the window settings in the corner. A "normal" temporal bone in a soft-tissue window, or a "normal" neck in a bone window, means only that the wrong window was used.

CHAPTER TWO

2 How the images are made

Reconstruction, slices and planes.

2.1 Bone and soft-tissue reconstructions

The raw data from the scanner can be reconstructed with different mathematical filters (*kernels*). A **bone (sharp, high-resolution) algorithm** gives crisp edges and fine detail at the cost of noise, and is used for the temporal bone, sinuses and facial skeleton. A **soft-tissue (smooth) algorithm** reduces noise and shows subtle density differences, and is used for the neck. A good temporal bone or sinus study is therefore reconstructed twice, in a bone algorithm for the bones and a soft-tissue algorithm for the soft tissues, and each is then viewed in its own window.

2.2 Slice thickness and multiplanar reformats

Modern multidetector scanners acquire a volume of very thin data (0.5 to 0.6 mm) in one pass, from which images can be rebuilt in any plane. **Temporal bone** studies use slices of about 0.5 to 0.6 mm, with axial and coronal reformats, and often oblique planes along the cochlea or the semicircular canals (Pöschl and Stenvers views). **Sinus** studies use about 1 mm in axial, coronal and sagittal planes; the coronal plane mirrors the surgeon's endoscopic view, and the sagittal plane shows the frontal recess. **Neck** studies use 2 to 3 mm soft-tissue slices with thinner bone images of the skull base, mandible and larynx where needed.

Because the volume can be reformatted, the patient needs to be scanned only once; direct coronal scanning with the neck extended is no longer needed and is avoided in trauma.

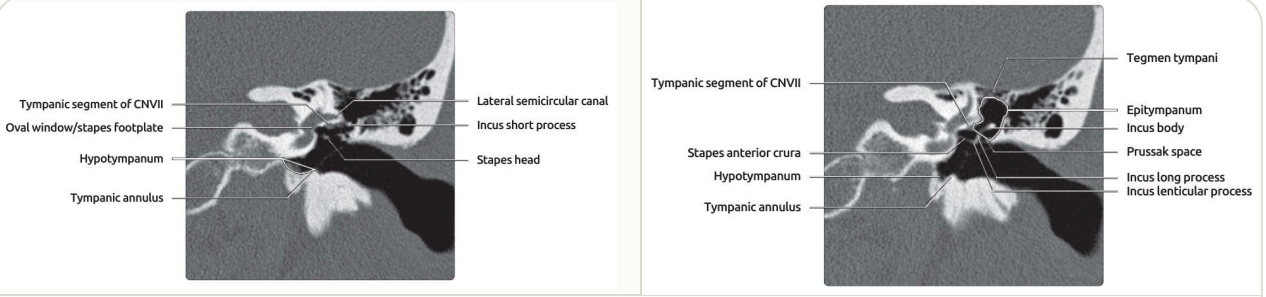


FIGURE 3 Coronal bone-algorithm CT of the middle ear in thin section: the tympanic segment of the facial nerve, the oval window and stapes, the scutum and Prussak's space are all visible. Detail like this needs submillimetre slices.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 84

2.3 Scrolling, not snapshots

On the workstation, anatomy is learnt by scrolling through the stack and following each structure from slice to slice, not by looking at single images. Follow the facial nerve from the internal auditory canal to the stylomastoid foramen, or the uncinate process from top to bottom, and the anatomy becomes three-dimensional.

3 CHAPTER THREE Contrast

When and why.

3.1 Iodinated contrast

Intravenous iodinated contrast opacifies vessels and enhances tissues with a blood supply. **Contrast-enhanced CT** is the standard for neck masses, cancer staging, nodes, abscesses and suspected vascular lesions. Many neck protocols use a **split bolus** or a delay of 60 to 90 seconds after injection, so that both arteries and mucosa enhance and tumours stand out from muscle. **CT angiography** times the scan to the arterial phase to show vessels, for trauma, paragangliomas, pulsatile tinnitus and bleeding.

Contrast is *not* needed for routine sinus or temporal bone imaging, where the question is about bone and air. Before contrast, ask about previous reactions, renal

function and metformin, and remember that iodine interferes with radioactive iodine treatment of thyroid cancer for weeks.

WHAT TO REQUEST

QUESTION	STUDY
Chronic rhinosinusitis before surgery	Non-contrast CT sinuses, thin bone algorithm, three planes (or cone-beam CT)
Cholesteatoma, conductive loss, trauma to the ear	Non-contrast thin-section CT temporal bones
Neck mass, cancer staging, nodes	Contrast-enhanced CT neck (or MRI)
Deep neck abscess or orbital complication	Contrast-enhanced CT
Pulsatile tinnitus, vascular tumour, trauma to vessels	CT angiography (and venography)
Facial fracture	Non-contrast thin CT with 3D reconstructions

4 CHAPTER FOUR Cone-beam CT and dose

Lower dose, fewer uses.

4.1 Cone-beam CT

Cone-beam CT uses a cone of X-rays and a flat detector rotating once around the head, usually with the patient seated. It gives excellent bony detail of the sinuses, temporal bone and teeth at a lower dose than conventional CT, and suffers less from metal artefact around cochlear implants. Its soft-tissue contrast is poor, Hounsfield values are unreliable, and motion degrades it, so it cannot replace conventional CT for soft tissue, infection or tumour.

4.2 Radiation

A sinus or temporal bone CT gives a modest effective dose, but the lens of the eye lies in the field of sinus and orbital scans, and children are more sensitive to

radiation. Use low-dose protocols for sinus CT, avoid repeat scans that will not change management, and prefer ultrasound or MRI in children when they can answer the question.

5 CHAPTER FIVE Artefacts

When the picture lies.

5.1 The common artefacts

CT ARTEFACTS IN HEAD AND NECK IMAGING

ARTEFACT	CAUSE	EFFECT AND REMEDY
Streak (beam hardening)	Dental amalgam, implants, dense bone	Dark and bright streaks across the oral cavity and oropharynx; angle the gantry, use metal-reduction software, or MRI
Motion	Swallowing, breathing, a restless child	Blurred or doubled edges, especially the larynx; repeat with breath-hold or quiet breathing
Partial volume	A thick slice containing two tissues	Thin bony plates (lamina papyracea, tegmen, stapes) appear absent; use thin slices before calling a dehiscence
Noise	Low dose, large patient	Grainy images; thicker reconstructions or a higher dose
Positioning	Head tilted	Asymmetry that mimics disease; compare like with like and reformat

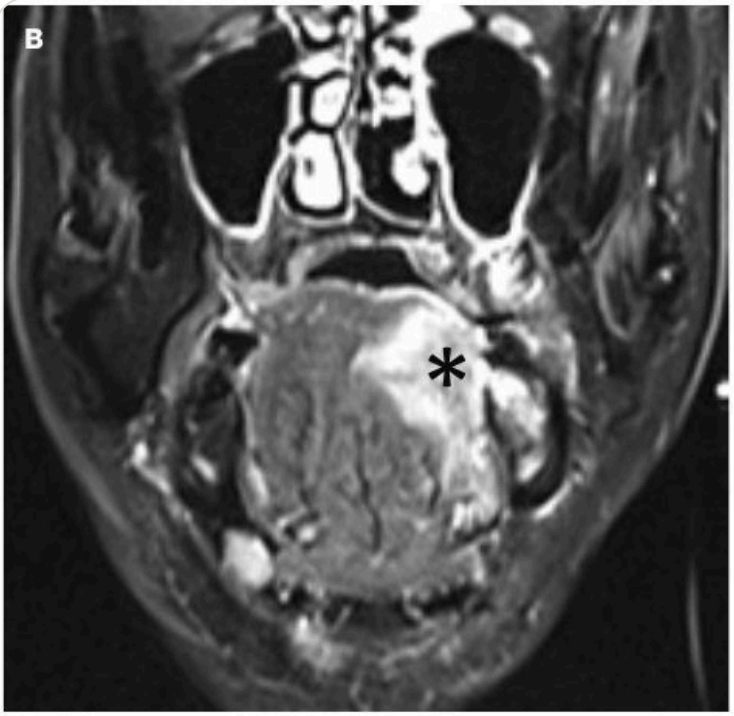


FIGURE 4 Dental amalgam artefact hiding a left oral tongue cancer on CT; MRI showed the tumour.

ESR Modern Radiology, Head and Neck Imaging

DO NOT OVER-CALL A DEHISCENCE

Thin bony plates such as the tegmen, the bone over the superior semicircular canal and the lamina papyracea can disappear on CT through partial volume averaging. Confirm on thin slices in more than one plane, and ideally on reformats along the structure (Pöschl view for the superior canal), before diagnosing a dehiscence.

READING ANY CT, EVERY TIME

1. **Check the basics**
   - → patient name and date, region, contrast or not, slice thickness, planes available
2. **Check the windows**
   - → bone window for bone and air
   - → soft-tissue window for soft tissue
   - → adjust if needed
3. **Scroll the whole stack**

→ follow structures slice to slice

→ compare left with right

4 **Use the planes**

→ axial, coronal and sagittal for every finding

5 **Measure**

→ Hounsfield units for fat, fluid, blood or enhancement

→ size in two or three planes

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Hounsfield units** of air, fat, water, soft tissue, blood and bone.
- **Windows** (level and width), and why every CT needs bone and soft-tissue viewing.
- **Bone versus soft-tissue reconstruction** kernels.
- **Slice thickness** and planes for temporal bone, sinus and neck studies.
- **When to give contrast,** and when it adds nothing.
- **Cone-beam CT:** advantages and limits.
- **Artefacts**, especially dental amalgam and partial volume.

6.2 Common pitfalls

COMMON PITFALLS

- Reading a temporal bone in a soft-tissue window, or a neck in a bone window.

- Requesting contrast for chronic sinusitis, or forgetting it for a neck abscess.
- Calling a dehiscence on one thick slice.
- Missing an oral cavity cancer hidden by dental artefact without asking for MRI.

Written for this app from Dunnebier, Imaging for Otolaryngologists ch. 1, ESR Modern Radiology, Head and Neck Imaging, Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), overview chapters. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 2

MRI made simple

MRI does not measure density. It shows how hydrogen nuclei, mostly in water and fat, behave in a magnetic field, and each *sequence* weights the picture towards a different property. The result is a family of images of the same slice, each telling a different part of the story. Learning MRI is mostly learning which sequence answers which question, and the handful of tissues whose brightness gives the diagnosis away: fat, fluid, blood, pus, cholesteatoma and enhancing tumour.

AT A GLANCE

- **T1**: fat bright, fluid dark; the best anatomy and the baseline for enhancement.
- **T2**: fluid bright; fat suppression (STIR or fat-sat) makes oedema and tumour stand out.
- **Post-gadolinium T1 fat-sat**: tumour, inflammation and perineural spread.
- **DWI/ADC**: restricted diffusion in pus, cholesteatoma (non-EPI DWI) and cellular tumour such as lymphoma.
- **CISS/FIESTA**: thin heavily T2 images of nerves in CSF, the labyrinth and the IAC.

1 CHAPTER ONE The basic sequences

What is bright, what is dark.

1.1 T1 and T2

On a **T1-weighted** image, **fat is bright** and **fluid is dark**. T1 gives the best anatomy, shows fat planes clearly (so tumour infiltrating fat stands out as grey replacing white), shows marrow, and is the baseline against which enhancement is judged. Bright signal on T1 before contrast has a short list: fat, subacute blood (methaemoglobin), proteinaceous fluid, melanin and gadolinium.

On a **T2-weighted** image, **fluid is bright**. CSF, oedema, cysts, retained secretions and most inflammation are bright; most tumours are intermediate; fibrous tissue, calcification and air are dark. The quickest way to tell which is which: find the CSF. If it is white, the image is T2; if it is dark, it is T1.

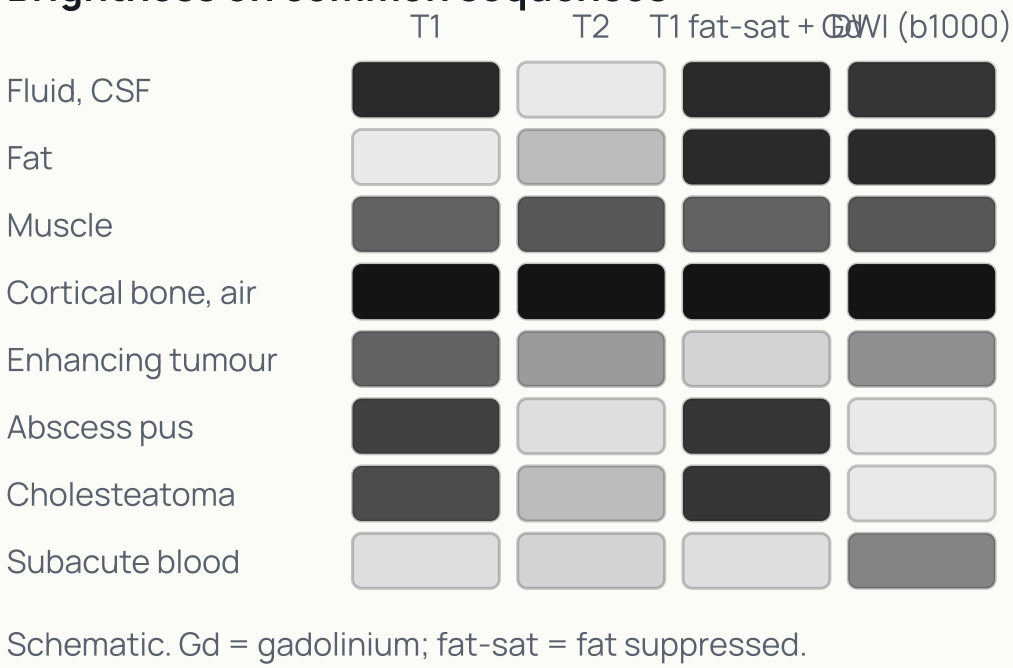


FIGURE 1 Schematic brightness of common tissues on T1, T2, post-gadolinium T1 with fat suppression, and diffusion. Abscess and cholesteatoma are the two classic "bright on diffusion" lesions in ENT.

Drawn for ENT Atlas

1.2 Fat suppression

Fat is bright on T1 and fairly bright on T2, so it can hide oedema and enhancing tumour lying in fat (in the orbit, neck, skull base marrow). **Fat suppression** removes its signal. **STIR** (short tau inversion recovery) is a robust fat-suppressed T2 for oedema and inflammation; **fat-saturated T2** and **fat-saturated post-contrast T1** are standard in head and neck protocols, because enhancing tumour and perineural spread show as bright against dark fat.

1.3 Gadolinium

Intravenous gadolinium shortens T1, so tissues with a blood supply or a broken blood–brain barrier become bright on T1. **Post-gadolinium T1 with fat suppression** is the workhorse for tumours, inflammation, meningeal and **perineural spread**, and for schwannomas in the IAC. Always compare with the pre-contrast T1: something bright *before* contrast is not enhancing.

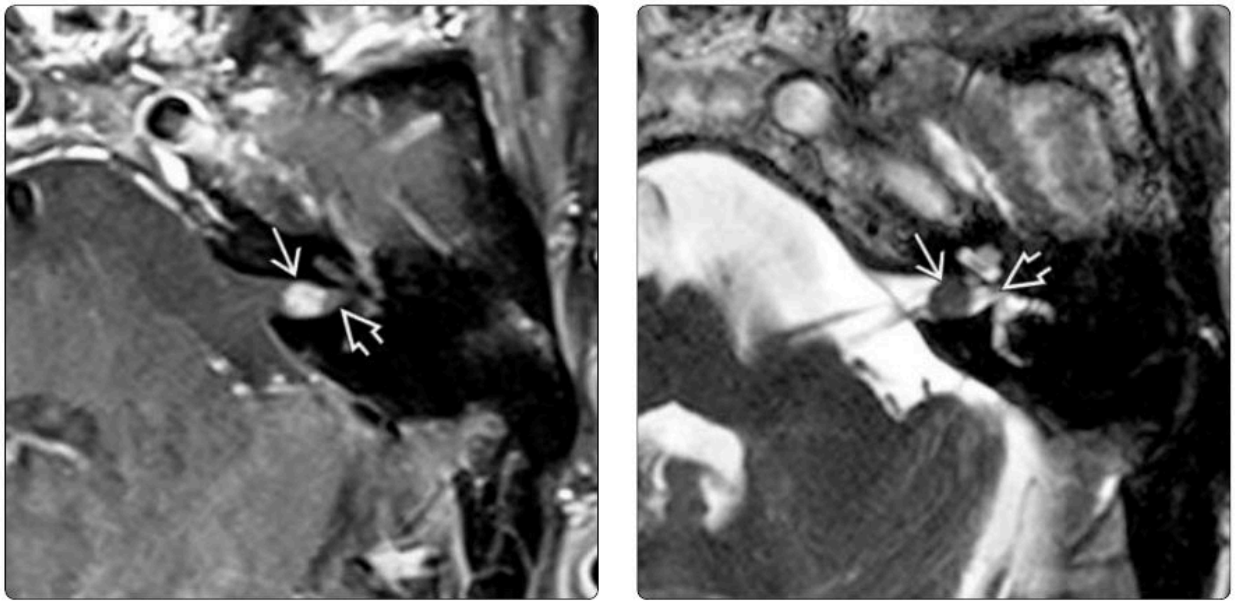


FIGURE 2 A small vestibular schwannoma in the internal auditory canal. Left: post-gadolinium T1 with fat suppression, the tumour enhances, with a cap of CSF lateral to it at the fundus. Right: CISS, the tumour is a dark filling defect in bright CSF; the screening sequence for asymmetric hearing loss.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1225

CHAPTER TWO

2 Special sequences for ENT

The ones you will meet.

2.1 Heavily T2-weighted thin-section images

CISS, **FIESTA** and **SPACE** (names differ between manufacturers) give submillimetre heavily T2-weighted images in which CSF and labyrinthine fluid are white and nerves, vessels and tumours are dark. They show the facial, cochlear and two vestibular nerves in the IAC, the fluid-filled cochlea and canals (absent in labyrinthitis ossificans, abnormal in malformations), and small schwannomas. Many centres screen asymmetric hearing loss with a non-contrast high-resolution T2 study, adding gadolinium when it is abnormal.

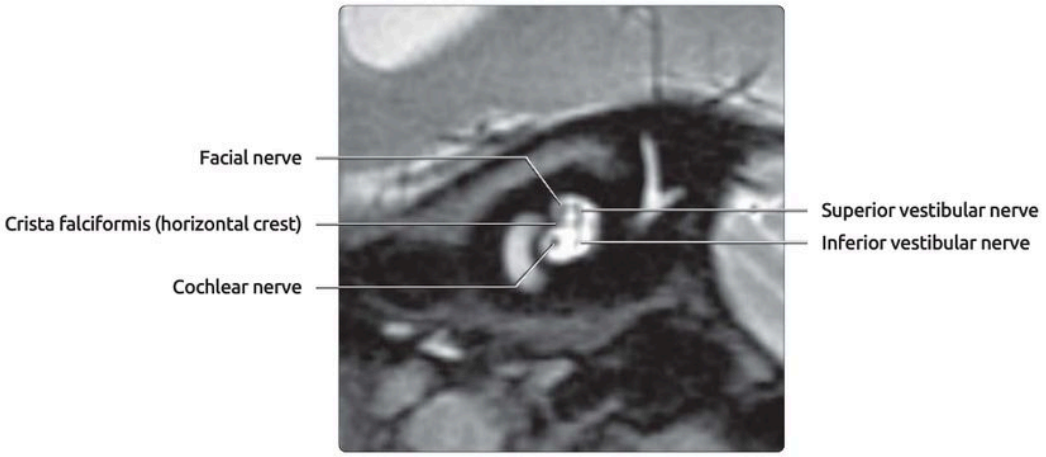


FIGURE 3 Oblique sagittal high-resolution T2 across the internal auditory canal. The four nerves sit in bright CSF: facial anterosuperior, cochlear anteroinferior, and the superior and inferior vestibular nerves posteriorly ("seven up, Coke down").

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 105

2.2 Diffusion-weighted imaging

DWI shows how freely water molecules move. Water that cannot move (in thick pus, keratin, or densely packed tumour cells) is **bright on DWI** and **dark on the ADC map**; checking the ADC map confirms true *restricted diffusion* rather than "T2 shine-through". In ENT, restricted diffusion means:

- **Abscess**, including orbital, intracranial, neck and periauricular collections.
- **Cholesteatoma**: **non-echo-planar (non-EPI) DWI** has few artefacts at the skull base and detects residual or recurrent cholesteatoma down to about 3 mm.
- **Highly cellular tumours**, especially **lymphoma** and undifferentiated carcinoma, and epidermoid cysts.

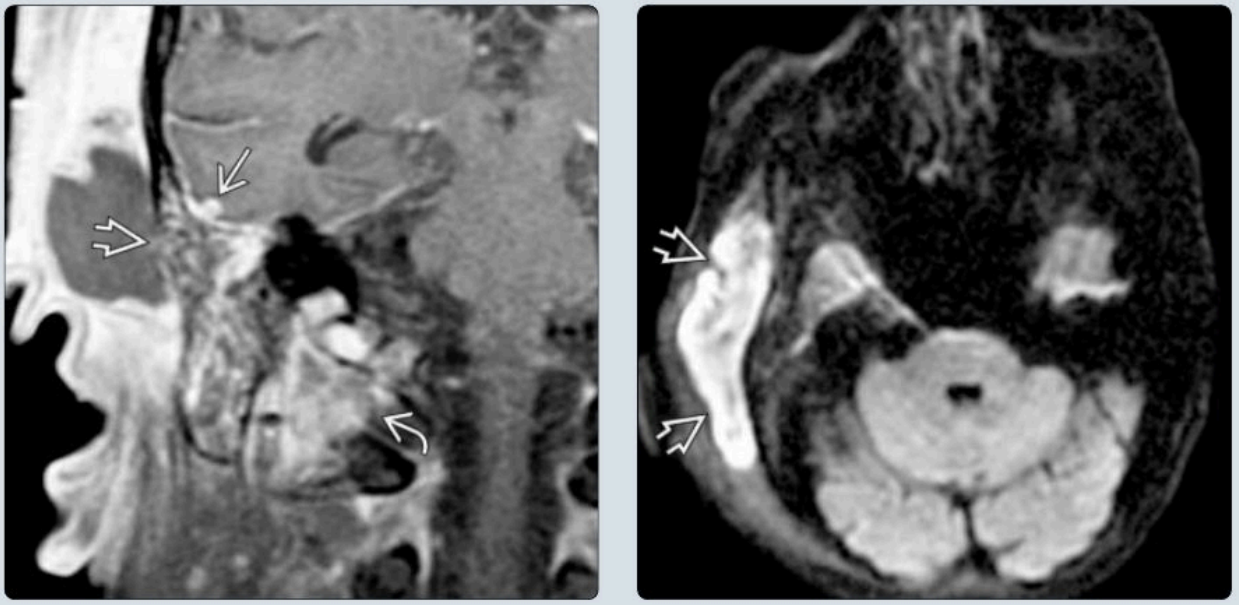


FIGURE 4 Coalescent otomastoiditis with a periauricular abscess. Left: post-gadolinium T1 with fat suppression, an enhancing mastoid with thickened enhancing meninges and a dehiscent outer cortex. Right: DWI, the abscess is bright (restricted diffusion).

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1042

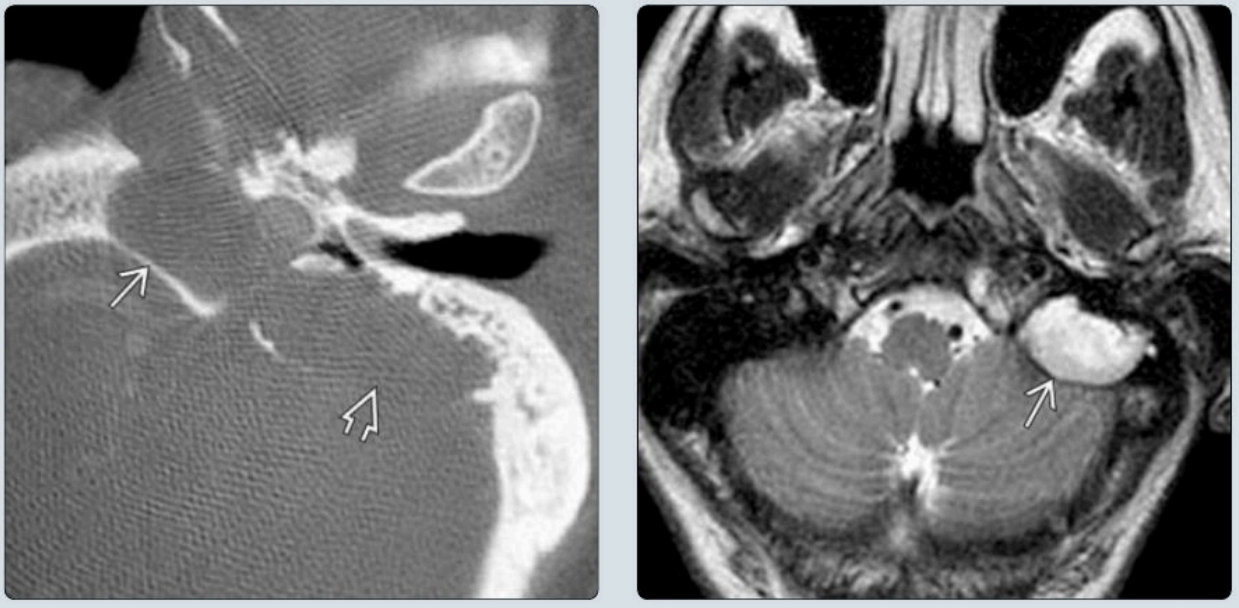


FIGURE 5 A congenital cholesteatoma read across modalities, part 1. Left: CT, a smooth expansile lesion of the mastoid and lateral clivus. Right: T2, bright and sharply marginated.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1030

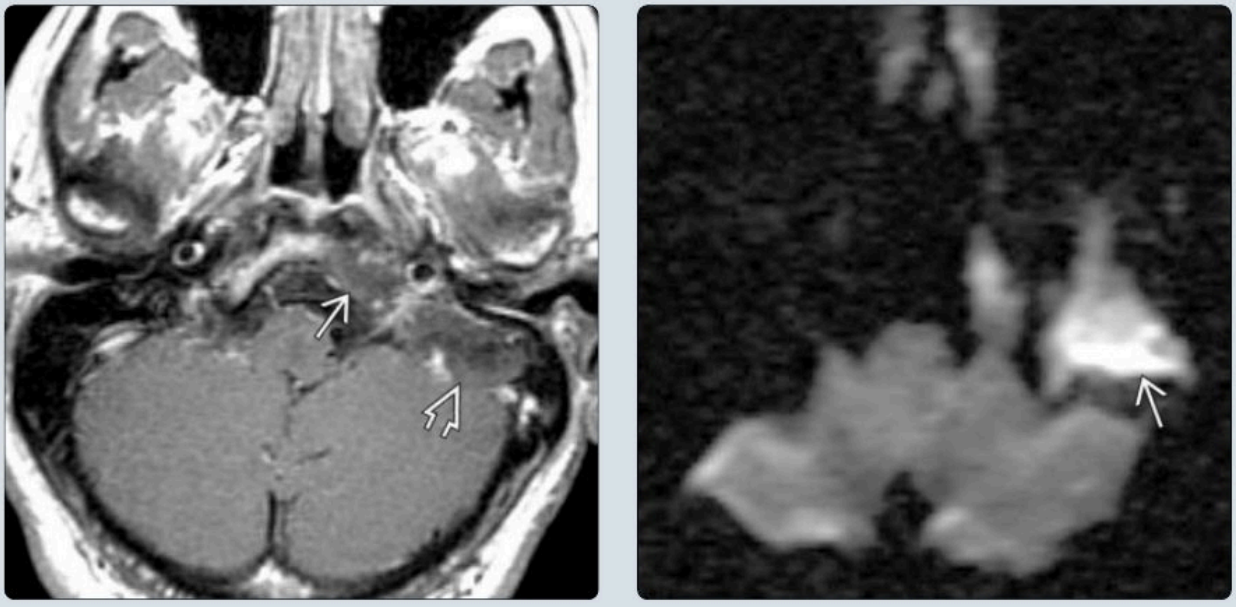


FIGURE 6 Part 2, the same lesion. Left: post-gadolinium T1, no enhancement. Right: DWI, bright. Expansion, no enhancement and restricted diffusion together mean cholesteatoma.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1030

2.3 Vessels and blood

MR angiography (time-of-flight or contrast-enhanced) and **MR venography** show arteries and venous sinuses without radiation, for pulsatile tinnitus, paragangliomas and sinus thrombosis. **Susceptibility sequences** (T2* gradient echo or SWI) show old blood as dark blooming, as in superficial siderosis or microhaemorrhage in a schwannoma.

3 CHAPTER THREE Reading an MRI study

A routine.

3.1 Find each sequence first

Open the series list and identify every sequence: T1, T2 (with or without fat suppression), post-contrast T1, DWI with ADC, and any thin T2. Then for each lesion ask the same questions in order.

CHARACTERISING A LESION ON MRI

1 T1 before contrast

- → **dark**
  fluid or most tumours
- → **bright**
  fat, subacute blood, protein, melanin

2 T2

- → **very bright**
  fluid, cyst, oedema, retained secretions
- → **intermediate**
  most tumours
- → **dark**
  fibrosis, calcification, air, dense secretions, fungus

3 After gadolinium

- → **none**
  cyst, cholesteatoma, fluid
- → **rim only**
  abscess or necrotic node
- → **solid**
  tumour or inflammation
- → **along a nerve**
  perineural spread

4 DWI and ADC

- → **bright DWI, dark ADC**
  pus, keratin (cholesteatoma, epidermoid), lymphoma
- → **no restriction**
  most other lesions

5 Put it together

- → location, margins, effect on bone (CT) and neighbours, then the short differential

SIGNAL OF LESIONS COMMONLY MET IN ENT

LESION	T1	T2	GADOLINIUM	DWI
Retained secretions	Dark (bright if thick)	Bright (dark if very thick or fungal)	Rim of mucosa only	Variable
Mucocoele	Variable with protein	Variable	Rim	Usually no restriction
Cholesteatoma	Dark	Bright	None (rim if inflamed)	Restricted
Cholesterol granuloma	Bright	Bright	None	No restriction
Abscess	Dark	Bright	Rim	Restricted
Schwannoma	Intermediate	Bright, may be cystic	Avid	No restriction
Squamous cell carcinoma	Intermediate	Intermediate	Moderate	Moderately restricted
Lymphoma	Intermediate	Intermediate, homogeneous	Mild, uniform	Strongly restricted
Lipoma or dermoid fat	Bright	Bright, suppresses with fat-sat	None	No restriction

4 CHAPTER FOUR Practicalities

Safety and limitations.

4.1 When MRI is not the answer

MRI is slow, so motion degrades it, and children often need sedation. It shows **cortical bone poorly**, so fine bony detail (ossicles, the otic capsule, sinus walls, fractures) belongs to CT; the two are complementary in the skull base and in cancer. Check for pacemakers, some cochlear implants and other implants, metal foreign bodies in the orbit, and claustrophobia. Cochlear implant magnets cause large

artefacts and may need removal or a special protocol. Gadolinium is avoided in pregnancy and in severe renal impairment.

DO NOT MISS

- Calling a bright-on-T1 lesion "enhancing" without looking at the pre-contrast T1.
- Calling high DWI signal "restricted" without checking the ADC map.
- Missing perineural spread because post-contrast images were not fat-suppressed.

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **T1 versus T2:** how to tell them apart, and what is bright on each.
- **Fat suppression** and why it matters in the head and neck.
- **Gadolinium:** what enhances; compare with the pre-contrast T1.
- **DWI and ADC:** abscess, cholesteatoma (non-EPI) and lymphoma.
- **CISS or FIESTA** for the IAC and the labyrinth.
- **CT versus MRI:** bone versus soft tissue.

5.2 Common pitfalls

COMMON PITFALLS

- Thinking that dark on T2 in a sinus excludes disease: thick secretions and fungus are dark.
- Ordering MRI to look at ossicles or a fracture.
- Forgetting cochlear implants and other implants on the safety form.

Written for this app from Dunnebier, Imaging for Otolaryngologists ch. 1, ESR Modern Radiology, Head and Neck Imaging, Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), Imaging Anatomy: Head and Neck (Chapman and Koontz). Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 3

Choosing the test

The best scan is the one that answers the clinical question with the least harm, delay and cost. Requests such as "CT head and neck please" waste radiation and time; requests such as "unilateral conductive loss with an attic retraction: is there cholesteatoma, and are the ossicles, tegmen and lateral canal intact?" get the right protocol and a useful report. This lesson matches the common ENT questions to the test that answers them.

AT A GLANCE

- Start from the clinical question, not the modality.
- Ultrasound first for thyroid, salivary glands, superficial lumps and nodes in the neck, with FNA.
- CT for bone (temporal bone, sinuses, fractures) and for acute infection; MRI for soft tissue, nerves and skull base spread.
- PET-CT for unknown primary, advanced stage and response after chemoradiotherapy (at about 12 weeks).
- Children: ultrasound and MRI before CT where possible.

1 CHAPTER ONE Start from the question

What do you need to know?

1.1 Bone, soft tissue, vessel or function?

Most ENT imaging questions fall into four kinds. **Bone and air** questions (sinus anatomy before surgery, ossicles, otic capsule, fractures, erosion by cholesteatoma) belong to **CT**. **Soft tissue** questions (the extent of a tumour, perineural or intracranial spread, the IAC, the labyrinth's fluid, cholesteatoma recurrence) belong to **MRI**. **Superficial lumps and glands** (thyroid, salivary glands, neck nodes, a lump in a child) belong to **ultrasound**, with fine-needle aspiration in the same sitting. **Function** questions (the swallow, aspiration, a pouch, metabolic activity of a tumour) belong to **contrast swallow, videofluoroscopy, nuclear medicine or PET-CT**.

COMMON ENT QUESTIONS AND THE TEST THAT ANSWERS THEM

QUESTION	FIRST TEST	WHY
Chronic rhinosinusitis, planning sinus surgery	Non-contrast CT sinuses (or cone-beam CT)	Bony anatomy, variants, skull base and orbit
Unilateral sinus opacification or suspected tumour	CT and contrast MRI	CT for bone; MRI separates tumour from secretions and shows orbit and dura
Orbital or intracranial complication of sinusitis	Contrast-enhanced CT; MRI if intracranial	Abscess, cavernous sinus, empyema
Chronic otitis media, cholesteatoma, conductive loss	Non-contrast thin-section CT temporal bones	Ossicles, scutum, tegmen, lateral canal, facial canal
Residual or recurrent cholesteatoma	Non-EPI diffusion MRI	Detects keratin down to about 3 mm
Asymmetric sensorineural loss or tinnitus	MRI of the internal auditory meatus (high-resolution T2 ± gadolinium)	Vestibular schwannoma and other CPA lesions
Pulsatile tinnitus	CT temporal bone with CT angiography and venography, or MRI with MRA and MRV	Paraganglioma, dehiscent jugular bulb, sinus stenosis, dural fistula
Facial palsy, atypical or not recovering	Contrast MRI from brainstem to parotid; CT for the canal	Tumour, perineural spread, fracture
Neck lump in an adult	Ultrasound with FNA, then contrast CT or MRI if malignant	Characterise, sample, stage
Neck lump in a child	Ultrasound	No radiation; most are nodes or congenital cysts
Thyroid nodule	Ultrasound (U or TI-RADS grading) with FNA	Risk grading and sampling
Salivary lump	Ultrasound with FNA or core; MRI for deep lobe or malignancy	Site, extent, facial nerve relation
Head and neck cancer staging	Contrast CT or MRI of primary and neck, chest CT; PET-CT for advanced disease	Local extent, nodes, distant spread

QUESTION	FIRST TEST	WHY
Unknown primary	PET-CT before panendoscopy	Finds the primary in a proportion
Response after chemoradiotherapy	PET-CT at about 12 weeks	Avoids unnecessary neck dissection (PET-NECK)
Dysphagia or suspected pouch	Barium swallow; videofluoroscopy for aspiration	Structure and function
Deep neck infection	Contrast-enhanced CT	Abscess versus phlegmon, airway, mediastinum
Facial or laryngeal fracture	Non-contrast thin CT	Fracture lines and displacement

CHAPTER TWO

2 The tests one by one

Strengths and limits.

2.1 Ultrasound

Ultrasound is quick, cheap, free of radiation and dynamic, and in skilled hands is the most accurate way to characterise thyroid nodules, salivary lumps and neck nodes and to guide FNA or core biopsy. A normal lymph node is oval with a bright fatty hilum; an abnormal one is round, loses its hilum, and may show necrosis, microcalcification or peripheral vascularity. Ultrasound cannot see behind bone or air, so it does not show the deep lobe of the parotid well, the retropharyngeal space, the larynx or the skull base, and it depends on the operator.

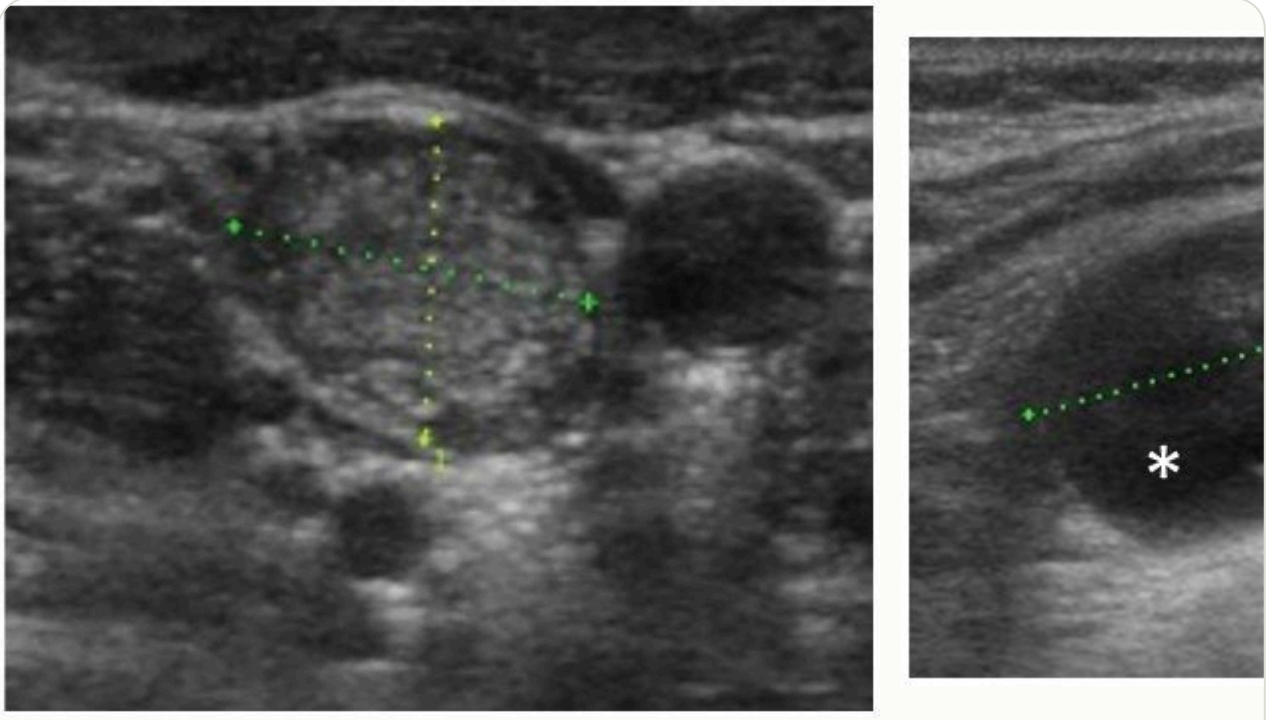


FIGURE 1 Abnormal neck nodes on ultrasound. Left: round, without a fatty hilum and with microcalcifications, typical of papillary thyroid cancer metastasis. Right: cystic change and loss of the hilum; fine-needle aspiration confirmed metastasis.

ESR Modern Radiology, Head and Neck Imaging

2.2 Plain films

Plain radiographs have a small, specific role: a lateral soft-tissue neck film for a radio-opaque foreign body or for the thumb sign of supraglottitis (only if the airway is safe and it will not delay care), an orthopantomogram for dental disease, osteoradionecrosis and mandibular fractures, and a chest film. Sinus X-rays are no longer used to diagnose sinusitis.

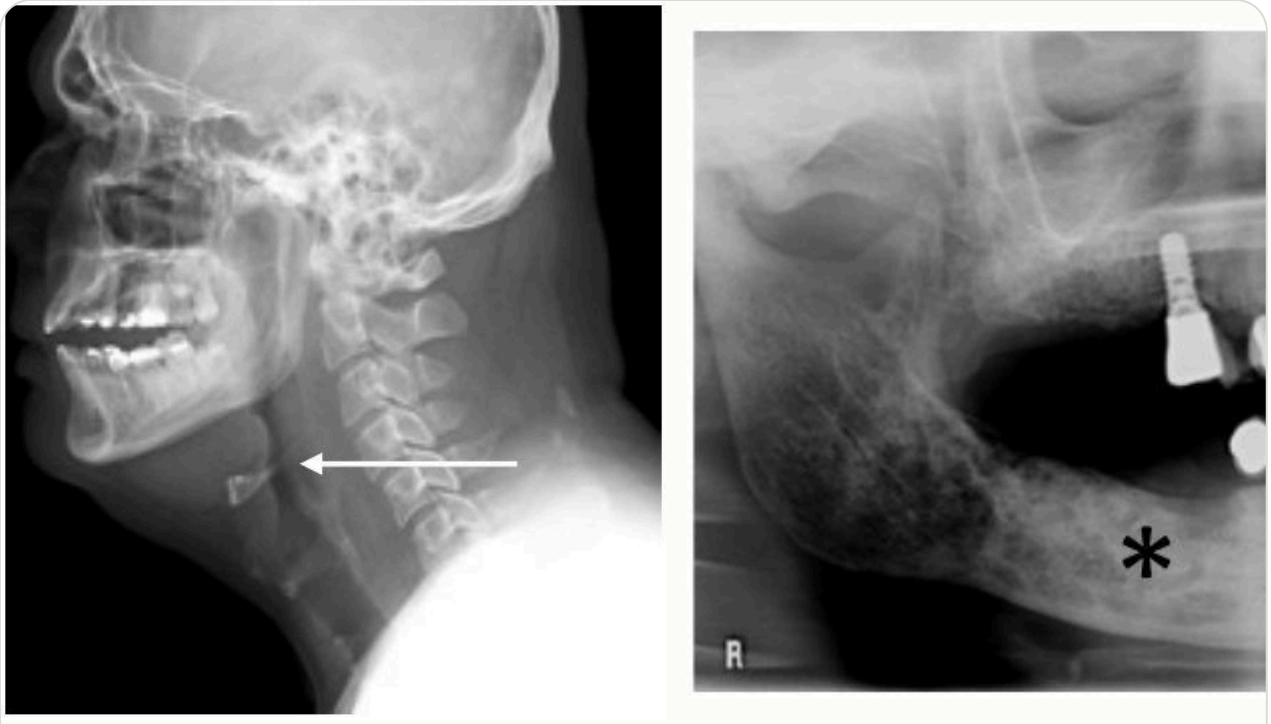


FIGURE 2 Plain films that still help. Left: lateral neck in acute supraglottitis, the swollen epiglottis narrowing the airway. Right: orthopantomogram showing osteoradionecrosis of the mandible.
ESR Modern Radiology, Head and Neck Imaging

2.3 CT and MRI

CT and MRI are covered in their own lessons. In short, CT is fast, widely available and best for bone and air, acute infection and trauma; MRI is best for soft-tissue extent, nerves, the labyrinth, the skull base and recurrence, but is slow, sensitive to motion and contraindicated with some implants. In cancer of the skull base, sinuses and nasopharynx the two are complementary.

2.4 PET-CT

PET-CT combines a CT with fluorodeoxyglucose (FDG) uptake, which is high in most cancers but also in inflammation, infection, recent surgery, brown fat and normal lymphoid tissue (tonsils, adenoids). It is used for advanced-stage cancer and distant metastases, for the unknown primary, for suspected recurrence, and to assess response at about **12 weeks** after chemoradiotherapy: the PET-NECK trial showed that surveillance by PET-CT, operating only on residual uptake, was as safe as planned neck dissection and spared most patients the operation. Too early a scan gives false positives from radiation inflammation.

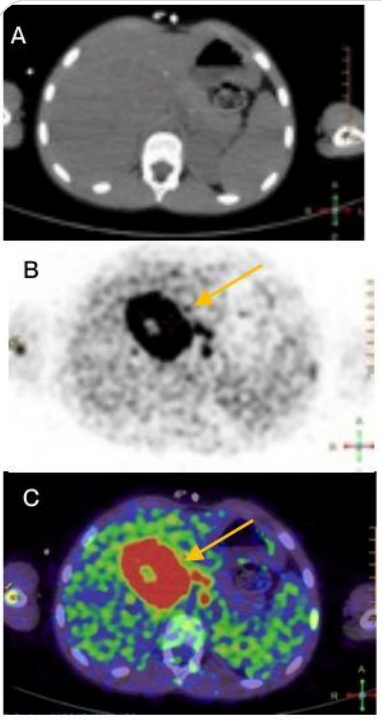


FIGURE 3 PET shows widespread disease in a child with lymphoma: the metabolic picture that CT alone would miss.

ESR Modern Radiology, Head and Neck Imaging

2.5 Swallow studies

A **barium swallow** shows the anatomy of the pharynx and oesophagus: pouch, web, stricture, ring, achalasia and tumour. **Videofluoroscopy** (modified barium swallow) with a speech therapist shows the dynamics of the oral and pharyngeal phases and aspiration with different textures. Water-soluble contrast is used when perforation or a leak is suspected.

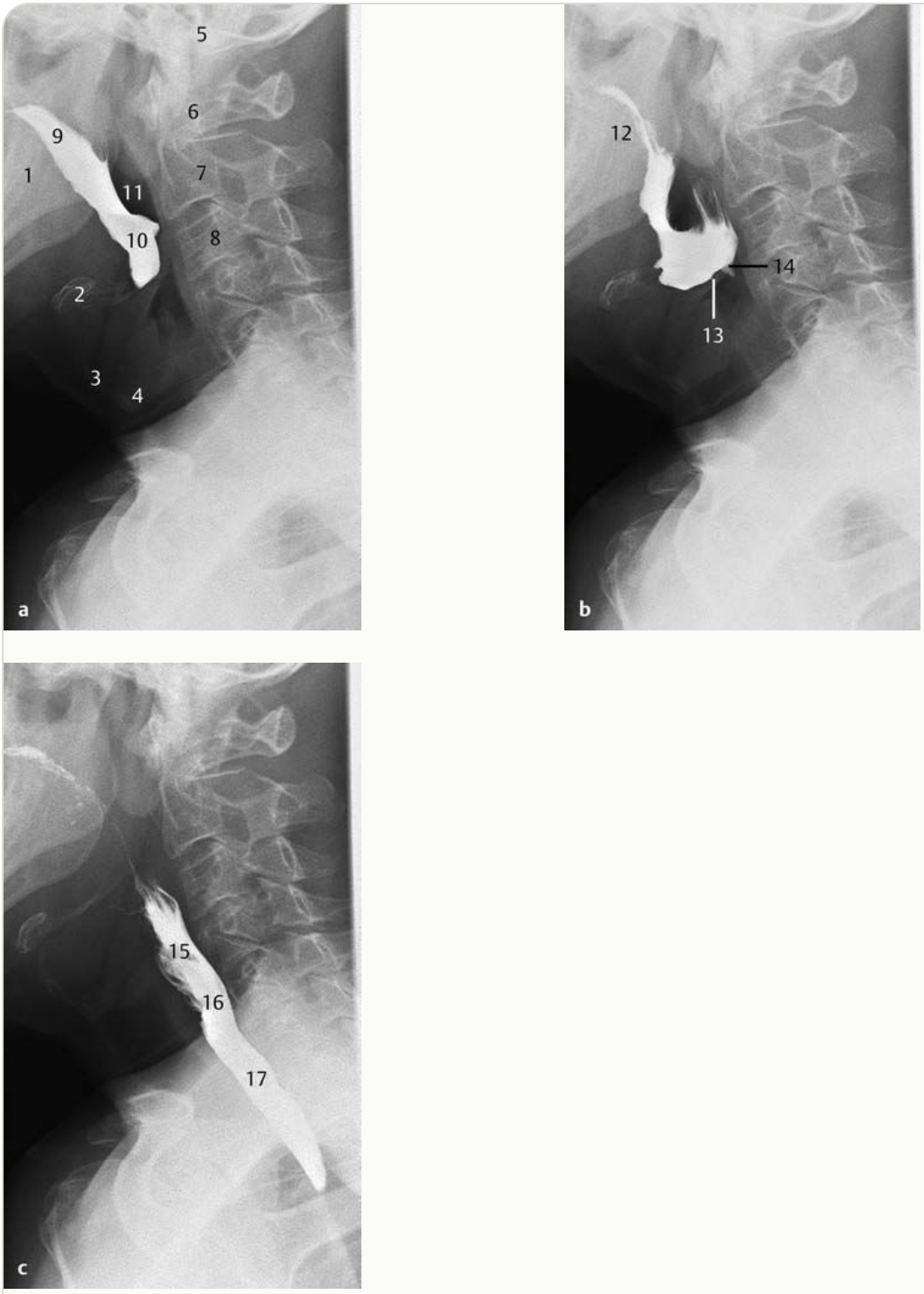


FIGURE 4 Contrast swallow images of the pharynx and oesophagus: a dynamic study shows both structure and transit.

Dunnebier, Imaging for Otolaryngologists

2.6 Angiography and nuclear medicine

Catheter angiography is now used mainly for treatment: embolisation of juvenile angiofibroma, paraganglioma and epistaxis, and balloon occlusion testing before

carotid sacrifice. **Nuclear medicine** offers sestamibi and SPECT-CT for parathyroid adenoma, iodine and technetium scans for the thyroid, and salivary scintigraphy for function.

3 CHAPTER THREE Safety and stewardship

Fewer, better scans.

3.1 Radiation, contrast and children

Each CT adds radiation, most important in children and for the lens of the eye. Ultrasound and MRI are preferred in children where they can answer the question; MRI in young children may need sedation or a "feed and wrap" approach in infants. Iodinated contrast is checked against allergy, renal function and planned radioactive iodine; gadolinium against renal function and pregnancy.

3.2 Writing a good request

State the clinical findings, the specific question, and anything that changes the protocol: suspected cholesteatoma (thin bone CT), pulsatile tinnitus (add angiography), cancer staging (contrast, include chest), cochlear implant or other metal (MRI safety). A precise question gets a precise answer.

CHOOSING THE TEST

1 What do you need to see?

- → **bone, air, ossicles, fracture**
  CT
- → **soft tissue, nerve, skull base, recurrence**
  MRI
- → **superficial lump or gland**
  ultrasound ± FNA
- → **swallow or aspiration**
  barium or videofluoroscopy

2 Is it urgent?

- → **airway, abscess, complication, trauma**
  contrast CT now
- → otherwise plan the best test

3 Is it a child?

- → prefer ultrasound or MRI
- → CT for bone questions and emergencies

4 Is it cancer?

- → primary and neck by CT or MRI
- → chest CT
- → PET-CT for advanced, unknown primary or response at 12 weeks

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Match the test to the question:** bone, soft tissue, superficial gland or function.
- **Ultrasound** features of an abnormal node.
- **PET-CT:** indications, false positives, and timing after chemoradiotherapy (PET-NECK).
- **Barium swallow versus videofluoroscopy.**
- **Imaging children** with the least radiation.
- **A good imaging request.**

4.2 Common pitfalls

COMMON PITFALLS

- Ordering a CT for a thyroid nodule or a child's neck lump instead of ultrasound.
- Requesting PET-CT too soon after radiotherapy.
- Ordering MRI for a fracture or ossicular detail.

Written for this app from Dunnebier, Imaging for Otolaryngologists, ESR Modern Radiology, Head and Neck Imaging, Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.). Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 4

Describing a lesion and reading a report

Radiology is a language. A good description lets a colleague picture the lesion without seeing the scan, and the act of describing it in a fixed order is what stops you missing things. The authors of *Diagnostic Imaging: Head and Neck* teach one idea above all others: **find the space or structure the lesion comes from, and the differential diagnosis follows.** This lesson gives you the order and the words, and a way of reading a radiology report critically.

AT A GLANCE

- Say **where** first: the space or structure of origin.
- Then **what**: size, shape, margins, density or signal, enhancement, calcification, fat.
- Then **what it does**: bone remodelling versus destruction, perineural and vascular spread, airway, nodes.
- Finish with a short **differential** in order of likelihood and what would settle it.

1 CHAPTER ONE Where is it?

Site comes first.

1.1 Space or structure of origin

Every lesion is described first by where it arises: a mucosal surface (pharynx, larynx, sinus lining), a deep space of the neck, a gland, a bone, a nerve, a vessel or a node. In the neck, the deep spaces are separated by layers of deep cervical fascia, and each space contains a limited set of structures, so each has a short list of possible lesions. The *neck by spaces* lesson teaches the map; here, the principle is enough: a mass is assumed to arise from the space it is centred in, and the way it displaces the surrounding fat and muscle confirms the space.

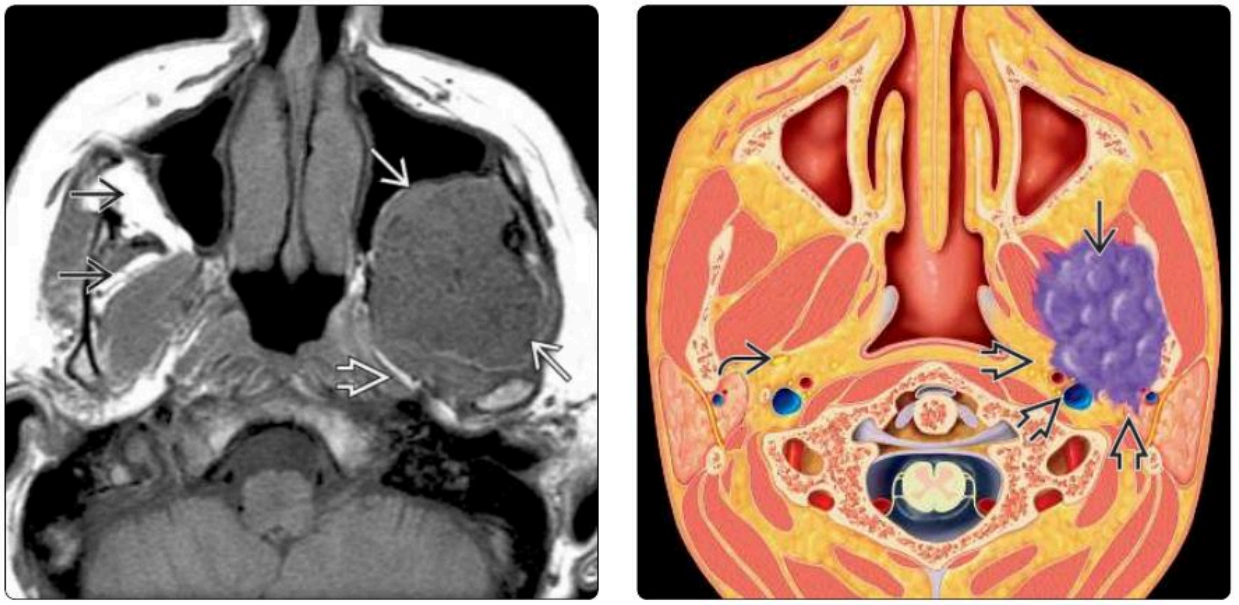


FIGURE 1 A masticator space tumour. Left: T1 MRI, the mass replaces the fat planes of the muscles of mastication and pushes the parapharyngeal fat posteromedially. Right: the same pattern drawn: knowing which way the fat moves tells you where the mass began.
Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 43

1.2 Laterality, size and extent

Give the side, the maximum size in three planes, and the borders the lesion reaches or crosses: the midline, the skull base, the orbit, the larynx's framework, the carotid. Size matters for staging, and extent for surgery.

2 CHAPTER TWO What is it like?

Internal characteristics.

2.1 Margins and shape

Well defined, smooth margins suggest a slow-growing or benign lesion (cyst, schwannoma, pleomorphic adenoma). **Ill-defined, infiltrative** margins suggest malignancy or infection. Lobulated, dumbbell or tubular shapes along a nerve suggest a nerve sheath tumour.

2.2 Density, signal and content

On CT describe density compared with muscle (low, equal, high), and look for **fat** (lipoma, dermoid), **fluid** (cyst, abscess, necrosis), **calcification** (chondroid tumours, thyroid, phleboliths in a venous malformation) and **air** (abscess, fistula). On MRI give T1 and T2 signal, fat suppression, and diffusion. Fluid–fluid levels suggest haemorrhage in a lymphatic malformation.

2.3 Enhancement

Describe whether the lesion enhances, how much and in what pattern: **none** (simple cyst, cholesteatoma), **rim** (abscess, necrotic node, infected cyst), **avid and uniform** (paraganglioma, haemangioma, schwannoma), **heterogeneous** (larger tumours with necrosis), **flow voids** (high-flow vascular tumours: the "salt and pepper" of paraganglioma).

3 CHAPTER THREE What is it doing?

Effect on neighbours.

3.1 Bone: remodelling or destruction

The behaviour of adjacent bone is one of the most useful clues on CT. **Remodelling** means the bone is expanded, thinned and bowed but still present, with a smooth margin; it implies slow growth (mucocoele, polyposis, schwannoma, juvenile angiofibroma, many benign tumours). **Destruction** means the bone is eroded and missing, with irregular, permeative or moth-eaten margins; it implies aggressive disease (carcinoma, aggressive infection such as invasive fungal sinusitis or necrotising otitis externa, metastasis). Smooth enlargement of a foramen suggests a nerve sheath tumour; irregular erosion of the foramen suggests malignancy.

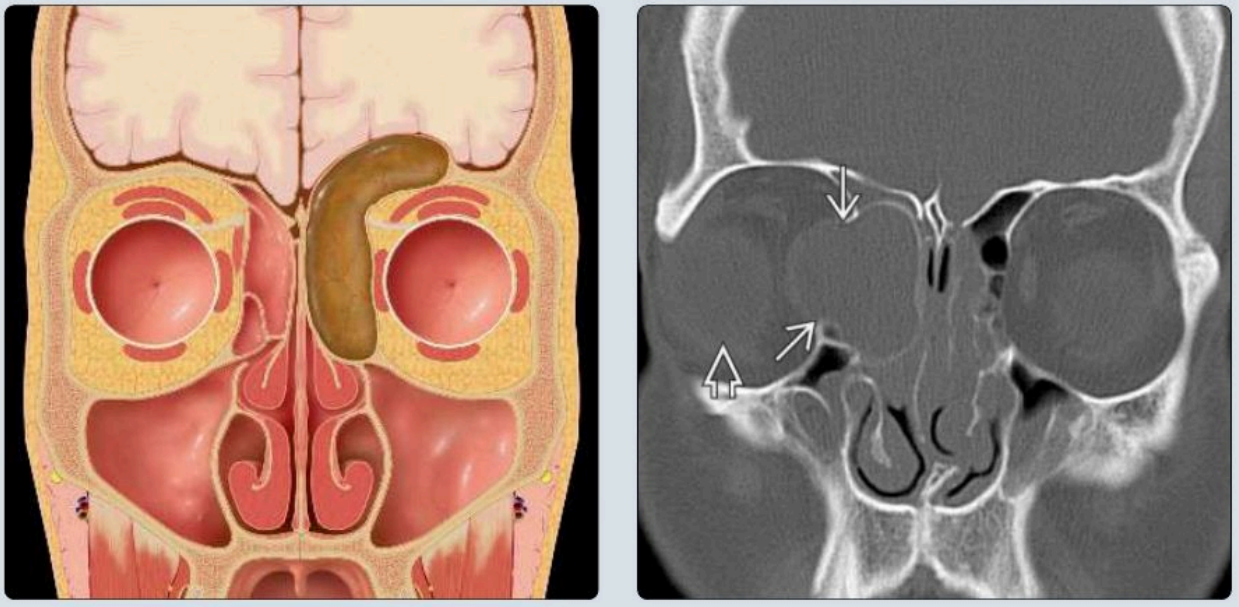


FIGURE 2 Remodelling. A large ethmoid mucocele expands the sinus and thins and bows its walls, including the lamina papyracea, without aggressive destruction.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 716

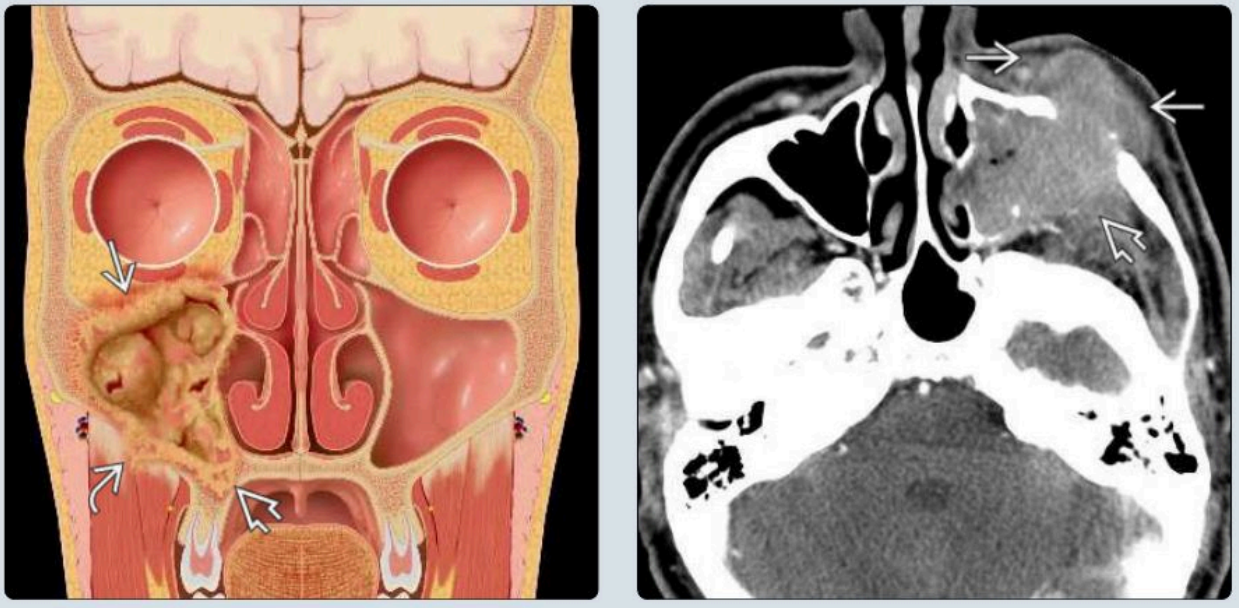


FIGURE 3 Destruction. Squamous cell carcinoma of the maxillary antrum breaks through the sinus walls into the cheek and the infratemporal fossa.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 754



FIGURE 4 Smooth enlargement of the foramen ovale by a trigeminal (V3) schwannoma, with an enhancing tubular tumour following the nerve. A benign nerve sheath tumour widens a foramen; carcinoma erodes it irregularly.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 56

3.2 Nerves, vessels and airway

Look for **perineural spread**: enlargement and enhancement of a nerve, loss of the fat pad at its foramen (pterygopalatine fossa, stylomastoid foramen, foramen ovale), widening of the foramen, and denervation changes in the muscles it supplies (atrophy and fatty change, or acute oedema and enhancement). Describe the **carotid**: displaced, abutted or encased (more than 270° of contact suggests encasement and usually unresectability). Note any narrowing of the **airway**.

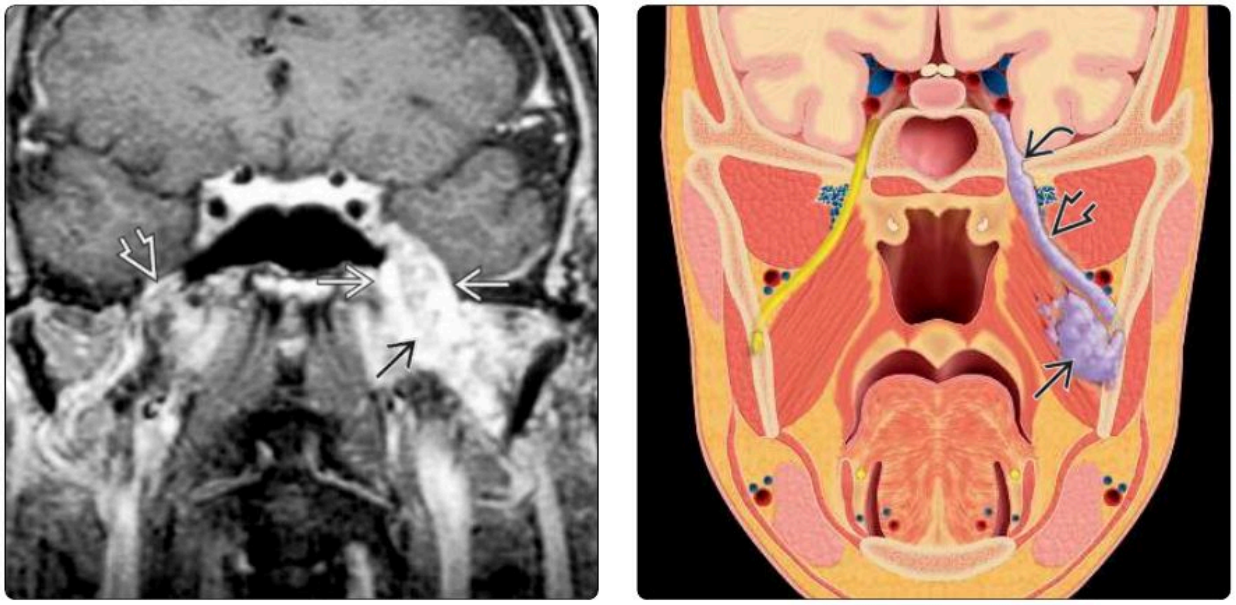


FIGURE 5 Perineural spread along the mandibular nerve. Left: post-gadolinium fat-suppressed T1, the nerve is thickened and enhancing and the foramen ovale widened. Right: the pattern drawn, tumour passing through the skull base along the nerve.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 43

3.3 Nodes

Every head and neck study includes the nodes. Name the level, size (short axis), shape, and features of malignancy: **necrosis** (the most reliable sign), round shape, loss of the fatty hilum, extranodal extension (irregular margins, fat stranding, invasion of muscle), and, in thyroid cancer, cystic change and calcification.

4 CHAPTER FOUR Building the differential

From description to diagnosis.

4.1 The three filters

A differential diagnosis is built by passing the findings through three filters: **location** (what normally lives in this space or structure), **imaging character** (cystic or solid, fat, calcification, enhancement pattern, bone behaviour) and **clinical context** (age, symptoms, immune status, known cancer, previous surgery or

radiotherapy). Give two to four diagnoses in order of likelihood, and say what would separate them: a further sequence, a biopsy, a blood test.

A DESCRIPTION IN ORDER

STEP	EXAMPLE
Site and space	Left parapharyngeal space, centred in the prestyloid compartment
Size and extent	3.4 × 2.8 × 4.1 cm, abutting the skull base, no intracranial extension
Margins and shape	Smooth, lobulated, well defined
Content	Low density on CT; bright on T2; no fat or calcification
Enhancement	Heterogeneous moderate
Effect on neighbours	Displaces the parapharyngeal fat medially; widens the stylomandibular gap; carotid displaced posteriorly, not encased; no bone erosion
Nodes	None enlarged
Differential	Deep lobe pleomorphic adenoma most likely; minor salivary tumour; schwannoma less likely as the carotid is displaced posteriorly

5 CHAPTER FIVE Reading a report

Critically, not passively.

5.1 The structure of a report

A radiology report gives the clinical details and question, the technique (modality, contrast, sequences), the comparison studies, the findings (usually organised by structure or space), and the **impression**, which should answer the clinical question and give the most important findings first. Standardised systems help: **TI-RADS** and the **U** grading for thyroid nodules, **NI-RADS** for the treated neck, **Lund–Mackay** scoring for sinus CT, and **TNM** stage for cancer.

5.2 Reading it as a surgeon

Read the images yourself before or alongside the report; you know the patient and the operation. Check that the question was answered, that the whole region was covered, and that the findings fit the clinical picture. Discrepancies should be discussed with the radiologist, ideally in a multidisciplinary meeting.

DESCRIBING ANY LESION

1. **Where**
   - → side
   - → space, structure or organ of origin
   - → size in three planes
   - → extent and borders crossed
2. **What**
   - → margins and shape
   - → density or signal
   - → fat, fluid, calcification, air
   - → enhancement pattern
   - → diffusion
3. **What it does**
   - → bone remodelled or destroyed
   - → nerves and foramina
   - → carotid and jugular
   - → airway
   - → orbit and dura

4 **Nodes**

→ levels, size, necrosis, extranodal extension

5 **Differential**

→ two to four diagnoses by likelihood

→ what would separate them

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Describe in order:** site, size, margins, content, enhancement, effect, nodes, differential.
- **Space of origin** as the key to the differential.
- **Remodelling versus destruction** of bone.
- **Perineural spread:** the signs.
- **Carotid encasement:** more than 270°.
- **Malignant node features:** necrosis, shape, hilum, extranodal extension.

6.2 Common pitfalls

COMMON PITFALLS

- Jumping to a diagnosis before saying where the lesion is.
- Calling bone "eroded" when it is smoothly remodelled.
- Forgetting the nodes and the nerves on a cancer scan.

- Reading the impression only.

Written for this app from Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), overview chapters, Imaging Anatomy: Head and Neck (Chapman and Koontz), ESR Modern Radiology, Head and Neck Imaging. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 5

The temporal bone on CT

The temporal bone packs the hearing and balance organs, the facial nerve, the carotid and the jugular bulb into a few cubic centimetres, and thin-section CT shows nearly all of it. Most beginners look at a temporal bone CT and see only noise. The fix is to learn one normal stack, slice by slice, until you know which structures should be on each image, and then to read every scan against the same checklist. This lesson walks through the normal ear in axial and coronal planes and then shows how the common diseases change it.

AT A GLANCE

- Thin **bone-algorithm** CT, no contrast, axial and coronal; Pöschl and Stenvers obliques when needed.
- Know the landmarks: the **ice-cream cone**, **Prussak's space**, the **scutum**, and the posterior wall trio of **sinus tympani, pyramidal eminence and facial recess**.
- Read lateral to medial: canal, drum and scutum, spaces, ossicles, tegmen and canals, mastoid, inner ear, IAC and facial nerve, vessels, petrous apex.
- Cholesteatoma: **tegmen, fistula, facial canal, sinus tympani, ossicles**. CT shows bone; non-EPI DWI shows keratin.
- Otosclerosis: lucency at the **fissula ante fenestram**; fractures: otic capsule **sparing or violating**.

1 CHAPTER ONE The study

What a good temporal bone CT looks like.

1.1 Protocol

A temporal bone CT is a **non-contrast**, thin-section study: slices of 0.5 to 0.6 mm through each ear separately, reconstructed with a **bone algorithm** and viewed in a **wide bone window** (width about 4000 HU). Axial images are made parallel to the lateral semicircular canal, and coronal images are reformatted from the same volume. Oblique reformats help with particular questions: the **Pöschl** plane, parallel to the superior canal, for superior canal dehiscence, and the **Stenvers** plane, perpendicular to the petrous ridge, for the cochlea and the vestibular aqueduct.

Contrast adds nothing to the bone question; when soft tissue matters (a mass, the IAC, the labyrinth's fluid, recurrent cholesteatoma), MRI is the complementary study.

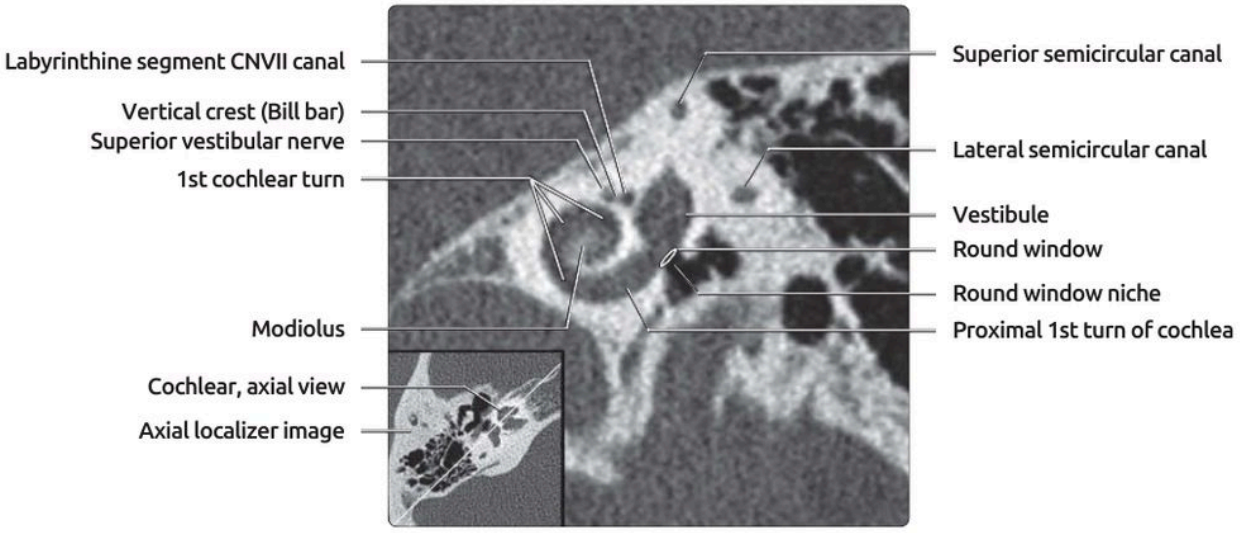


FIGURE 1 Oblique reformat along the long axis of the cochlea: the whole basal turn in one image, with the superior vestibular nerve canal and the labyrinthine facial nerve just above it. Oblique planes rescue structures that run obliquely to the axial plane.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 90

1.2 How to scroll

Read each ear alone, then compare the two sides at the same level. Follow tubular structures from slice to slice rather than looking for them on a single image: the facial nerve from the IAC to the stylomastoid foramen, the ossicular chain from the malleus head to the stapes, each semicircular canal around its loop. The normal tympanic membrane is barely visible; the ossicles, scutum, and the thin bony plates (tegmen, sigmoid plate, the bone over the superior canal) should be crisp.

2 CHAPTER TWO The normal ear in axial

Superior to inferior.

2.1 Upper levels: the canals and the attic

Start at the top. The first structure seen is the **superior semicircular canal** rising under the arcuate eminence. Moving down, the **aditus ad antrum** joins the epitympanum to the mastoid antrum, and the **lateral semicircular canal** comes into view, with the porus of the **internal auditory canal** medially and the vestibular aqueduct on the posterior surface of the petrous bone.

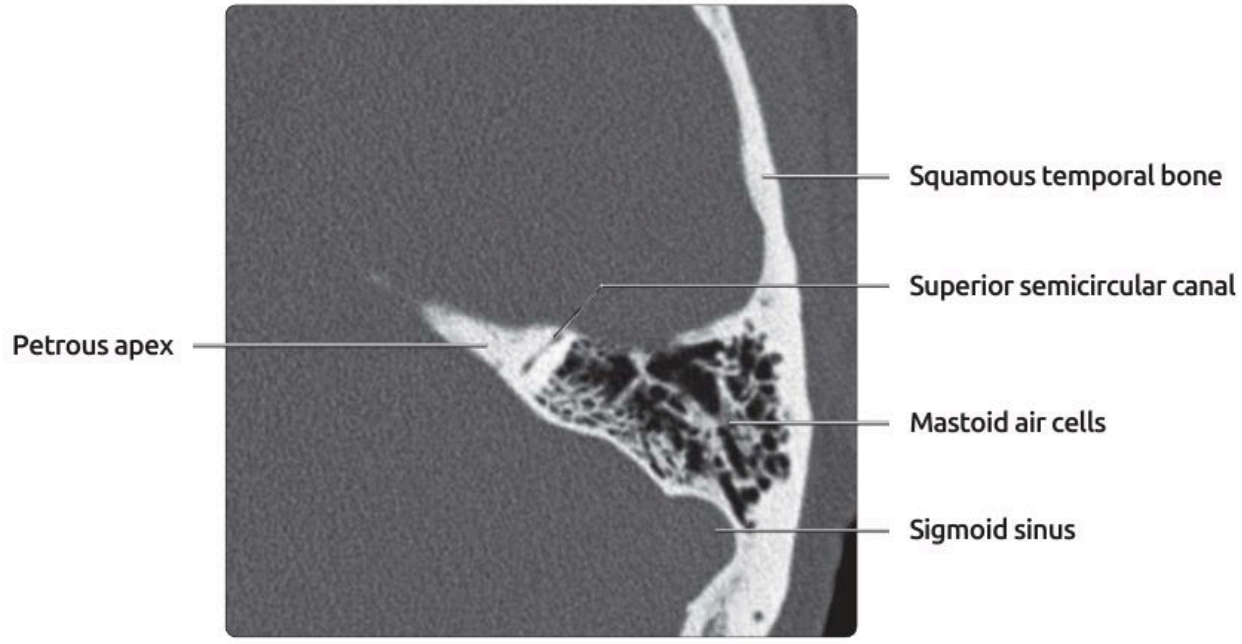


FIGURE 2 Highest axial level: the superior semicircular canal projects upwards from the inner ear under the arcuate eminence.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 54

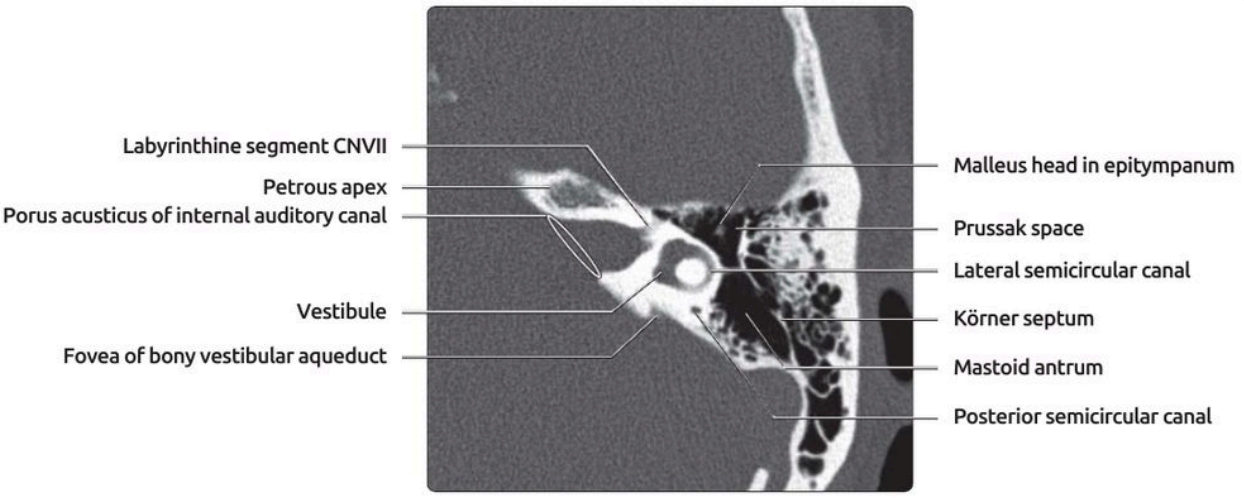


FIGURE 3 At the lateral semicircular canal: the porus of the internal auditory canal, the vestibule and the fovea of the vestibular aqueduct medially; the malleus head in the attic, Prussak's space, the antrum and Körner's septum laterally.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 54

2.2 The attic and the facial nerve

At the level of the **labyrinthine segment** of the facial nerve, just above the cochlea, the attic shows the **ice-cream cone**: the malleus head (the ice cream) sitting on the body and short process of the incus (the cone). Lateral to the ossicles lies **Prussak's space**, the lateral epitympanic recess, which is where the common pars flaccida cholesteatoma begins. A little lower, the **tympanic segment** runs from the geniculate ganglion anteromedially, beneath the lateral canal, to the posterior genu, and the **cog**, a bony ridge, crosses the anterior attic.

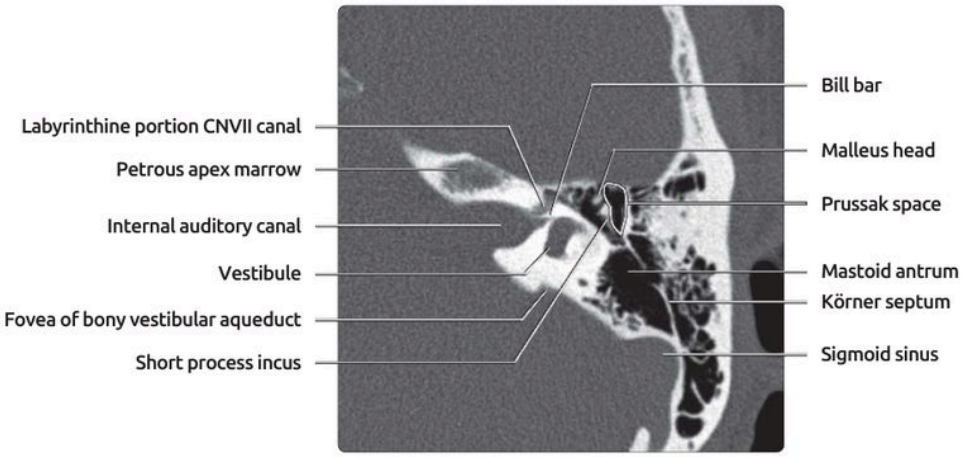


FIGURE 4 Through the labyrinthine facial nerve: malleus head and short process of incus (the ice-cream cone), Prussak's space lateral to them, the mastoid antrum and Körner's septum behind.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 55

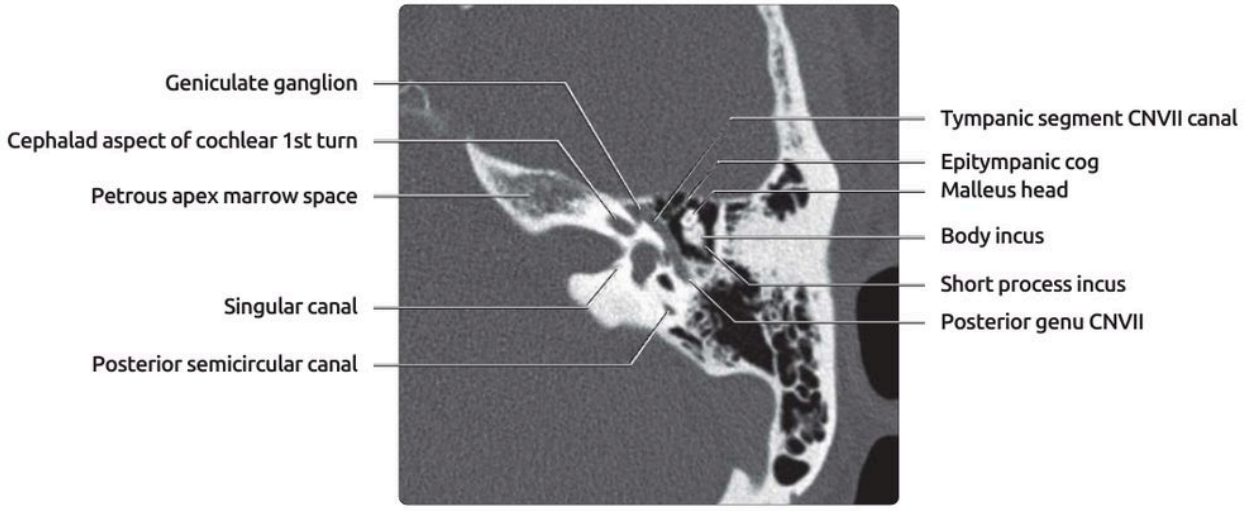


FIGURE 5 The tympanic facial nerve runs from the geniculate ganglion to the posterior genu. The cog crosses the anterior epitympanum.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 55

2.3 The posterior wall: three recesses

At the oval window level, the posterior wall of the middle ear shows three landmarks from medial to lateral: the **sinus tympani**, the **pyramidal eminence** (with the stapedius) and the **facial recess**, with the mastoid facial nerve just behind. The sinus tympani is hidden from the surgeon's view through a standard mastoidectomy, so disease there must be reported.

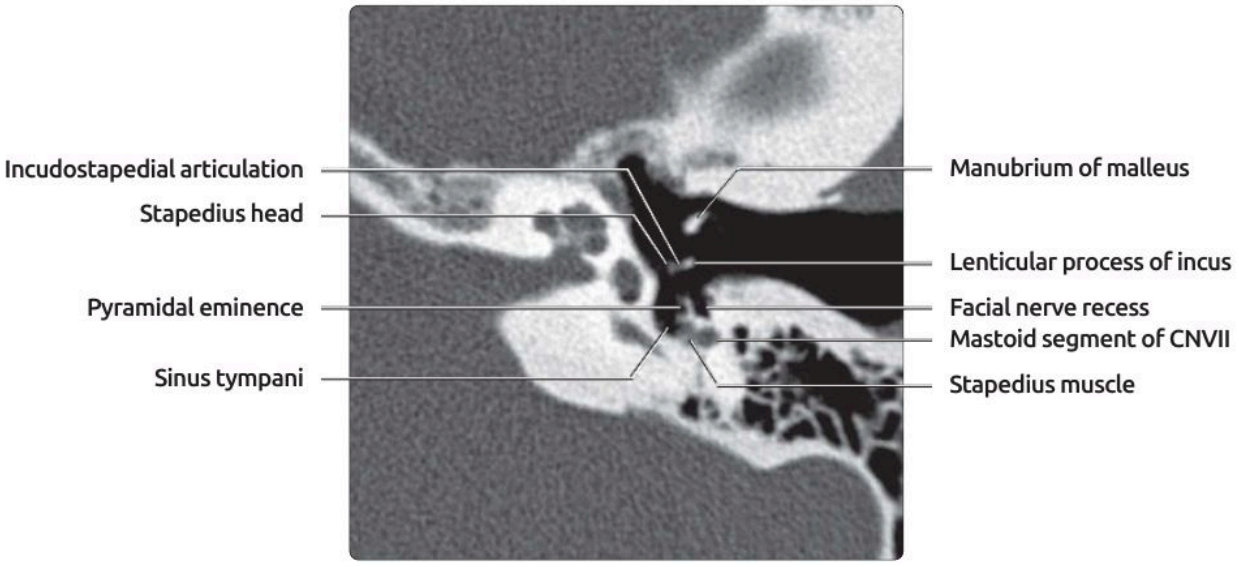


FIGURE 6 Magnified axial view of the posterior mesotympanum: sinus tympani, pyramidal eminence and facial recess, with the mastoid segment of the facial nerve and the stapedius behind; the incudostapedial joint and manubrium in front.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 83

2.4 Lower levels: cochlea, carotid and jugular

At the mid-cochlear level, the **incudostapedial joint** and the tensor tympani tendon from the cochleariform process are seen. Lower still, the **sigmoid plate** separates mastoid air cells from the sigmoid sinus, the **horizontal petrous carotid** runs anteromedially, and the **jugular foramen** sits posteromedially. At the bottom, the **stylomastoid foramen** lies just anteromedial to the mastoid tip, where the facial nerve leaves the skull.

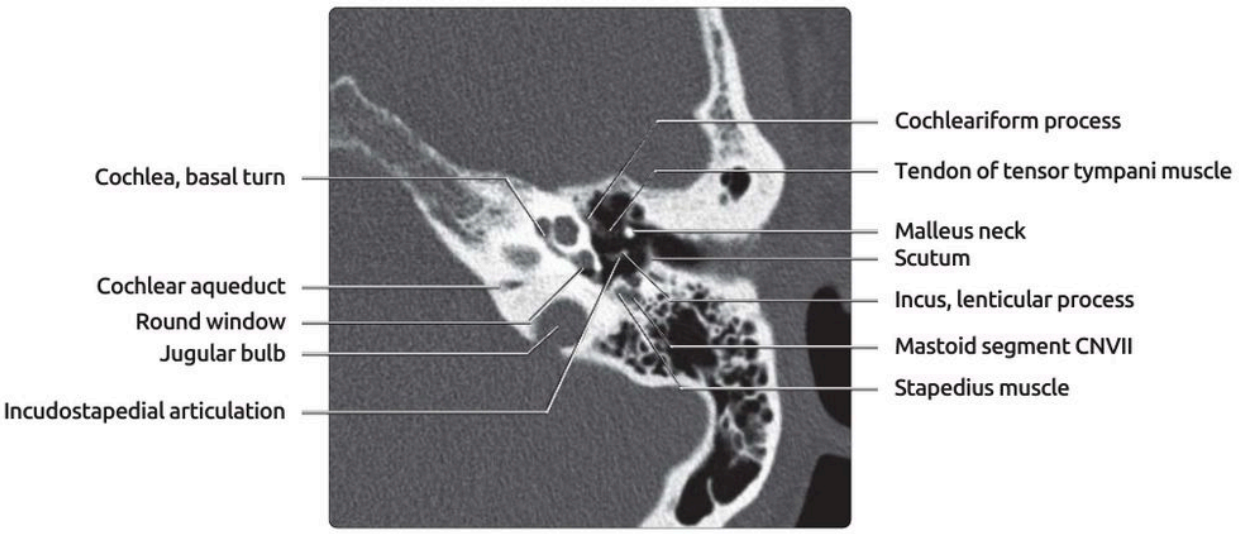


FIGURE 7 Mid-cochlear level: cochleariform process and tensor tympani tendon, and the incudostapedial joint between the lenticular process and stapes head.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 56

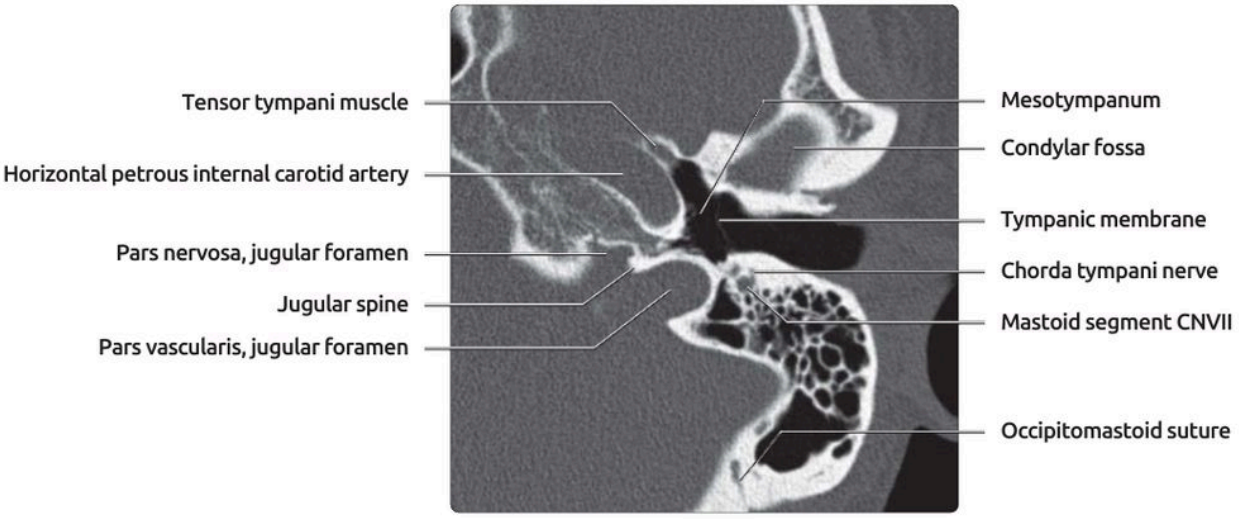


FIGURE 8 Mesotympanum: the faint normal tympanic membrane, the horizontal petrous carotid running anteromedially, and the jugular foramen.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 57

3 CHAPTER THREE
The normal ear in coronal

Posterior to anterior.

3.1 The planes that show the roof and the walls

Coronal images show the **tegmen** (the roof of the middle ear and mastoid), the **scutum** (the sharp upper attachment of the tympanic membrane), the **tympanic annulus** below, the **oval window** with the stapes, the **lateral canal** bulging into the attic, and the tympanic facial nerve under it. These are the structures cholesteatoma erodes, so the coronal plane is where the cholesteatoma questions are answered.

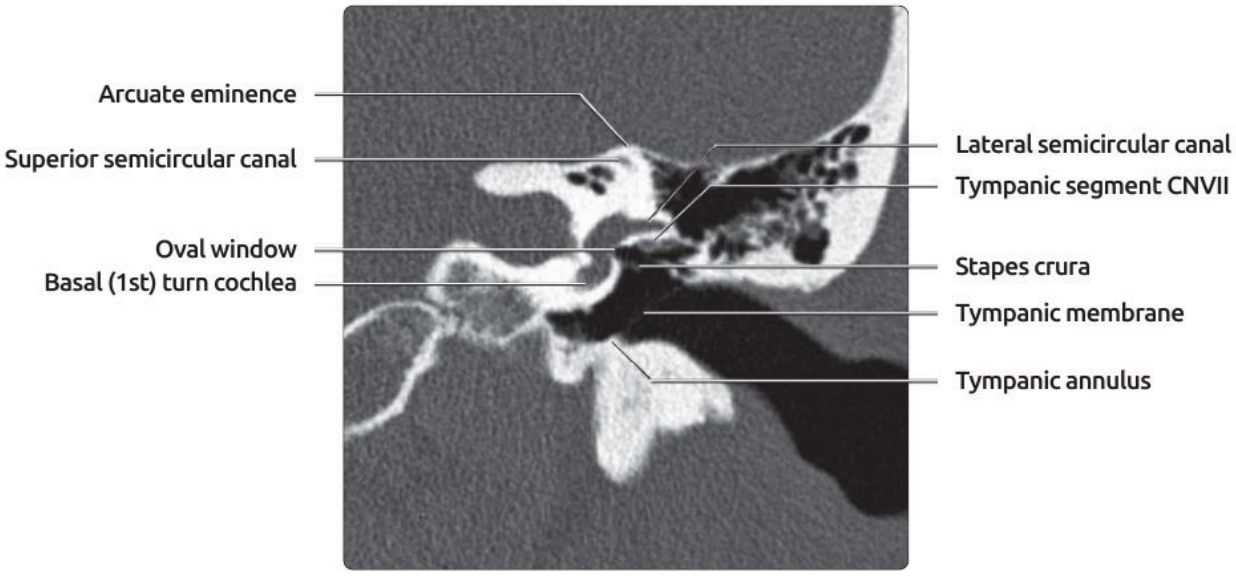


FIGURE 9 At the oval window: the basal turn of the cochlea, the barely visible tympanic membrane, and the tympanic annulus as the lower landmark between middle ear and canal.
Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 59

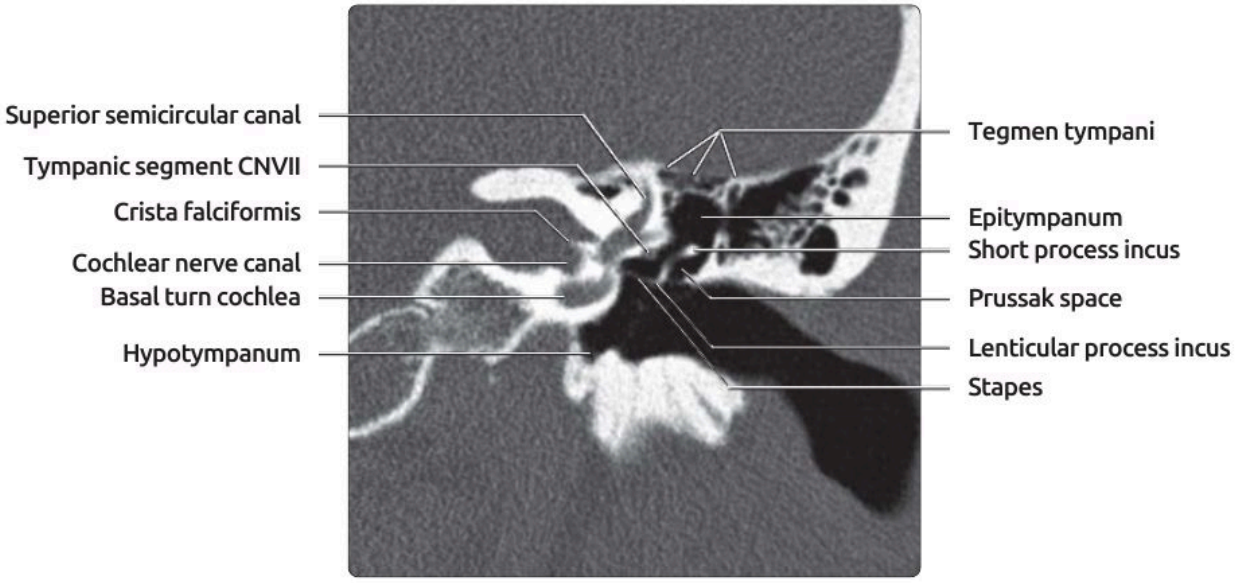


FIGURE 10 Mid mesotympanum: Prussak's space bounded by the lateral attic wall and scutum laterally and the ossicles medially. A sharp scutum is normal; a blunted one is the earliest bony sign of attic cholesteatoma.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 60

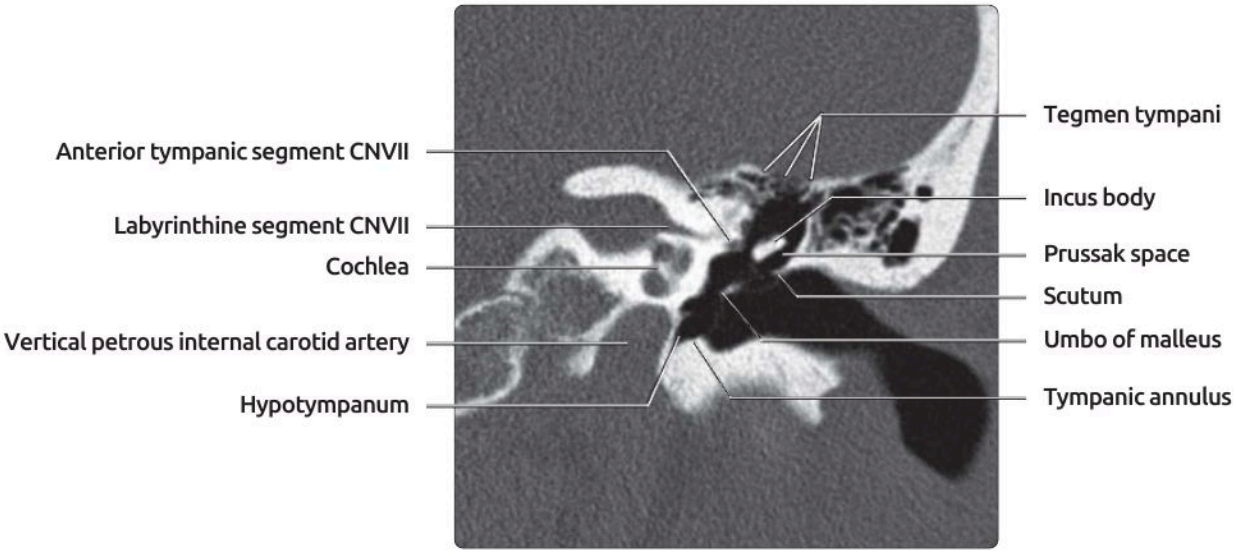


FIGURE 11 The tegmen tympani roofing the attic, with the scutum, Prussak's space and the incus body beneath it; the tympanic annulus below, and the labyrinthine and anterior tympanic facial nerve above the cochlea.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 60

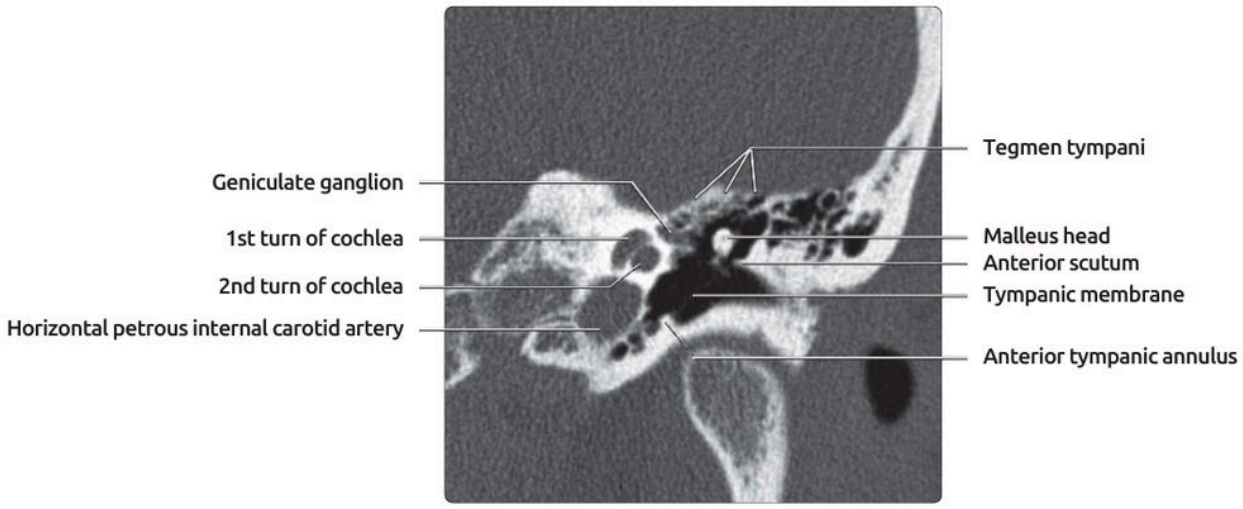


FIGURE 12 Anteriorly: a thick, well-defined tegmen, the geniculate ganglion on the cochlea and the horizontal carotid beneath it.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 61

CHAPTER FOUR

4 A checklist for every temporal bone

Read the same way every time.

4.1 Lateral to medial

READING A TEMPORAL BONE CT

1. External canal
   - → patency and width
   - → soft tissue
   - → bone erosion or new bone (exostoses, osteoma)
   - → atresia

2 Tympanic membrane and scutum

- → thickening, retraction
- → scutum sharp or blunted

3 Middle ear spaces

- → aeration of attic, mesotympanum, hypotympanum
- → Prussak's space
- → sinus tympani and facial recess

4 Ossicles

- → malleus head and incus body (ice-cream cone)
- → long process and lenticular process of incus
- → stapes
- → erosion, dislocation, fixation

5 Walls and covers

- → tegmen tympani and tegmen mastoideum
- → lateral canal (fistula)
- → facial canal (dehiscence)
- → sigmoid plate

6 Mastoid

- → pneumatisation
- → opacification
- → septal erosion (coalescence)
- → outer cortex

7 Inner ear

- → cochlea (two and a half turns)
- → vestibule
- → three canals and their bony covers
- → vestibular aqueduct width
- → lucency of the otic capsule

8 Nerves and canals

- → IAC width and symmetry
- → facial nerve course from IAC to stylomastoid foramen

9 Vessels

- → carotid canal wall
- → jugular bulb (high, dehiscent, eroded)
- → sigmoid sinus

10 Petrous apex

- → air, marrow or fluid
- → expansion or erosion
- → compare the two sides

ALWAYS COMPARE SIDES

Asymmetry is often the first clue. Pneumatisation of the petrous apex, height of the jugular bulb and width of the vestibular aqueduct vary between people, so a finding is judged against the other ear before it is called abnormal.

5 CHAPTER FIVE Common patterns

What disease looks like.

5.1 Opacification: fluid, granulation or cholesteatoma?

CT shows soft tissue in the middle ear well but cannot tell its nature: fluid, inflamed mucosa, granulation, cholesterol granuloma and cholesteatoma all look the same density. What separates them on CT is **what the tissue does to bone**. Opacification with intact ossicles, scutum and septa favours effusion or granulation; a soft-tissue mass with **blunting of the scutum, ossicular erosion and medial displacement of the ossicles** favours cholesteatoma. Where doubt remains, non-EPI diffusion MRI shows keratin as bright.

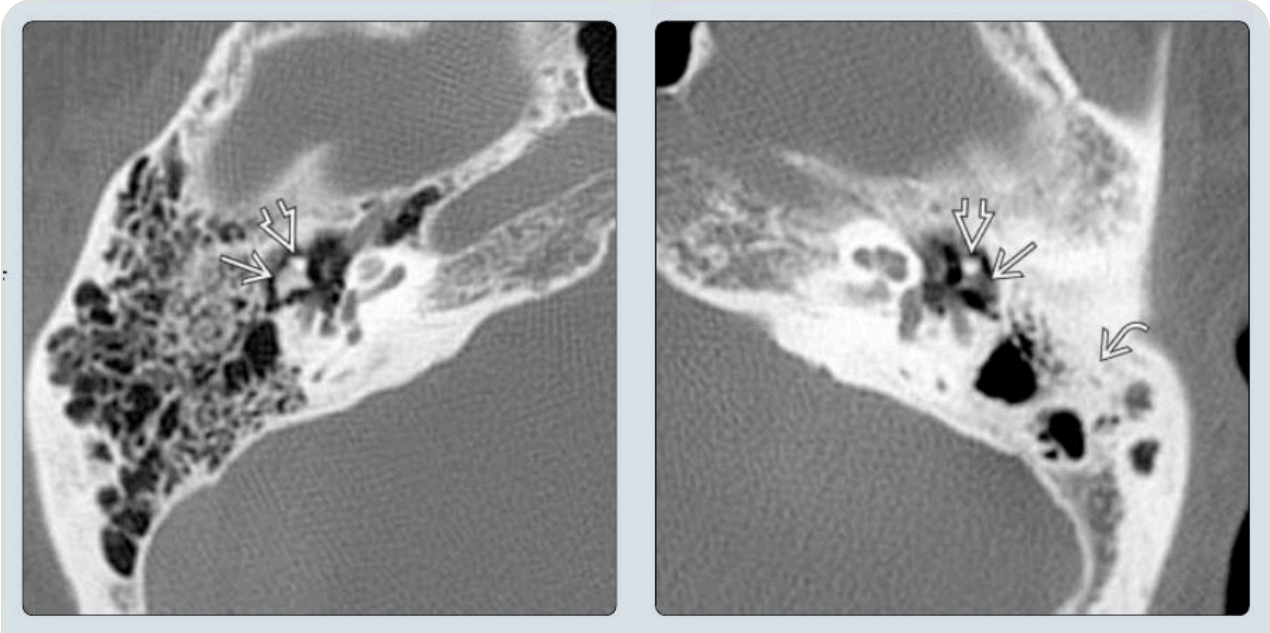


FIGURE 13 Chronic otitis media. Left: a normal short process of the incus and malleus head. Right: the other ear, with deossification of the short process of the incus and an underpneumatised mastoid.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1046

5.2 Cholesteatoma

The **pars flaccida** (attic) cholesteatoma, about 80% of acquired cases, fills Prussak's space, blunts the scutum and pushes the malleus head and incus

medially, then spreads back through the aditus into the antrum. The **pars tensa** cholesteatoma begins in the posterosuperior mesotympanum, erodes the long process of the incus and the stapes, and fills the sinus tympani and facial recess. The report should answer five surgical questions: is the **tegmen** intact, is there a **labyrinthine fistula** (usually the lateral canal), is the **facial canal** dehiscent, is the **sinus tympani** involved, and what has happened to the **ossicles**.

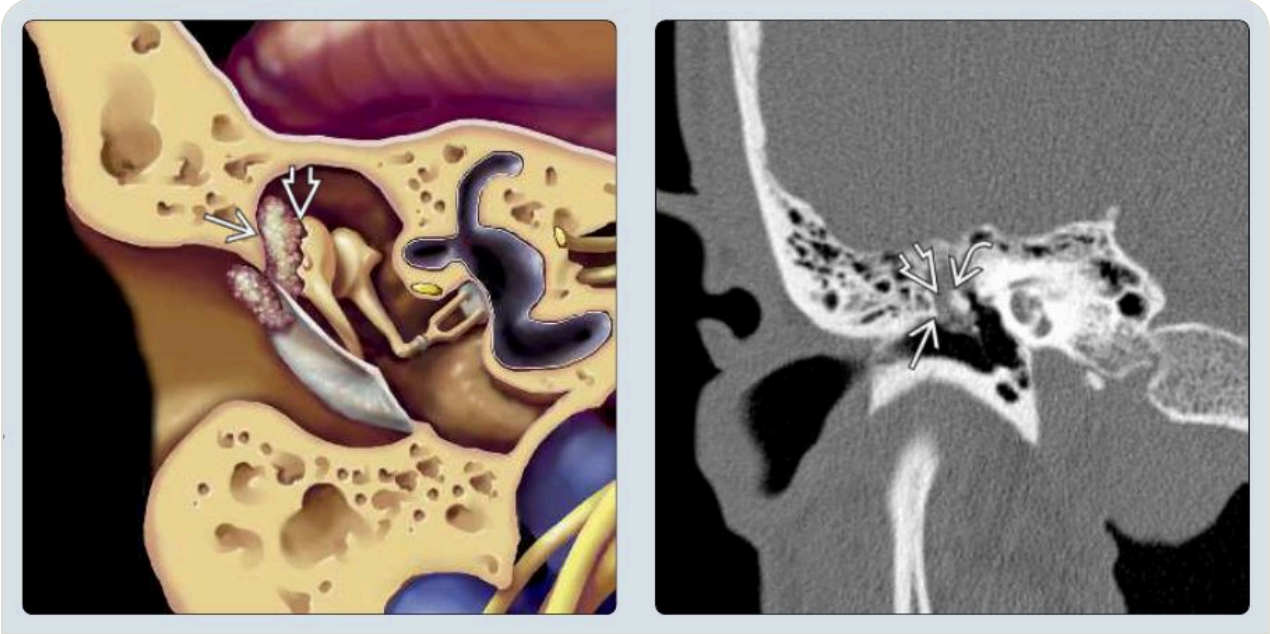


FIGURE 14 A small pars flaccida cholesteatoma. Left: drawing of the attic sac in Prussak's space. Right: coronal CT, soft tissue in Prussak's space, a blunted scutum and a mildly eroded, medially displaced malleus head.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1050

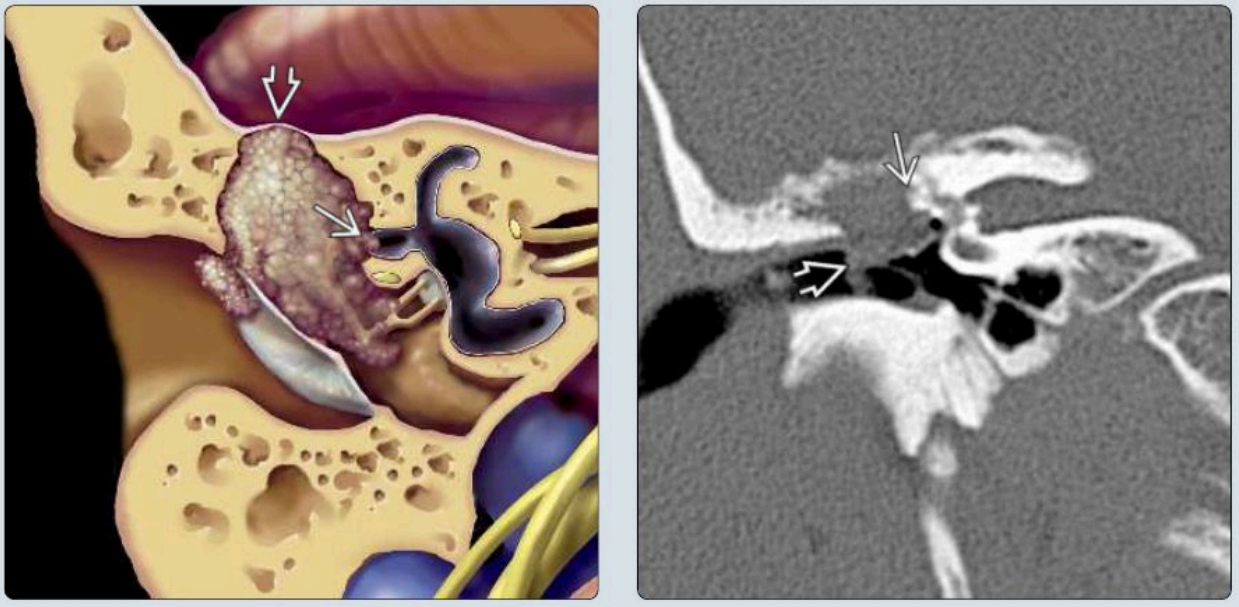


FIGURE 15 A large pars flaccida cholesteatoma. Left: drawing of ossicular erosion, lateral canal fistula and tegmen scalloping. Right: coronal CT, a mass filling the middle ear and mastoid with a fistula into the lateral semicircular canal.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1050

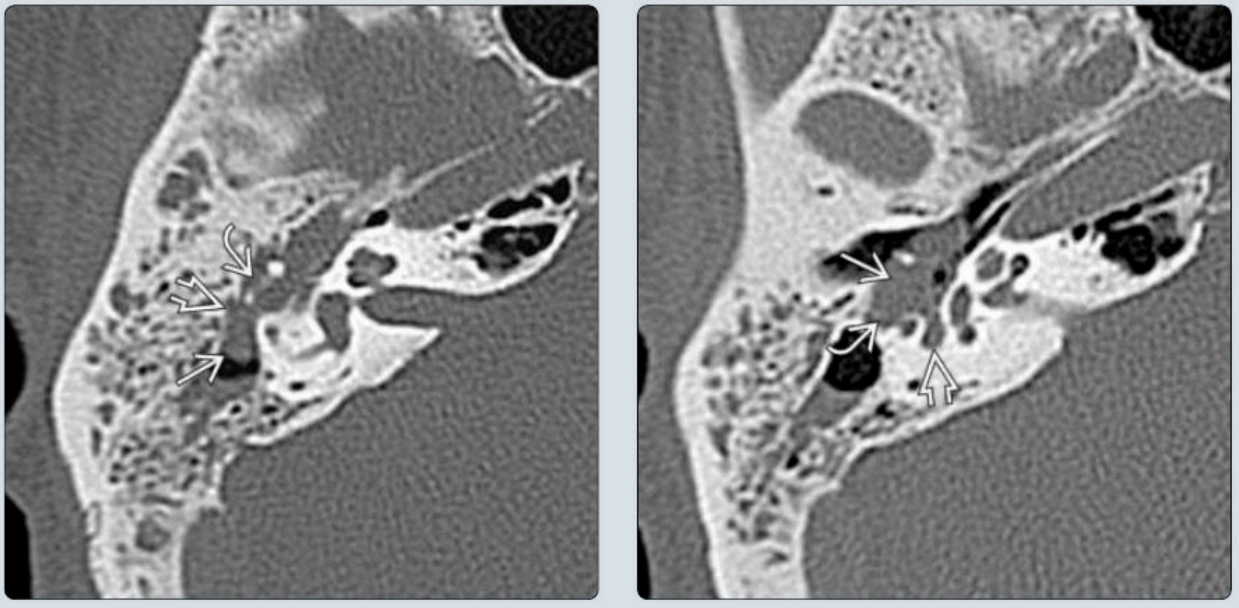


FIGURE 16 Pars tensa cholesteatoma. Left: erosion of the short process of the incus and spread through the aditus into the antrum. Right: erosion of the incus and stapes, with disease in the facial recess and the sinus tympani, the blind spot for the surgeon.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1054

5.3 Otosclerosis

The otic capsule is normally uniformly dense, among the densest bone in the body. **Fenestral otosclerosis** appears as a lucent (otospongiotic) focus at the **fissula ante fenestram**, just anterior to the oval window, sometimes thickening the footplate. **Cochlear otosclerosis** adds lucent foci around the cochlea, the "double ring" or halo. The changes are subtle: look anterior to the oval window on every scan of a patient with conductive loss and an intact drum.

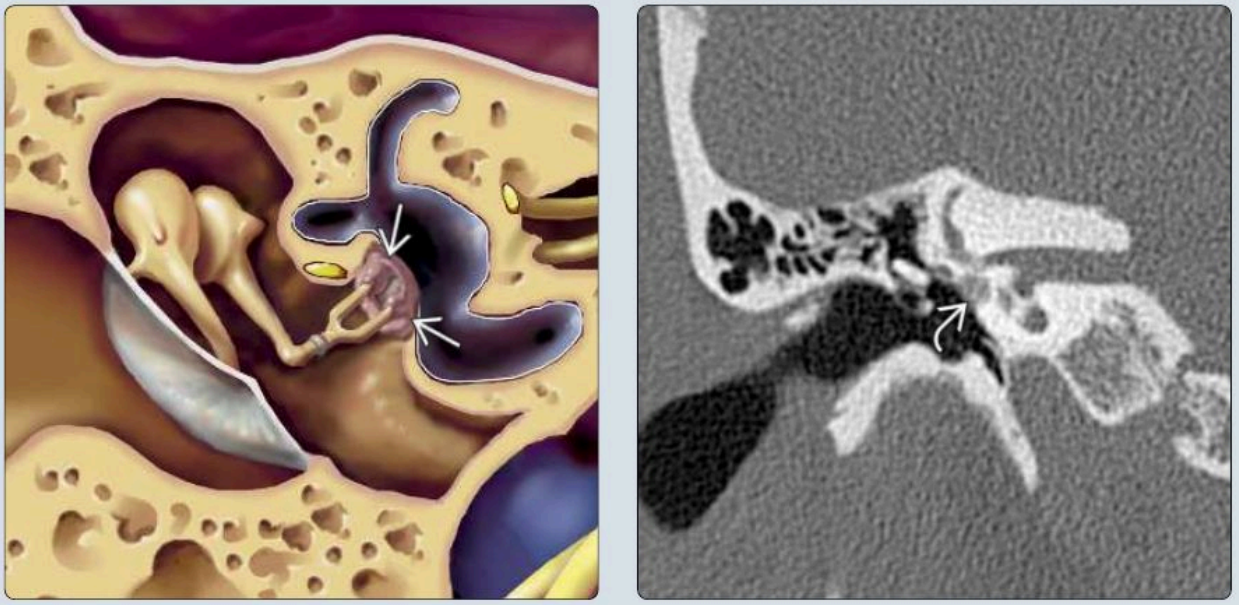


FIGURE 17 Fenestral otosclerosis. Left: drawing of the plaque around the footplate. Right: coronal CT, a lucent focus just anterior to the oval window.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1120

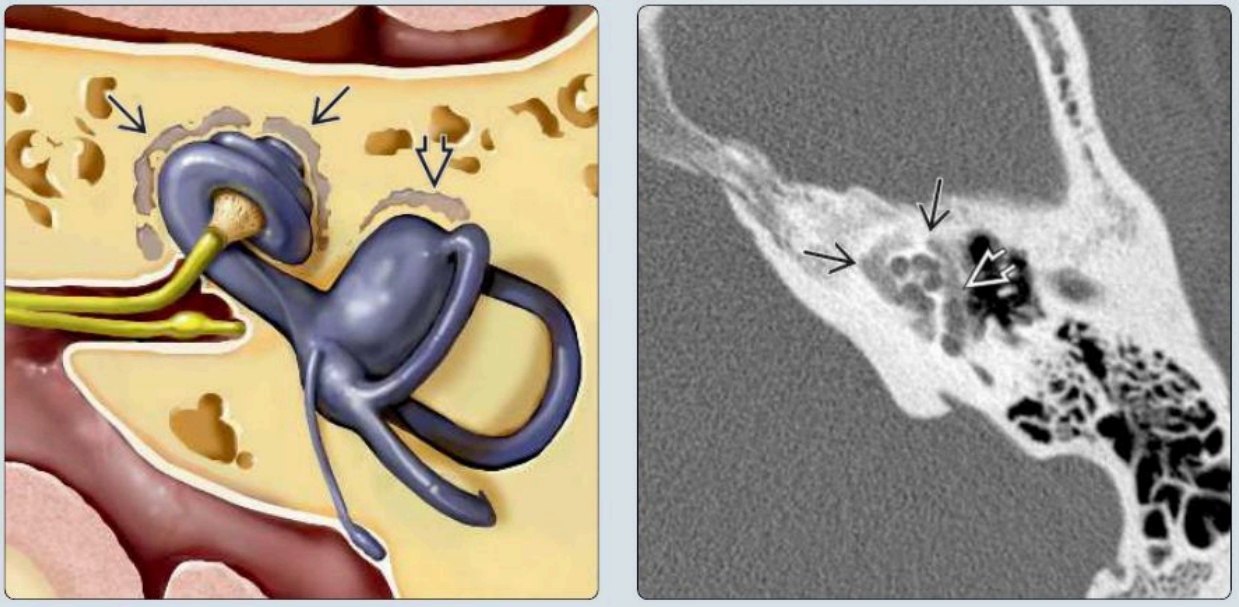


FIGURE 18 Cochlear otosclerosis. Left: drawing of the pericochlear halo. Right: axial CT, lucent foci around the cochlea with the fenestral focus at the fissula.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1120

5.4 Superior canal dehiscence

Absence of bone over the superior canal produces a "third window", with autophony, sound- or pressure-induced vertigo and an air–bone gap with normal stapedial reflexes. It must be confirmed on thin slices in the **Pöschl** plane, because partial volume can make thin normal bone disappear.

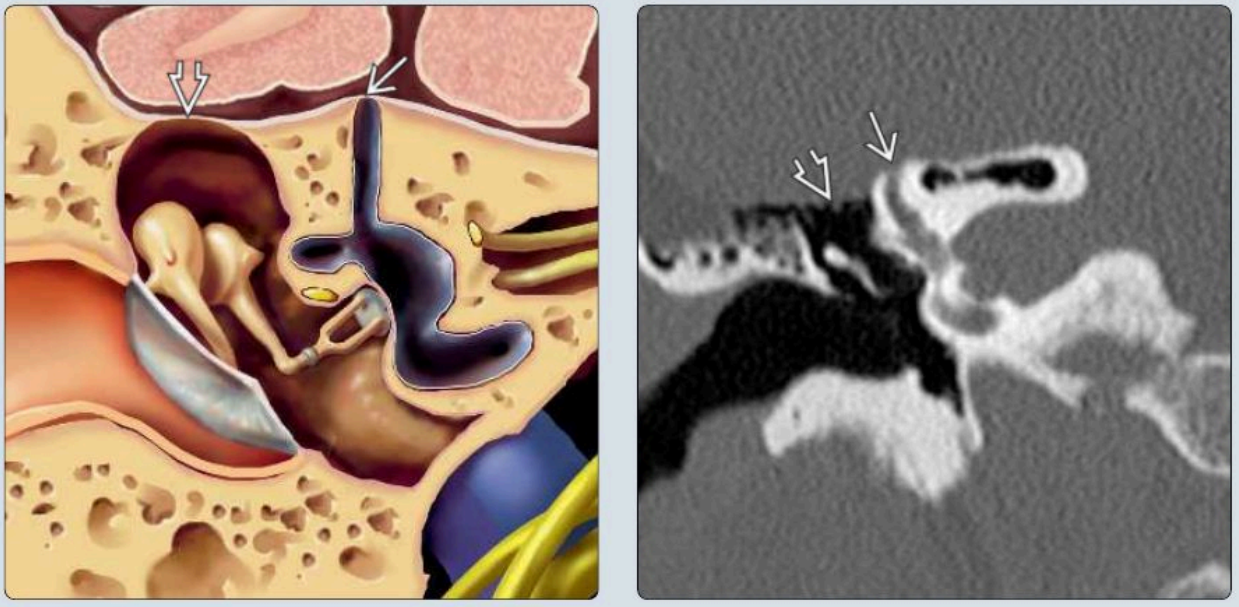


FIGURE 19 Superior semicircular canal dehiscence. Left: drawing of the uncovered canal and a thin tegmen. Right: coronal CT, no bone over the roof of the canal.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1134

5.5 Fractures

Temporal bone fractures are described by their relation to the petrous ridge and, more usefully, by whether they **spare or violate the otic capsule**. **Longitudinal** fractures run along the petrous ridge, through the canal and middle ear, and disrupt the ossicles (incus dislocation most often), giving conductive loss. **Transverse** fractures cross the ridge and the otic capsule, and cause sensorineural loss, vertigo and facial palsy. In every fracture look at the ossicles, the facial canal (the geniculate region is the commonest site of injury), the tegmen (CSF leak), the carotid canal and the jugular foramen. Sutures, fissures and small canals can mimic a fracture.

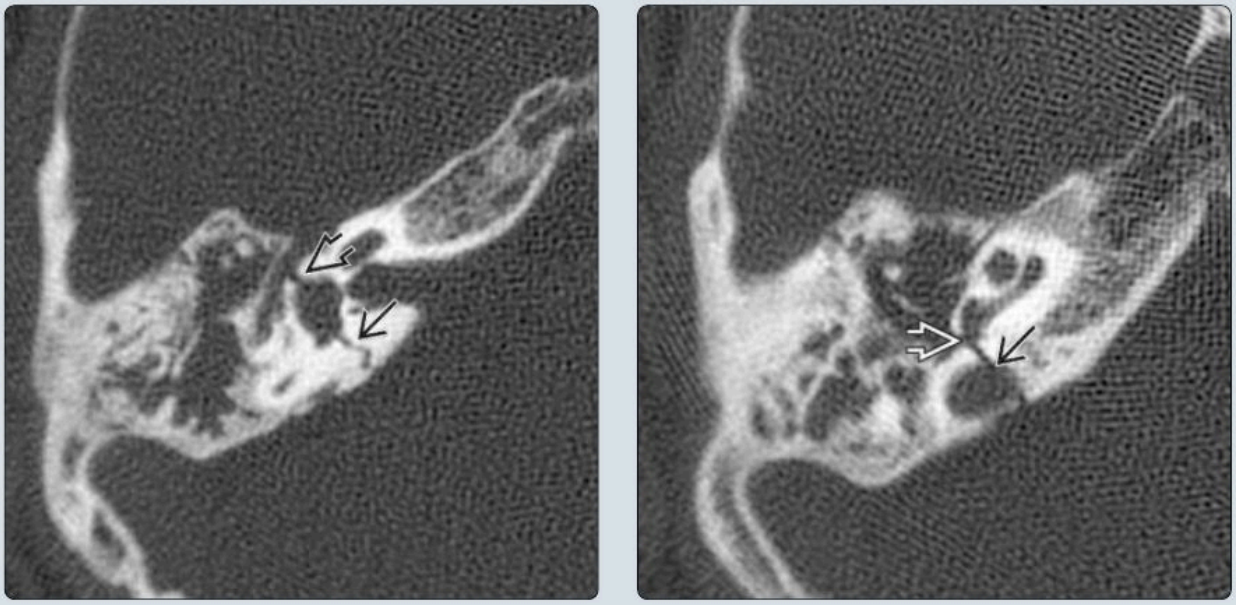


FIGURE 20 A transverse, otic capsule-violating fracture. Left: the line crosses the vestibule and reaches the tympanic facial nerve canal. Right: lower down, it extends across the jugular foramen into the round window niche.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 968

6 CHAPTER SIX The congenital ear

Questions for atresia.

6.1 Before atresia surgery

In congenital aural atresia, CT answers the questions that decide whether reconstruction is feasible: the **thickness and type** of the atresia plate (bony or membranous), the **size of the middle ear**, the **ossicles** (usually fused malleus and incus), the **course of the facial nerve** (often displaced anteriorly and laterally over the atresia plate), the **oval and round windows**, any **congenital cholesteatoma** behind the plate, and the **inner ear**, which is usually normal because it develops separately. Scoring systems such as the Jahrsdoerfer grade combine these findings.

6.2 The vestibular aqueduct

A **large vestibular aqueduct** is the commonest imaging finding in children with sensorineural loss. The aqueduct is measured at its midpoint on axial images (the Cincinnati criteria: more than 1 mm at the midpoint or more than 2 mm at the operculum) and is often associated with an incomplete partition of the cochlea.

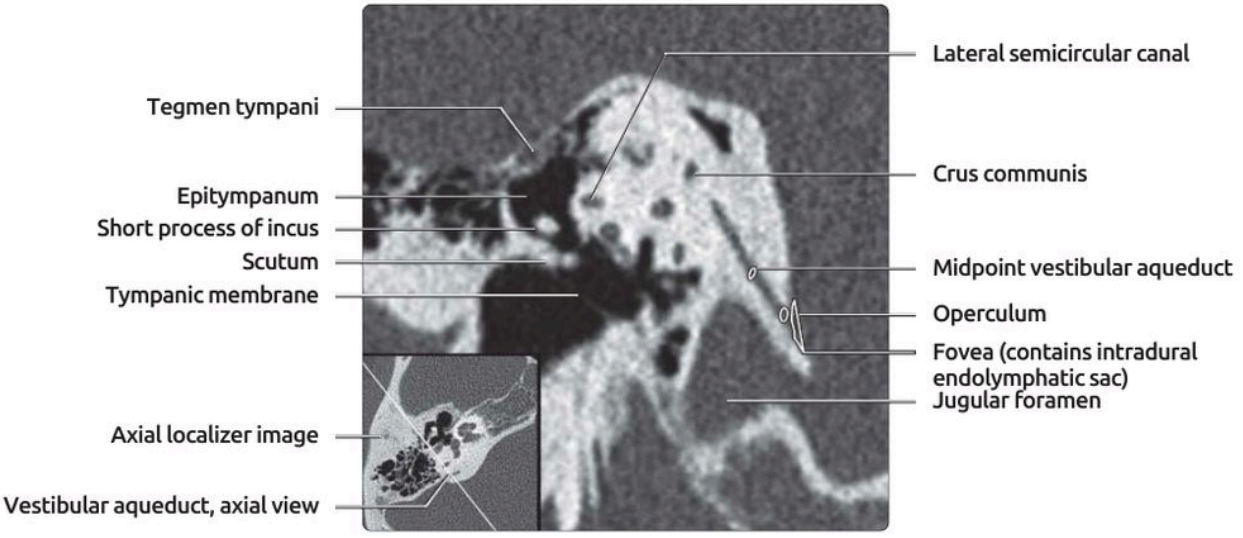


FIGURE 21 Transverse oblique reformat profiling the vestibular aqueduct along its length, the plane in which its width is best judged.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 90

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Protocol:** thin bone-algorithm CT, no contrast, axial and coronal, oblique planes when needed.
- **Name the structures** on an axial slice: ice-cream cone, Prussak's space, sinus tympani, facial recess, pyramidal eminence.
- **The five cholesteatoma questions:** tegmen, fistula, facial canal, sinus tympani, ossicles.

- **Scutum blunting** as the early sign of attic cholesteatoma.
- **Fenestral otosclerosis** at the fissula ante fenestram.
- **Fractures:** otic capsule-sparing versus violating, and the structures to check.
- **Atresia** checklist and the large vestibular aqueduct.

7.2 Common pitfalls

COMMON PITFALLS

- Calling middle ear opacification "cholesteatoma" without bony evidence; CT cannot tell its nature.
- Missing disease in the sinus tympani.
- Diagnosing a dehiscence of the tegmen, facial canal or superior canal on a single slice.
- Overlooking subtle fenestral otosclerosis in a patient with conductive loss and a normal drum.
- Calling a suture or a small canal a fracture.

Written for this app from Imaging Anatomy: Head and Neck (Chapman and Koontz), temporal bone chapters, Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), temporal bone overview and diagnoses, Dunnebier, Imaging for Otolaryngologists. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 6

The inner ear, IAC and CPA on MRI

The commonest MRI an ENT surgeon requests is the scan for asymmetric sensorineural loss or unilateral tinnitus. Most are normal, and the radiologist's report says so, but you will be asked to look at the images yourself in clinic and in the multidisciplinary meeting, and you should be able to find the four nerves in the internal auditory canal, the fluid of the labyrinth and the cerebellopontine angle cistern, and to recognise the handful of lesions that account for nearly every positive scan.

AT A GLANCE

- **High-resolution T2** (CISS or FIESTA) screens; **post-gadolinium fat-suppressed T1** characterises.
- "**Seven up, Coke down**": facial anterosuperior, cochlear anteroinferior, vestibular nerves posterior.
- About 90% of positive scans for unilateral SNHL are **vestibular schwannomas**: centred on the porus, acute angle, ice cream on a cone.
- **Meningioma**: broad dural base, obtuse angle, dural tail. **Epidermoid**: bright on DWI; arachnoid cyst is not.
- Labyrinthitis **enhances**; ossification **removes the T2 fluid signal**. Enhancement of the IAC or labyrinthine facial nerve is always abnormal.

1 CHAPTER ONE The study

Two sequences do most of the work.

1.1 Heavily T2-weighted thin sections

A **high-resolution heavily T2-weighted** sequence (CISS, FIESTA, SPACE or DRIVE, depending on the manufacturer) in submillimetre slices makes CSF and labyrinthine fluid white and the nerves, vessels and tumours dark. It is the screening sequence for asymmetric hearing loss in many centres, and it shows the inner ear's anatomy in children before cochlear implantation.

1.2 Post-gadolinium T1 with fat suppression

A thin **post-gadolinium T1** shows enhancing lesions: schwannomas, meningiomas, labyrinthitis, facial neuritis, perineural tumour and metastases. It is added when the T2 screen is abnormal or equivocal, and used from the start when the question is inflammatory (facial palsy, sudden loss with vertigo) or when the T2 sequence would miss an intralabyrinthine lesion. Always look at the pre-contrast T1: blood and fat are bright before contrast and can mimic enhancement.

WHO NEEDS WHAT

For unilateral sensorineural loss in an otherwise healthy adult, a lesion is found in fewer than 5% of scans, and about 90% of those are vestibular schwannomas. Meningioma, epidermoid and aneurysm make up most of the rest. In a child the emphasis changes: the scan looks for malformations, an absent cochlear nerve and post-meningitic ossification.

2 CHAPTER TWO The normal anatomy

Nerves, fluid and the cistern.

2.1 The four nerves: "seven up, Coke down"

On an oblique sagittal image across the mid internal auditory canal, four round dark nerves sit in white CSF. The **facial nerve** is anterosuperior, the **cochlear nerve** anteroinferior (and normally the largest of the four), and the **superior and inferior vestibular nerves** are posterior. At the fundus, the horizontal **crista falciformis** separates the facial and superior vestibular nerves above from the cochlear and inferior vestibular nerves below, and the vertical crest (**Bill's bar**) separates the facial nerve from the superior vestibular nerve. Closer to the porus, the cochlear and vestibular nerves merge into the vestibulocochlear nerve, which wraps around the facial nerve like a **catcher's mitt around a ball**.

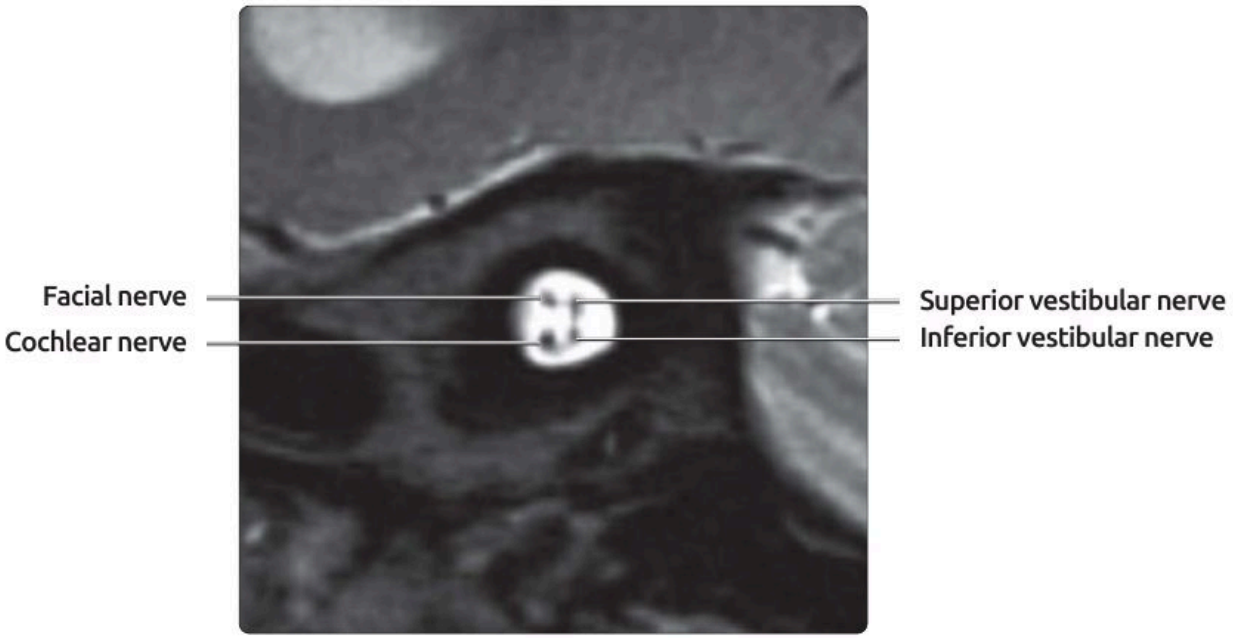


FIGURE 1 Oblique sagittal high-resolution T2 in the mid internal auditory canal: four nerves in bright CSF. The cochlear nerve, anteroinferior, is normally slightly larger than the facial nerve above it.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 62

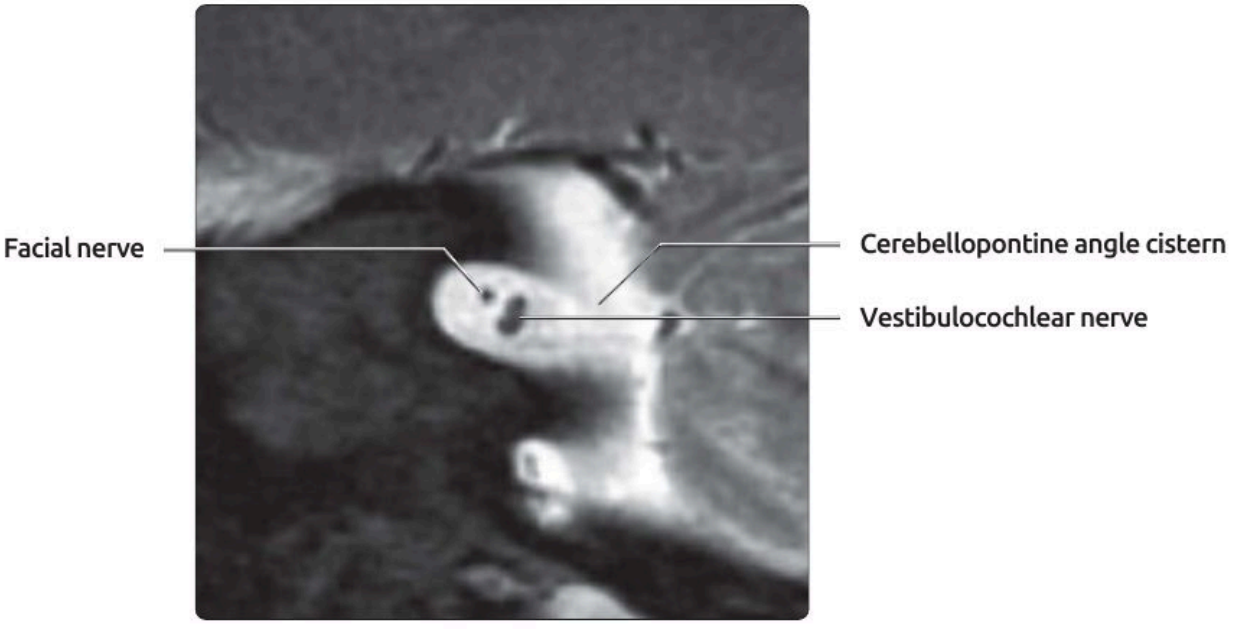


FIGURE 2 At the porus: the vestibulocochlear nerve curves around the facial nerve, the "catcher's mitt and ball".

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 62

2.2 The labyrinth

Axial T2 shows the inner ear as white fluid within dark bone: the **cochlea** with its basal, middle and apical turns, divided by the osseous spiral lamina into scala tympani and scala vestibuli, the **modiolus** as a grey hub, the **vestibule**, and the **semicircular canals**. The **cochlear nerve** can be followed from the anteroinferior IAC through the cochlear nerve canal into the modiolus. Normal fluid signal in every part of the labyrinth is what you check; a dark filling defect means tumour, fibrosis or bone.

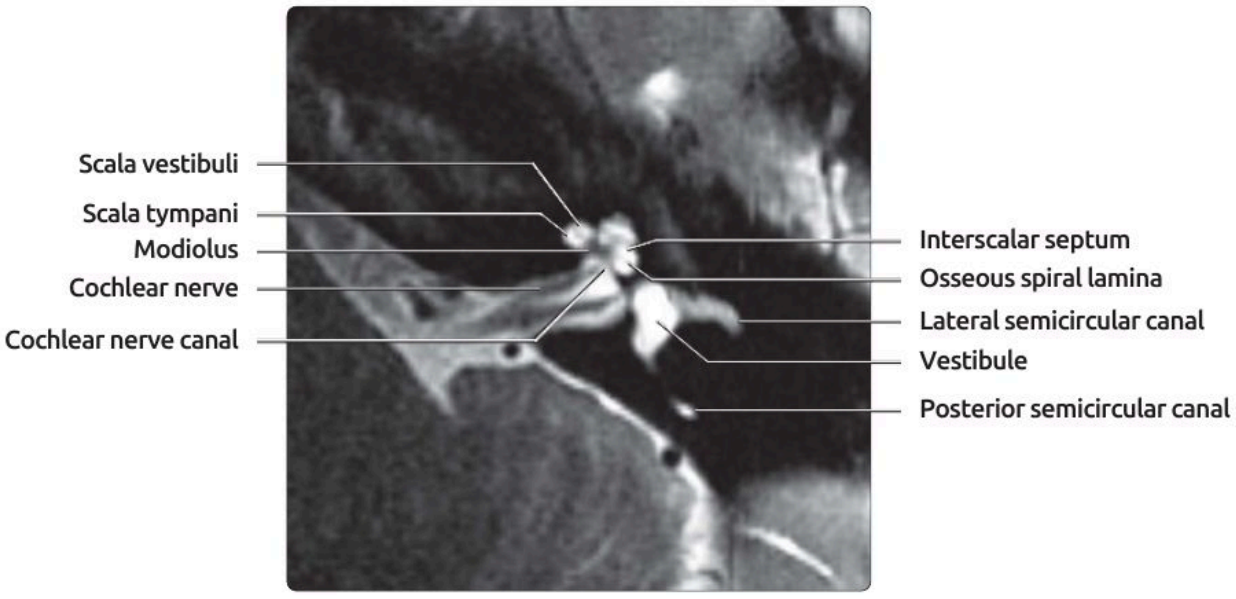


FIGURE 3 Axial high-resolution T2 through the mid cochlea: scala tympani and scala vestibuli, the modiolus, the cochlear nerve entering through the cochlear nerve canal, the vestibule and the canals.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 89

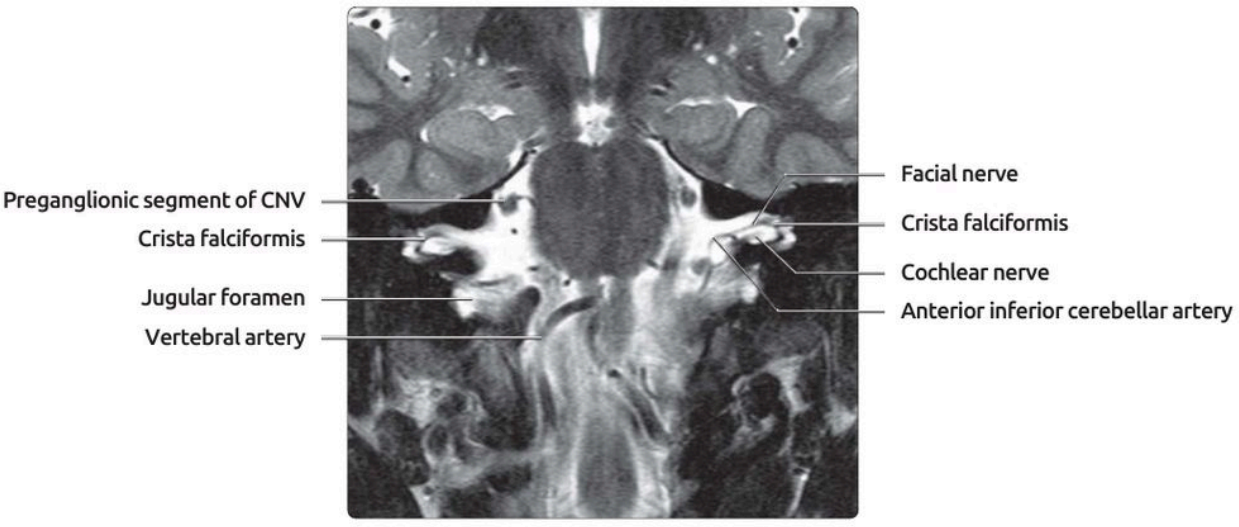


FIGURE 4 Coronal T2 through the IAC fundus: the crista falciformis divides the canal into an upper compartment (facial and superior vestibular nerves) and a lower one (cochlear and inferior vestibular nerves).

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 107

2.3 The cerebellopontine angle cistern

The CPA cistern contains the facial and vestibulocochlear nerves running from the brainstem to the porus, the **anterior inferior cerebellar artery** looping near or into the canal, and, further up, the trigeminal nerve. Vascular loops touching the eighth nerve are very common in people without symptoms and should not be over-interpreted. Normal **flocculus** and **choroid plexus** can look like a small mass; compare the two sides.

3 CHAPTER THREE A checklist

Read every IAC MRI the same way.

READING AN IAC AND CPA MRI

1. **Identify sequences**
   - → thin T2
   - → T1 before and after gadolinium

- → DWI
- → FLAIR if the brain is included

2 **Each IAC on T2**

- → four nerves visible
- → any filling defect in CSF
- → fundal cap of fluid
- → porus width and symmetry

3 **Each labyrinth on T2**

- → bright fluid in cochlea (all turns), vestibule and three canals
- → modiolus
- → vestibular aqueduct and sac

4 **After gadolinium**

- → enhancement in IAC, fundus, labyrinth, facial nerve segments
- → compare with pre-contrast T1

5 **CPA cistern**

- → mass
- → its shape and angle with the petrous bone
- → dural tail
- → brainstem and fourth ventricle

6 **The rest**

- → brainstem and cerebellum
- → venous sinuses
- → superficial siderosis

→ mastoid and petrous apex

4 CHAPTER FOUR The common lesions

What a positive scan usually shows.

4.1 Vestibular schwannoma

The vestibular schwannoma arises in the IAC (usually from a vestibular nerve) and grows medially into the CPA, taking the shape of an **ice cream on a cone**: the cone in the canal, the ball in the cistern. It is centred on the porus, widens it, and meets the petrous bone at an acute angle. On T2 it is a dark filling defect in white CSF, often with a bright **fundal cap** of fluid lateral to it when it does not reach the fundus; it enhances avidly, often heterogeneously, and larger tumours become cystic or bleed. Report the size (intracanalicular and extracanalicular), the fundal cap, brainstem compression and the fourth ventricle (the basis of the Koos grade), and whether the cochlear nerve canal is involved, which matters for hearing preservation.

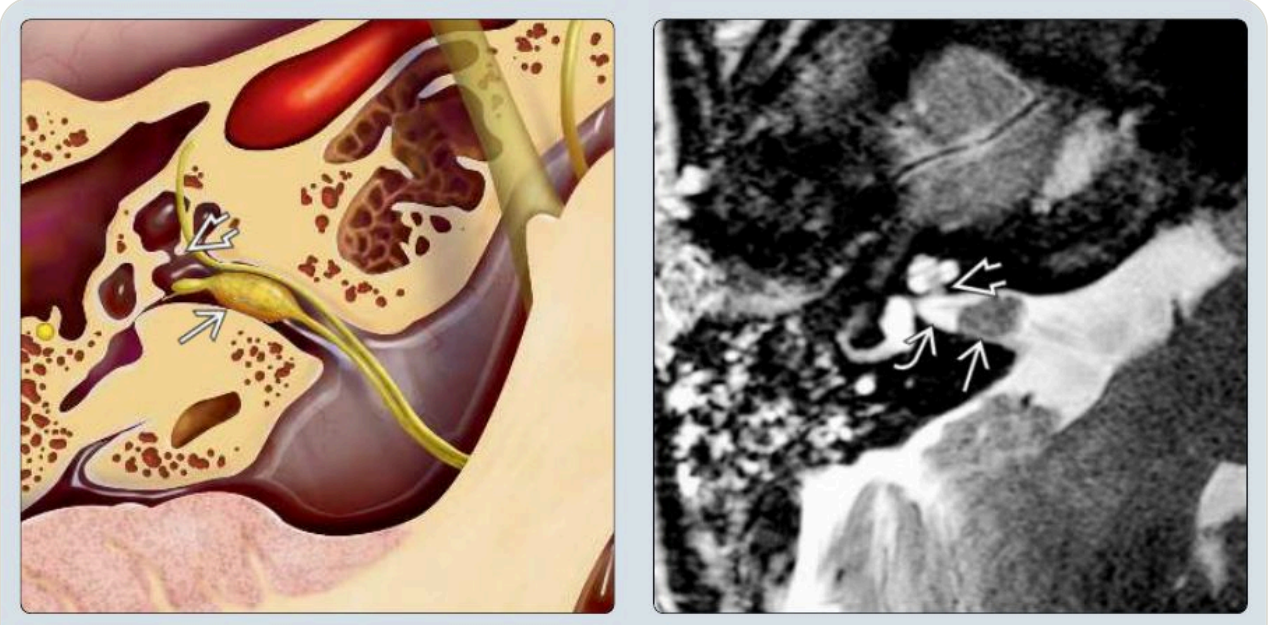


FIGURE 5 A small intracanalicular vestibular schwannoma. Left: drawing of the tumour on the superior vestibular nerve. Right: axial T2, a dark mass in bright CSF, with a fluid cap at the fundus and a free cochlear nerve canal.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1222

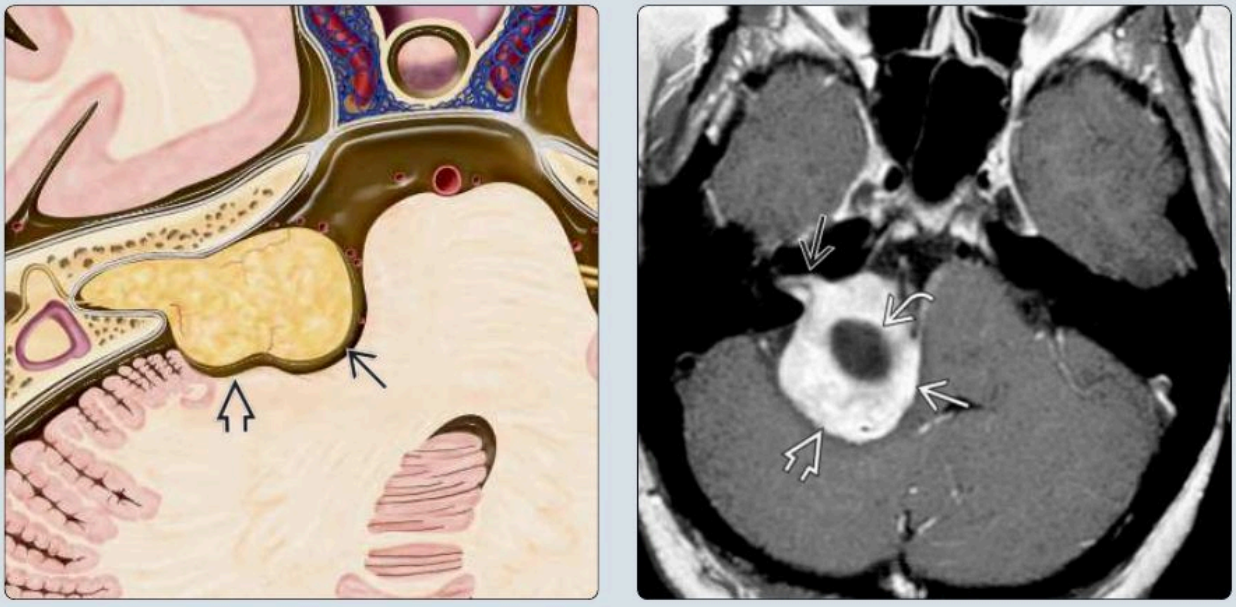


FIGURE 6 A large vestibular schwannoma. Left: drawing of the ice-cream-on-cone shape compressing the middle cerebellar peduncle. Right: post-gadolinium T1, enhancing tumour in the canal and cistern, with a large intramural cyst.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1222

4.2 Meningioma

The CPA meningioma is the second commonest CPA mass. It has a **broad dural base** on the petrous bone, meets it at an **obtuse angle**, is usually **eccentric to the porus**, and seldom enters the canal far. A **dural tail**, calcification, and hyperostosis of the adjacent bone favour it. It enhances avidly and uniformly.

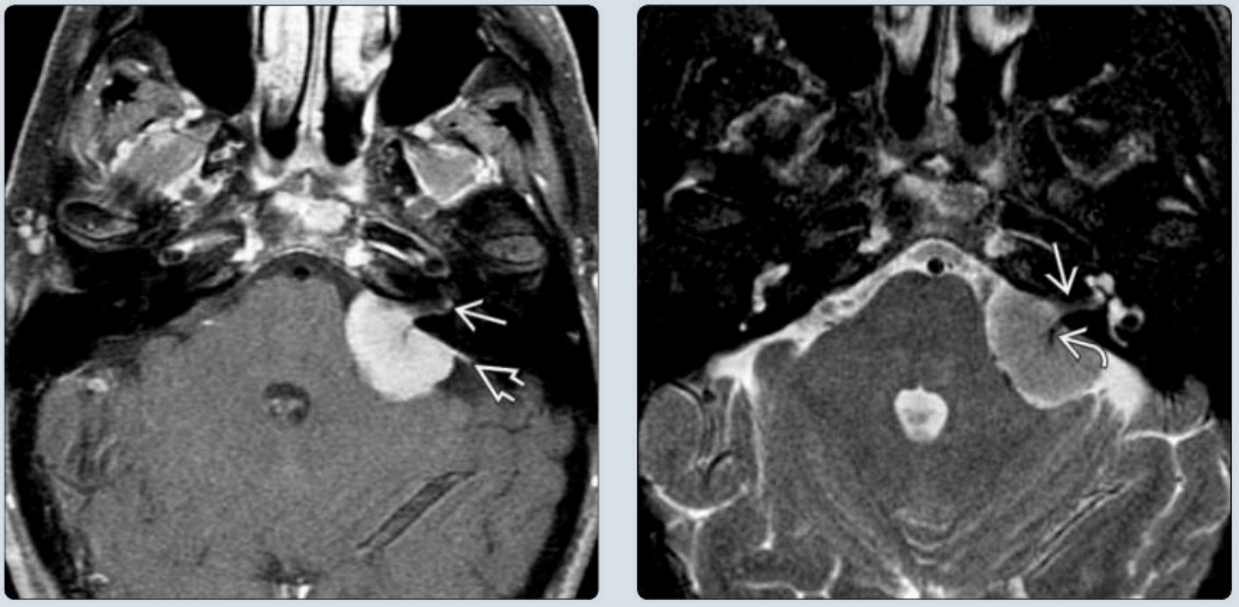


FIGURE 7 A CPA meningioma. Left: post-gadolinium T1, a mass over the porus with a dural tail along the posterior temporal bone and a dot of enhancement in the fundus. Right: T2, a dural artery feeder as a flow void, and dark tumour inside the canal.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1228

4.3 Epidermoid and arachnoid cysts

An **epidermoid cyst** follows CSF on T1 and T2, does not enhance, and insinuates around nerves and vessels rather than displacing them. It does not suppress fully on FLAIR and is **bright on diffusion**. An **arachnoid cyst** follows CSF on every sequence including FLAIR and diffusion, and displaces structures. Diffusion separates the two at a glance.

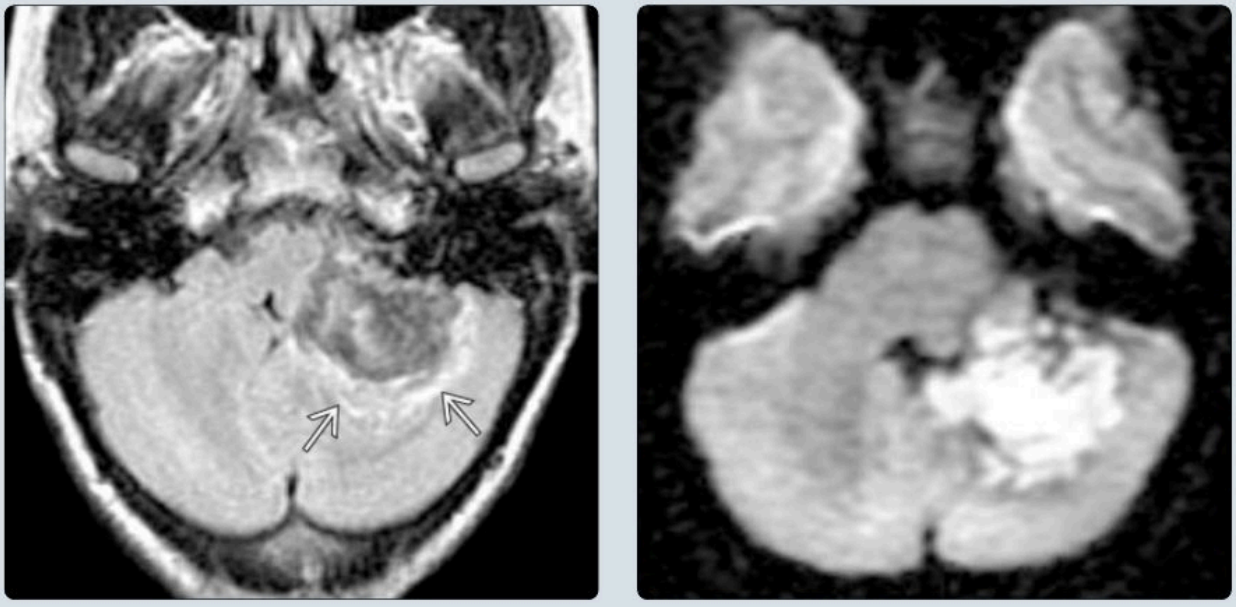


FIGURE 8 A CPA epidermoid. Left: FLAIR, incomplete suppression of the lesion. Right: DWI, bright restricted diffusion, which distinguishes it from an arachnoid cyst.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1202

4.4 Lipoma and others

A **lipoma** is bright on T1 before contrast and disappears with fat suppression; it can be mistaken for an enhancing tumour if the pre-contrast T1 is not checked. **Facial nerve schwannoma** may mimic a vestibular one but extends along the labyrinthine segment to the geniculate ganglion. **Aneurysm**, metastasis and sarcoidosis are rare.

5 CHAPTER FIVE The labyrinth and the facial nerve

Inflammation and small tumours.

5.1 Labyrinthitis and labyrinthine ossification

In **labyrinthitis** (viral, bacterial or after meningitis) the fluid on T2 is normal or slightly reduced but the labyrinth **enhances** after gadolinium. After bacterial meningitis, fibrous tissue and then bone replace the fluid (**labyrinthitis ossificans**): the T2 signal in the cochlea is lost before CT shows bone. In a child

deafened by meningitis, early MRI and urgent cochlear implantation are needed before the cochlea ossifies.

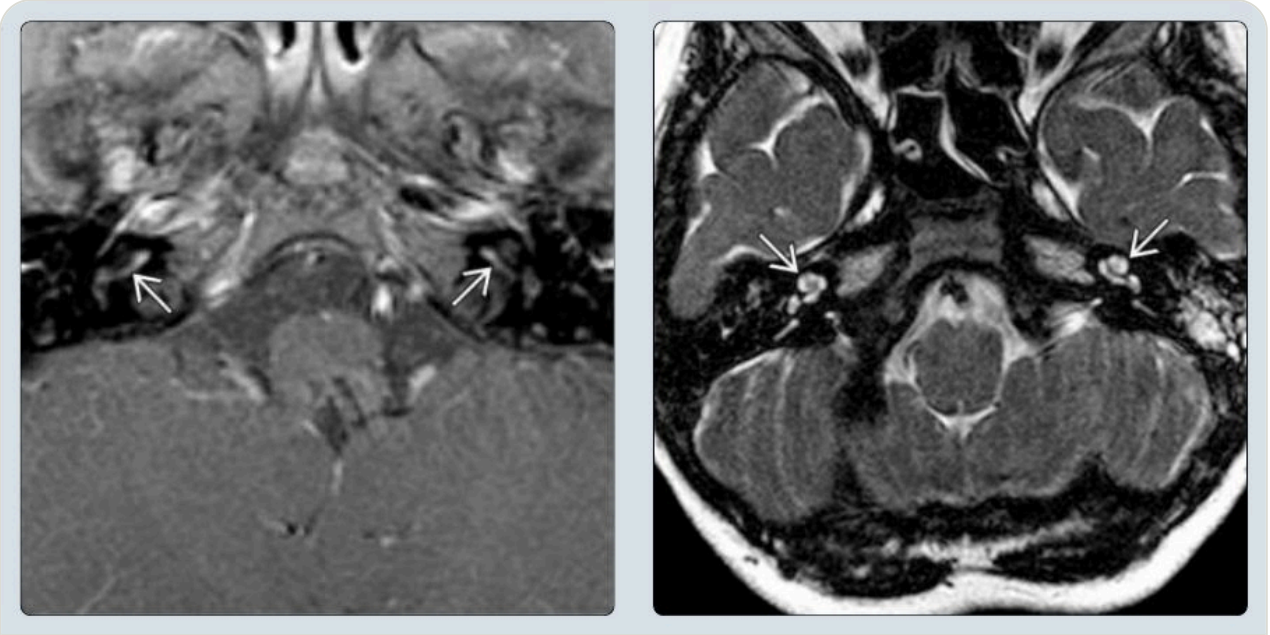


FIGURE 9 Labyrinthitis. Left: post-gadolinium T1 after meningitis, abnormal enhancement of both basal turns. Right: T2 in labyrinthitis, the cochlear fluid is still normal; enhancement, not loss of fluid, is the sign.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1112

5.2 Intralabyrinthine schwannoma

A small schwannoma inside the cochlea or vestibule appears as a **dark filling defect** on T2 and **focal enhancement** after gadolinium. Haemorrhage (bright before contrast) and labyrinthitis (enhancement without a filling defect) are the mimics.

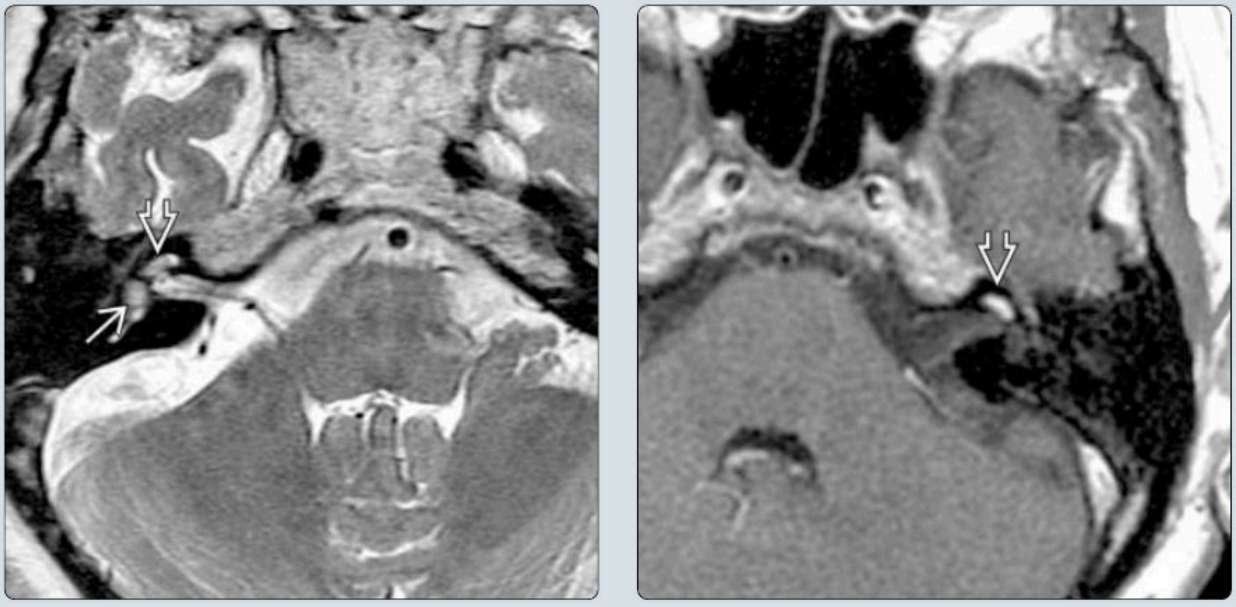


FIGURE 10 Intralabyrinthine schwannoma. Left: T2, a dark filling defect in the bright cochlear fluid. Right: post-gadolinium T1, focal enhancement in the cochlea.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1126

5.3 Facial nerve enhancement

The facial nerve has a venous plexus around its **geniculate, tympanic and upper mastoid segments**, so mild enhancement there can be normal. Enhancement in the **IAC and labyrinthine segments** is always abnormal. In **Bell's palsy**, the nerve enhances smoothly and linearly, typically with a "fundal tuft" at the lateral IAC and along the labyrinthine segment, without enlargement. Nodular or thickened enhancement, extension beyond the temporal bone, or a palsy that does not recover point to tumour or perineural spread from the parotid.

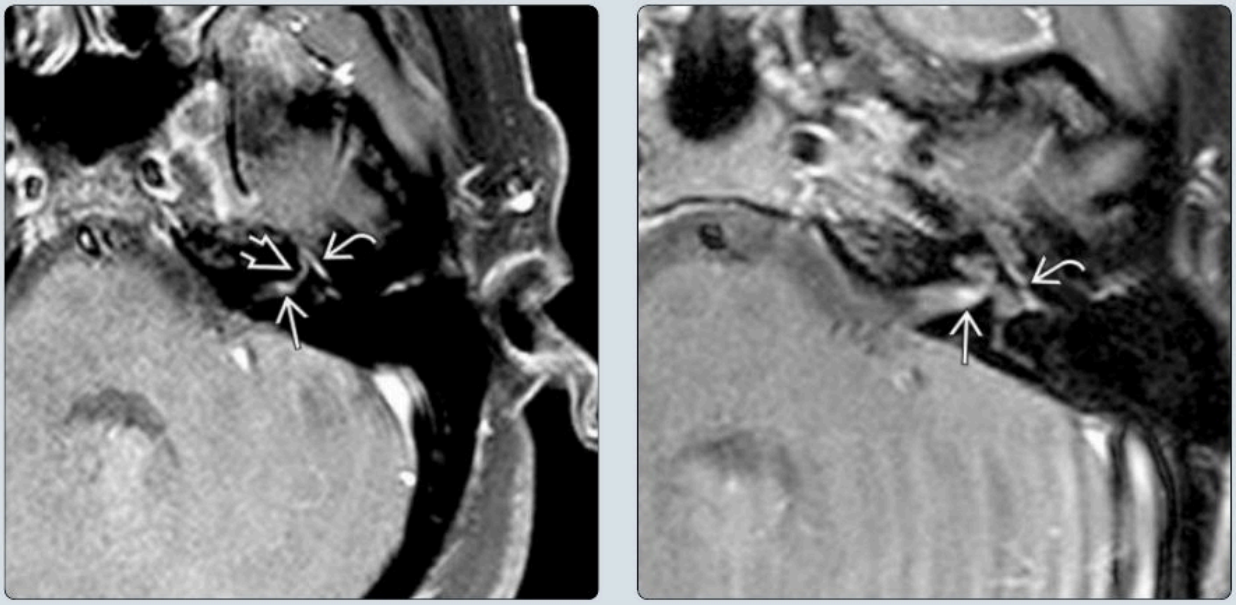


FIGURE 11 Bell's palsy. Left and right: post-gadolinium T1 with fat suppression, a tuft of enhancement at the IAC fundus and enhancement of the labyrinthine and tympanic segments, smooth and not enlarged.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1166

5.4 The child before cochlear implantation

Before implantation, MRI (with CT in malformations) confirms that the **cochlear nerve** is present on oblique sagittal T2, that the **cochlea is patent** and fluid-filled (ossification after meningitis), and describes any **malformation** (common cavity, incomplete partition, large vestibular aqueduct). A small IAC with only one nerve usually means an absent cochlear nerve; that nerve is usually the facial.

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Sequences:** high-resolution T2 and post-gadolinium fat-suppressed T1, and when each is needed.

- **"Seven up, Coke down"** in the IAC; crista falciformis and Bill's bar.
- **Schwannoma versus meningioma:** centred on the porus and acute angle, versus broad dural base, obtuse angle and dural tail.
- **Epidermoid versus arachnoid cyst:** diffusion.
- **Labyrinthitis ossificans:** loss of fluid signal on T2; urgency of implantation after meningitis.
- **Normal versus abnormal facial nerve enhancement.**

6.2 Common pitfalls

COMMON PITFALLS

- Calling a lipoma or haemorrhage "enhancing" without the pre-contrast T1.
- Reporting a normal vascular loop as the cause of tinnitus or hearing loss.
- Missing an intralabyrinthine schwannoma by looking only at the IAC.
- Calling normal geniculate enhancement abnormal.
- Mistaking the flocculus or choroid plexus for a small CPA mass.

Written for this app from Imaging Anatomy: Head and Neck (Chapman and Koontz), internal auditory canal and inner ear, Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), CPA-IAC overview and diagnoses, Dunnebier, Imaging for Otolaryngologists. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 7

The skull base and its foramina

The skull base is where ENT, neurosurgery and head and neck oncology meet, and its scans frighten trainees because of the number of holes. In practice a small set of foramina carries the disease that matters to an otolaryngologist: the cribriform plate and ethmoid roof for CSF leaks and sinonasal tumours, the pterygopalatine fossa and foramina rotundum and ovale for perineural spread, the jugular foramen and hypoglossal canal for the lower cranial nerves, and the petrous apex and clivus for a handful of characteristic lesions. Learn those, and the rest falls into place.

AT A GLANCE

- **Bone CT** for foramina and matrix; **MRI** for extent, marrow and perineural spread. Read the **pre-contrast T1** first.
- The **pterygopalatine fossa** should contain fat; it connects nose, infratemporal fossa, orbit, middle cranial fossa and palate.
- Perineural spread: enlarged enhancing nerve, **lost fat at the foramen**, widened foramen, denervated muscle.
- Petrous apex by signal: marrow suppresses, **cholesterol granuloma bright on T1 and T2**, cholesteatoma bright on DWI.
- **Chordoma** midline clivus; **chondrosarcoma** petrooccipital fissure. Jugular foramen: paraganglioma permeative, schwannoma smooth, meningioma sclerotic.

1 CHAPTER ONE The study

CT for bone, MRI for everything else.

1.1 Partners, not rivals

Thin-section bone CT (1 mm or less, with multiplanar reformats) shows the foramina, the fine bony walls, erosion, remodelling and the matrix of a lesion (chondroid rings, ground glass, calcification). **MRI** shows the soft-tissue extent and the marrow: the **pre-contrast T1** is the key sequence, because normal adult skull base marrow is fatty and bright, so tumour replacing marrow shows as grey against white. Post-gadolinium fat-suppressed T1 shows enhancement, dural spread and

perineural tumour; DWI suggests epidermoid; MRA and MRV show the carotid and the venous sinuses.

READ THE PRE-CONTRAST T1 OF THE SKULL BASE FIRST

Normal marrow in the clivus, petrous apex, sphenoid and crista galli is fatty and bright on T1 in adults. A clivus that is darker than the pons on T1, or an asymmetric dark area of marrow, is the first sign of infiltration by tumour, myeloma or metastasis. In children the marrow is still red and darker, and the sphenooccipital synchondrosis stays open until about 20 years of age.

CHAPTER TWO

2 The three floors

Anterior, central and posterior.

2.1 Anterior skull base

The floor of the anterior cranial fossa is formed by the orbital plates of the frontal bone, the **cribriform plate** and ethmoid roof, and the planum sphenoidale. The cribriform plate sits lower than the **fovea ethmoidalis** (the ethmoid roof), and the two are joined by the thin **lateral lamella**, the weakest point and a common site of iatrogenic and spontaneous CSF leaks. Below it lie the nose, sinuses and orbits, so most anterior skull base lesions start there: sinonasal carcinoma, olfactory neuroblastoma, mucocele, osteoma and meningioma.

2.2 Central skull base

The central skull base is the sphenoid bone and the temporal bone in front of the petrous ridge: the **sella**, the clivus above the synchondrosis, and the greater wings with their foramina. Below it lie the masticator, parotid and pharyngeal mucosal spaces, which is why nasopharyngeal carcinoma and masticator space tumours reach the middle cranial fossa through it.

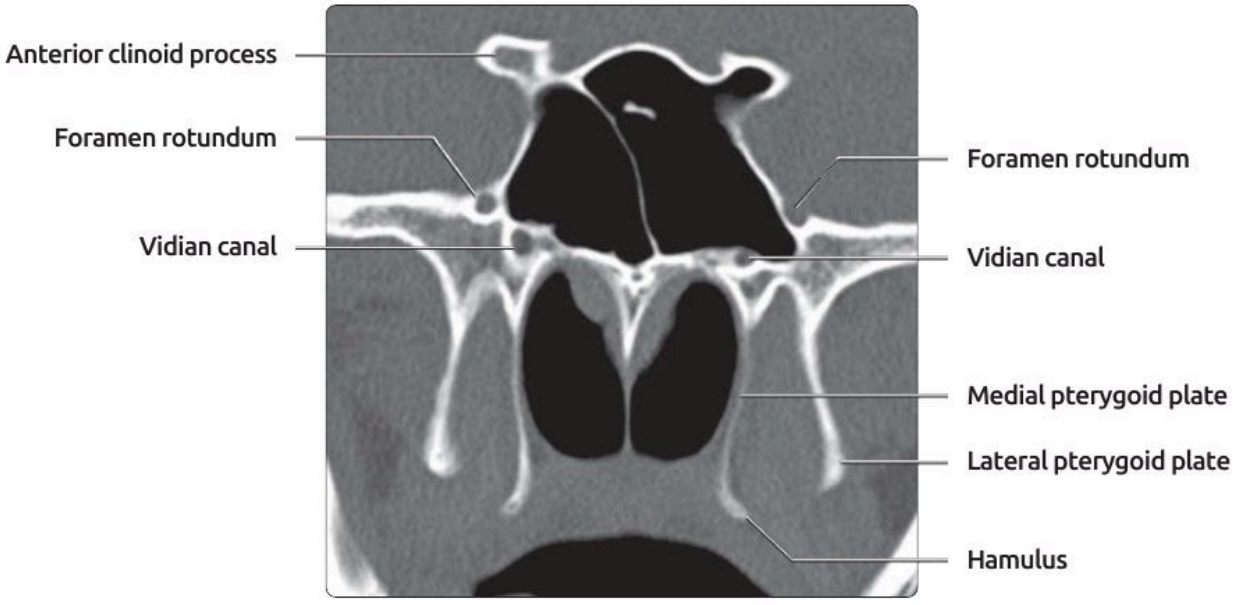


FIGURE 1 Coronal bone CT through the sphenoid: the foramen rotundum (maxillary nerve) above and lateral to the vidian canal (nerve of the pterygoid canal), with the pterygoid plates below. Two round holes in the sphenoid on each side.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 36

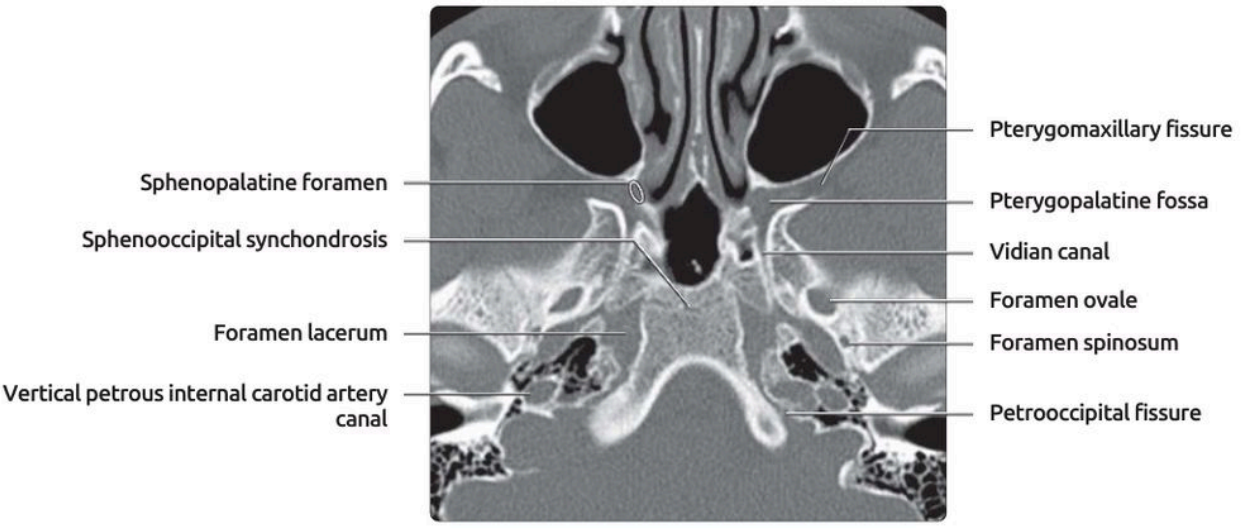


FIGURE 2 Axial bone CT: the foramen ovale in the greater wing of the sphenoid, the route of the mandibular nerve and of perineural spread from the masticator space.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 35

2.3 Posterior skull base

The posterior skull base is the bowl of the posterior fossa: the clivus below the synchondrosis, the occipital condyles and the posterior surface of the petrous bone. It contains the **internal auditory canal**, the **jugular foramen** (an

anteromedial **pars nervosa** with the glossopharyngeal nerve and a larger posterolateral **pars vascularis** with the jugular bulb, vagus and accessory nerves), the **hypoglossal canal** below and medial to it, and the foramen magnum. On coronal CT, the jugular tubercle between the jugular foramen and the hypoglossal canal forms the "beak" of an eagle's head.

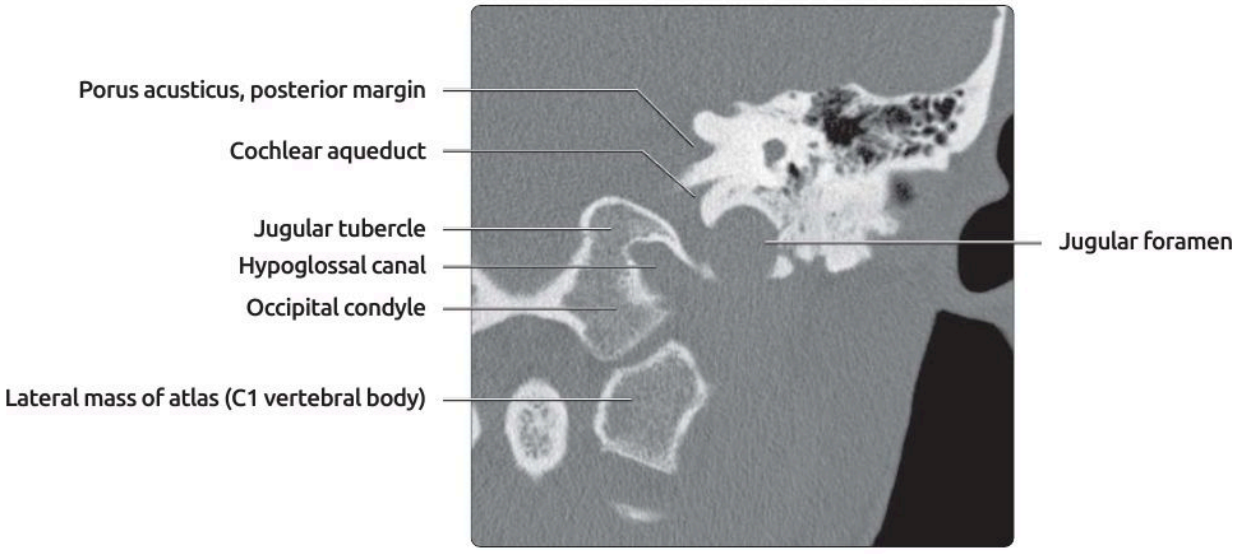


FIGURE 3 Coronal bone CT: the "eagle" of the posterior skull base. The jugular tubercle (the beak) separates the jugular foramen above from the hypoglossal canal below.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 46

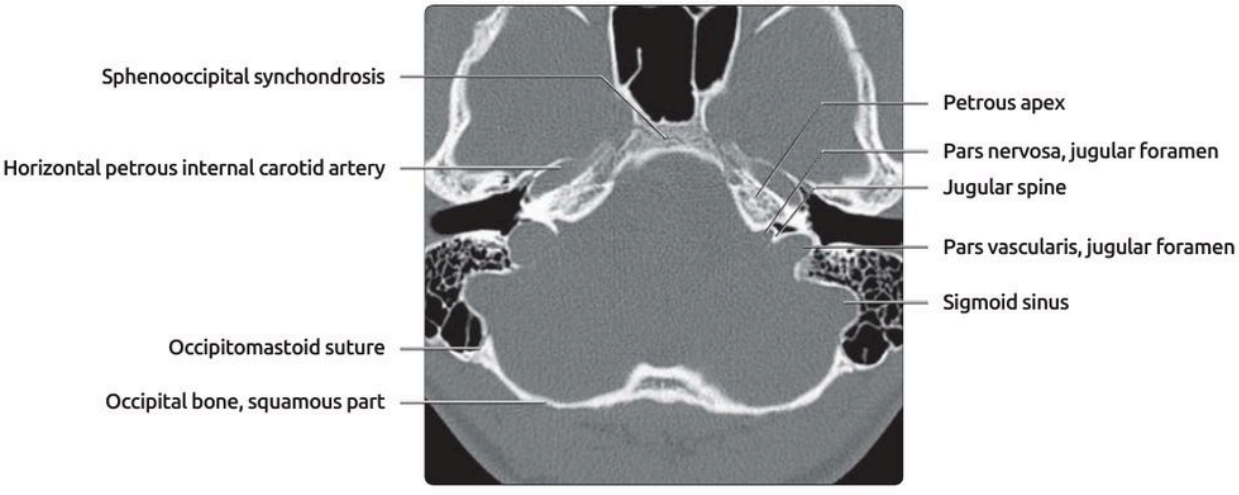


FIGURE 4 Axial bone CT at the mid jugular foramen: the smaller anteromedial pars nervosa and the larger posterolateral pars vascularis.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 43

3 CHAPTER THREE
The foramina that matter

Holes, contents and the disease that uses them.

THE ENT FORAMINA

FORAMEN OR FISSURE	MAIN CONTENTS	DISEASE THAT USES IT
Cribriform plate	Olfactory nerve fibres	CSF leak, encephalocele, olfactory neuroblastoma, sinonasal carcinoma
Optic canal	Optic nerve, ophthalmic artery	Fractures, sphenoid and orbital apex disease
Superior orbital fissure	III, IV, VI, V1, superior ophthalmic vein	Orbital apex and cavernous sinus spread
Inferior orbital fissure	Infraorbital nerve and vessels	Orbit to pterygopalatine fossa and infratemporal fossa
Foramen rotundum	Maxillary nerve (V2)	Perineural spread from the cheek, palate and sinus
Vidian canal	Nerve of the pterygoid canal	Juvenile angiofibroma; perineural spread to the petrous carotid
Foramen ovale	Mandibular nerve (V3)	Perineural spread from the masticator space; trigeminal schwannoma
Foramen spinosum	Middle meningeal artery	Landmark lateral to the foramen ovale
Foramen lacerum	Fibrocartilage; carotid passes above	Nasopharyngeal carcinoma's route to the cavernous sinus
Stylomastoid foramen	Facial nerve	Perineural spread from the parotid
Jugular foramen	IX, X, XI, jugular bulb	Paraganglioma, schwannoma, meningioma
Hypoglossal canal	Hypoglossal nerve	Schwannoma, metastasis, clival tumours

3.1 The pterygopalatine fossa: the crossroads

The pterygopalatine fossa is a small fat-filled space behind the maxillary antrum, and on axial CT and T1 MRI it should contain **fat**. It connects medially to the nose through the **sphenopalatine foramen**, laterally to the infratemporal fossa through the **pterygomaxillary fissure**, upwards to the orbit through the inferior orbital fissure, backwards to the middle cranial fossa through the **foramen rotundum** and to the foramen lacerum through the **vidian canal**, and downwards to the palate through the palatine canals. Loss of its fat on either side is a key sign of perineural tumour, juvenile angiofibroma, invasive fungal disease or lymphoma.

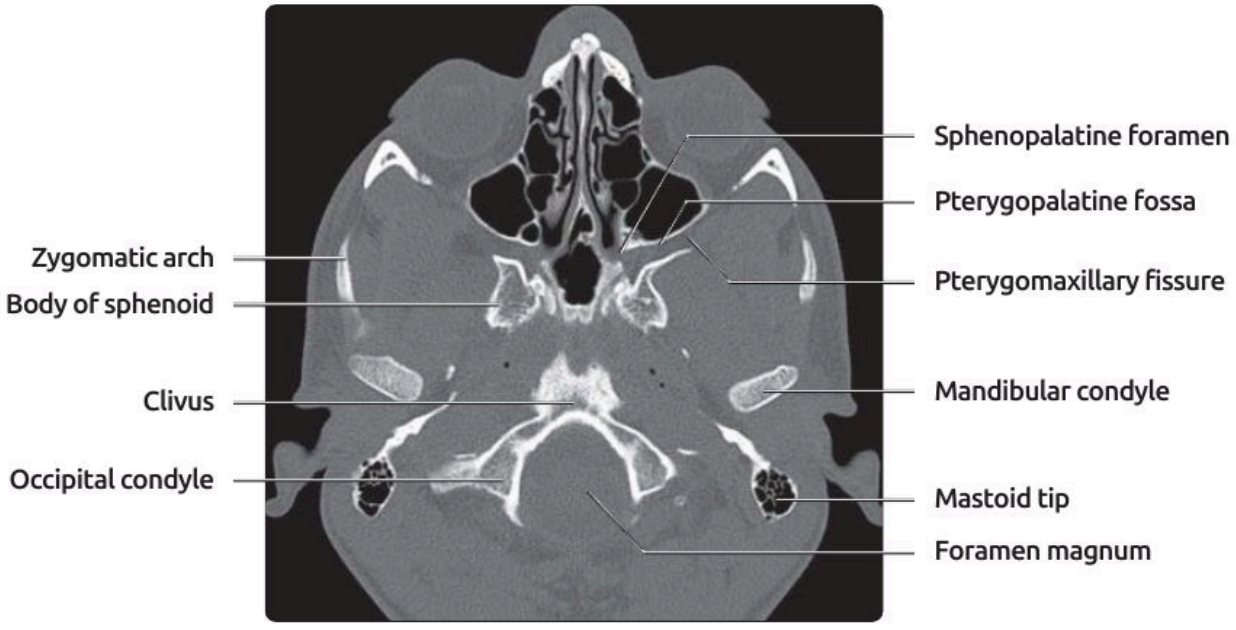


FIGURE 5 Axial bone CT: the pterygopalatine fossa behind the maxillary antrum, opening medially through the sphenopalatine foramen and laterally through the pterygomaxillary fissure.
Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 10

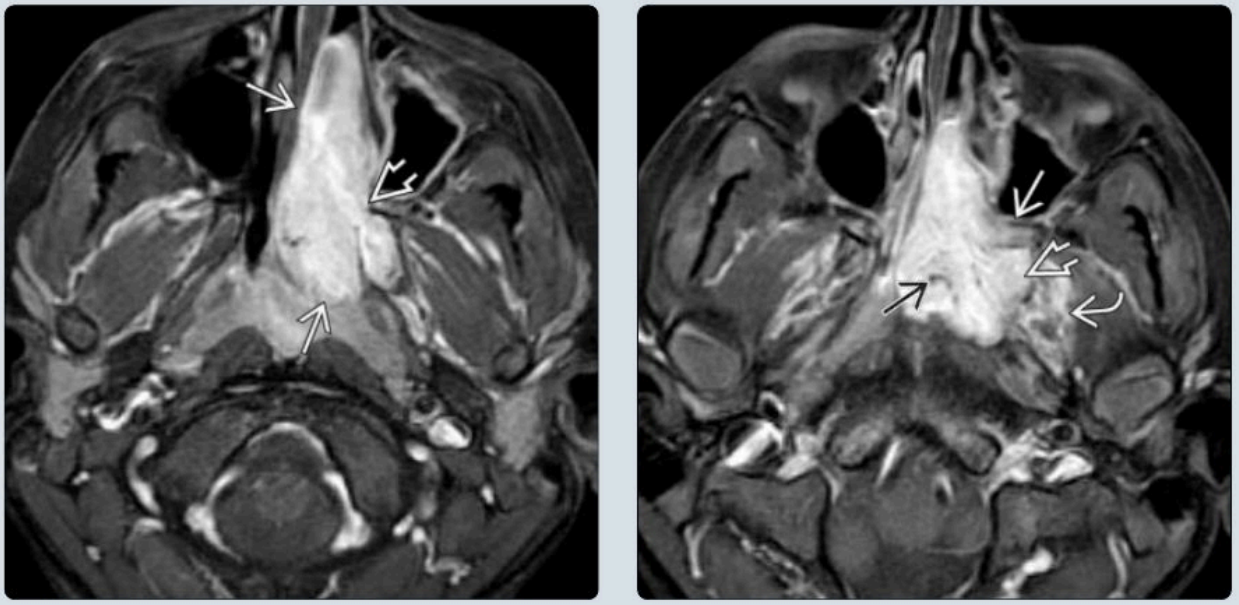


FIGURE 6 Juvenile angiofibroma. Left: post-gadolinium T1, an enhancing mass in the nasal cavity and nasopharynx arising near the sphenopalatine foramen. Right: the same tumour with flow voids, widening the pterygopalatine fossa, destroying the pterygoid base and entering the masticator space.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 740

3.2 Perineural spread

Tumour can travel along nerves through the skull base, often silently. Look for an enlarged, enhancing nerve, **loss of fat** at the opening of a foramen (pterygopalatine fossa, foramen ovale, stylomastoid foramen), **widening** of the foramen, filling of **Meckel's cave**, and denervation of the muscles supplied (atrophy of the muscles of mastication with V3, of the tongue with XII). Adenoid cystic carcinoma, squamous carcinoma of the skin and lip, and lymphoma are the common culprits.

4 CHAPTER FOUR The petrous apex

A short list, sorted by signal.

4.1 Normal variation first

The petrous apex is **pneumatised in about a third** of people, often asymmetrically, and otherwise contains **fatty marrow**. Asymmetric marrow is bright on T1 and can

be mistaken for a lesion; it suppresses with fat saturation and the bone is not expanded on CT. Fluid trapped in apical air cells after otitis is bright on T2 without expansion or bone loss. Neither needs treatment.

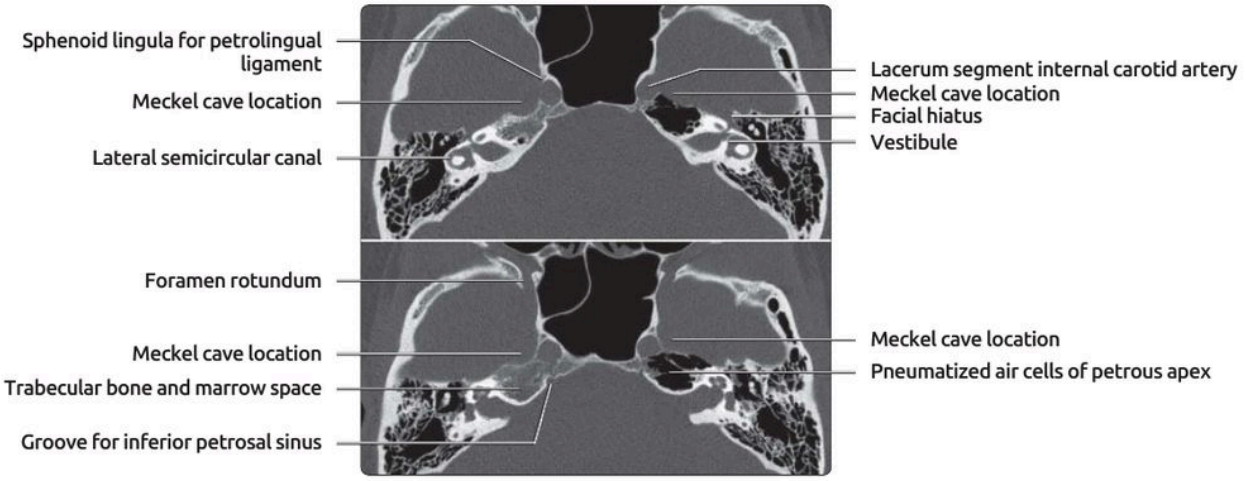


FIGURE 7 Axial CT: asymmetric pneumatisation of the petrous apices, aerated on one side and marrow-filled on the other. Variation, not disease.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 97

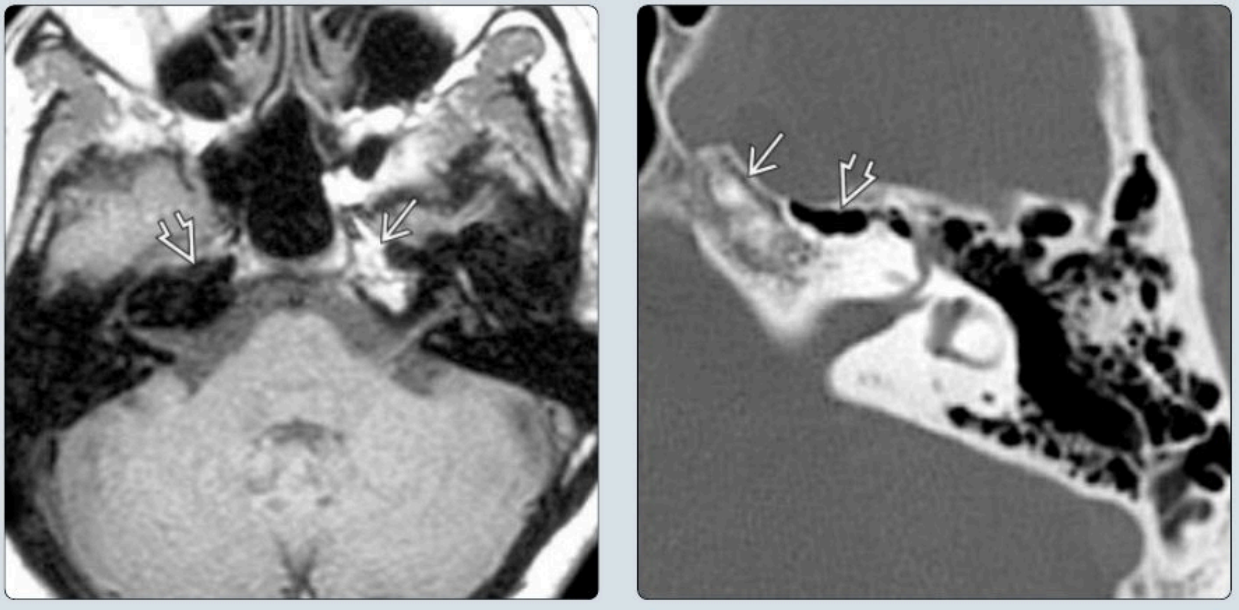


FIGURE 8 Asymmetric marrow. Left: T1, a bright fatty focus in one petrous apex and dark air in the other. Right: CT, the bright side is simply marrow, with no expansion.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1140

4.2 The lesions

PETROUS APEX LESIONS ON MRI AND CT

LESION	T1	T2	GADOLINIUM	DWI	CT
Asymmetric marrow	Bright, suppresses with fat-sat	Intermediate	None	No restriction	Normal, not expanded
Trapped fluid	Dark	Bright	None	No restriction	Opacified cells, intact septa, no expansion
Cholesterol granuloma	**Bright, does not suppress**	Bright	None	No restriction	Smooth expansion, thin cortex
Cholesteatoma or epidermoid	Dark	Bright	None	**Restricted**	Smooth expansion
Apicitis	Dark	Bright	Enhancing marrow and dura	Abscess restricts	Destruction of trabeculae and cortex
Mucocele	Variable	Bright	Rim	No restriction	Expanded air cell

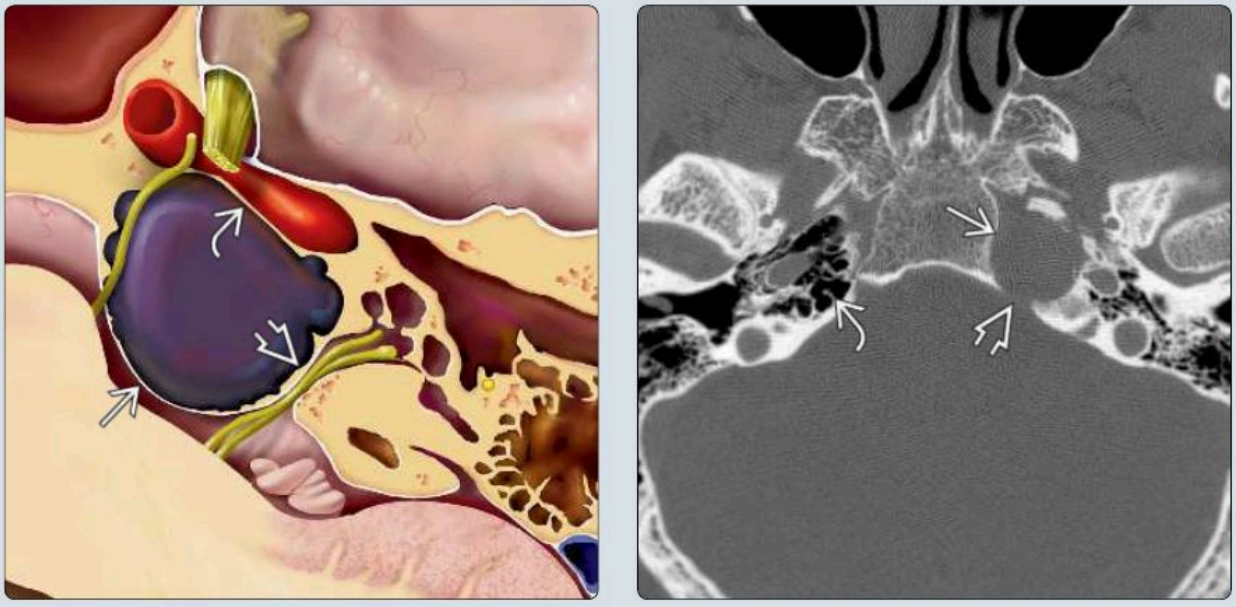


FIGURE 9 Cholesterol granuloma of the petrous apex. Left: drawing of an expansile cystic lesion. Right: axial CT, smooth expansion of the left apex with thinned, partly dehiscent cortex; the other apex is well pneumatised.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1152

4.3 Apicitis

Infection spreading from the middle ear into a pneumatised apex causes **petrous apicitis**, classically with Gradenigo's triad of otorrhoea, deep facial pain (V) and a sixth nerve palsy. CT shows destruction of the apical trabeculae, and MRI shows enhancing marrow and dura and any abscess.

5 CHAPTER FIVE The clivus and the jugular foramen

Signature lesions.

5.1 Chordoma and chondrosarcoma

Chordoma arises in the **midline clivus**, destroys bone, contains fragments of bone, is **very bright on T2** and often "thumbs" the pons. **Chondrosarcoma** arises **off the midline**, at the **petrooccipital fissure**, contains **chondroid rings and arcs** of calcification on CT, and is also very bright on T2. Position, more than signal, separates them.



FIGURE 10 Clival chordoma. Left: drawing of an expansile destructive midline mass indenting the pons. Right: sagittal T1, low-signal tumour replacing the clivus, with the classic "thumb" against the pons.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 874

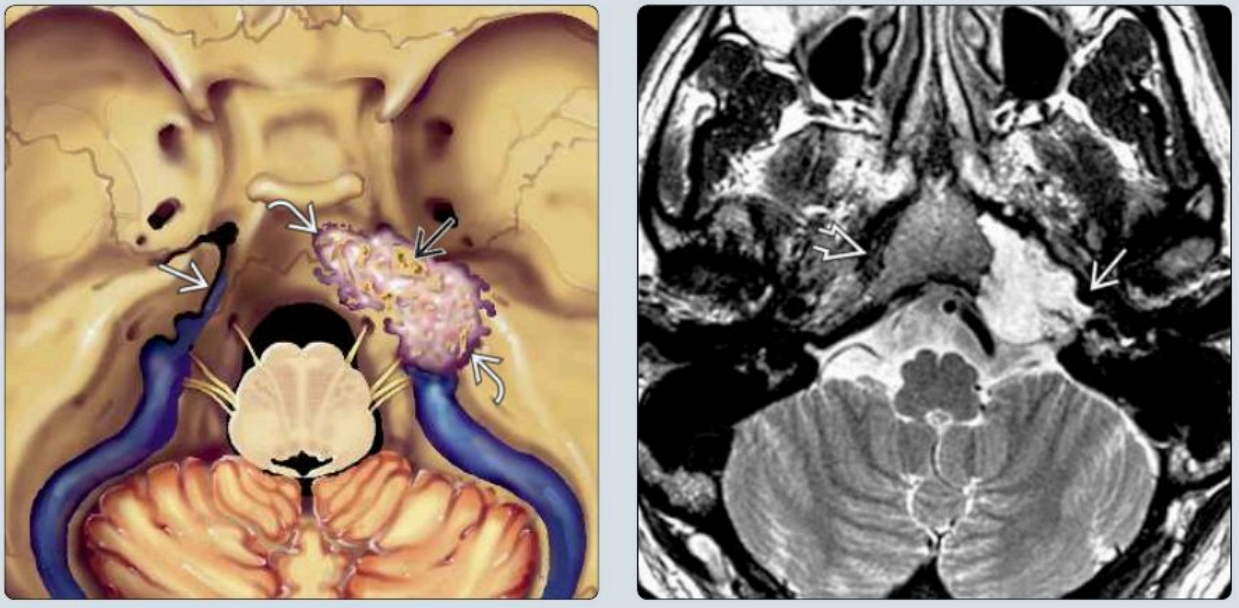


FIGURE 11 Chondrosarcoma. Left: drawing of the lesion centred on the petrooccipital fissure with chondroid calcification. Right: T2, a very bright off-midline mass compressing the petrous carotid.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 952

5.2 The jugular foramen

Three tumours make up nearly all jugular foramen masses, and the **bone margin** on CT separates them. A **paraganglioma** (glomus jugulare) gives **permeative, moth-eaten** erosion, spreads superolaterally into the middle ear, and on MRI shows **"salt and pepper"** (flow voids and slow flow) with intense enhancement. A **schwannoma smoothly enlarges and scallops** the foramen, often extending along the nerve, without flow voids. A **meningioma** causes **sclerotic, permeative** change or hyperostosis and spreads along the dura with a dural tail. A high or dehiscent jugular bulb and asymmetric foramina are normal variants.

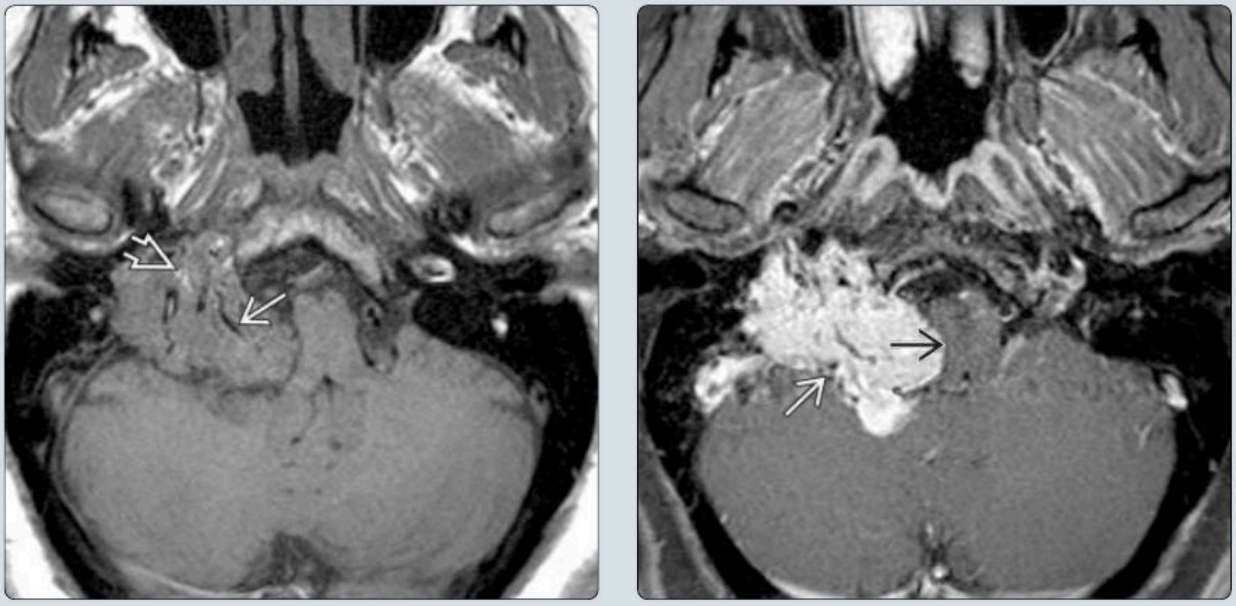


FIGURE 12 Glomus jugulare paraganglioma on MRI. Left: T1, a jugular foramen mass with salt-and-pepper flow voids. Right: post-gadolinium T1, intense enhancement with intracranial extension against the medulla.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 892

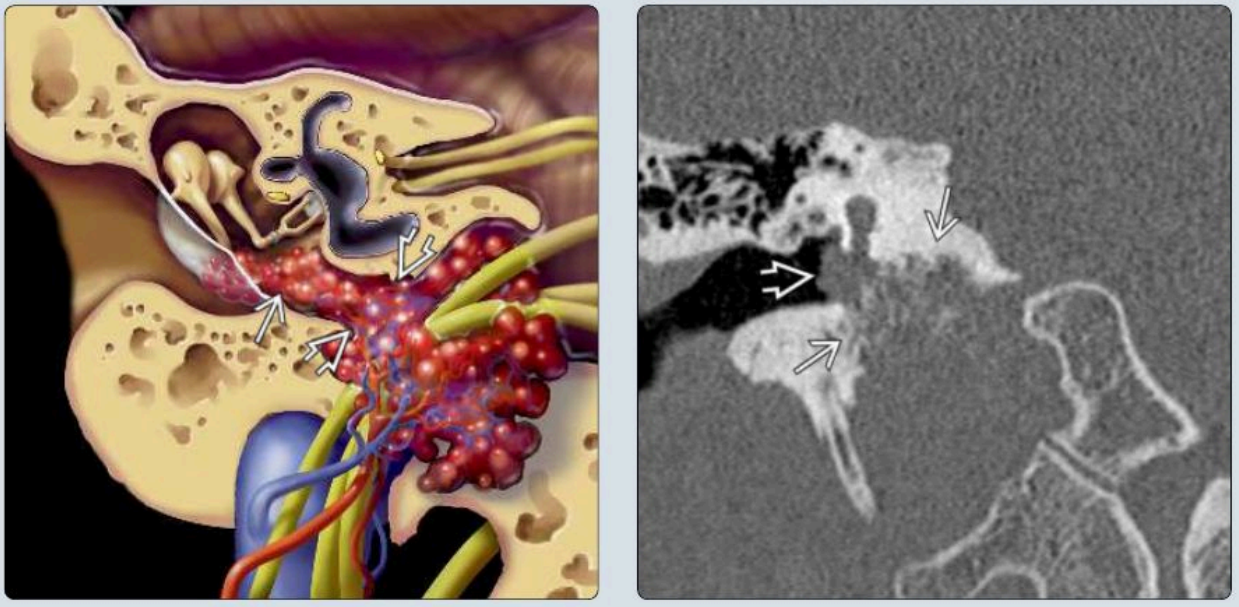


FIGURE 13 The same tumour on CT. Left: drawing of its spread from the jugular foramen into the middle ear. Right: coronal CT, permeative-destructive margins and extension up into the middle ear.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 892

CHAPTER SIX

6 CSF leaks and encephaloceles

Finding the hole.

6.1 Imaging a leak

A **thin-section bone CT** of the anterior and middle skull base finds most defects: the cribriform plate and lateral lamella, the ethmoid roof, the sphenoid (especially a pneumatised **lateral recess**), the frontal sinus, and the tegmen in the ear. **MRI** (heavily T2-weighted) then shows whether the defect contains CSF, meninges or brain (meningocele or **meningoencephalocele**), and separates fluid from retained mucus in an opacified sinus. CT cisternography is kept for difficult cases. In spontaneous leaks, look for signs of raised intracranial pressure: an empty sella, multiple defects and arachnoid pits.

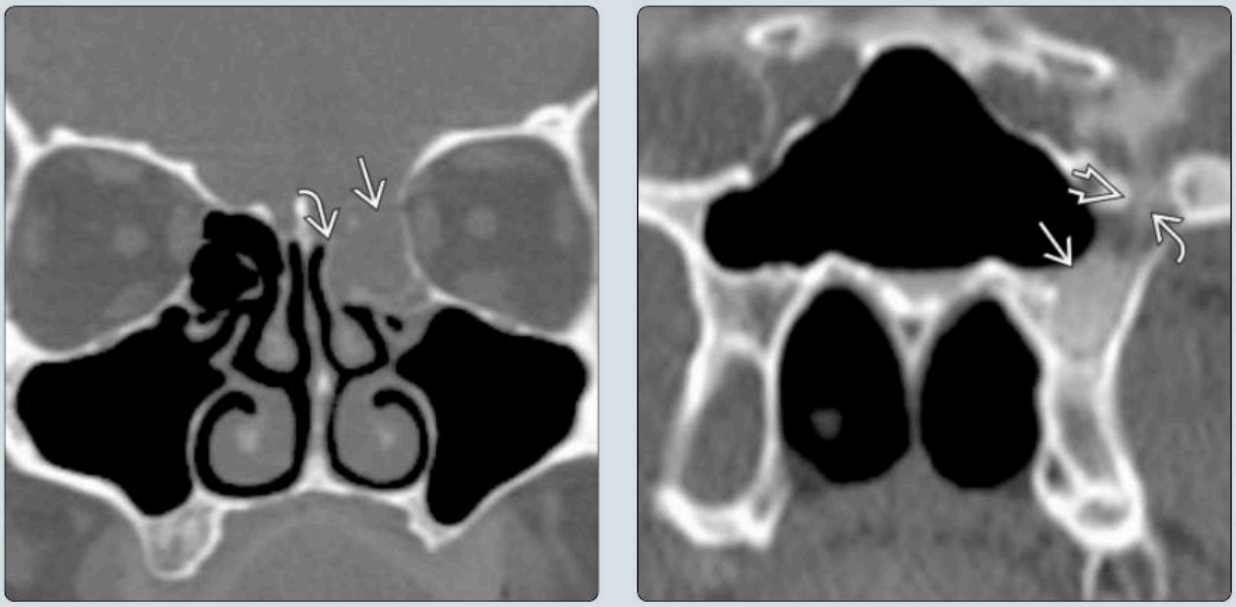


FIGURE 14 CSF leak. Left: coronal CT, a bony defect in the ethmoid roof lateral to the middle turbinate attachment with ethmoid opacification; MRI showed a meningoencephalocele. Right: CT after intrathecal contrast, contrast passing through a sphenoid defect.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 920

READING A SKULL BASE STUDY

1 Marrow

- → **pre-contrast T1**
  clivus, petrous apex, sphenoid
- → dark or asymmetric marrow
- → compare sides

2 Foramina

- → rotundum, ovale, vidian, stylomastoid, jugular, hypoglossal
- → widened, eroded or filled
- → fat at their openings

3 Pterygopalatine fossa

- → fat present on both sides
- → connections

→ widening

4 **Bone**

→ remodelled or destroyed

→ matrix (chondroid, ground glass, sclerosis)

5 **Lesion by site**

→ **anterior**
sinonasal, olfactory, meningioma

→ **clivus midline**
chordoma

→ **petrooccipital fissure**
chondrosarcoma

→ **jugular foramen**
paraganglioma, schwannoma, meningioma

6 **Intracranial**

→ dura, cavernous sinus, Meckel's cave, brain

→ carotid and venous sinuses

CHAPTER SEVEN

7 For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **CT versus MRI** for the skull base, and why the pre-contrast T1 matters.
- **Foramina and contents**, especially rotundum, ovale, vidian, stylomastoid and jugular.

- **The pterygopalatine fossa** and its connections.
- **Signs of perineural spread.**
- **Petrous apex:** marrow, fluid, cholesterol granuloma and cholesteatoma by signal.
- **Chordoma versus chondrosarcoma**, and the **three jugular foramen tumours** by their bone margins.
- **CSF leak:** CT for the defect, MRI for the content.

7.2 Common pitfalls

COMMON PITFALLS

- Calling asymmetric marrow or a non-pneumatised apex a lesion.
- Missing marrow infiltration by reading only post-contrast images.
- Missing loss of fat in the pterygopalatine fossa.
- Calling fat in the crista galli a dermoid, or an open synchondrosis a fracture in a teenager.

Written for this app from Imaging Anatomy: Head and Neck (Chapman and Koontz), skull base chapters, Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), skull base overview and diagnoses, Dunnebier, Imaging for Otolaryngologists. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 8

Sinus CT before surgery

The sinus CT is the scan an ENT surgeon reads more often than any radiologist does, and it is read differently: not only "is there disease?" but "where does it drain, what is in my way, and where are the dangers?". A surgeon who reads the CT systematically before every endoscopic sinus operation knows the uncinate, the frontal recess, the height of the skull base, the lamina papyracea, the carotid and optic nerves, and the anterior ethmoid artery before the endoscope goes in. This lesson builds that habit.

AT A GLANCE

- Thin **non-contrast** CT in **coronal, axial and sagittal** planes, ideally after medical treatment.
- **Ostiomeatal unit** drains maxillary, anterior ethmoid and frontal; **sphenoethmoidal recess** drains posterior ethmoid and sphenoid.
- Before surgery read **CLOSE**: cribriform (Keros), lamina papyracea, Onodi cell, sphenoid (optic nerve and carotid), ethmoid artery.
- **Lund–Mackay**: 0–2 per sinus, 0 or 2 per ostiomeatal complex, total 0–24.
- Hyperdense contents: **allergic fungal sinusitis** or fungal ball. **Unilateral** disease or bone erosion: **MRI**.

1 CHAPTER ONE The study

Thin, non-contrast, three planes.

1.1 Protocol

A sinus CT for chronic rhinosinusitis is a **non-contrast**, low-dose, thin-section (about 1 mm or less) study, reconstructed in a bone algorithm, viewed in a bone window, and reformatted in **coronal, axial and sagittal** planes. The **coronal** plane mirrors the endoscopic view and shows the ostiomeatal unit; the **sagittal** plane shows the frontal recess and its cells; the **axial** plane shows the sphenoid, the posterior ethmoid and the relation of the optic nerve and carotid. Cone-beam CT is an alternative at lower dose. Most centres also use the dataset for image guidance. Contrast and MRI are added when a complication, a tumour or invasive fungal disease is suspected.

SCAN AFTER MEDICAL TREATMENT, NOT DURING AN ACUTE ATTACK

A CT done during a cold or an acute exacerbation shows transient mucosal thickening that may not reflect chronic disease. Ideally the scan follows a course of maximal medical treatment, and it should be read with the patient's endoscopic findings in mind: up to a third of people without symptoms have some mucosal thickening.

2 CHAPTER TWO The drainage pathways

Where each sinus empties.

2.1 The ostiomeatal unit

The maxillary, anterior ethmoid and frontal sinuses all drain into the **middle meatus** through a common pathway called the **ostiomeatal unit**. On coronal CT, the **uncinate process** is a thin sickle of bone running up from the lateral nasal wall; lateral to it is the **ethmoid infundibulum**, into which the **maxillary ostium** opens; above it is the **ethmoid bulla**, the largest anterior ethmoid cell; and the gap between the free edge of the uncinate and the bulla is the **hiatus semilunaris**, opening into the middle meatus. A small obstruction here blocks three sinuses at once.

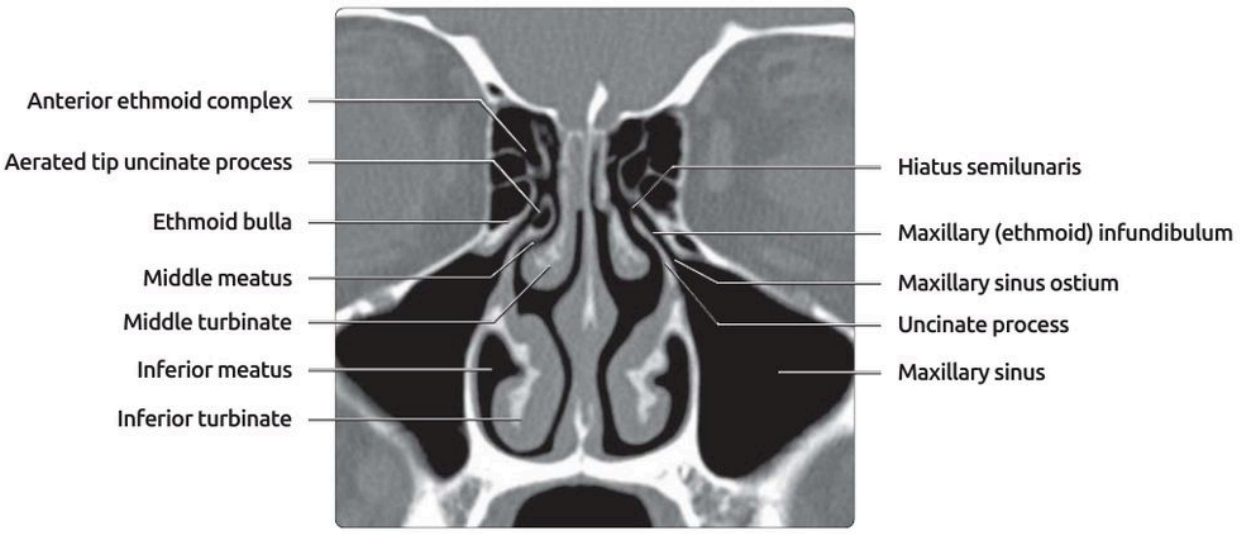


FIGURE 1 Coronal CT through the ostiomeatal unit: the uncinate process, the maxillary infundibulum and ostium, the ethmoid bulla, the hiatus semilunaris and the middle meatus beneath the middle turbinate.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 250

2.2 The frontal recess

The frontal sinus drains through its ostium into the **frontal recess**, an hourglass-shaped space bounded by the cells around it rather than by a true duct: the **agger nasi** cell in front, the **ethmoid bulla** (or a suprabullar cell) behind, and the lamina papyracea laterally. Where the uncinate attaches superiorly (to the lamina papyracea, the skull base or the middle turbinate) decides whether the frontal recess drains medial or lateral to it. The **sagittal** view is the best way to understand it.

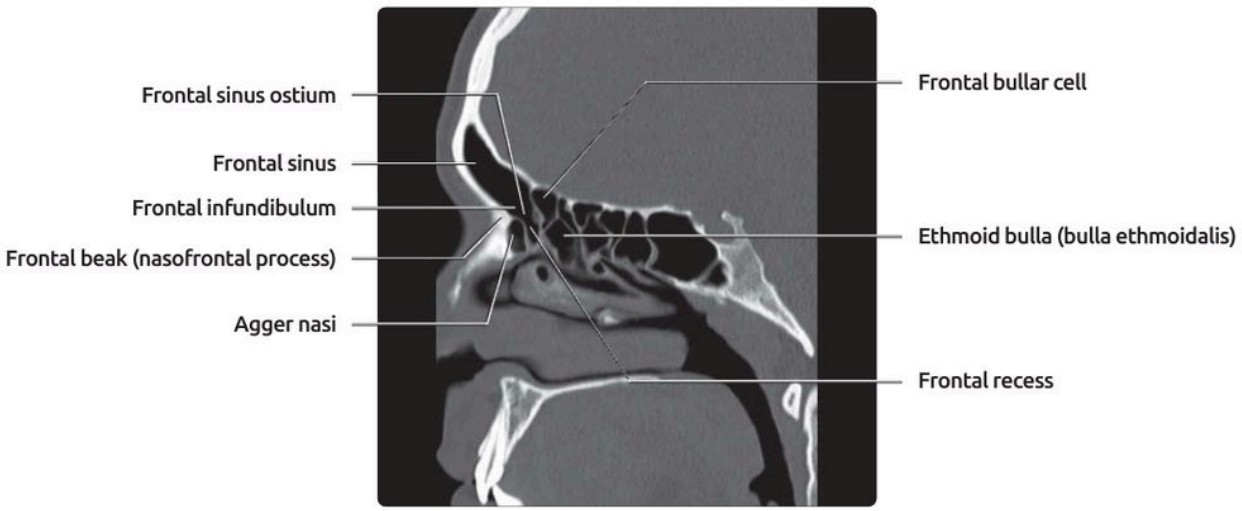


FIGURE 2 Sagittal CT of the frontal sinus drainage pathway: frontal sinus, ostium and infundibulum above, the frontal recess below, bounded by the agger nasi in front and the ethmoid bulla and a frontal bullar cell behind.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 266

2.3 The sphenoethmoidal recess

The posterior ethmoid and sphenoid sinuses drain into the **sphenoethmoidal recess**, medial to the superior turbinate, behind the ostiomeatal unit. Disease confined to the posterior ethmoid and sphenoid means the obstruction is here, not at the middle meatus.

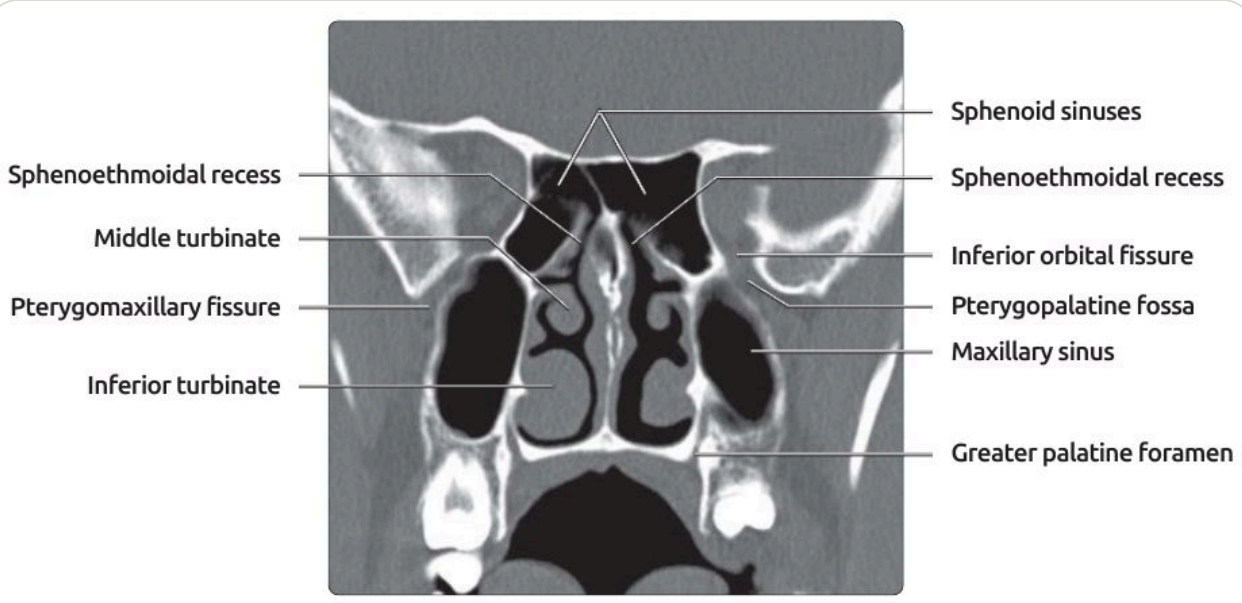


FIGURE 3 Axial CT: the sphenoethmoidal recess, a narrow vertical slit in the posterior nose into which the posterior ethmoid and sphenoid sinuses drain.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 242

PATTERNS OF OBSTRUCTION

PATTERN	SINUSES INVOLVED	SITE OF THE BLOCK
Infundibular	Maxillary alone	Maxillary ostium or infundibulum
Ostiomeatal unit	Maxillary, anterior ethmoid, frontal on one side	Middle meatus
Sphenoethmoidal recess	Posterior ethmoid and sphenoid	Sphenoethmoidal recess
Sinonasal polyposis	All sinuses, both sides, with polyps in the nose	Diffuse mucosal disease
Sporadic	Scattered, no drainage logic	Retention cysts, mucosal thickening

3 CHAPTER THREE Variants that change surgery

Know them before you start.

3.1 Cells and turbinates

Variants matter when they narrow a drainage pathway or put a structure at risk. A **concha bullosa** (a pneumatised middle turbinate) may narrow the middle meatus. A **paradoxical** middle turbinate curves laterally. A **Haller cell** (infraorbital ethmoid cell) along the orbital floor narrows the maxillary infundibulum. A large **agger nasi** or other **frontal cells** narrow the frontal recess; these are now named by the International Frontal Sinus Anatomy Classification (agger nasi, supra agger, supra agger frontal, supra bulla, supra bulla frontal, supraorbital ethmoid and frontal septal cells), which has largely replaced the Kuhn types used in older texts.

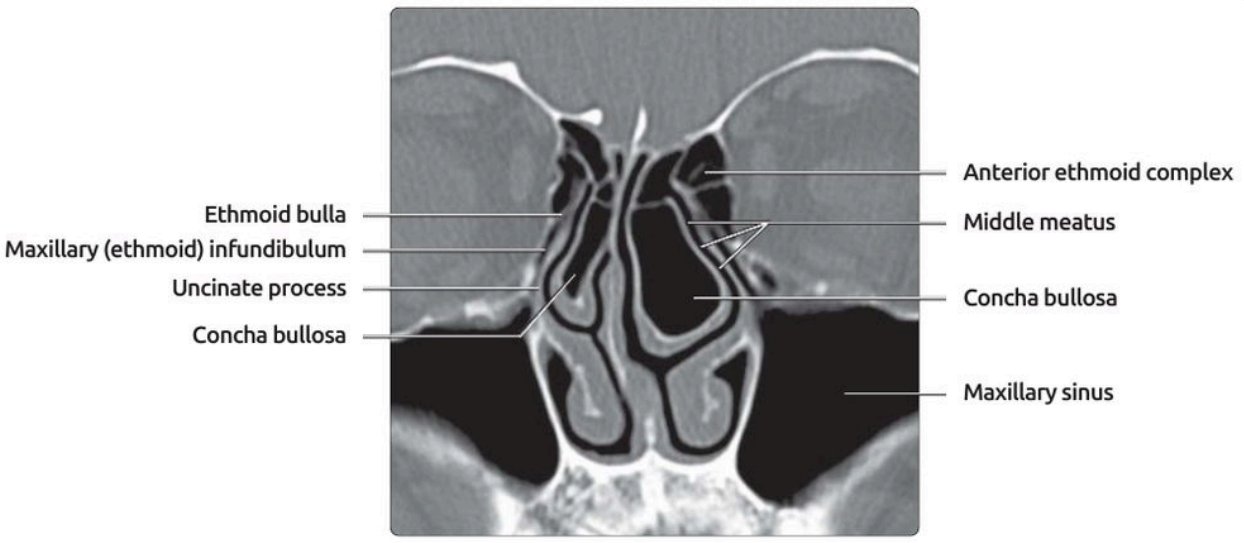


FIGURE 4 Coronal CT: bilateral aerated middle turbinates (concha bullosa), each narrowing its middle meatus.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 250

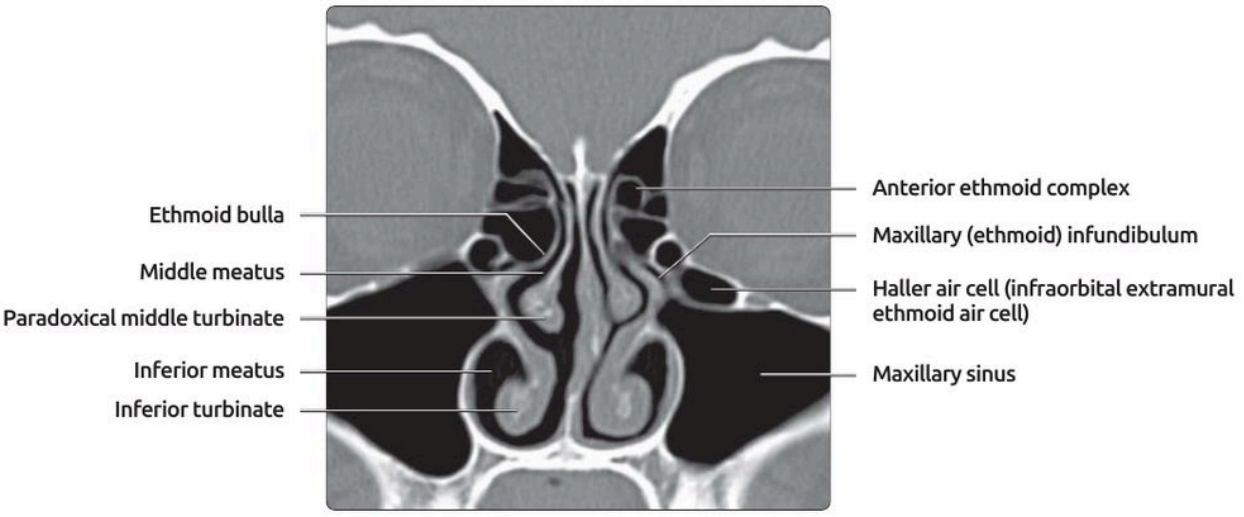


FIGURE 5 Coronal CT: a Haller (infraorbital ethmoid) cell along the orbital floor, at the roof of the maxillary infundibulum.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 250

3.2 The skull base: Keros

The **cribriform plate** sits lower than the **fovea ethmoidalis** (ethmoid roof), joined by the thin **lateral lamella**. The depth of the olfactory fossa is graded by **Keros**: type I 1 to 3 mm, type II 4 to 7 mm, type III 8 to 16 mm. The deeper the fossa, the longer and thinner the lateral lamella and the greater the risk of a CSF leak. Asymmetry between the two sides is common and should be noted.

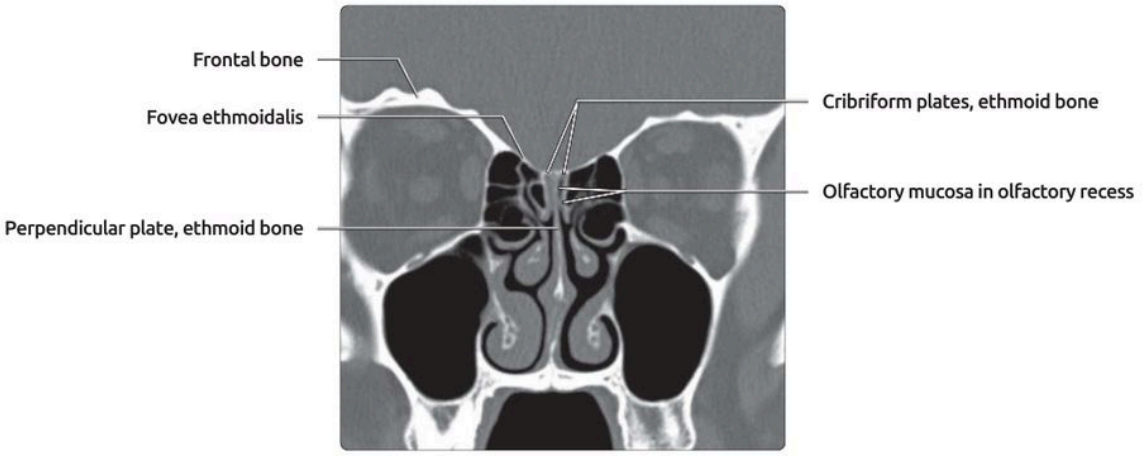


FIGURE 6 Coronal CT: the fovea ethmoidalis forming the ethmoid roof, stepping down to the lower cribriform plates either side of the perpendicular plate.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 23

3.3 The anterior ethmoid artery

The **anterior ethmoid artery** crosses the ethmoid roof from the orbit to the olfactory fossa, marked on coronal CT by a small notch in the medial orbital wall. When a **supraorbital ethmoid cell** is present, the artery may hang below the skull base in a bony mesentery and is at greater risk.

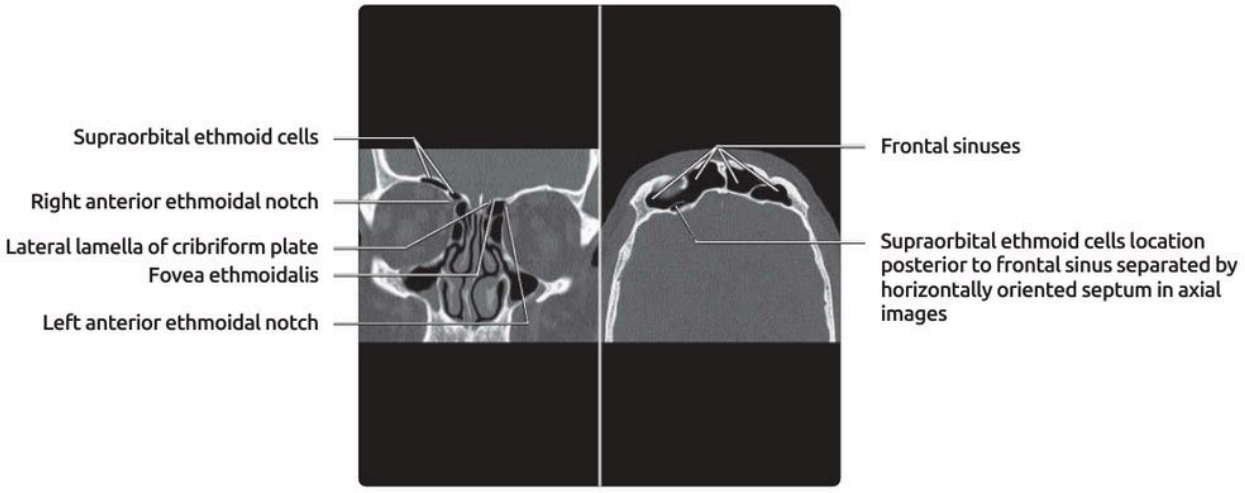


FIGURE 7 Supraorbital ethmoid cells. Left: coronal CT, cells above the anterior ethmoidal notch, with the lateral lamella and fovea ethmoidalis. Right: axial CT, the cells lying behind the frontal sinus.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 268

3.4 The sphenoid and the posterior ethmoid

An **Onodi (sphenoethmoidal) cell** is a posterior ethmoid cell that extends lateral and above the sphenoid, carrying the optic nerve in its wall; the surgeon who mistakes it for the sphenoid may injure the nerve. Look on axial and coronal images for **dehiscence** of the bone over the **optic nerve** and the **internal carotid artery** in the lateral wall of the sphenoid, and for an **intersinus septum** that inserts onto the carotid canal: twisting it can tear the artery.

3.5 The lamina papyracea

The lamina papyracea is the thin lateral wall of the ethmoid. A **dehiscence** (congenital or after old trauma) lets orbital fat bulge medially and puts the orbit at risk. Disease that is pushing it outwards (mucocele, allergic fungal sinusitis) is also noted.

THE PRE-OPERATIVE SINUS CT CHECKLIST (CLOSE)

1. **C, cribriform plate**
   - → Keros depth
   - → asymmetry
   - → lateral lamella
   - → any defect
2. **L, lamina papyracea**
   - → intact or dehiscent
   - → orbital fat herniation
3. **O, Onodi cell**
   - → present
   - → optic nerve in its wall
4. **S, sphenoid**

- → optic nerve and carotid dehiscence
- → septum onto carotid
- → pneumatisation

5 **E, ethmoid artery**

- → anterior ethmoid artery notch
- → hanging below skull base
- → supraorbital cells

6 **Drainage**

- → uncinate attachment
- → infundibulum and maxillary ostium
- → frontal recess cells
- → sphenoethmoidal recess

7 **Disease**

- → pattern of opacification
- → polyps
- → osteitis
- → hyperdensity
- → bone erosion

CHAPTER FOUR

4 Reading the disease

What the opacification means.

4.1 Scoring: Lund–Mackay

The **Lund–Mackay** score grades each sinus (maxillary, anterior ethmoid, posterior ethmoid, sphenoid, frontal) on each side as 0 (clear), 1 (partial) or 2 (complete opacification), and each ostiomeatal complex as 0 or 2, for a total of 0 to 24. It is used for research and audit and correlates poorly with symptoms, but it gives a common language.

4.2 Chronic rhinosinusitis and polyposis

Chronic rhinosinusitis shows mucosal thickening and opacification in a pattern that follows the drainage pathways, often with **osteitis**: thickened, sclerotic sinus walls from long-standing inflammation. **Polyposis** fills the nose and all sinuses on both sides, often with widening of the ethmoid labyrinth.

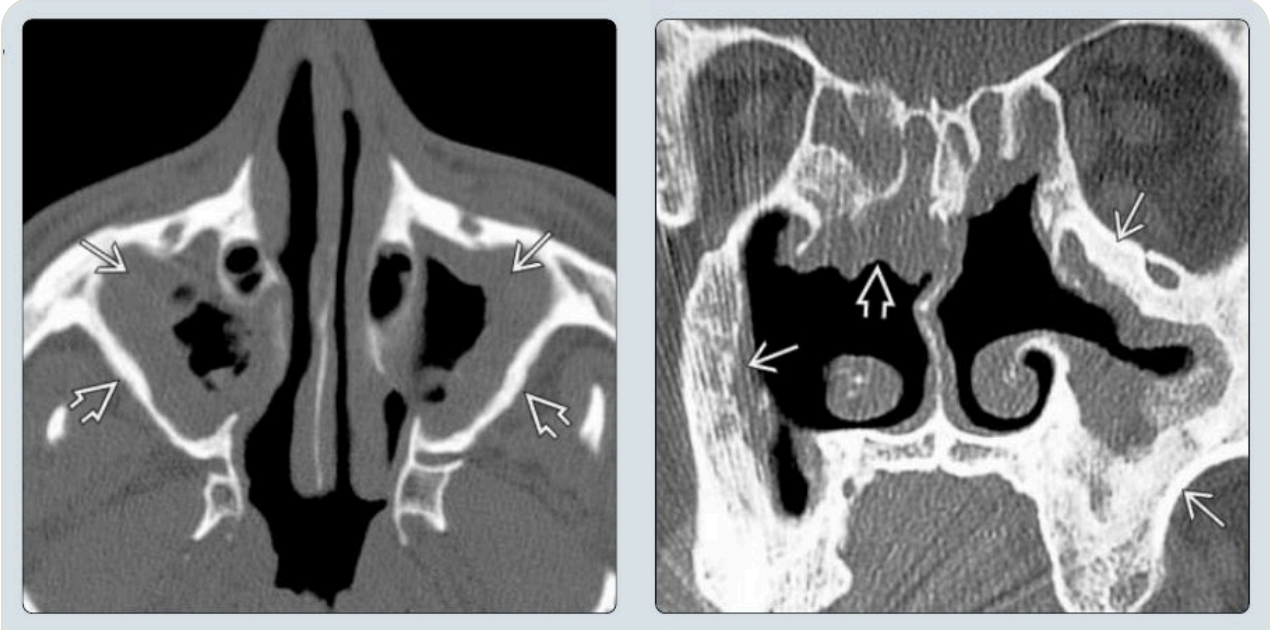


FIGURE 8 Chronic rhinosinusitis. Left: axial CT in cystic fibrosis, maxillary mucosal thickening with thickened sinus walls. Right: coronal CT, diffuse osteitis of the sinus walls.
Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 692



FIGURE 9 Sinonasal polyposis. Left and right: coronal CT, polyps filling the nasal cavity and occluding both middle meatuses, with opacified sinuses.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 708

4.3 Hyperdensity: think fungus or inspissated secretions

Dense material within an opacified sinus means thick, inspissated secretions, fungus or calcification. **Allergic fungal sinusitis** expands and fills several sinuses, usually on one side, with **central high density** and a low-density rim of mucosa ("double density"), and is **dark on T2**. A **fungal ball** fills a single sinus (usually the maxillary) with dense material and fine calcifications, and the walls are often sclerotic.

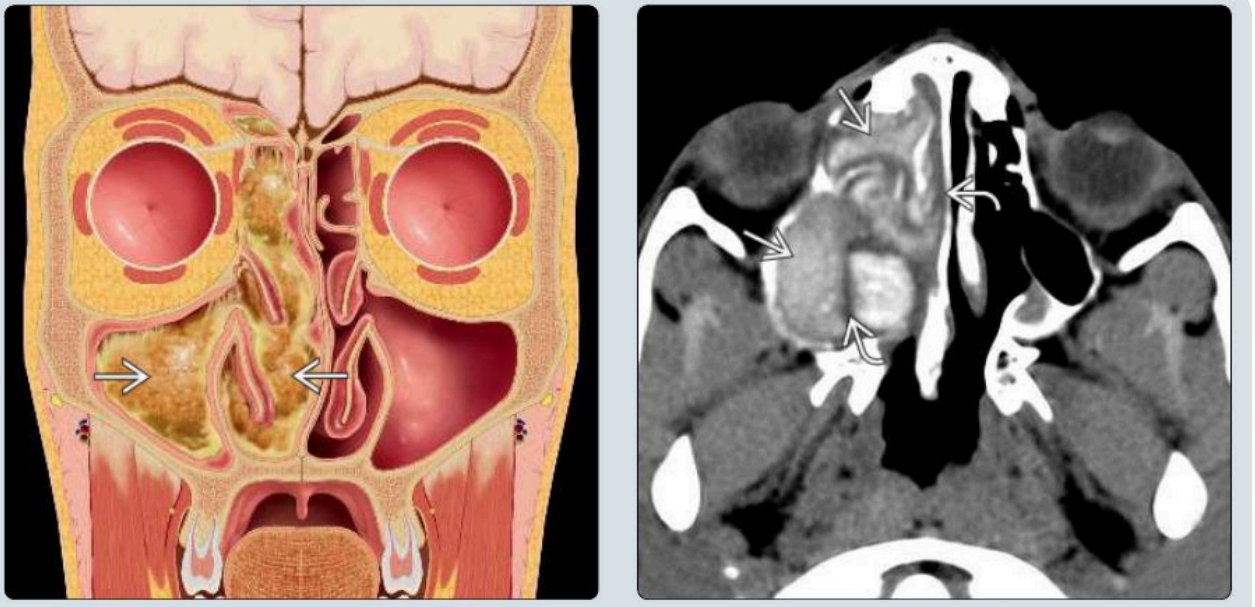


FIGURE 10 Allergic fungal sinusitis. Left: drawing of expanded sinuses filled with allergic mucin. Right: axial CT, unilateral expanded sinuses with central dense material and a low-density rim.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 700

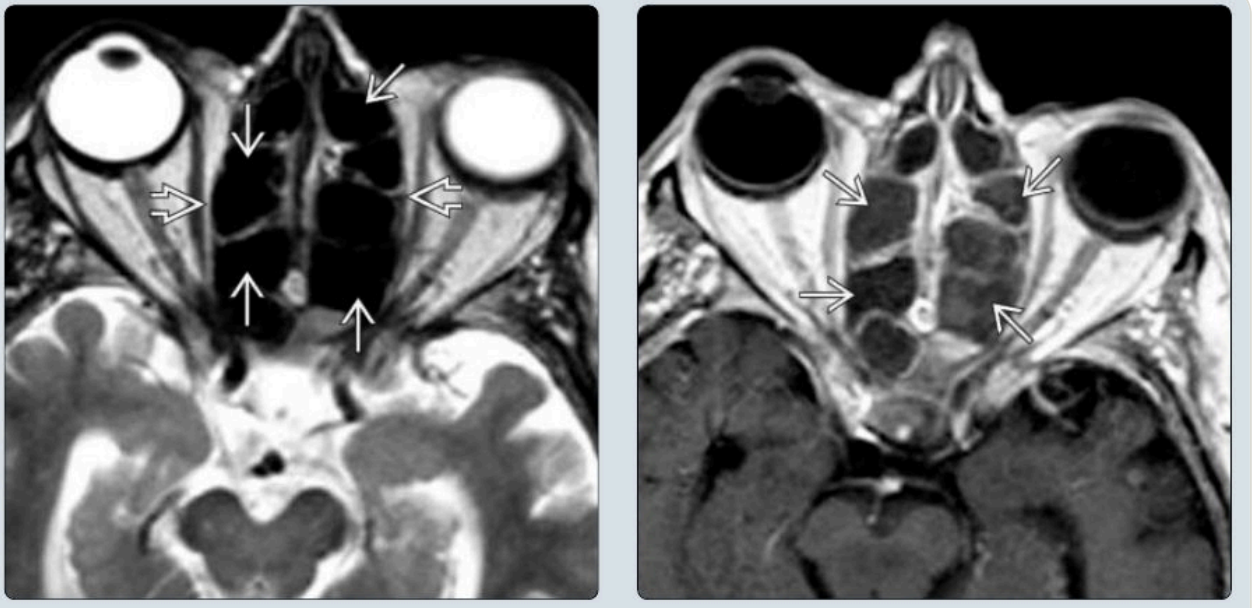


FIGURE 11 The same disease on MRI. Left: T2, the ethmoid contents are very dark and could be mistaken for air. Right: post-gadolinium T1, only the mucosa enhances; the sinuses are full.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 700

4.4 Unilateral disease: exclude a tumour

Opacification of one side with a mass in the nose, **bone destruction**, or **focal hyperostosis** at one point of the sinus wall needs MRI. **Inverted papilloma** typically

arises from the lateral nasal wall near the middle meatus, often with a focal bony spur at its attachment, and has a **convoluted, cerebriform** pattern on T2 and post-contrast MRI. MRI separates tumour (intermediate T2, solid enhancement) from obstructed secretions (bright T2, rim enhancement).

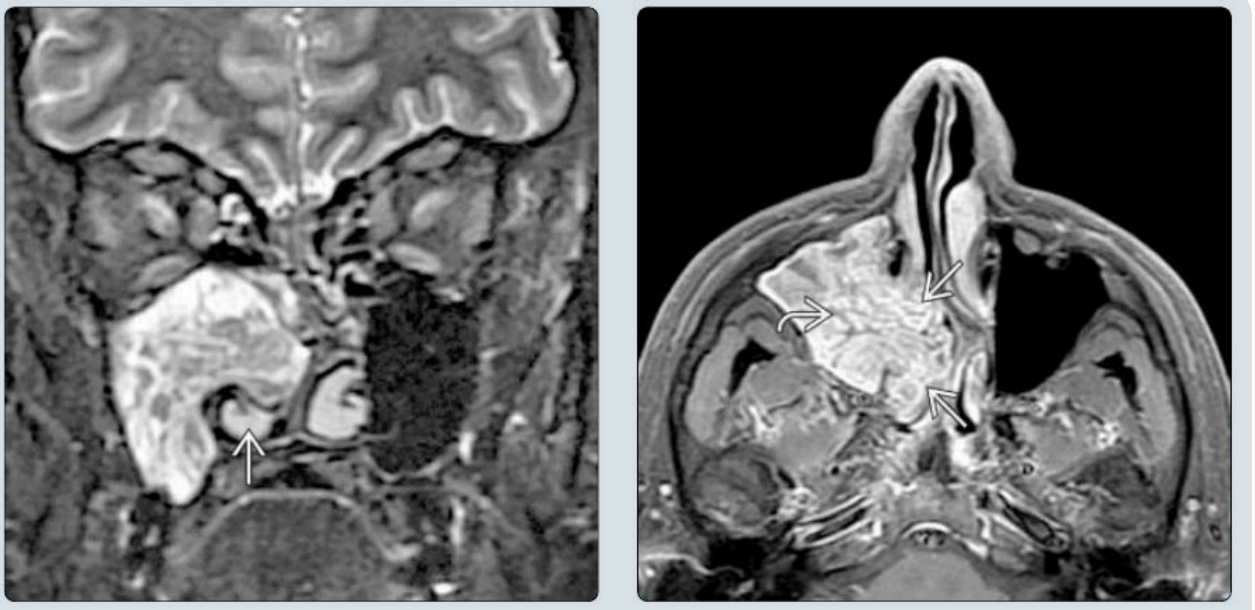


FIGURE 12 Inverted papilloma on MRI. Left: T2 fat-suppressed, the convoluted cerebriform pattern. Right: post-gadolinium T1, the tumour centred on the middle meatus with nasal and antral parts, separate from the obstructed secretions.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 744

4.5 Emergencies

Acute invasive fungal sinusitis in an immunocompromised patient may show little on CT at first: unilateral mucosal thickening, **stranding of the fat** around the sinus (in front of the maxilla, in the pterygopalatine fossa, in the orbit) and subtle bone erosion. On MRI, devitalised mucosa **does not enhance** (the "black turbinate"). Endoscopy and biopsy should not wait for the scan. Orbital and intracranial complications of sinusitis need contrast-enhanced CT and often MRI; they are covered in the orbit lesson.

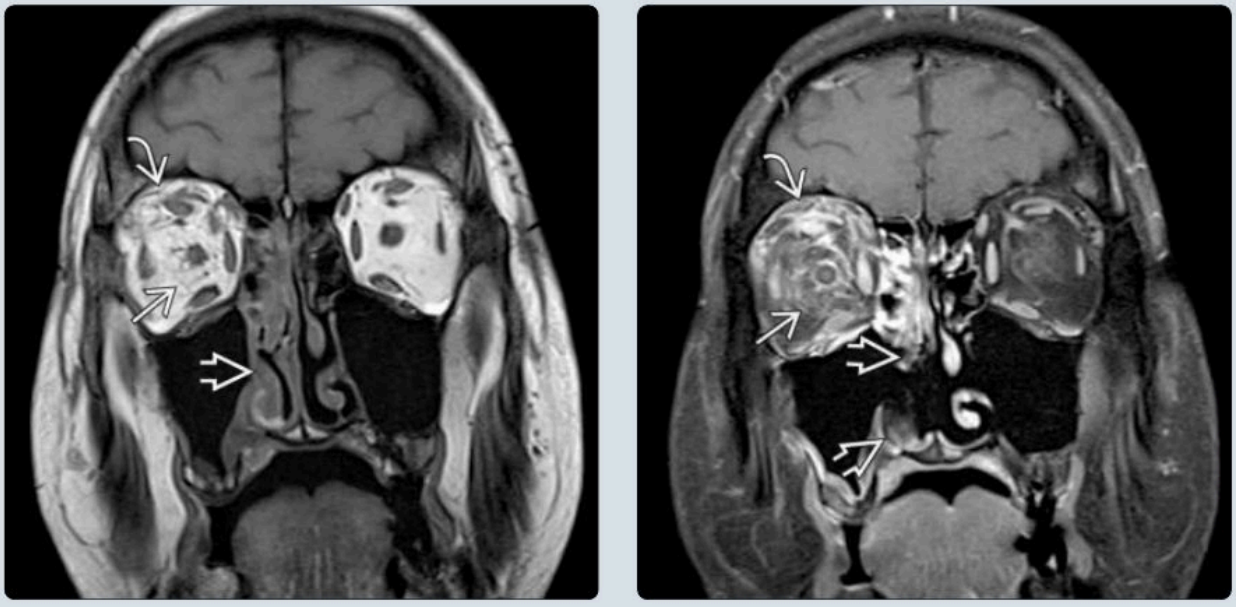


FIGURE 13 Acute invasive fungal sinusitis in leukaemia. Left: coronal T1, soft tissue in the right nasal cavity. Right: post-gadolinium T1, the inferior and middle turbinates do not enhance: dead mucosa.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 704

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Protocol:** thin non-contrast CT in three planes, after medical treatment.
- **The ostiomeatal unit** and the three drainage pathways.
- **Variants:** concha bullosa, Haller cell, agger nasi and frontal cells, Onodi cell.
- **Keros** classification and the lateral lamella.
- **The CLOSE checklist** before surgery.
- **Lund–Mackay** score.
- **Hyperdense sinus:** allergic fungal sinusitis versus fungal ball.
- **Unilateral disease** needs MRI; inverted papilloma features.

5.2 Common pitfalls

COMMON PITFALLS

- Operating without reading the CT in all three planes.
- Mistaking an Onodi cell for the sphenoid.
- Calling dark T2 signal in a sinus "air" when it is fungal mucin.
- Reassuring a unilateral opacification without endoscopy or MRI.
- Waiting for imaging when invasive fungal sinusitis is suspected.

Written for this app from Imaging Anatomy: Head and Neck (Chapman and Koontz), ostiomeatal unit and frontal recess, Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), sinonasal overview and diagnoses, Dunnebier, Imaging for Otolaryngologists. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 9

The orbit

The orbit sits between three sinuses, and much of the orbital imaging an ENT surgeon sees is about the sinuses getting in: periorbital infection, a subperiosteal abscess, a tumour breaching the lamina papyracea, a fracture of the floor or medial wall, or the decompression of thyroid eye disease. The orbit is easy to read once you think of it as a cone of fat divided by the muscles and the orbital septum into compartments, because each compartment has its own short list of disease.

AT A GLANCE

- Contrast **CT** for infection and trauma; fat-suppressed **MRI** for the optic nerve, apex, cavernous sinus and tumours.
- Think in compartments: **preseptal or postseptal**; **intraconal, conal or extraconal**.
- **Subperiosteal abscess**: lentiform rim-enhancing collection along the lamina papyracea, displacing the medial rectus.
- **Cavernous sinus thrombosis**: enlarged superior ophthalmic vein with a filling defect.
- **Blowout fracture**: look for muscle entrapment; a child's trapdoor fracture may look almost normal. **Thyroid eye disease** spares the tendons.

1 CHAPTER ONE The study

Which scan, and why.

1.1 CT and MRI

Contrast-enhanced CT with thin bone images is the first test for orbital infection, for trauma (non-contrast, thin bone algorithm in three planes) and for any process involving the sinus walls. **MRI** with fat suppression (STIR or fat-suppressed T2, and fat-suppressed post-gadolinium T1) is best for the optic nerve, the orbital apex, the cavernous sinus, tumours and intracranial spread. Fat suppression matters more here than anywhere: orbital fat is bright on T1 and hides enhancing tissue unless it is suppressed.

2 CHAPTER TWO The compartments

Septum, cone and apex.

2.1 Preseptal and postseptal

The **orbital septum**, a fibrous sheet from the orbital rim to the tarsal plates, divides the eyelids and soft tissues in front (**preseptal**) from the orbit proper behind (**postseptal**). Infection limited to the preseptal tissues swells the lids but leaves the orbital fat clear; postseptal infection streaks the fat and may push the globe forwards.

2.2 Intraconal, conal and extraconal

The four rectus muscles and their fascia form a **cone** from the annulus of Zinn at the apex to the globe. Inside the cone (**intraconal**) are the optic nerve and its sheath, the ophthalmic artery and the superior ophthalmic vein; lesions here (cavernous venous malformation, optic nerve tumours) push the eye straight forwards. **Conal** disease involves the muscles themselves (thyroid eye disease, myositis, lymphoma). The **extraconal** space between the cone and the bony walls holds the lacrimal gland and is where sinus disease arrives first: a subperiosteal abscess or sinonasal tumour displaces the globe away from the involved wall.

For personal study. Follow local protocols and senior advice for patients.

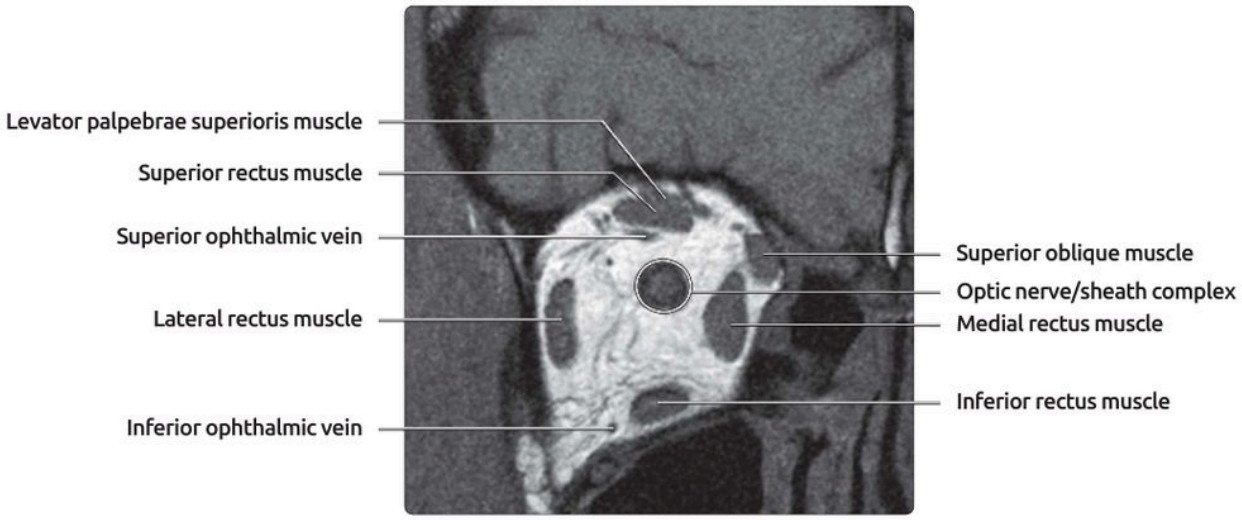


FIGURE 1 Coronal T1 in the mid orbit: the four rectus muscles and the superior oblique form the cone around the optic nerve-sheath complex in bright fat, with the superior ophthalmic vein beneath the superior rectus.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 213

2.3 The bony walls and the apex

The **medial wall** is the paper-thin lamina papyracea over the ethmoid sinuses; the **floor** is the roof of the maxillary antrum, with the infraorbital canal running through it; the **roof** borders the anterior cranial fossa and the frontal sinus. At the apex, the **optic canal** (optic nerve and ophthalmic artery) lies superomedially, the **superior orbital fissure** (III, IV, VI, V1, superior ophthalmic vein) between the greater and lesser wings, and the **inferior orbital fissure** opening to the pterygopalatine and infratemporal fossae. The **nasolacrimal duct** runs down from the lacrimal sac in the medial canthus to the inferior meatus.

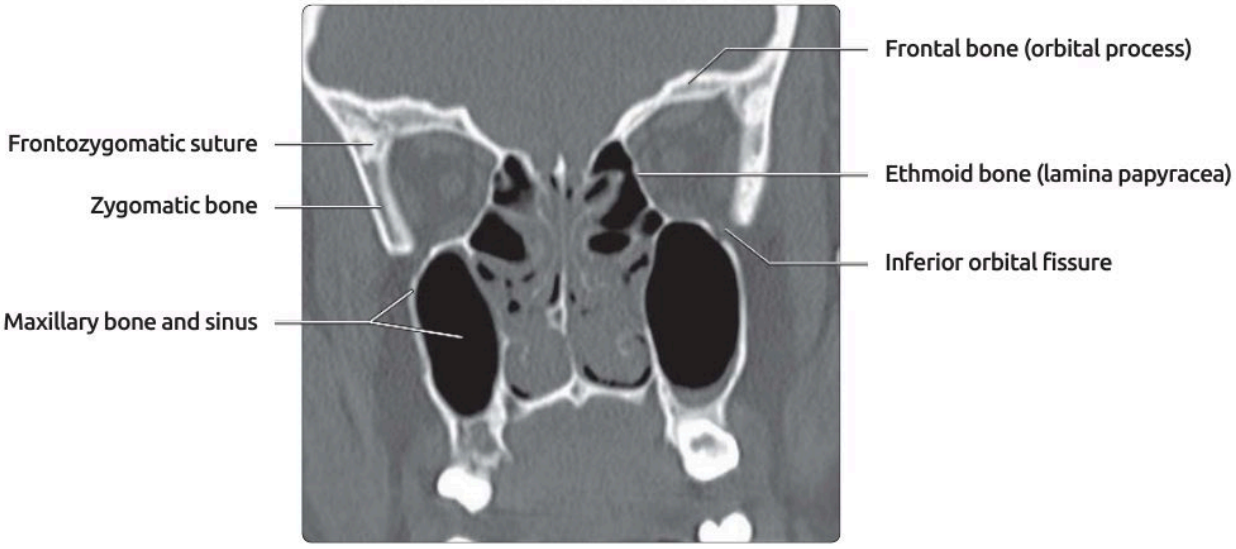


FIGURE 2 Coronal bone CT in the mid orbit: the thin lamina papyracea, the orbital floor and the infraorbital canal. The coronal plane shows wall integrity best.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 219

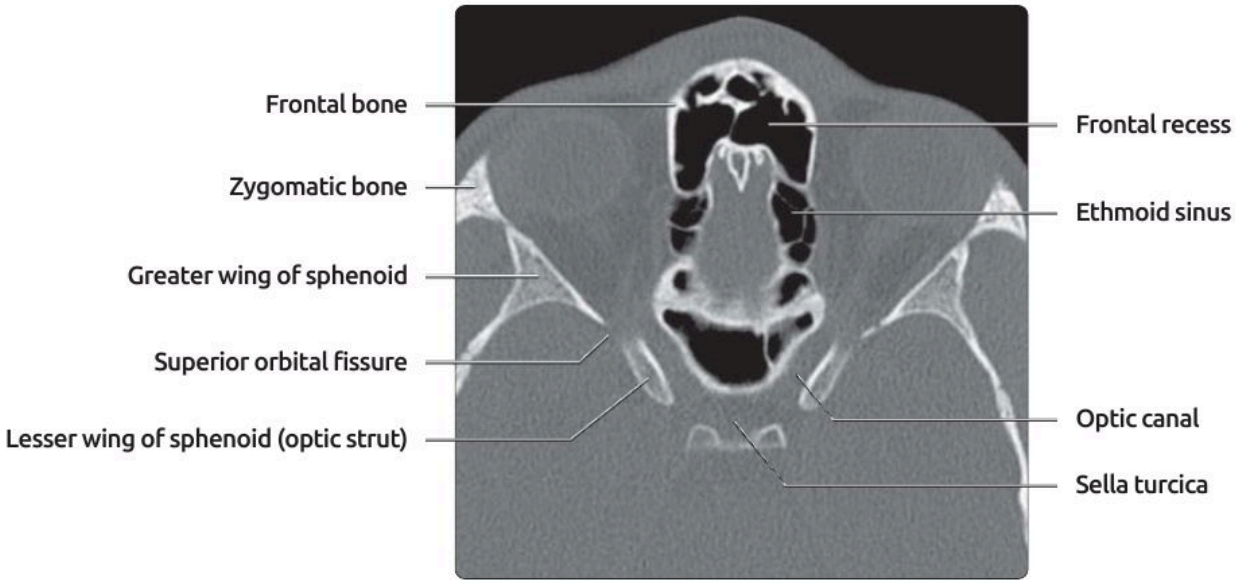


FIGURE 3 Axial bone CT of the upper orbit: the optic canals angling back towards the chiasm, with the superior orbital fissure lateral to them.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 218

3 CHAPTER THREE Infection

The complications of sinusitis.

3.1 Chandler's stages on imaging

ORBITAL COMPLICATIONS OF SINUSITIS

STAGE	IMAGING	WHAT TO LOOK FOR
I, preseptal cellulitis	Swelling and enhancement in front of the septum; orbital fat clear	Usually children; may follow skin infection
II, orbital cellulitis	Stranding of postseptal fat; no collection	Proptosis, restricted movement
III, subperiosteal abscess	Lentiform, rim-enhancing collection along a wall, usually medial; displaced medial rectus	Size, extent backwards, vision
IV, orbital abscess	Rim-enhancing collection within the orbital fat	Needs drainage
V, cavernous sinus thrombosis	Enlarged superior ophthalmic vein with a filling defect; non-enhancing, bulging cavernous sinus	Bilateral signs, cranial nerve palsies

3.2 The subperiosteal abscess

A subperiosteal abscess sits between the bone and the periorbita, usually along the **lamina papyracea** next to opacified ethmoid cells. It is **lentiform**, rim-enhancing, displaces the medial rectus and the globe laterally, and restricts on diffusion. Features that favour surgical drainage include a large or posteriorly extending collection, visual compromise, age over about 10 years, a frontal sinus origin, and failure to improve on antibiotics.

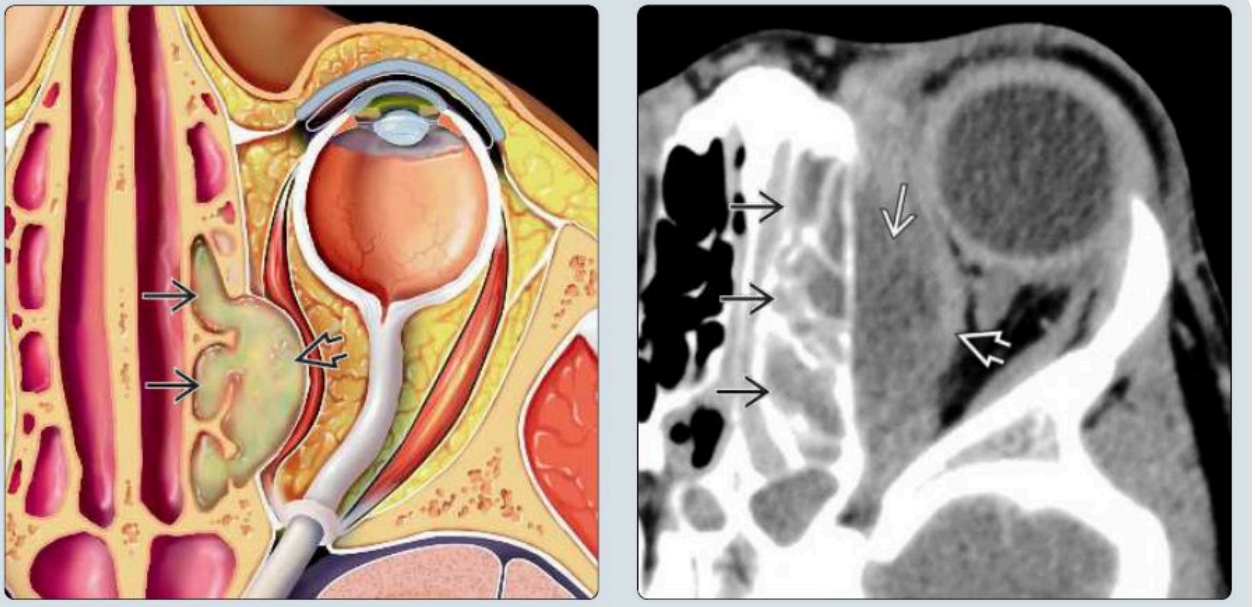


FIGURE 4 Subperiosteal abscess. Left: drawing of ethmoid infection passing through the lamina papyracea to form a collection displacing the cone. Right: contrast CT, opacified ethmoid cells and a large medial extraconal abscess displacing the medial rectus.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 810



FIGURE 5 Left: preseptal cellulitis after facial impetigo, swelling in front of the septum with clear orbital fat and clear sinuses. Right: STIR MRI, preseptal and postseptal cellulitis from ethmoid sinusitis, without an abscess.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 814

3.3 Cavernous sinus thrombosis

Look for an **enlarged superior ophthalmic vein** containing a filling defect, an expanded cavernous sinus with filling defects or poor enhancement, and narrowing of the cavernous carotid. MRI with venography confirms it and shows intracranial complications.



FIGURE 6 Contrast CT in septic cavernous sinus thrombosis: filling defects in both superior ophthalmic veins and the cavernous sinuses, with extension to the basilar venous plexus and superior petrosal sinus.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 235

4 CHAPTER FOUR Trauma

Floors, walls and the optic nerve.

4.1 Blowout fractures

A **pure blowout fracture** breaks the floor or medial wall without involving the rim. Coronal CT shows the fracture, herniation of fat and of the inferior or medial rectus, and the volume of the defect. A rounded, displaced inferior rectus or muscle caught in the fracture, especially the **trapdoor** fracture of a child with little visible on CT but marked restriction and pain on upgaze, is a surgical emergency. Large defects risk enophthalmos.



FIGURE 7 Orbital floor blowout fractures. Left: coronal CT, a mildly depressed floor fracture. Right: an older floor fracture with herniation of fat and part of the inferior rectus through the defect.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 980

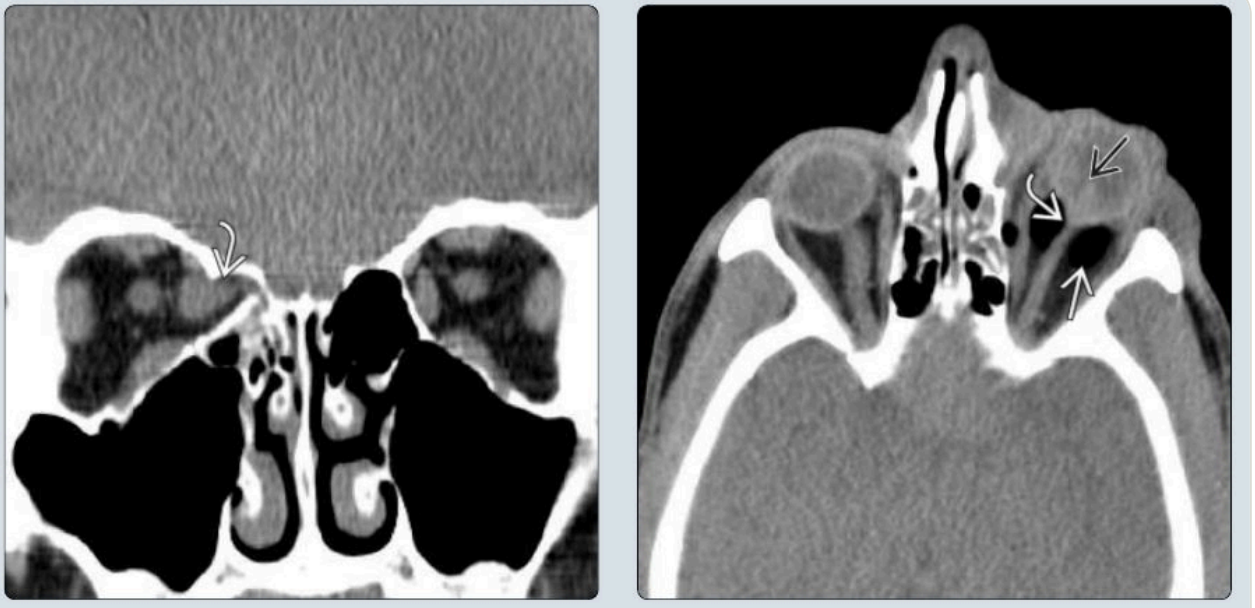


FIGURE 8 Left: a lamina papyracea fracture with the medial rectus deformed into the defect. Right: retrobulbar air and blood with proptosis and tenting of the back of the globe, the warning sign of a compartment syndrome.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 980

4.2 Retrobulbar haemorrhage and the optic canal

Proptosis with **tenting** of the posterior globe (the globe drawn into a cone at the optic nerve insertion) means rising orbital pressure and threatens sight; it is treated clinically by lateral canthotomy and cantholysis without waiting for imaging. In high-energy trauma, look for fractures through the **optic canal** and the anterior clinoid.

CHAPTER FIVE

5 The muscles and masses

Conal and intraconal disease.

5.1 Thyroid eye disease

Thyroid eye disease enlarges the **muscle bellies** while sparing their tendons, usually on both sides, most often the inferior and medial recti (the mnemonic "I'M SLow": inferior, medial, superior, lateral). It also increases orbital fat. Crowding of the enlarged muscles at the **apex** compresses the optic nerve and is an indication for urgent treatment, sometimes endoscopic decompression of the medial wall and floor. STIR signal in the muscles suggests active inflammation.

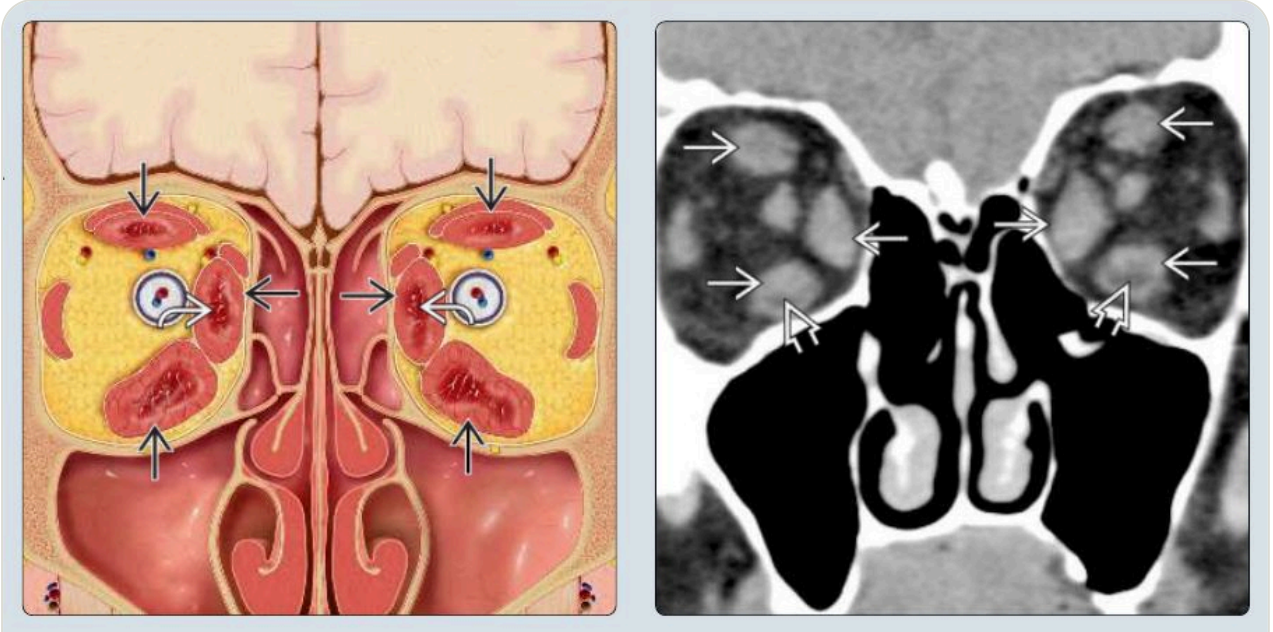


FIGURE 9 Thyroid eye disease. Left: drawing of symmetric muscle enlargement. Right: coronal CT, enlarged inferior, medial and superior recti on both sides.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 822

5.2 Idiopathic orbital inflammation and lymphoma

Idiopathic orbital inflammation (pseudotumour) is usually unilateral and painful, involves the muscle **including its tendon**, the fat, the lacrimal gland or the sclera, and responds to steroids. **Lymphoma** is painless, moulds itself around the globe and orbital structures without destroying them, enhances uniformly and restricts strongly on diffusion; the lacrimal gland is a common site. Biopsy is often needed to separate them.

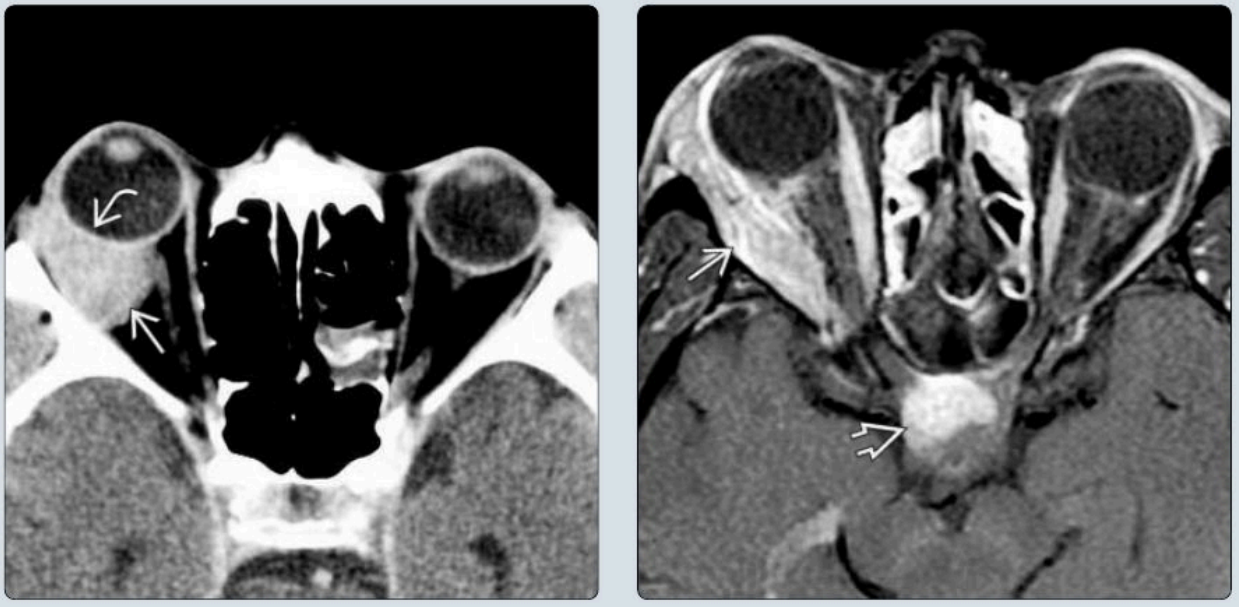


FIGURE 10 Orbital lymphoma. Left: contrast CT, a lobulated retrobulbar mass moulding to the shape of the globe. Right: post-gadolinium T1, uniformly enhancing tumour in the lacrimal gland extending back along the lateral rectus.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 854

5.3 Cavernous venous malformation

The commonest benign orbital mass in adults is the **cavernous venous malformation** (formerly cavernous haemangioma): a well-defined, oval, intraconal mass that is bright on T2 and fills in with contrast progressively from a point. It pushes the globe forwards and seldom threatens vision unless it sits at the apex.

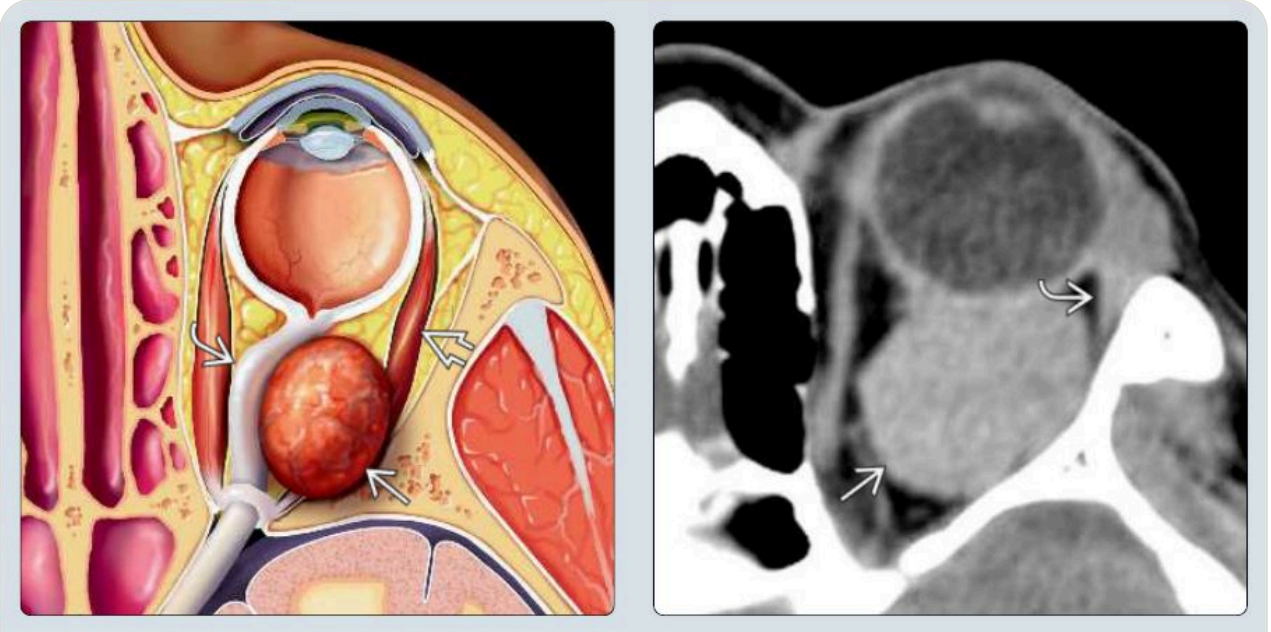


FIGURE 11 Cavernous venous malformation. Left: drawing of an oval intraconal mass displacing the optic nerve. Right: CT, a well-defined, slightly dense intraconal mass with the lateral rectus draped around it.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 804

5.4 Sinonasal tumours and the orbit

For a sinonasal malignancy the key question is whether the **periorbita** has been breached. Thinning or loss of the lamina papyracea alone does not mean invasion; obliteration of the extraconal fat next to the tumour, nodular contact with the medial rectus, or tumour within the muscle does. MRI answers this best, and it often decides between preserving and removing the eye.

READING AN ORBIT SCAN

1. **Septum**
   - → preseptal or postseptal
   - → lids and fat
2. **Walls**
   - → lamina papyracea, floor, roof
   - → fracture, erosion, remodelling

- → adjacent sinuses

3 **Extraconal**

- → collection, mass, lacrimal gland
- → periorbita

4 **Muscles**

- → size, belly or tendon, symmetry
- → entrapment

5 **Intraconal**

- → optic nerve and sheath
- → superior ophthalmic vein
- → mass

6 **Globe and apex**

- → proptosis
- → tenting
- → optic canal
- → superior orbital fissure
- → cavernous sinus

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Compartments:** preseptal and postseptal; intraconal, conal and extraconal.
- **Chandler's stages** and what each looks like on CT.
- **Subperiosteal abscess:** imaging features and indications for drainage.
- **Cavernous sinus thrombosis:** the superior ophthalmic vein sign.
- **Blowout fracture:** muscle entrapment and the trapdoor fracture in children.
- **Thyroid eye disease:** muscle bellies, apical crowding, decompression.
- **Periorbital invasion** by sinonasal tumour.

6.2 Common pitfalls

COMMON PITFALLS

- Calling preseptal cellulitis "orbital" without looking at the fat behind the septum.
- Missing a small medial subperiosteal abscess on a non-contrast scan.
- Reassuring a child with a trapdoor fracture because the CT looks almost normal.
- Calling invasion of the orbit from erosion of the lamina papyracea alone.

Written for this app from Imaging Anatomy: Head and Neck (Chapman and Koontz), orbit chapters, Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), orbit and trauma diagnoses, Dunnebier, Imaging for Otolaryngologists. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 10

The neck by spaces

The neck looks like a crowded jumble on CT until you see it as a set of tubes and compartments wrapped by the three layers of the deep cervical fascia. Each compartment, or **space**, contains a limited set of normal structures, and nearly every lesion in it comes from one of them. Harnsberger's insight, the backbone of *Diagnostic Imaging: Head and Neck*, is that if you can name the space a mass arises in, the differential diagnosis shrinks to a few lines. This lesson teaches the map and the one trick that tells you where a deep mass began.

AT A GLANCE

- Name the **space** first: its contents give the differential.
- Suprahyoid: four spaces (**mucosal, masticator, parotid, carotid**) around the **parapharyngeal fat**; retropharyngeal and perivertebral behind.
- Fat pushed **laterally**: mucosal; **posteriorly**: masticator; **medially**: deep parotid; **anteriorly**: carotid.
- Carotid body paraganglioma **splays** the ICA and ECA; vagal tumours push the ICA **forwards**.
- Cellulitis, phlegmon and abscess differ on contrast CT; only an abscess is drained. The danger space reaches the mediastinum.

1 CHAPTER ONE The fascia and the spaces

A map of the neck.

1.1 Three layers, many spaces

The **superficial layer** of the deep cervical fascia invests the neck: it surrounds the masticator and parotid spaces above the hyoid and the sternocleidomastoid, trapezius and strap muscles below. The **middle layer** lines the deep surface of the pharyngeal mucosa above and wraps the visceral space (larynx, trachea, thyroid, oesophagus) below. The **deep layer** surrounds the perivertebral space, and a slip of it, the **alar fascia**, separates the retropharyngeal space from the danger space behind it. All three contribute to the **carotid sheath**.

1.2 The suprahyoid neck

Above the hyoid, a pair of fat-filled **parapharyngeal spaces** sits at the centre, surrounded by four key spaces: the **pharyngeal mucosal space** medially, the **masticator space** in front, the **parotid space** laterally and the **carotid space** behind. In the midline behind the pharynx are the **retropharyngeal space** and the **perivertebral space**.

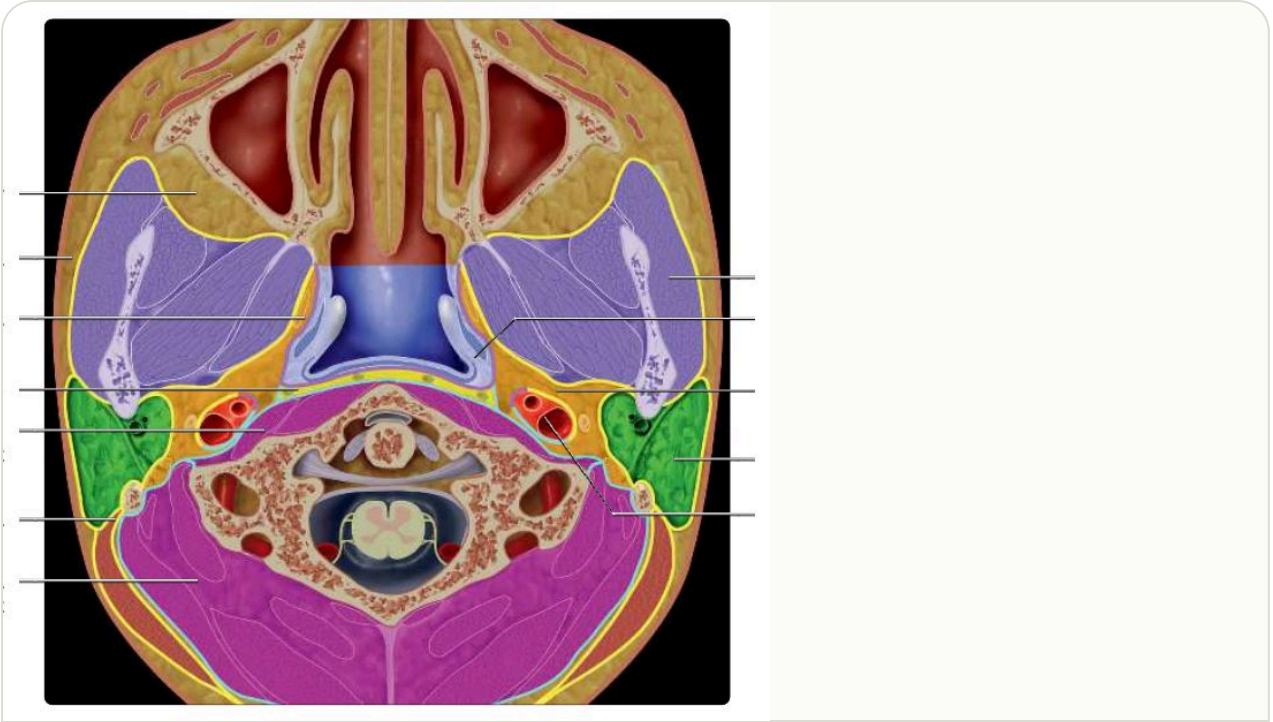


FIGURE 1 The spaces of the suprahyoid neck at the level of the nasopharynx, colour-coded: pharyngeal mucosal space (blue), masticator spaces (purple), parotid spaces (green), carotid spaces with the artery and vein (red), the fat of the parapharyngeal spaces (yellow-orange) in the middle of them, and the perivertebral space (magenta) behind. The fascial layers are drawn as coloured lines.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 6

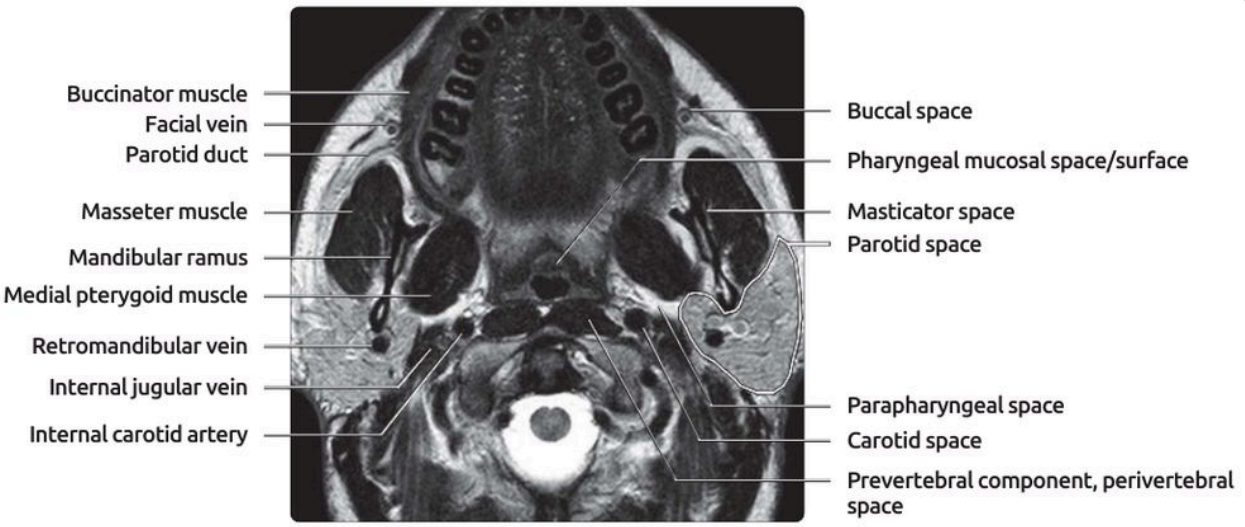


FIGURE 2 The same spaces on axial T2 MRI: the pharyngeal mucosal space, masticator space, parotid space (outlined) and carotid space arranged around the parapharyngeal fat, with the prevertebral muscles behind.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 284

1.3 The infrahyoid neck

Below the hyoid, the **visceral space** in the centre holds the larynx and hypopharynx, the trachea and oesophagus, the thyroid and parathyroid glands, and the recurrent laryngeal nerves. On either side are the **carotid spaces**, behind them the **posterior cervical spaces** (fat, the accessory nerve and level V nodes), and in the midline behind are the retropharyngeal and perivertebral spaces. The **anterior cervical space** is the fat between the carotid space and the visceral space.

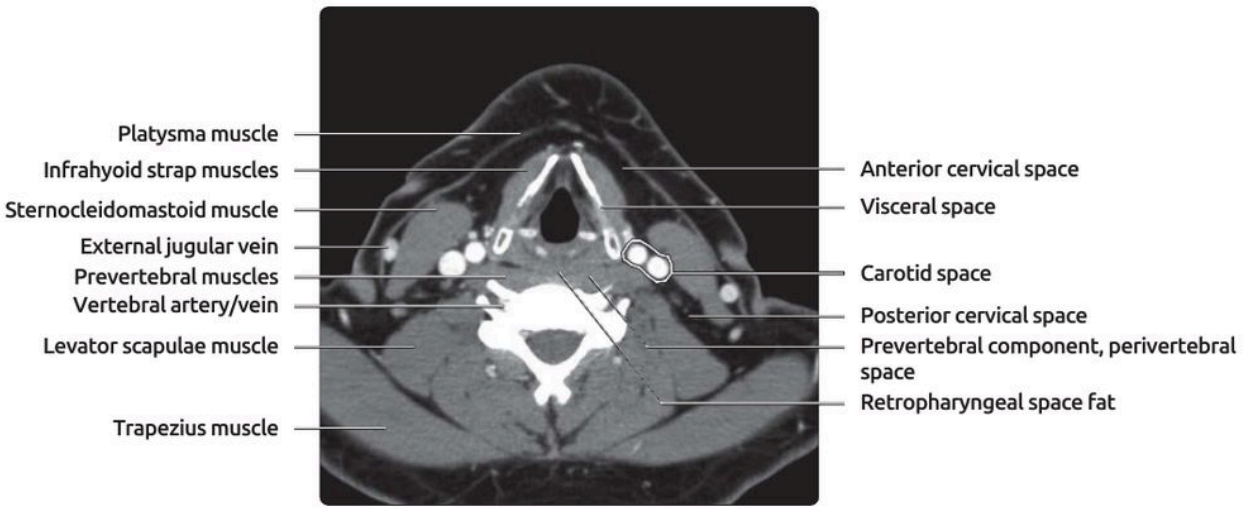


FIGURE 3 Axial contrast CT at the glottis: the visceral space with the larynx and strap muscles, the carotid spaces behind it on either side, the posterior cervical spaces and the perivertebral space.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 279

CHAPTER TWO

2 What each space contains

And therefore what arises there.

THE SPACES, THEIR CONTENTS AND THEIR COMMON LESIONS

SPACE	NORMAL CONTENTS	COMMON LESIONS
Pharyngeal mucosal	Mucosa, lymphoid tissue (adenoids, tonsils), minor salivary glands, constrictor muscles	Squamous cell carcinoma, lymphoma, tonsillar abscess, minor salivary tumour
Parapharyngeal	Fat, occasional minor salivary rests	Rare primary lesions; mostly displaced by its neighbours. Pleomorphic adenoma of salivary rests, lipoma
Masticator	Mandibular ramus and body, muscles of mastication, V3, pterygoid venous plexus	Odontogenic abscess, sarcoma, perineural tumour on V3, benign masseter hypertrophy
Parotid	Parotid gland, facial nerve, intraparotid nodes, retromandibular vein	Pleomorphic adenoma, Warthin tumour, carcinoma, nodes

SPACE	NORMAL CONTENTS	COMMON LESIONS
Carotid	Internal carotid, internal jugular vein, IX to XII (only X below the hyoid), sympathetic chain	Paraganglioma, schwannoma, thrombosis, nodes alongside
Retropharyngeal	Fat; lateral (Rouvière) and medial retropharyngeal nodes above the hyoid	Nodal metastasis, suppurative node, abscess, oedema
Perivertebral	Vertebrae, prevertebral and scalene muscles, brachial plexus, vertebral artery	Discitis and osteomyelitis, metastasis, chordoma, brachial plexus schwannoma
Visceral	Larynx, hypopharynx, trachea, oesophagus, thyroid, parathyroids	Carcinoma of each, thyroid and parathyroid lesions
Posterior cervical	Fat, accessory nerve, level V nodes	Nodes, lymphatic malformation, lipoma, schwannoma

CHAPTER THREE

3 Where did it start?

The parapharyngeal fat tells you.

3.1 The displacement rule

The parapharyngeal space is mostly fat and rarely the origin of disease, so it acts as a marker. A mass growing in one of the four surrounding spaces pushes the fat away from itself:

- a **pharyngeal mucosal** mass pushes the fat **laterally**;
- a **masticator** space mass pushes it **posteriorly** (and medially);
- a **deep lobe parotid** mass pushes it **medially** (and widens the stylomandibular gap);
- a **carotid space** mass pushes it **anteriorly**, and lifts the styloid process forwards;
- a lateral **retropharyngeal** mass pushes it anteriorly without displacing the styloid.

A mass completely surrounded by fat arises in the parapharyngeal space itself, usually from salivary rests.

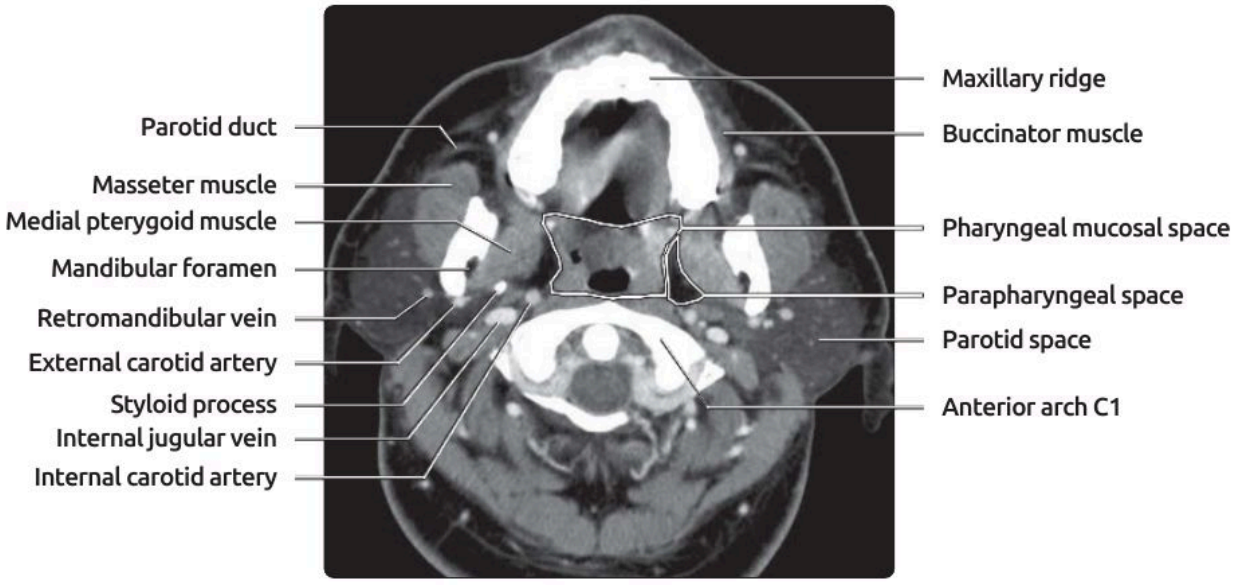


FIGURE 4 Axial contrast CT at the level of the maxillary ridge: the pharyngeal mucosal space outlined between the paired parapharyngeal fat pads, with the masticator spaces in front and the carotid spaces behind.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 277



FIGURE 5 A true parapharyngeal space pleomorphic adenoma. Left: CT, a well-defined mass completely surrounded by parapharyngeal fat. Right: T1 MRI, the mass separate from the deep lobe of the parotid, with a rim of fat all round.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 14

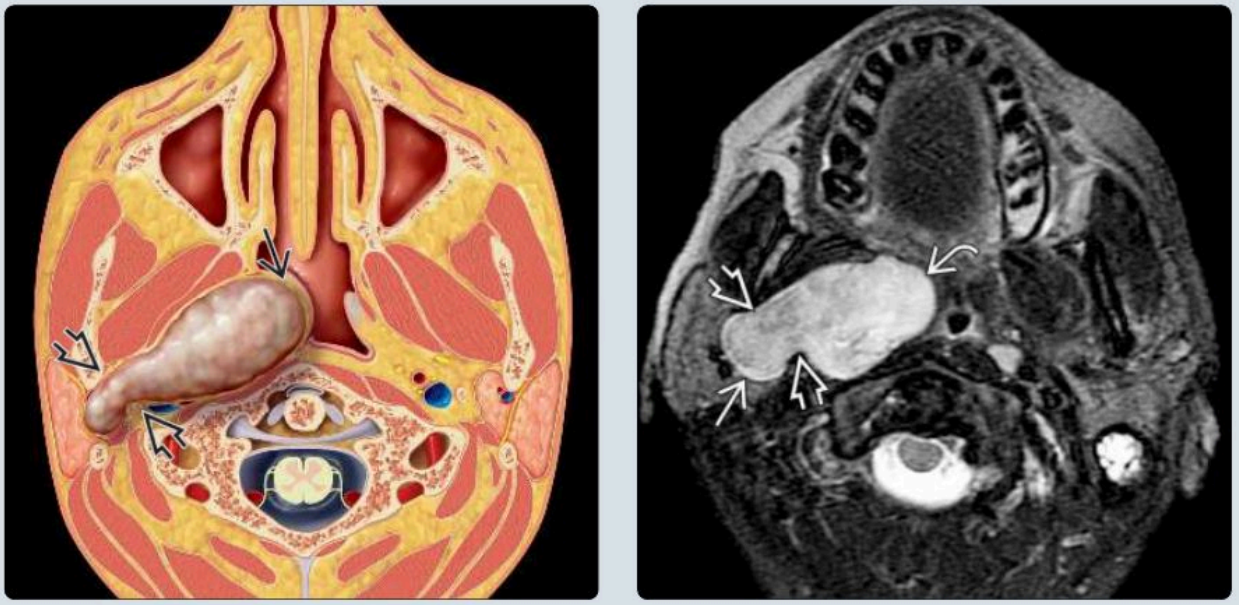


FIGURE 6 A deep lobe parotid pleomorphic adenoma. Left: drawing of a pear-shaped tumour pushing the parapharyngeal fat medially. Right: T2, the bright tumour passing through a widened stylomandibular notch and displacing the fat and the tonsil medially.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 88

3.2 The carotid space and the vessels

In the carotid space, the position of the vessels refines the diagnosis. A **carotid body paraganglioma** sits in the bifurcation and **splays** the internal and external carotid arteries apart. A **vagal** paraganglioma or schwannoma, higher up, pushes the internal carotid **anteriorly** (and medially). A **sympathetic** schwannoma, medial to the vessels, separates neither. Paragangliomas enhance intensely and show flow voids ("salt and pepper"); schwannomas enhance less and may be cystic.

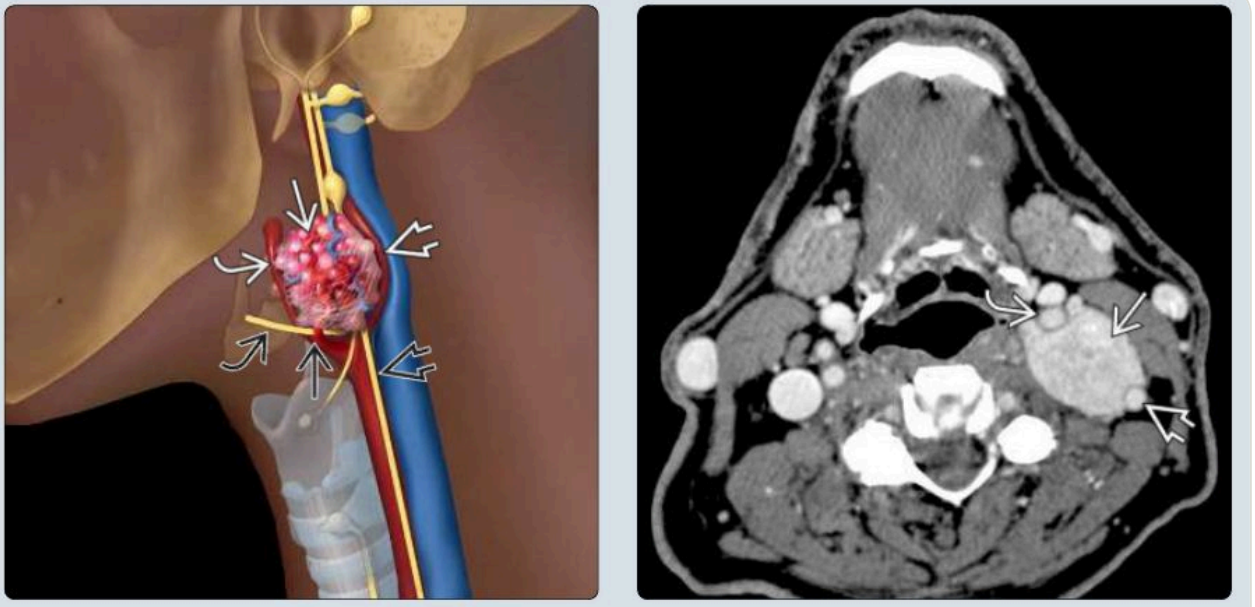


FIGURE 7 Carotid body paraganglioma. Left: drawing of the tumour in the bifurcation, splaying the internal and external carotids. Right: contrast CT, an avidly enhancing mass sitting in the notch between the two arteries.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 140

4 CHAPTER FOUR Spaces as pathways

How disease spreads.

4.1 Infection

The spaces also explain how infection travels. A **peritonsillar abscess** can rupture through the superior constrictor into the parapharyngeal fat and on to the masticator or carotid spaces. Dental infection enters the masticator, submandibular and sublingual spaces. The **retropharyngeal space** extends down to about T1–T6, and the **danger space** behind it to the diaphragm, so infection there threatens the mediastinum. A suppurative retropharyngeal node in a child (a rim-enhancing node, often lateral) is not the same as a true retropharyngeal abscess (fluid distending the space from side to side, often with mass effect on the airway).

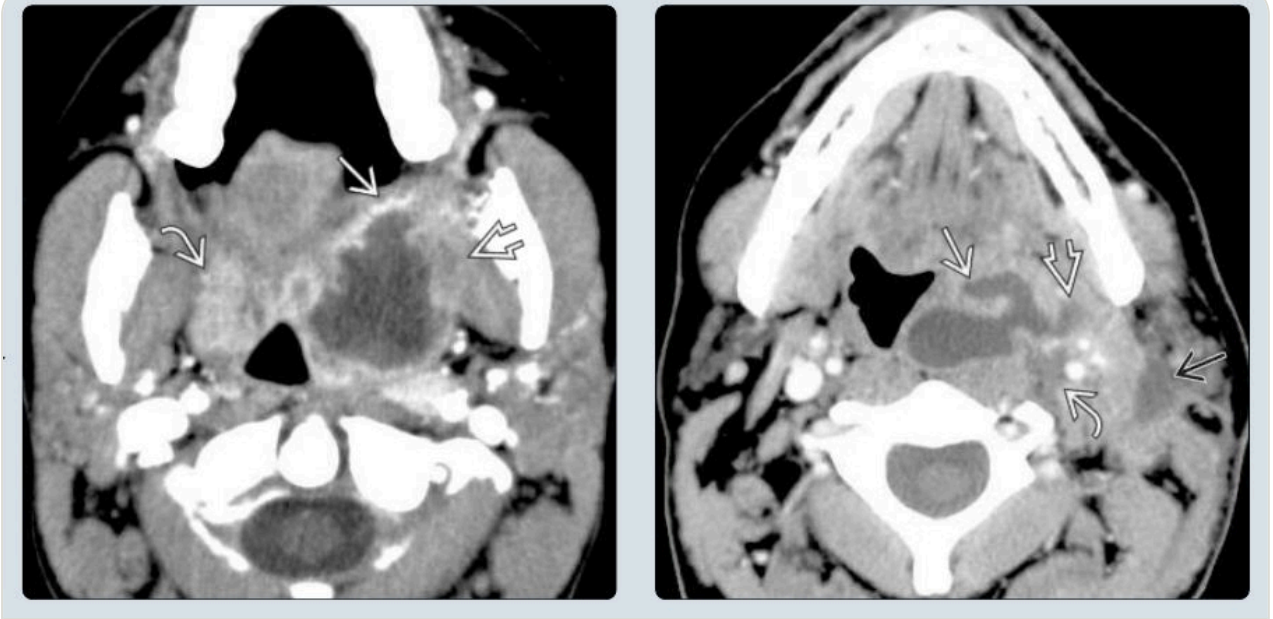


FIGURE 8 Tonsillar abscess complications. Left: a tonsillar abscess extending through the capsule into the buccal space and the medial pterygoid. Right: a large abscess that has ruptured into the carotid space and the submandibular space.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 28

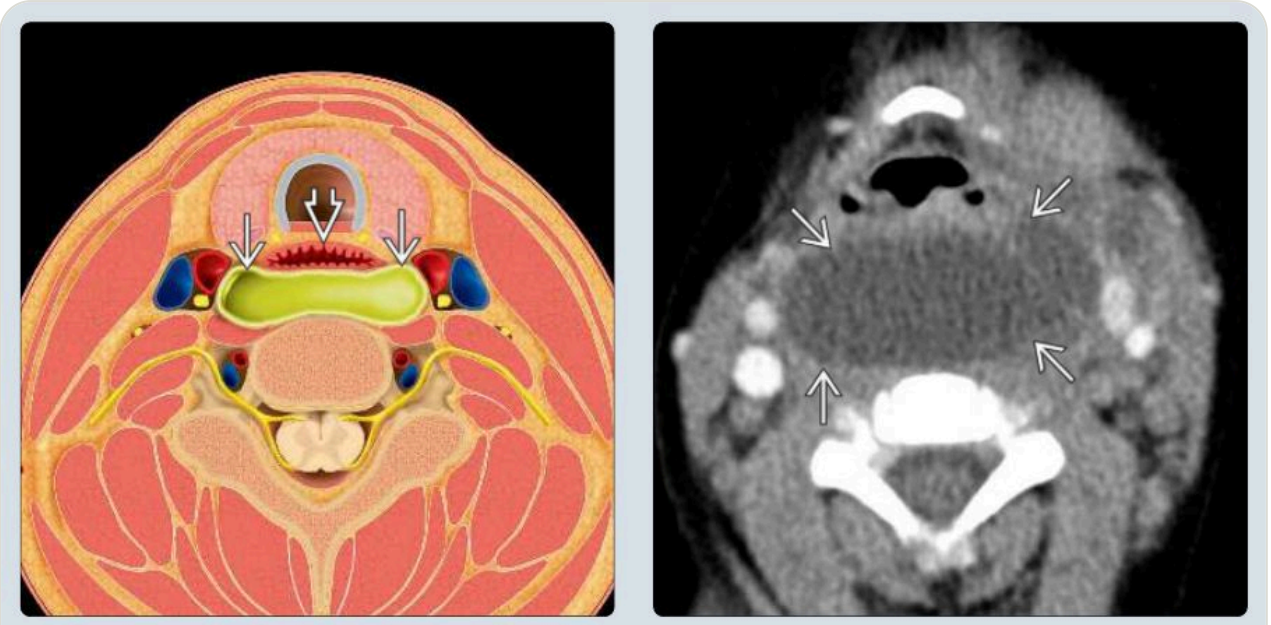


FIGURE 9 Retropharyngeal abscess. Left: drawing of the collection behind the pharynx. Right: contrast CT in an infant, a low-density collection distending the retropharyngeal space and pushing the pharynx forwards.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 168

4.2 Abscess, phlegmon or cellulitis?

On contrast CT, **cellulitis** is stranding of the fat; **phlegmon** is ill-defined, enhancing soft tissue without a fluid centre; an **abscess** is a fluid collection with a well-formed enhancing rim. Only a mature abscess is drained. Look also for the airway, the internal jugular vein (thrombosis, Lemierre syndrome), the carotid, the mediastinum and the cause (tooth, stone, tonsil, foreign body).

PLACING A DEEP NECK MASS

1. Above or below the hyoid
   - → **suprahyoid** — find the parapharyngeal fat
   - → **infrahyoid** — visceral, carotid, posterior cervical
2. Which way is the fat pushed
   - → **laterally** — mucosal
   - → **posteriorly** — masticator
   - → **medially** — deep parotid
   - → **anteriorly** — carotid or retropharyngeal
   - → **surrounded** — parapharyngeal itself
3. What normally lives there
   - → match to the contents table
4. Look at the vessels
   - → **splayed** — carotid body

→ **ICA pushed forwards**
vagal

→ **stylomandibular gap widened**
deep parotid

5 Character

→ cyst, fat, enhancement, flow voids, bone

6 Differential

→ two to four diagnoses by space and character

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The three layers** of deep cervical fascia and what they wrap.
- **The suprahyoid spaces** around the parapharyngeal fat, and the **infrahyoid spaces**.
- **The displacement rule** for the parapharyngeal fat.
- **Contents of each space** and the lesions that arise there.
- **Carotid space masses:** carotid body versus vagal paraganglioma versus schwannoma.
- **Danger space** and mediastinal spread; abscess versus phlegmon.

5.2 Common pitfalls

COMMON PITFALLS

- Describing a deep neck mass without naming its space.

- Calling a deep lobe parotid tumour "parapharyngeal" when the fat is pushed medially.
- Draining a phlegmon.
- Calling a suppurative retropharyngeal node a retropharyngeal abscess.

Written for this app from Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), suprahyoid and infrahyoid neck overview, Imaging Anatomy: Head and Neck (Chapman and Koontz), neck spaces chapters, ESR Modern Radiology, Head and Neck Imaging. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 11

The larynx and hypopharynx

The endoscope shows the surface of the larynx and hypopharynx; imaging shows what lies beneath it. The two are complementary, and for laryngeal and hypopharyngeal cancer the stage depends on things only a scan can see: the fat of the pre-epiglottic and paraglottic spaces, the cartilages, the tissues outside the larynx and the nodes. This lesson teaches you to find the level of each slice, to know the normal thickness of each wall, and to read the scan as a surgeon planning between voice-preserving treatment and laryngectomy.

AT A GLANCE

- Contrast CT in **quiet breathing** plus a **breath-hold or phonation** pass; MRI when cartilage invasion is uncertain.
- False cords have **fat** beneath them; true cords have **muscle**, the arytenoids and the top of the cricoid.
- Commissures and subglottic mucosa **≤1 mm**; postcricoid soft tissue **≤1 cm**.
- Pre-epiglottic or paraglottic invasion is at least **T3**; inner cortex erosion T3, through the outer cortex **T4a**.
- Cord paralysis: medially rotated arytenoid, dilated ventricle and pyriform sinus. Scan from skull base to the **aortopulmonary window**.

1 CHAPTER ONE The study

Fast, thin and contrast-enhanced.

1.1 Protocol

Contrast-enhanced CT is the standard: thin slices (about 2.5 to 3 mm or thinner) from the skull base or mandible to the clavicles, during **quiet breathing** so the cords are abducted and the anterior commissure is open, with axial images angled along the plane of the cords and coronal and sagittal reformats. A second pass during a **breath hold, phonation or modified Valsalva** adducts the cords and opens the pyriform sinuses, showing whether a cord moves and whether a pyriform tumour is real or just collapsed mucosa. **MRI** is slower and degraded by swallowing, but it is

more sensitive for **cartilage invasion** and helps when CT is equivocal. PET-CT helps for nodal disease, second primaries and recurrence after treatment.

2 CHAPTER TWO Finding your level

Which slice am I on?

2.1 The landmarks

Read a laryngeal CT from the hyoid down, naming each level from its landmarks.

- **Supraglottis, upper:** the free epiglottis, the valleculae in front of it, and the C-shaped, fat-filled **pre-epiglottic space** behind the hyoid.
- **Supraglottis, lower (false cords):** the aryepiglottic folds, the pyriform sinuses behind them, and **fat** in the paraglottic space beneath the false cords.
- **Glottis (true cords):** the **arytenoid cartilages** with their vocal processes, the top of the **cricoid lamina**, and **muscle** (thyroarytenoid) rather than fat beneath the cords. The anterior commissure, where the cords meet, should be thin.
- **Subglottis:** the complete **cricoid ring** with no arytenoids, and a thin rim of mucosa inside it.

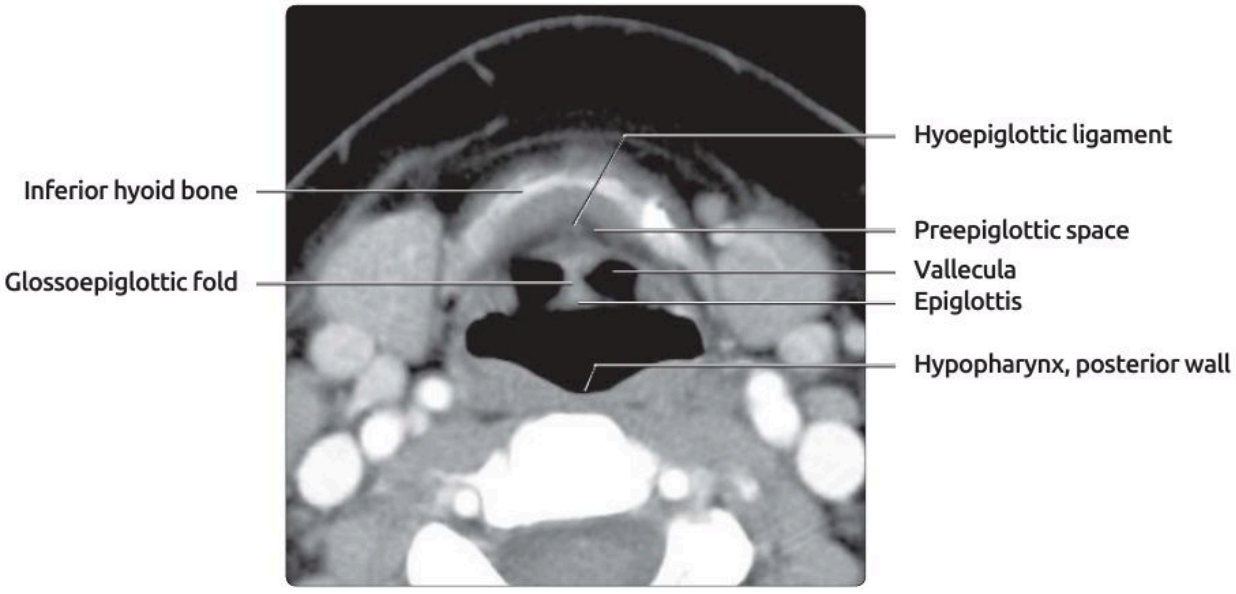


FIGURE 1 Axial contrast CT of the high supraglottis: the C-shaped pre-epiglottic space filled with fat, a common place for tumour to hide.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 379

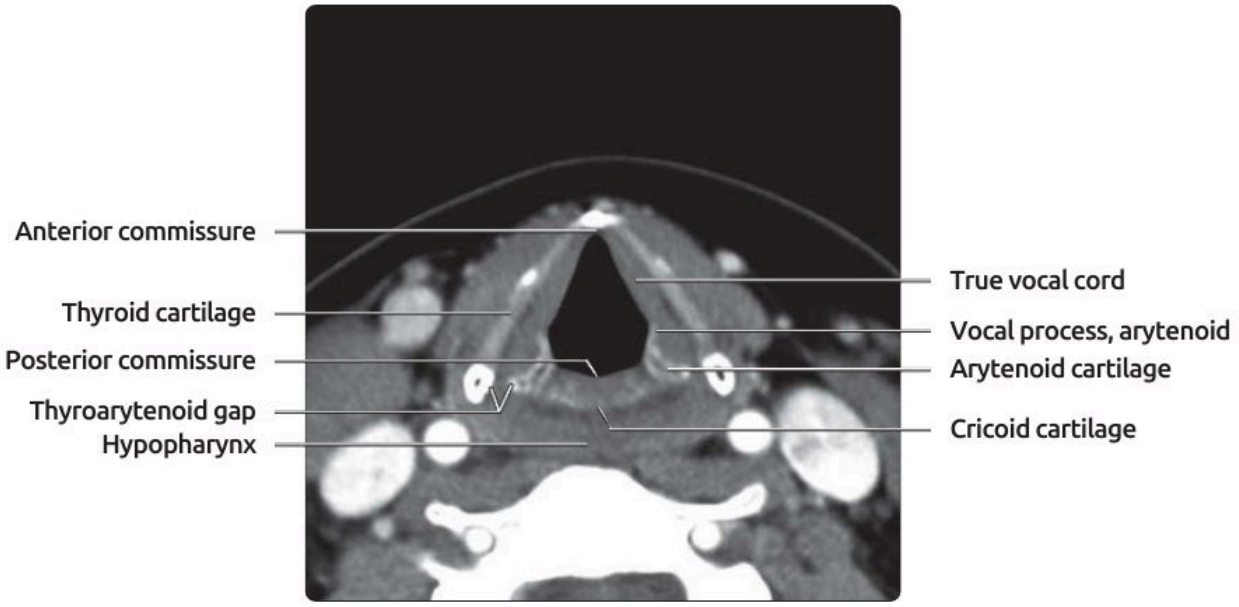


FIGURE 2 The glottic level in quiet breathing: the true cords abducted, the vocal processes of the arytenoids, the thin anterior and posterior commissures, and the cricoid lamina behind.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 380

2.2 Normal thickness

Some measurements are worth remembering. The mucosa of the **anterior commissure**, the **posterior commissure** and the **subglottis** is normally 1 mm or less; any visible soft tissue there in a patient with a laryngeal tumour suggests spread. The soft tissue between the cricoid and the spine in the **postcricoid** region (postcricoid hypopharynx and posterior hypopharyngeal wall together) is normally no more than about 1 cm.

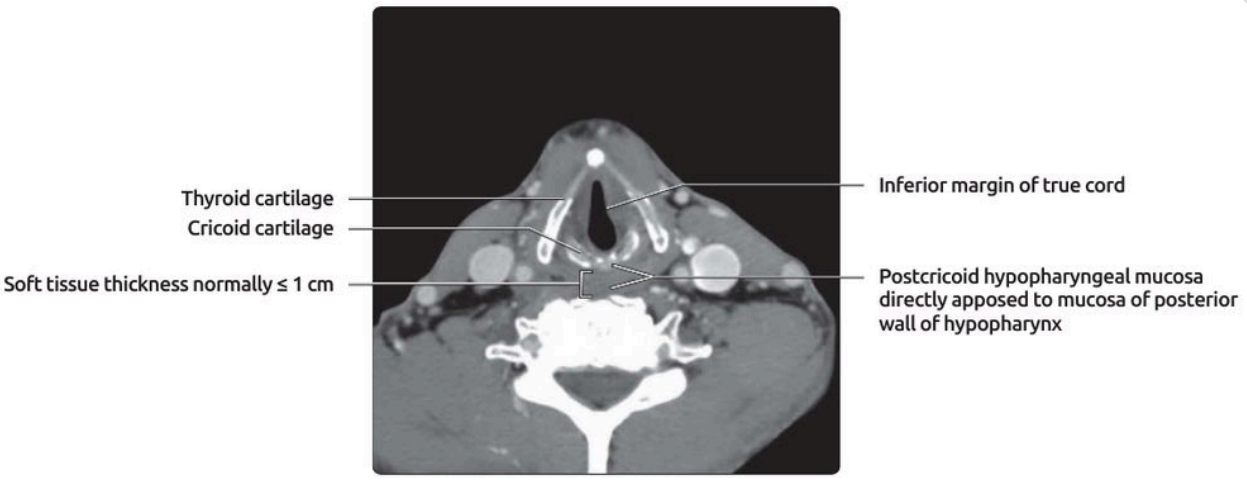


FIGURE 3 Axial contrast CT at the upper cricoid: the inferior margin of the true cords in front, and the postcricoid hypopharyngeal mucosa against the posterior pharyngeal wall behind. The tissue between cricoid and spine is normally 1 cm or less.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 400

CHAPTER THREE

3 The deep spaces and the cartilages

The clinical blind spots.

3.1 Pre-epiglottic and paraglottic spaces

The **pre-epiglottic space** (in front of the epiglottis) and the **paraglottic spaces** (lateral to the ventricle and cords) are continuous fat-filled compartments with no fascia between them. Tumour can spread through them beneath intact mucosa, from the epiglottis to the false cord and down to the glottis, crossing the ventricle (**transglottic** spread). Invasion of either space makes a supraglottic or glottic cancer at least **T3**. On CT, look for tumour replacing the fat; on MRI, for intermediate signal replacing bright T1 fat.

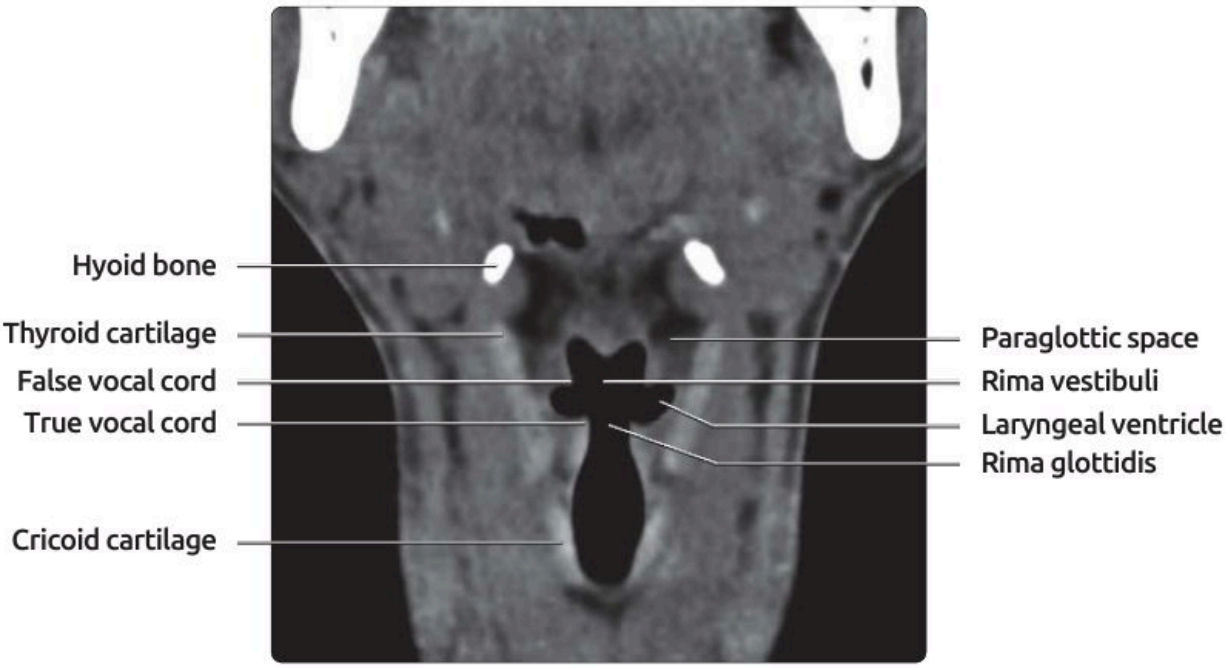


FIGURE 4 Coronal CT: the laryngeal ventricle, the air space between the false and true cords. A tumour that crosses it is transglottic.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 384

3.2 Cartilage

The thyroid, cricoid and arytenoid cartilages ossify irregularly with age, and unossified cartilage looks like soft tissue, so cartilage invasion is one of the hardest calls in head and neck imaging. **Sclerosis** next to a tumour is non-specific (it may be reactive); **erosion or lysis** is more specific; tumour **outside** the cartilage in the strap muscles or the thyroid is definite. On MRI, cartilage that is bright on T2 and enhances like the tumour suggests invasion, while much brighter T2 signal than tumour suggests inflammation. The distinction matters: erosion of the inner cortex makes a tumour T3, extension through the outer cortex makes it T4a, and full-thickness invasion usually means laryngectomy rather than voice-preserving treatment.

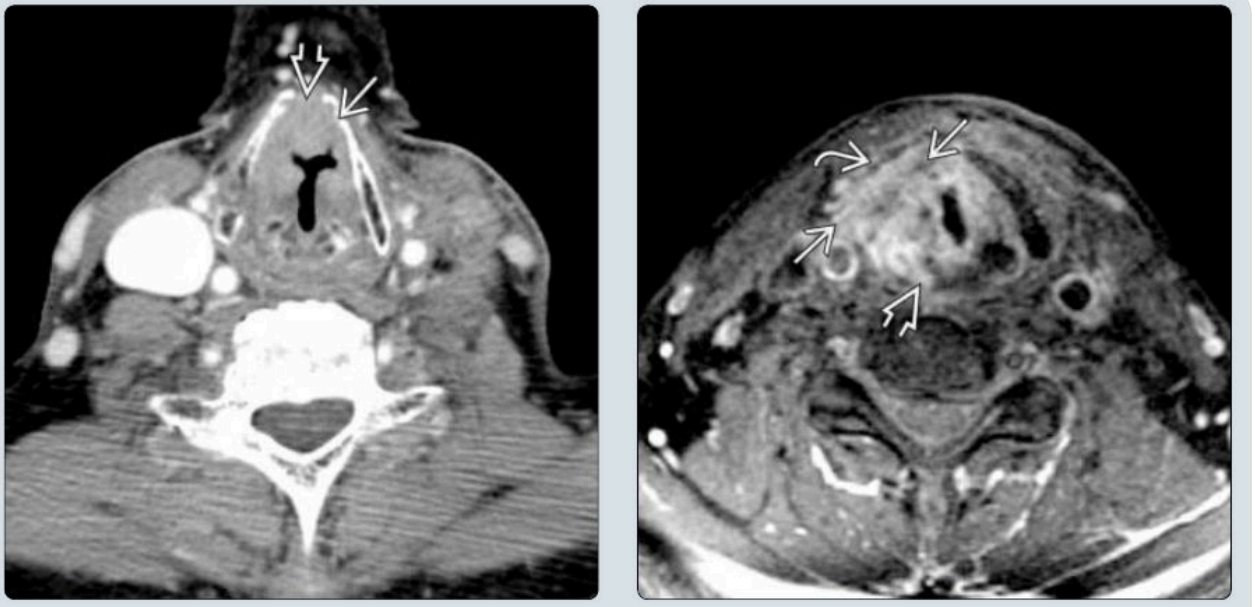


FIGURE 5 Glottic carcinoma and cartilage. Left: a bulky anterior tumour across both cords and the anterior commissure, with erosion of the inner cortex of the thyroid cartilage. Right: post-gadolinium MRI of recurrent tumour invading the cricoid and penetrating the thyroid cartilage.
Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 530

4 CHAPTER FOUR Cancer by site

What to report.

4.1 Supraglottic, glottic and subglottic

Supraglottic cancers have a rich lymphatic supply and often present with nodes, frequently on both sides. Look at the pre-epiglottic and paraglottic fat, the vallecula and tongue base, the aryepiglottic fold and pyriform sinus, and the cartilage. **Glottic** cancers seldom spread to nodes early; report the anterior commissure (and spread across it to the other cord), the posterior commissure, subglottic extension, the paraglottic space, cord movement on the breath-hold images, and the cartilage. **Subglottic** cancers are rare, spread to paratracheal nodes and are often advanced at presentation.

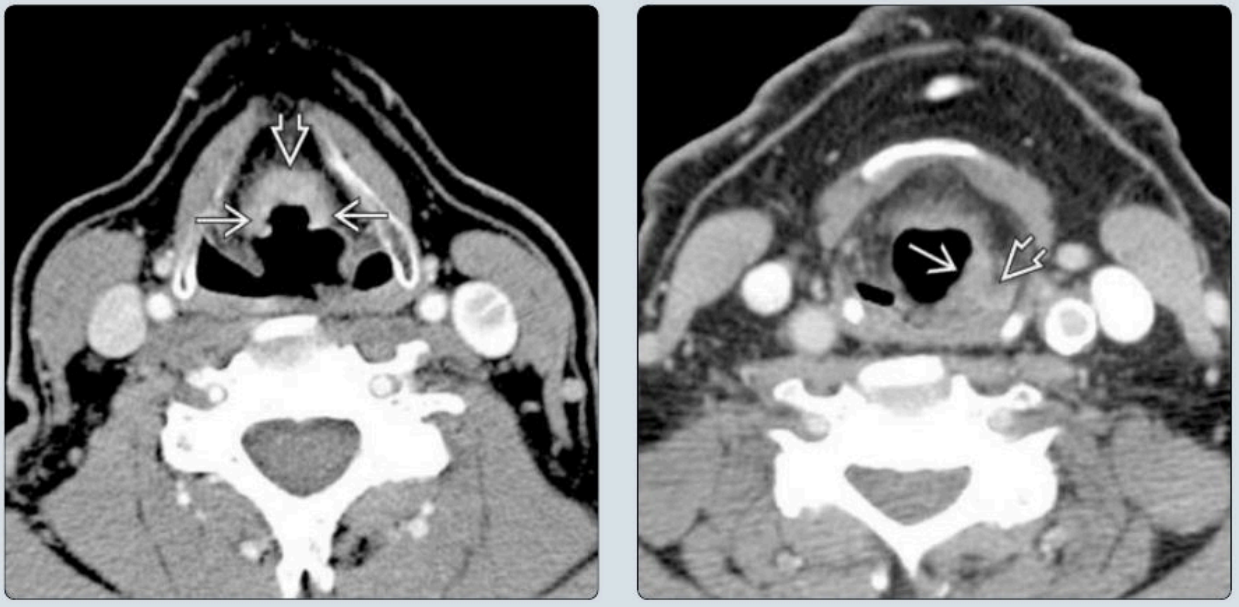


FIGURE 6 Supraglottic carcinoma. Left: a small epiglottic tumour staged T3 because it invades the pre-epiglottic fat; the epiglottis should never be this thick. Right: a small aryepiglottic fold tumour with a collapsed pyriform sinus but no hypopharyngeal tumour.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 526

4.2 Hypopharynx

The hypopharynx has three subsites: the **pyriform sinuses**, the **posterior pharyngeal wall** and the **postcricoid** region. The **pyriform apex** reaches down to the level of the cricoid and is a clinical blind spot, a common hiding place for the primary when a patient presents with a neck node. Hypopharyngeal cancers spread submucosally, invade the thyroid and cricoid cartilages, the thyroid gland and the cervical oesophagus, and present with nodes. Report the lowest extent of tumour (oesophageal involvement changes the operation) and the relation to the prevertebral fascia (invasion makes it unresectable).

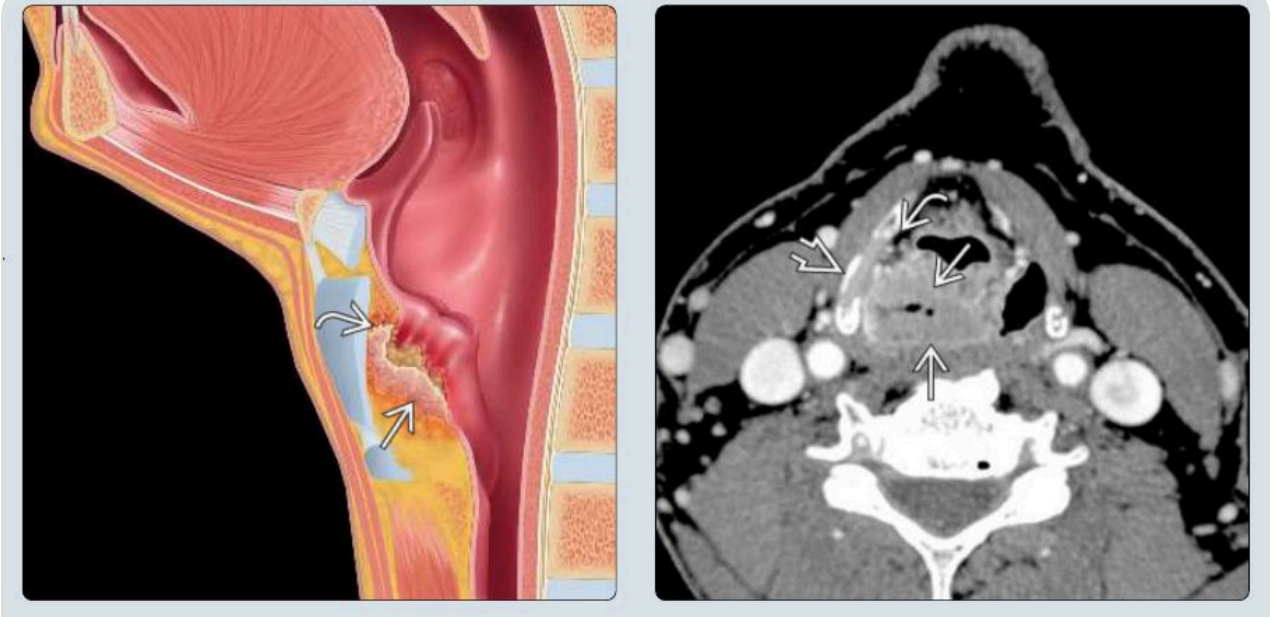


FIGURE 7 Pyriform sinus carcinoma. Left: drawing of the tumour in the pyriform sinus. Right: contrast CT, a mass filling the right pyriform sinus and spreading along all its walls.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 518

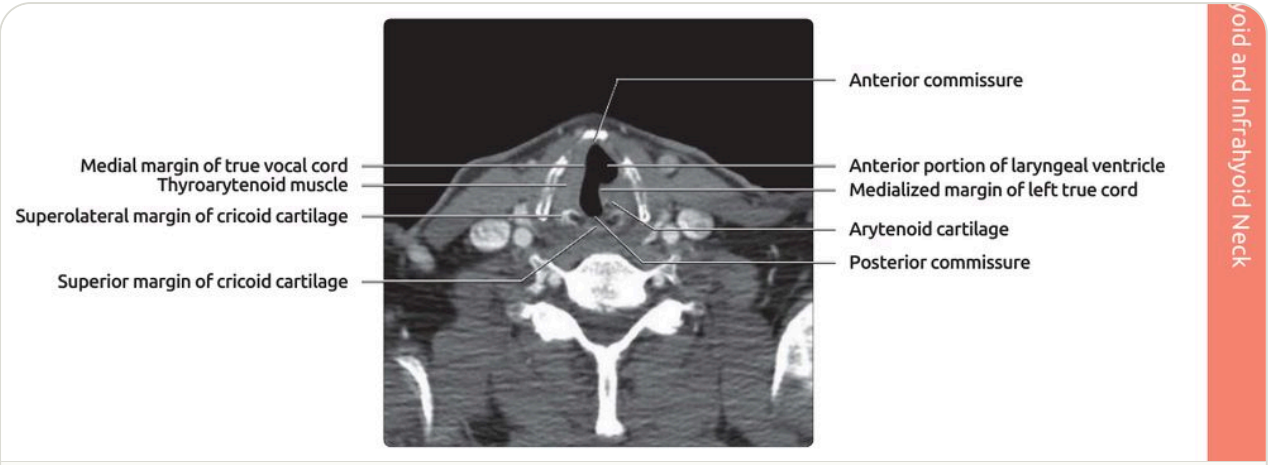


FIGURE 8 Axial CT in left vocal cord paralysis: the left arytenoid is rotated medially; the left pyriform sinus and ventricle are widened.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 389

5 CHAPTER FIVE
Other common questions

Palsy, laryngocele, trauma, stenosis.

5.1 Vocal cord paralysis

A paralysed cord produces a recognisable picture on CT: the ipsilateral **arytenoid rotates medially**, the **ventricle** and **pyriform sinus** dilate, the **aryepiglottic fold** thickens and moves medially, and the posterior cricoarytenoid muscle atrophies. The search for the cause follows the nerve: from the **skull base** (jugular foramen) down the carotid space, through the thyroid bed, to the **aortopulmonary window** on the left and the subclavian on the right. A scan that stops at the clavicles misses a left-sided cause in the chest.

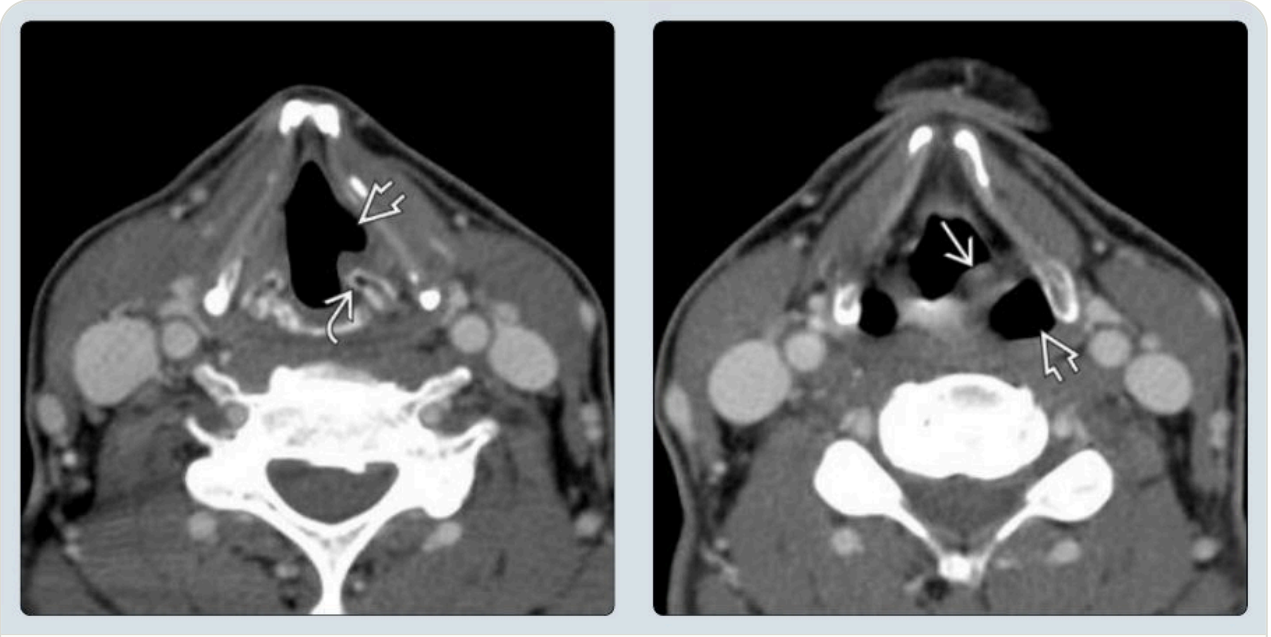


FIGURE 9 Left vocal cord paralysis. Left: a medially rotated left arytenoid with a dilated ventricle. Right: an enlarged left pyriform sinus and a thickened, medially placed aryepiglottic fold.
Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 296

5.2 Laryngocele

A **laryngocele** is a dilated saccule of the ventricle, filled with air or fluid. An **internal** one stays within the paraglottic space; an **external** or mixed one passes through the thyrohyoid membrane into the neck. Because a carcinoma at the ventricle can

block the saccule, every laryngocele in an adult needs the ventricle examined for tumour.



FIGURE 10 Laryngocele. Left: drawing of a laryngocele squeezing through the thyrohyoid membrane into the neck. Right: coronal CT of a mixed laryngocele with internal and external components.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 292

5.3 Trauma and stenosis

In **laryngeal trauma**, thin non-contrast CT shows fractures of the thyroid and cricoid cartilages, arytenoid dislocation, haematoma and air; a displaced fracture or one involving the cricoid usually needs repair. **Subglottic and tracheal stenosis** after intubation or tracheostomy is measured on CT with sagittal and coronal reformats: its level below the cords, its length and its narrowest diameter.

READING A LARYNGEAL AND HYPOPHARYNGEAL CT

1 **Level each slice**

- → hyoid and pre-epiglottic fat
- → false cords and paraglottic fat
- → true cords and arytenoids
- → cricoid ring

2 **Mucosal surfaces**

- → epiglottis, aryepiglottic folds, false and true cords
- → commissures ≤1 mm
- → subglottis ≤1 mm

3 **Deep spaces**

- → pre-epiglottic and paraglottic fat
- → transglottic spread

4 **Cartilage**

- → sclerosis, erosion, lysis
- → tumour outside the larynx

5 **Hypopharynx**

- → pyriform sinuses to the apex
- → posterior wall
- → postcricoid ≤1 cm
- → oesophagus below

6 **Movement**

- → cords on the breath-hold images
- → signs of paralysis

7 **Beyond**

- → thyroid gland, strap muscles, prevertebral fascia, carotid
- → nodes both sides

CHAPTER SIX

6 For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Protocol:** contrast CT in quiet breathing plus a breath-hold or phonation pass; MRI for cartilage.
- **Levels:** how to tell false cord from true cord on axial CT.
- **Pre-epiglottic and paraglottic spaces** as blind spots, and their effect on T stage.
- **Cartilage invasion:** sclerosis versus erosion versus extralaryngeal spread.
- **Normal thickness:** commissures and subglottis ≤1 mm, postcricoid ≤1 cm.
- **Pyriform apex** and the unknown primary.
- **Vocal cord paralysis** signs, and scanning to the aortopulmonary window.

6.2 Common pitfalls

COMMON PITFALLS

- Calling a collapsed pyriform sinus a tumour without the breath-hold images.
- Calling cartilage invasion from sclerosis alone, or from unossified cartilage.

- Missing tumour in the pre-epiglottic fat under a normal-looking epiglottis.
- Stopping a scan for left cord palsy at the clavicles.
- Missing a ventricular carcinoma behind a laryngocele.

Written for this app from Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), hypopharynx, larynx and trachea overview, Imaging Anatomy: Head and Neck (Chapman and Koontz), larynx and hypopharynx, ESR Modern Radiology, Head and Neck Imaging. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 12

The oral cavity and oropharynx

You can see and feel most of the oral cavity and oropharynx, so the scan is rarely needed to find a lesion. It is needed to measure what you cannot feel: how deep a tongue cancer goes, whether it crosses the midline or reaches the mandible, whether a floor of mouth infection has spread below the mylohyoid, and where a cystic neck node in a young adult has come from. This lesson covers the four areas of the oral cavity, the oropharynx, and the questions each scan should answer.

AT A GLANCE

- **MRI** for oral and oropharyngeal cancer (less amalgam, depth, marrow, nerves); **contrast CT** for infection.
- The **mylohyoid** divides the **sublingual** space above from the **submandibular** space below.
- Tongue cancer: **depth of invasion**, the **lingual septum**, both **neurovascular bundles**.
- Mandible: cortex on **CT**, marrow on **MRI**; follow the inferior alveolar canal.
- A **cystic level II node** in an adult is metastasis (often HPV oropharynx) until proven otherwise.

1 CHAPTER ONE The study

MRI for tumour, CT for infection.

1.1 Choosing the test

MRI with fat-suppressed post-gadolinium T1 and T2 is the best study for oral and oropharyngeal cancer: it is far less affected by **dental amalgam** than CT, shows the depth of a tongue tumour and the marrow of the mandible, and detects perineural spread. **Contrast CT** is quicker, images the neck and chest in one pass, and is the first test for **infection**, with bone windows of the mandible to find the tooth, the stone or the osteomyelitis responsible. Puffed-cheek or open-mouth techniques separate the cheek, gum and tongue surfaces. CT or cone-beam CT shows the cortex of the mandible; MRI shows its marrow.

2 CHAPTER TWO The anatomy

Four areas of the oral cavity.

2.1 Mucosa, sublingual space, submandibular space and root of tongue

The oral cavity is divided by the **mylohyoid muscle**, a sling from one side of the mandible to the other, into two compartments.

- The **oral mucosal surface**: the lips, buccal mucosa, alveolar ridges, retromolar trigone, hard palate, floor of mouth and oral tongue. Squamous cell carcinoma arises here.
- The **sublingual space**, above and medial to the mylohyoid and lateral to the genioglossus: the sublingual gland, the deep part of the submandibular gland and its duct, the lingual nerve, artery and vein, and the hypoglossal nerve. The two sides connect anteriorly beneath the frenulum.
- The **submandibular space**, below the mylohyoid: the submandibular gland, level I nodes, the facial vein and the anterior belly of the digastric. It opens posteriorly into the sublingual space at the back edge of the mylohyoid and upwards into the parapharyngeal space.
- The **root of the tongue**: the genioglossus and geniohyoid muscles and the **lingual septum** in the midline, the landmark for midline crossing.

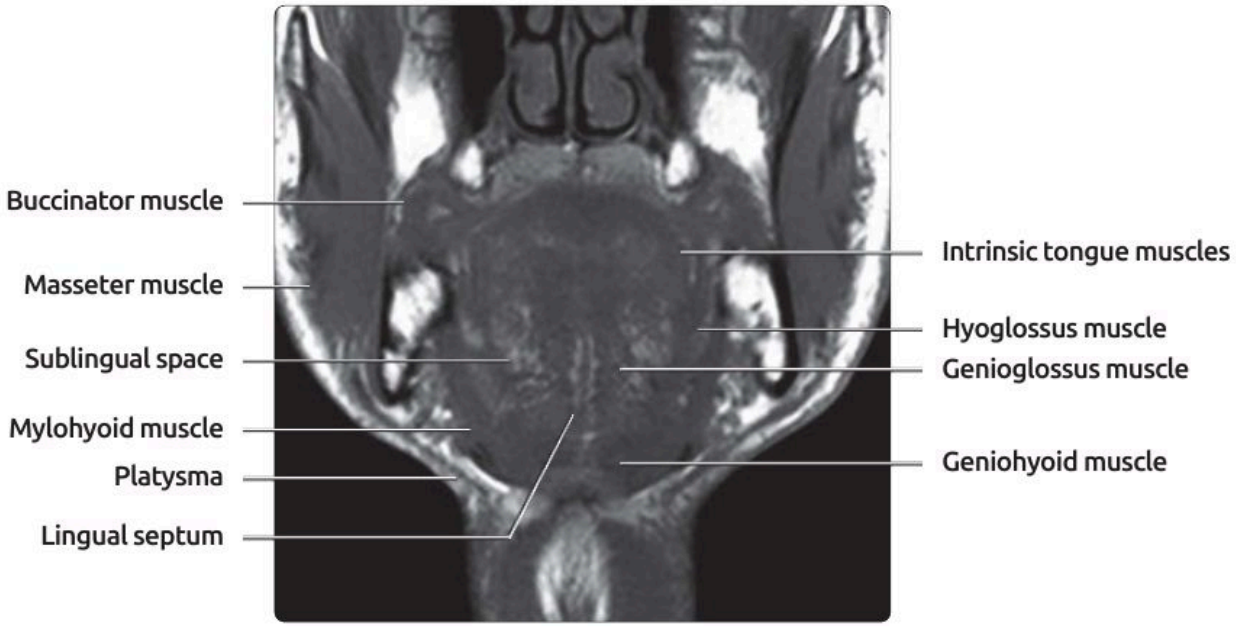


FIGURE 1 Coronal T1 through the oral cavity: the mylohyoid sling separating the sublingual space above from the submandibular space below, with the genioglossus, hyoglossus and intrinsic tongue muscles and the midline lingual septum.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 483

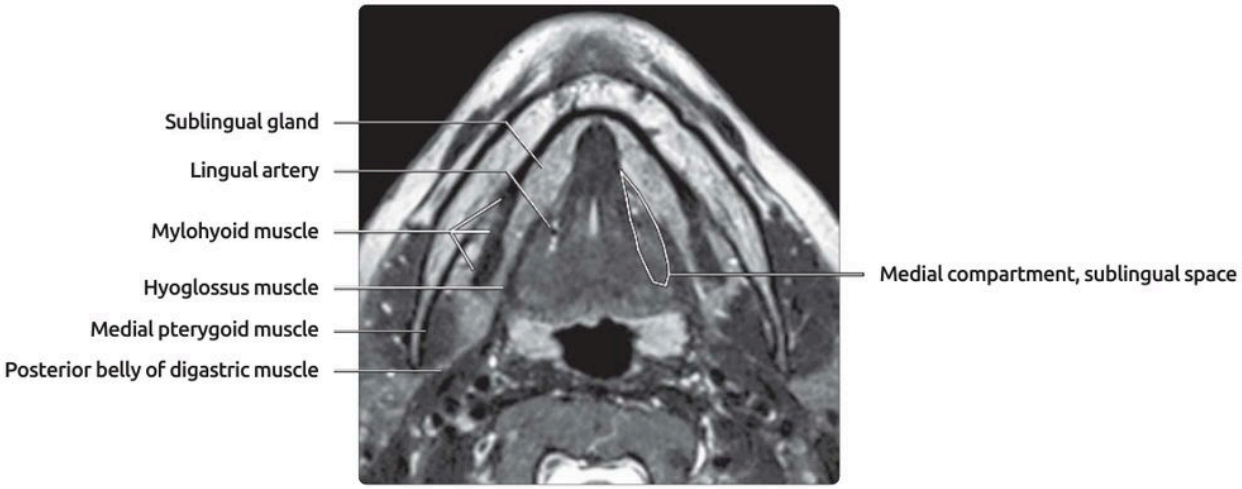


FIGURE 2 Axial T2 fat-suppressed: the sublingual space between the mylohyoid laterally and the hyoglossus and genioglossus medially, with the sublingual gland and lingual artery inside it.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 466

2.2 The oropharynx

Behind the circumvallate papillae and the anterior tonsillar pillars lies the oropharynx: the **base of tongue** with the lingual tonsil, the **palatine tonsils** in the lateral walls, the **soft palate** and the posterior pharyngeal wall. The lymphoid tissue

of the tonsils and tongue base is normally bulky and asymmetric, so a small tumour may be invisible on the scan; asymmetry, enhancement and restricted diffusion help.

3 CHAPTER THREE Oral cancer

What the surgeon needs.

3.1 Oral tongue

Report the **depth of invasion** (or tumour thickness on MRI), which drives the T stage and the decision to treat the neck; whether the tumour **crosses the lingual septum**; its relation to the **neurovascular bundle** (lingual artery and hypoglossal nerve) on each side, because involvement of both bundles rules out tongue preservation; extension to the floor of mouth, tongue base and mandible; and the nodes. Tongue tumours often spread to level II and III and may skip levels.

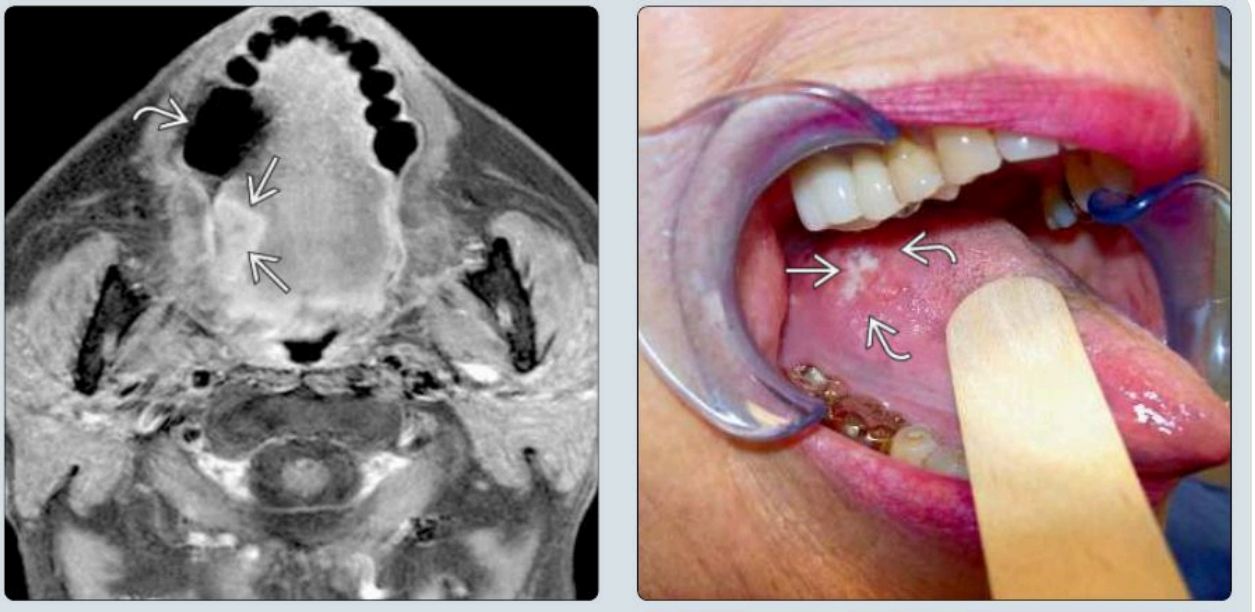


FIGURE 3 Oral tongue carcinoma. Left: post-gadolinium MRI, a wedge-shaped enhancing ulcer at the lateral tongue, well seen despite the amalgam. Right: the clinical lesion. Imaging measured its depth and showed no nodes.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 504

3.2 Floor of mouth and gum

Floor of mouth cancers involve the sublingual gland and the submandibular duct (obstructing the gland), cross the midline easily, and reach the inner cortex of the mandible early. **Alveolar** and **retromolar trigone** cancers lie on the bone: the question is whether the **mandible** is invaded and how far. Cortical erosion is seen on CT or cone-beam CT; marrow replacement (loss of bright fat on T1) on MRI, which is sensitive but can overcall. Spread along the **inferior alveolar canal** is perineural spread to the skull base. The retromolar trigone also leads along the pterygomandibular raphe up to the masticator space and skull base.



FIGURE 4 Floor of mouth carcinoma. Left: post-gadolinium MRI, a subtle anterior lesion involving the sublingual gland, with loss of the dark cortex of the mandible beside it. Right: the tumour at operation.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 508

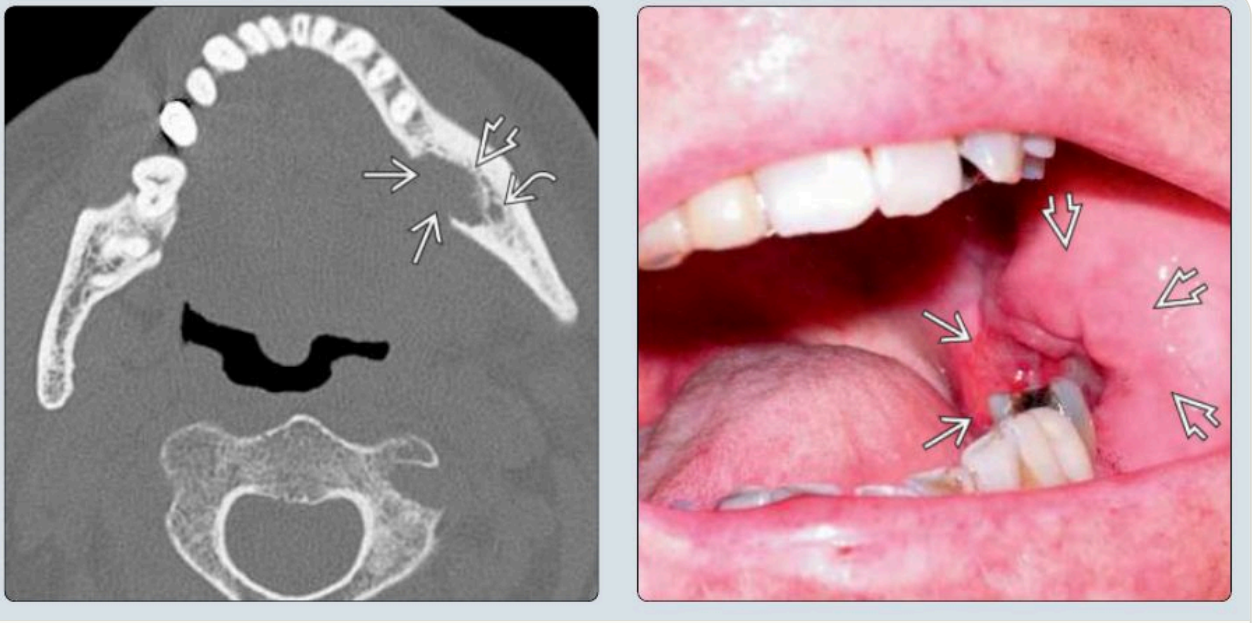


FIGURE 5 Retromolar trigone carcinoma. Left: bone CT, a lytic defect in the mandible close to the inferior alveolar canal. Right: the mucosal tumour behind the last molar.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 512

4 CHAPTER FOUR Oropharyngeal cancer

Small primaries, cystic nodes.

4.1 HPV and the cystic node

HPV-related oropharyngeal carcinoma typically affects younger patients, has a **small primary** in the tonsil or tongue base that may be invisible on CT, and presents with large, often **cystic** level II nodes. A cystic level II mass in an adult should be treated as a metastasis until proven otherwise, not as a branchial cyst. Look carefully at both tonsils and the tongue base for asymmetry, and use PET-CT before panendoscopy when the primary is not seen.

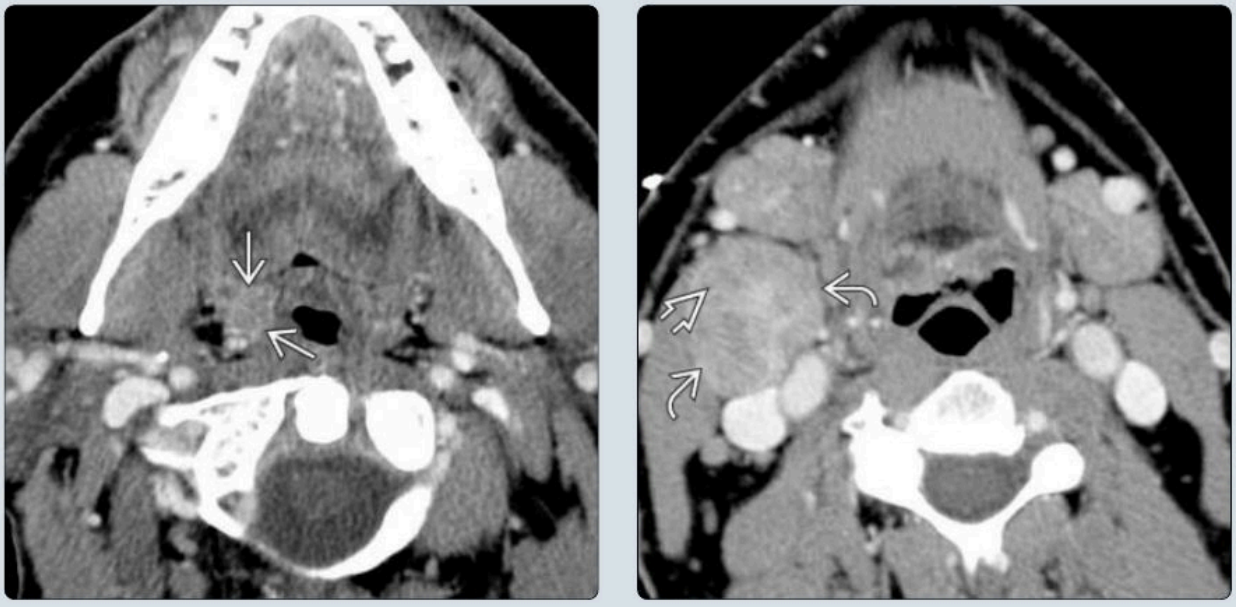


FIGURE 6 HPV-related tonsillar carcinoma. Left: a small enhancing mass in the right palatine tonsil, easily called normal. Right: a large necrotic level IIA node on the same side, the reason the patient presented.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 498

4.2 What to report

For tonsillar tumours report spread to the soft palate, tongue base, **parapharyngeal space**, medial pterygoid and mandible, and the relation to the internal carotid. For tongue base tumours report extension to the vallecula, pre-epiglottic space and larynx, and midline crossing. Encasement of the carotid or invasion of the prevertebral muscles or skull base makes the tumour T4b.

5 CHAPTER FIVE Infection and cysts

The floor of the mouth.

5.1 Abscess and Ludwig's angina

Most oral cavity infections start in a tooth. The roots of the first molar and the teeth in front of it lie **above** the mylohyoid line, so their infections enter the **sublingual** space; the roots of the second and third molars lie **below** it, so their infections enter the **submandibular** space. **Ludwig's angina** is a spreading cellulitis of both

sublingual and submandibular spaces on both sides that elevates the tongue and threatens the airway. Contrast CT shows the extent, any drainable collection, the airway, the source tooth and spread to the parapharyngeal space and mediastinum.



FIGURE 7 Sublingual space abscess. Left: drawing of a horseshoe-shaped collection in both sublingual spaces joined beneath the frenulum. Right: contrast CT, the rim-enhancing collection above the mylohyoid on both sides.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 408

5.2 Ranula

A **simple ranula** is a mucus retention cyst of the sublingual gland confined to the sublingual space. A **diving (plunging) ranula** passes behind the free edge of the mylohyoid, or through a gap in it, into the submandibular space, leaving a characteristic **tail** in the sublingual space that shows its origin.

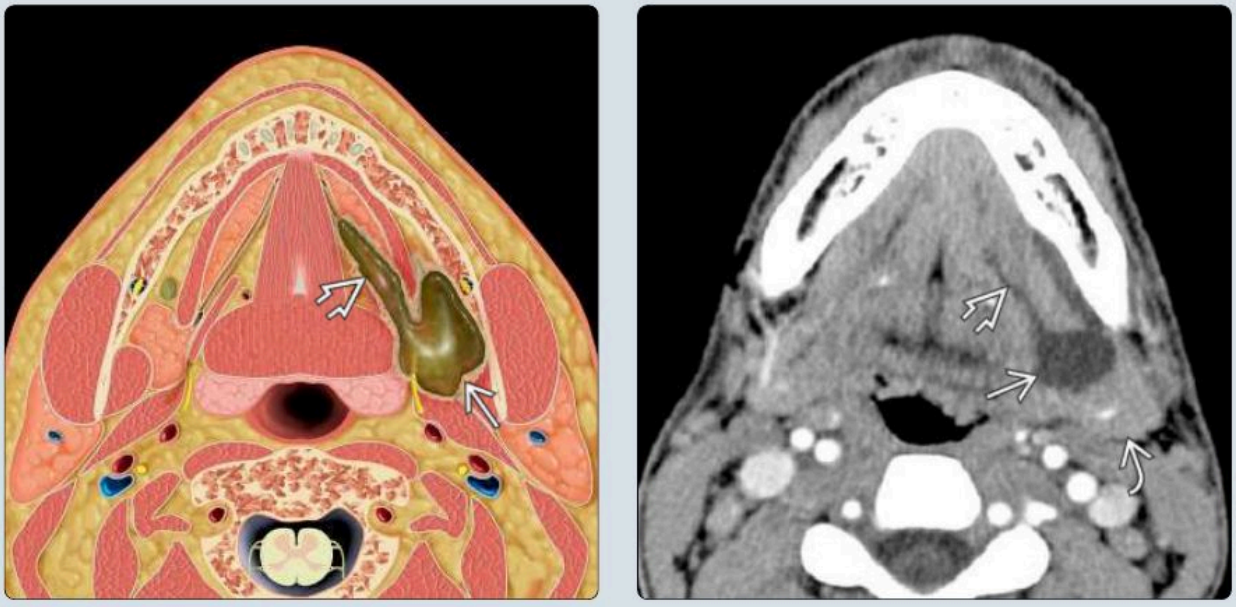


FIGURE 8 Diving ranula. Left: drawing of a ranula herniating backwards from the sublingual space. Right: contrast CT, a cystic mass in the submandibular space against the back of the mylohyoid, displacing the submandibular gland.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 400

5.3 The denervated tongue

After hypoglossal nerve injury, one half of the tongue changes: at first it is **swollen, bright on T2 and enhancing**; later it becomes **fatty and atrophic**, and the normal side can then look like a mass. The sharp midline border gives it away. Always follow the hypoglossal nerve back to the skull base to find the cause.

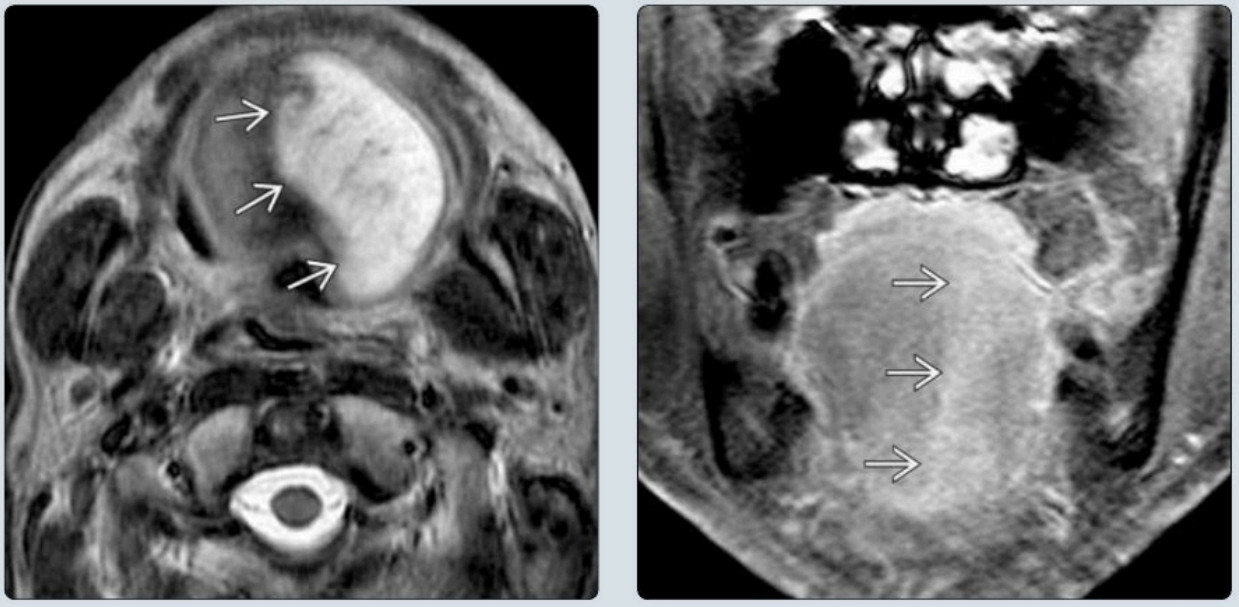


FIGURE 9 Acute and subacute denervation. Left: T2, a swollen, bright left half of the tongue, sharply limited at the midline. Right: post-gadolinium T1, enhancement of the denervated muscle.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 388

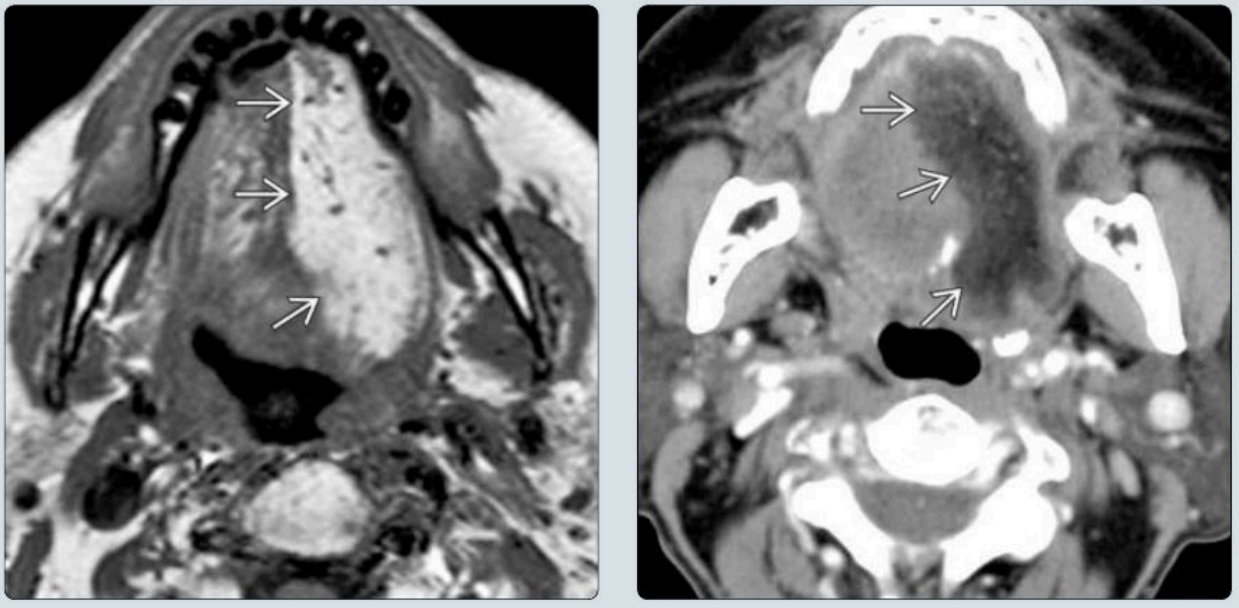


FIGURE 10 Chronic denervation. Left: T1, fatty replacement of the left half of the tongue. Right: CT, fatty atrophy; the normal right side could be mistaken for a tumour.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 388

READING AN ORAL CAVITY OR OROPHARYNX SCAN

1 Site

→ which mucosal subsite

- → sublingual, submandibular or root of tongue
- → oropharyngeal subsite

2 Tongue tumour

- → depth of invasion
- → lingual septum
- → neurovascular bundles
- → floor of mouth
- → tongue base

3 Bone

- → mandible cortex on CT
- → marrow on MRI
- → inferior alveolar canal
- → maxilla and palate

4 Deep spread

- → parapharyngeal space
- → pterygoids
- → pterygomandibular raphe
- → skull base
- → carotid

5 Infection

- → collection
- → above or below the mylohyoid
- → airway

→ source tooth or stone

→ mediastinum

6 Nodes

→ levels I to III both sides

→ cystic or necrotic nodes

6 CHAPTER SIX For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **MRI for oral cancer** (amalgam, depth, marrow, nerves); **CT for infection**.
- **The mylohyoid** and the sublingual and submandibular spaces.
- **Oral tongue:** depth of invasion, lingual septum, neurovascular bundles.
- **Mandible invasion:** CT for cortex, MRI for marrow; the inferior alveolar canal.
- **HPV-related cancer:** small primary, cystic level II node.
- **Dental infection** above or below the mylohyoid line; Ludwig's angina.
- **Diving ranula** and the denervated tongue.

6.2 Common pitfalls

COMMON PITFALLS

- Relying on CT for a tongue tumour hidden by amalgam.
- Calling a cystic level II node in an adult a branchial cyst.
- Mistaking a normal half-tongue next to an atrophic one for a tumour.

- Calling marrow invasion from MRI alone when the cortex is intact.

Written for this app from Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), oral cavity and squamous cell carcinoma chapters, Imaging Anatomy: Head and Neck (Chapman and Koontz), oral cavity, ESR Modern Radiology, Head and Neck Imaging. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 13

Lymph nodes

Every head and neck scan is also a scan of the lymph nodes, and the nodes often matter more than the primary: nodal metastasis is the single most important prognostic factor in head and neck squamous carcinoma, and a node is how many cancers first present. This lesson teaches the imaging levels from their landmarks, the features that make a node suspicious, the patterns that point to the primary, and the appearances of the non-cancerous causes you will meet in clinic.

AT A GLANCE

- Levels from landmarks: **hyoid** (II/III), **cricoid** (III/IV), **back of sternocleidomastoid** (V), **jugular vein** (IIA/IIB), **manubrium** (VI/VII).
- Malignant features together: **necrosis**, **round shape** and lost hilum, size, clustering, **extranodal extension**.
- Do not forget the **retropharyngeal** nodes: first station for the nasopharynx.
- A **cystic node** in an adult is metastasis until proven otherwise: HPV oropharynx or papillary thyroid.
- Carotid contact **≥270°** suggests encasement and unresectability.

1 CHAPTER ONE Levels from landmarks

Translating the neck dissection to the scan.

1.1 The landmarks

The surgical levels are drawn on imaging from fixed landmarks that can be found on any axial slice: the **mylohyoid** and the **posterior edge of the submandibular gland**, the **hyoid bone**, the lower edge of the **cricoid cartilage**, the **posterior edge of the sternocleidomastoid**, the **internal jugular vein**, the **carotid arteries** and the top of the **manubrium**.

IMAGING LEVELS

LEVEL	BOUNDARIES ON AXIAL IMAGING
IA	Submental: between the anterior bellies of the digastrics, below the mylohyoid

LEVEL	BOUNDARIES ON AXIAL IMAGING
IB	Submandibular: lateral to the anterior digastric, in front of the posterior edge of the submandibular gland
IIA	Skull base to the lower body of the hyoid, behind the submandibular gland, in front of the posterior edge of the sternocleidomastoid; in front of, beside or inseparable from the internal jugular vein
IIB	As IIA, but behind the internal jugular vein with a fat plane between
III	Lower hyoid to the lower cricoid, in front of the posterior edge of the sternocleidomastoid
IV	Lower cricoid to the clavicle, in front of the posterior edge of the sternocleidomastoid
VA	Behind the posterior edge of the sternocleidomastoid, skull base to the lower cricoid
VB	Behind the sternocleidomastoid, lower cricoid to the clavicle
VI	Between the carotids, hyoid to the manubrium (prelaryngeal, pretracheal, paratracheal)
VII	Between the carotids, below the top of the manubrium, above the brachiocephalic vein

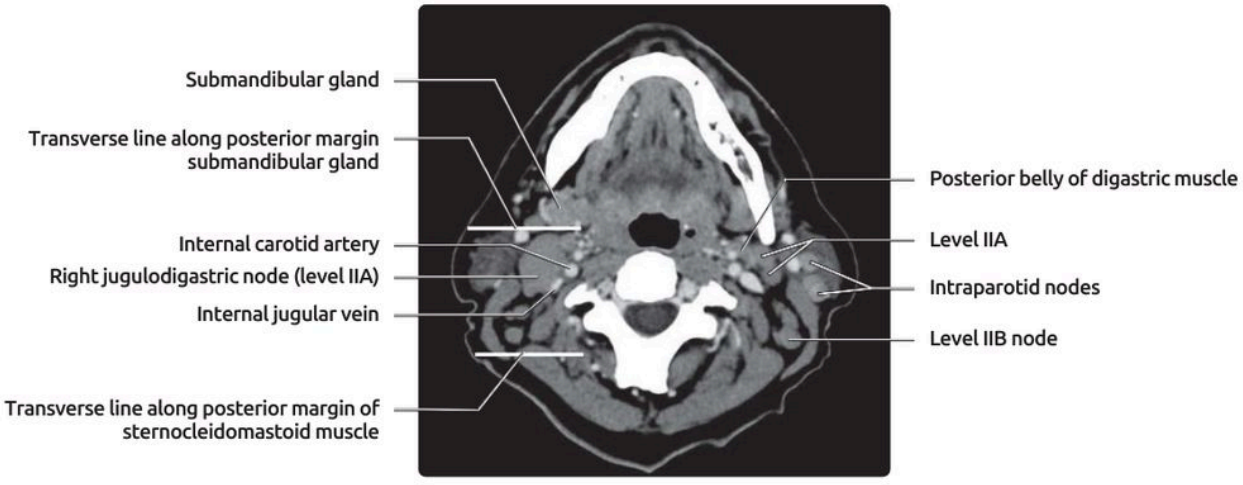


FIGURE 1 Axial contrast CT above the hyoid: lines along the posterior edges of the submandibular glands and sternocleidomastoids separate level I from level II, and level II from level V. The enlarged right level II node is the jugulodigastric node.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 430

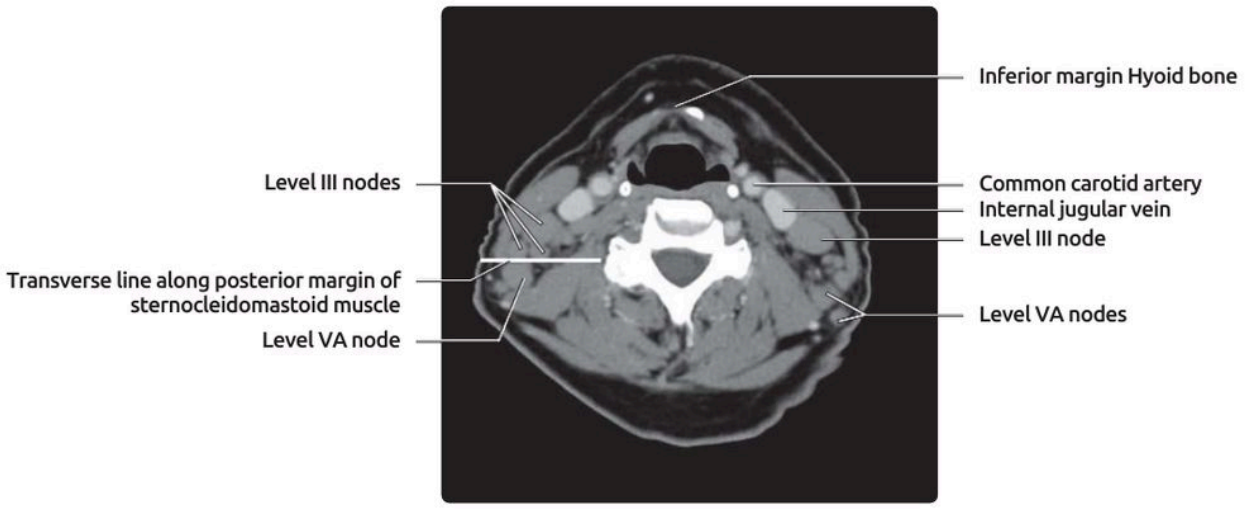


FIGURE 2 Just below the hyoid: level III nodes along the internal jugular vein in front of the line along the back of the sternocleidomastoid, level VA nodes behind it.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 431



FIGURE 3 At the top of the manubrium: the line between level VI above and level VII (the superior mediastinum) below.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 432

1.2 Nodes without a number

Some groups are named, not numbered, and are easily missed: the **retropharyngeal** nodes (lateral, or nodes of Rouvière, at the level of C1–C2 beside the longus capitis, and medial), the **parotid** and **facial** nodes, the **occipital** and **supraclavicular** nodes. Retropharyngeal nodes cannot be felt and are the first station for the nasopharynx and a common one for the oropharynx, hypopharynx and thyroid.

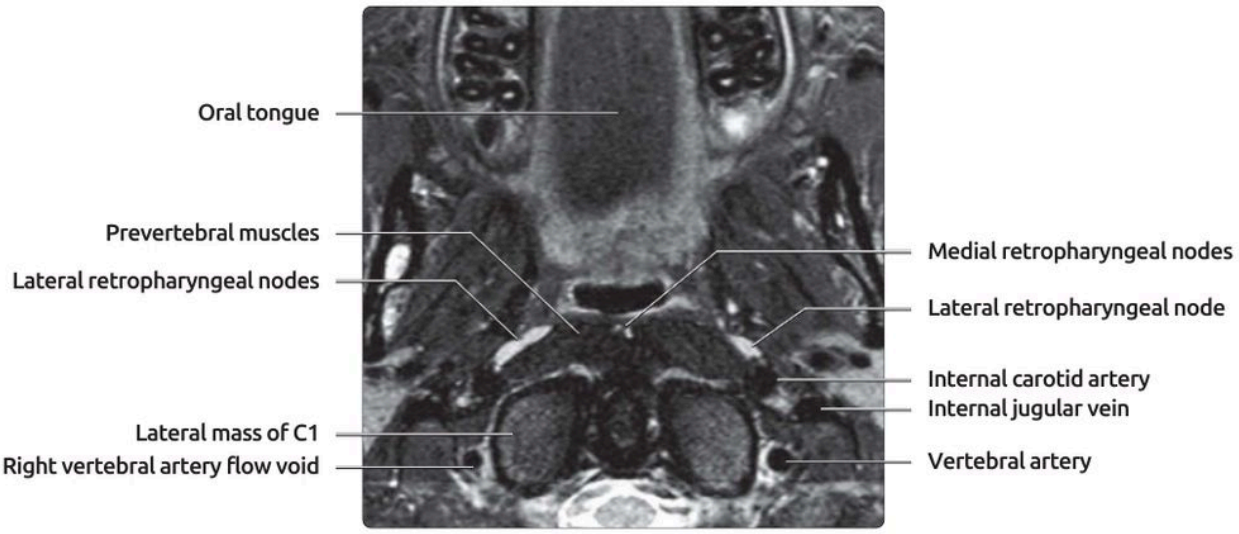


FIGURE 4 Axial fat-suppressed T2: the lateral retropharyngeal nodes (of Rouvière) anterolateral to the prevertebral muscles, and the medial retropharyngeal nodes.
Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 435

CHAPTER TWO

2 Is this node abnormal?

Use all the features together.

2.1 The malignant features

Size alone misclassifies one node in five or more, so use every feature together.

- **Necrosis**: a focal low-density defect in an untreated node is the most reliable sign of metastasis. Do not mistake the normal fatty hilum for necrosis.
- **Shape**: normal nodes are oval or kidney-shaped with a fatty hilum; malignant nodes become **round** (long-to-short axis ratio under 2) and lose the hilum.
- **Size**: thresholds vary; commonly used limits are about 10 mm in short axis (a little more for the jugulodigastric node) and 5 to 8 mm for a lateral retropharyngeal node. A **cluster** of three or more borderline nodes in the drainage area of a tumour is suspicious.
- **Extranodal extension**: an irregular, ill-defined margin, stranding of the surrounding fat, and invasion of muscle, vessel or skin. It changes the stage and the treatment.
- **Asymmetry and enhancement**: a node that enhances more than its neighbours, or that is new, is suspicious.

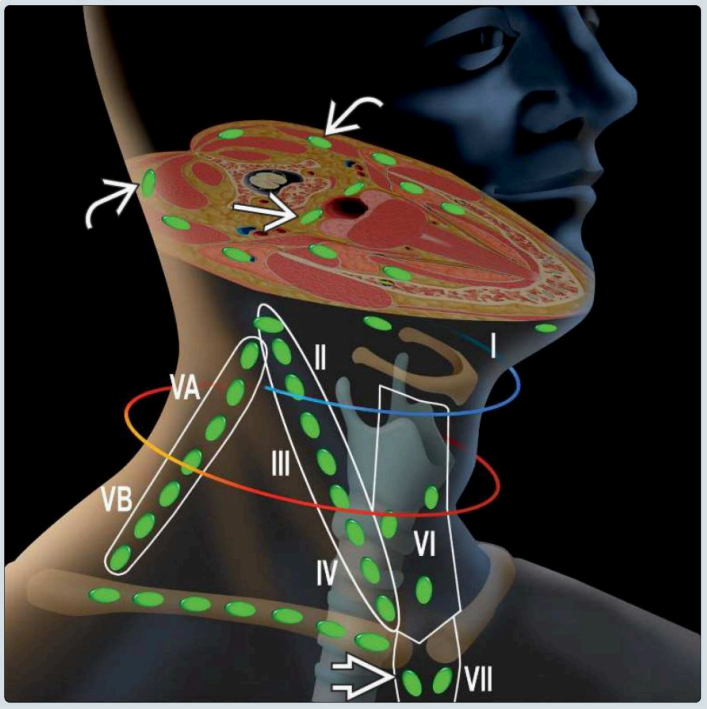


FIGURE 5 The nodal network of the neck, drawn with the levels outlined, the hyoid plane (blue) and cricoid plane (orange-red) that divide them, and, in the axial slice above, the retropharyngeal and superficial nodes (arrows). Level II is the commonest site of metastasis from squamous carcinoma.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 540

2.2 Ultrasound, MRI and PET

Ultrasound shows the hilum and its vessels best: a normal node is oval with an **echogenic hilum** and **hilar vascularity**; a metastatic node is round, loses the hilum and shows **peripheral** vascularity, and can be sampled by FNA in the same sitting. On **MRI**, metastatic squamous carcinoma has a low ADC on diffusion. **PET-CT** improves detection in advanced disease and the unknown primary, but misses small and cystic nodes and gives false positives with reactive nodes.

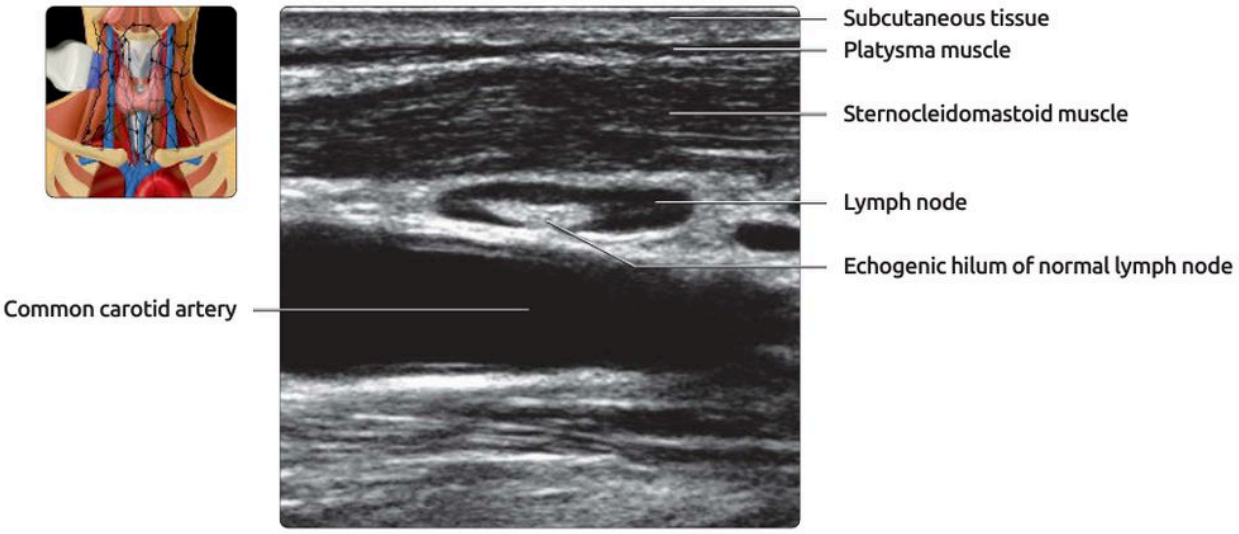


FIGURE 6 Ultrasound of a normal cervical node: an elliptical, hypoechoic node with an echogenic fatty hilum, in front of the common carotid artery.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 436

2.3 Unresectability

A nodal mass that **encases the carotid** (contact of 270° or more), invades the **prevertebral muscles** or the skull base, or infiltrates the skin widely is usually unresectable. Contact under 180° usually means the artery can be peeled free.

3 CHAPTER THREE Where is the primary?

Patterns of spread.

3.1 Reading the nodes backwards

Each primary site drains to predictable levels, so the level of an abnormal node points to the primary. Nodes outside the usual pattern should prompt a search for a second primary.

THE NODE AND THE LIKELY PRIMARY

NODES	LIKELY PRIMARIES
IA and IB	Lip, anterior oral cavity, floor of mouth, skin of the face

NODES	LIKELY PRIMARIES
IIA (jugulodigastric)	Oropharynx (tonsil, tongue base), oral tongue, supraglottis, nasopharynx
Retropharyngeal and VA	Nasopharynx; also oropharynx and hypopharynx
III and IV	Larynx, hypopharynx, thyroid
VI	Thyroid, subglottis, cervical oesophagus
Parotid and preauricular	Skin of the scalp, ear and face (squamous carcinoma, melanoma)
Left supraclavicular (Virchow)	Chest and abdomen

3.2 The cystic node

A cystic or largely necrotic node in an adult is a **metastasis until proven otherwise**. A **thick-walled** necrotic node most often comes from the tonsil or tongue base; a cystic node with a thin wall in a young adult suggests HPV-related oropharyngeal cancer; a cystic node with **calcification**, bright T1 signal (thyroglobulin) or avid enhancement suggests **papillary thyroid carcinoma**, even when the thyroid looks normal.



FIGURE 7 Papillary thyroid carcinoma in a node. Left: contrast CT, a large heterogeneous level III node displacing the jugular vein, with calcification lower down. Right: ultrasound, marked internal heterogeneity; FNA confirmed papillary carcinoma.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 340

CHAPTER FOUR

4 Not every abnormal node is cancer

The common mimics.

4.1 Reactive, suppurative, tuberculous and lymphoma

Reactive nodes are enlarged but oval, with a preserved hilum and no necrosis, often in several levels on both sides. **Suppurative** nodes, usually in children, have a thick enhancing rim around a fluid centre with **inflammatory stranding** of the fat and skin around them; they may progress to an abscess. **Tuberculous** nodes are often matted, necrotic and rim-enhancing, in the posterior triangle and lower neck, with less surrounding inflammation than their necrosis suggests, and calcify later. **Lymphoma** produces large, homogeneous, **non-necrotic** nodes in many chains, often with involvement of Waldeyer's ring, and restricts strongly on diffusion.

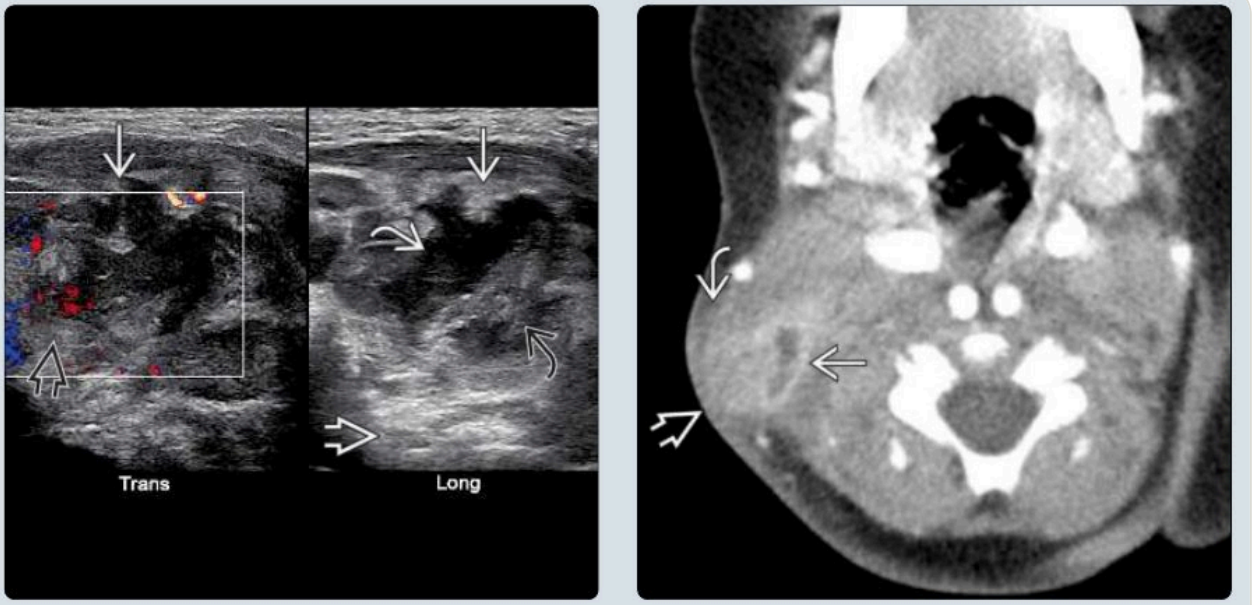


FIGURE 8 Suppurative lymphadenitis. Right: contrast CT, a rim-enhancing, centrally necrotic level II node with cellulitis and oedema of the overlying skin and fat.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 314

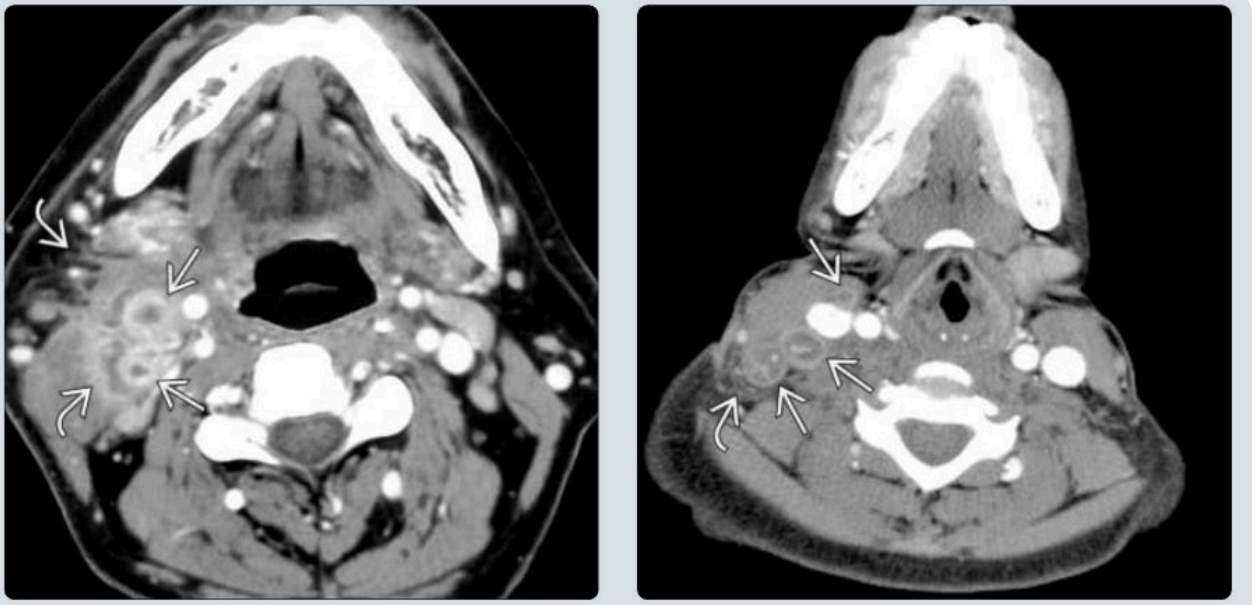


FIGURE 9 Tuberculous lymphadenitis. Left: multiple necrotic, rim-enhancing nodes with extranodal inflammation. Right: matted necrotic nodes that had not responded to treatment.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 318

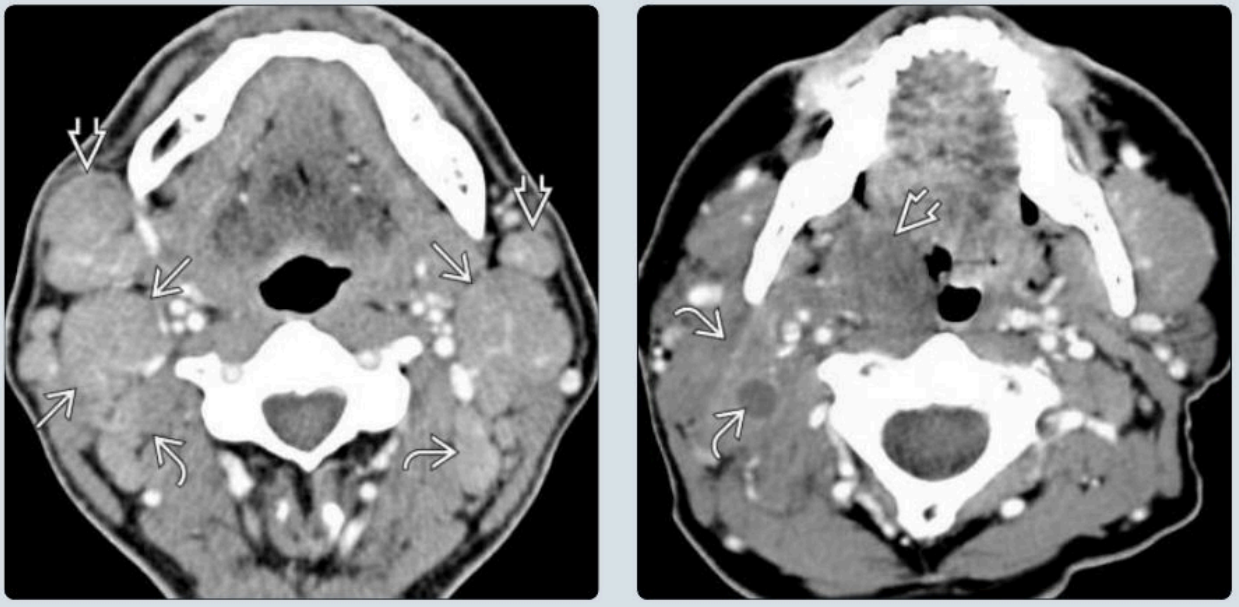


FIGURE 10 Non-Hodgkin lymphoma. Left: large, round, solid nodes in many chains on both sides without necrosis. Right: nodal and tonsillar lymphoma, with a partly necrotic level II mass.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 336

READING THE NODES ON ANY HEAD AND NECK SCAN

1 Every level, both sides

- → I to VII
- → retropharyngeal
- → parotid, facial, occipital, supraclavicular

2 Each suspicious node

- → level
- → short axis
- → shape and hilum
- → necrosis or cystic change
- → calcification
- → margins (extranodal extension)

3 Relations

- → carotid contact in degrees
- → jugular vein
- → prevertebral muscles
- → skin

4 Pattern

- → expected drainage of the primary
- → contralateral nodes
- → nodes outside the pattern

5 Next step

- → ultrasound and FNA
- → PET-CT for unknown primary or advanced disease

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Imaging levels** and their landmarks: hyoid, cricoid, submandibular gland, sternocleidomastoid, jugular vein, manubrium.
- **Retropharyngeal nodes** and the nasopharynx.
- **Malignant features:** necrosis, shape, size, extranodal extension.
- **Ultrasound** features of normal and abnormal nodes.
- **Node to primary** patterns; the cystic node in an adult.
- **Carotid encasement** and unresectability.

- **Mimics:** reactive, suppurative, tuberculous, lymphoma.

5.2 Common pitfalls

COMMON PITFALLS

- Using size alone to call a node benign or malignant.
- Calling the fatty hilum necrosis.
- Forgetting the retropharyngeal nodes.
- Calling a cystic level II mass in an adult a branchial cyst.
- Missing a thyroid primary behind a calcified or cystic node.

Written for this app from Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), lymph node overview and diagnoses, Imaging Anatomy: Head and Neck (Chapman and Koontz), cervical lymph nodes, ESR Modern Radiology, Head and Neck Imaging. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 14

Thyroid and salivary glands

The thyroid, parotid and submandibular glands lie close to the skin, so **ultrasound** is the first test for almost every lump in them, ideally with a needle in the same sitting. CT and MRI come next, for what ultrasound cannot see: the deep lobe of the parotid, the retrosternal goitre, invasion of the airway, perineural spread and the nodes behind the pharynx. This lesson covers the normal glands, the ultrasound features that grade risk, and the cross-sectional signs that change an operation.

AT A GLANCE

- **Ultrasound first**, with FNA, for thyroid and salivary lumps.
- Suspicious nodule: **solid, hypoechoic, taller than wide, irregular, microcalcifications**. Grade by **U** or **TI-RADS**.
- **CT** for retrosternal goitre and invasive cancer; remember iodine delays radioactive iodine.
- The facial nerve lies **lateral to the retromandibular vein**; deep lobe tumours widen the **stylomandibular gap**.
- Pleomorphic adenoma: very **bright on T2**. Warthin: parotid **tail**, multiple, smokers. Low T2 and ill-defined margins suggest malignancy.

1 CHAPTER ONE The thyroid

Ultrasound first.

1.1 The normal gland and its relations

The thyroid wraps the front of the trachea from the level of the cricoid down, with the isthmus across the second to fourth tracheal rings. On ultrasound it is homogeneous and brighter than the strap muscles; on CT it is dense even before contrast because of its iodine, and enhances avidly. The **carotid** and **jugular vein** lie laterally, the **oesophagus** behind the left lobe, and the **tracheoesophageal grooves** hold the recurrent laryngeal nerves, the paratracheal (level VI) nodes and the parathyroid glands.

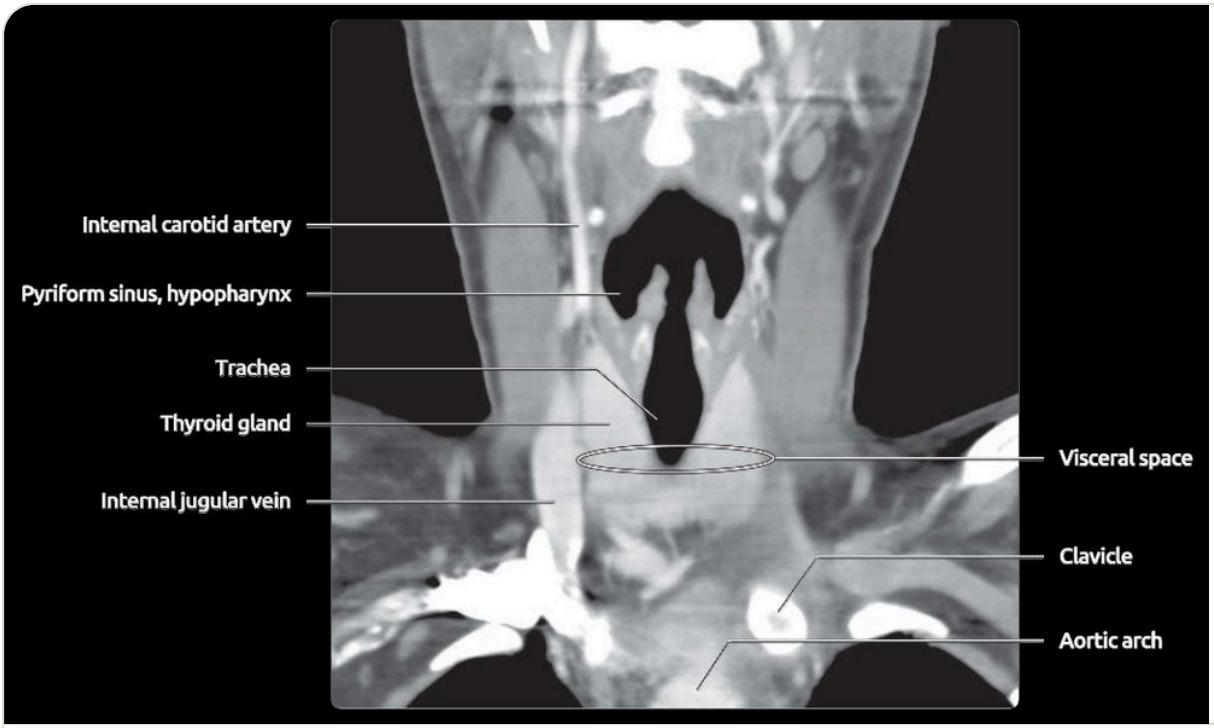


FIGURE 1 Coronal contrast CT: the chevron-shaped thyroid in the visceral space, wrapping the trachea below the larynx.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 369

1.2 Grading a nodule

Ultrasound grades a nodule's risk by its features. The UK **U grading** (U1 normal, U2 benign, U3 indeterminate, U4 suspicious, U5 malignant) and the American **TI-RADS** (TR1 to TR5, scored by points) use the same features:

ULTRASOUND FEATURES OF THYROID NODULES

FEATURE	REASSURING	SUSPICIOUS
Composition	Cystic, spongiform	Solid
Echogenicity	Iso- or hyperechoic, with a halo	Hypoechoic, especially markedly hypoechoic
Shape	Wider than tall	**Taller than wide** on transverse view
Margin	Smooth, well defined	Lobulated, irregular, extrathyroidal extension
Echogenic foci	Comet-tail artefact (colloid), eggshell rim	**Microcalcifications**, interrupted rim calcification

FEATURE	REASSURING	SUSPICIOUS
Nodes	Normal	Round, cystic or calcified nodes in levels III, IV, VI

Under the British Thyroid Association guidance, U3 to U5 nodules are sampled by FNA; under ACR TI-RADS the size threshold for FNA falls as the score rises (for example, 1 cm for TR5 and 2.5 cm for TR3). Cytology is reported in the Thy (UK) or Bethesda categories.



FIGURE 2 Differentiated thyroid carcinoma on ultrasound: a well-defined nodule with many tiny hyperechoic microcalcifications. FNA showed papillary carcinoma.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 238

1.3 When to add CT or MRI

Ultrasound cannot see behind the sternum or the trachea. **CT** (or MRI) is added for a **retrosternal goitre** (its lowest extent, tracheal narrowing and deviation, and whether it can be delivered through the neck), for a cancer that may invade the **trachea, larynx, oesophagus or recurrent nerve**, and for nodes in the mediastinum or behind the pharynx. In differentiated thyroid cancer, iodinated contrast delays radioactive iodine treatment by weeks, so the need for contrast is discussed with the team; MRI avoids the problem.

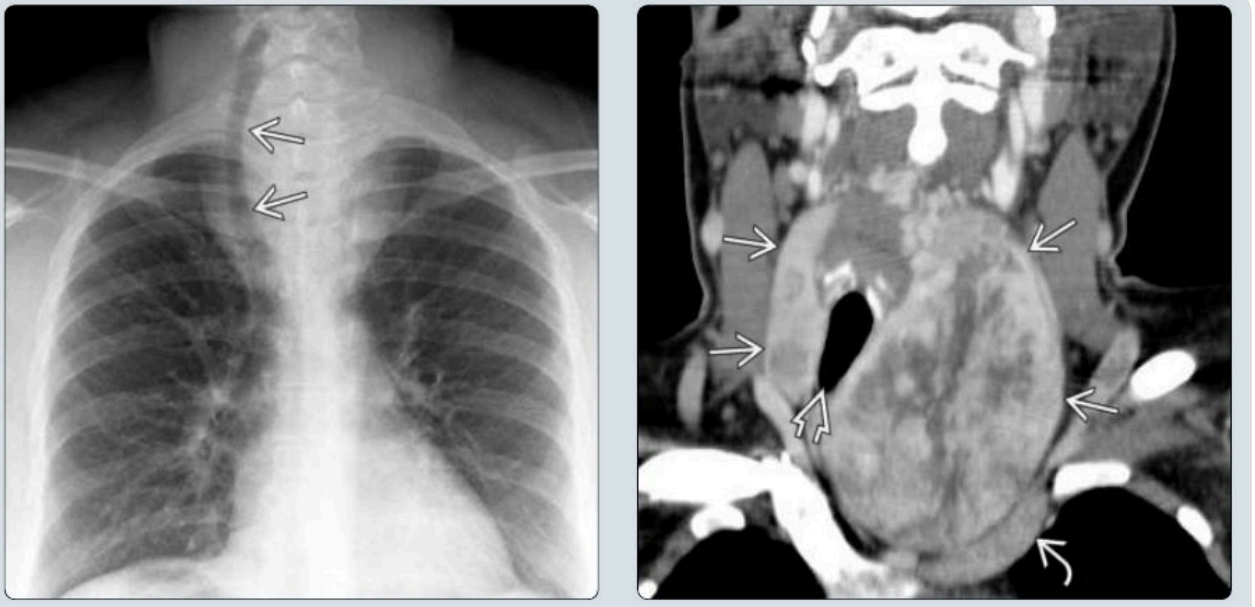


FIGURE 3 Multinodular goitre. Left: a chest film, the trachea displaced by a mass in the neck and upper mediastinum. Right: coronal contrast CT, heterogeneous enlargement of both lobes, the left extending into the mediastinum.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 226

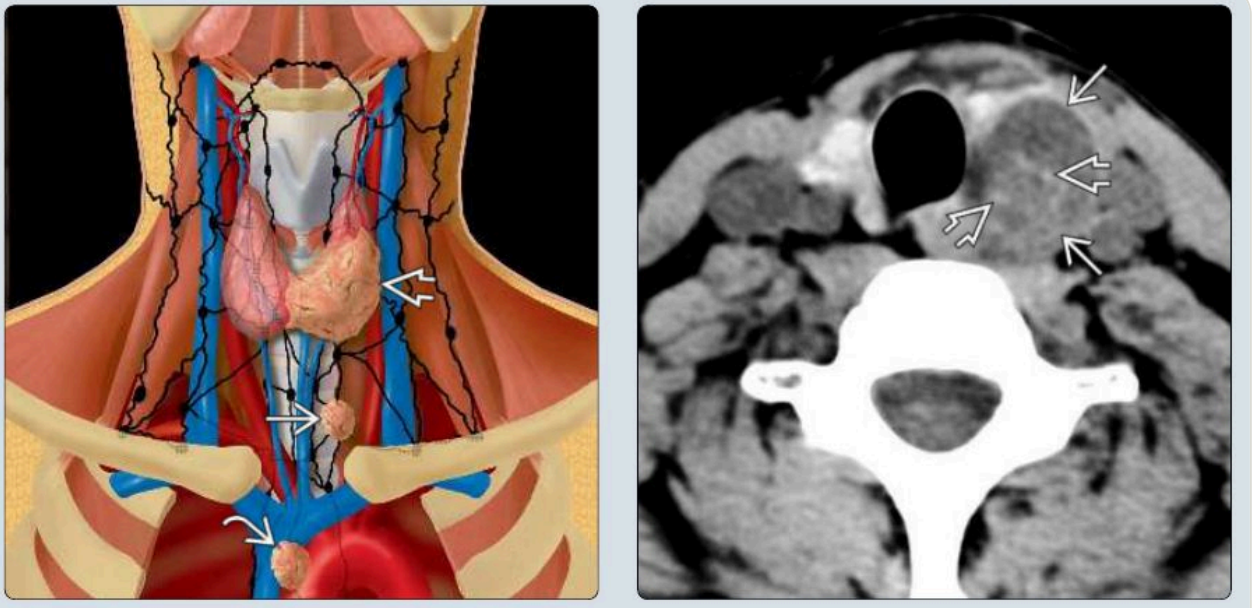


FIGURE 4 Differentiated thyroid carcinoma on CT. Left: drawing of a tumour in the left lobe. Right: non-contrast CT, a well-defined mass in the left lobe with fine, speckled microcalcifications.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 238

1.4 The incidental nodule and the parathyroids

Thyroid nodules are often found by chance on CT, MRI or PET. Uptake of FDG in a focal nodule carries a significant risk of cancer and needs ultrasound; a small, non-

avid nodule on CT in an older patient usually does not. For **primary hyperparathyroidism**, ultrasound and **sestamibi** scintigraphy (often with SPECT-CT) localise most adenomas, and **four-dimensional CT** helps when they disagree or after failed surgery. An adenoma is a small, oval, hypoechoic nodule behind the thyroid, often with a feeding vessel at one pole.

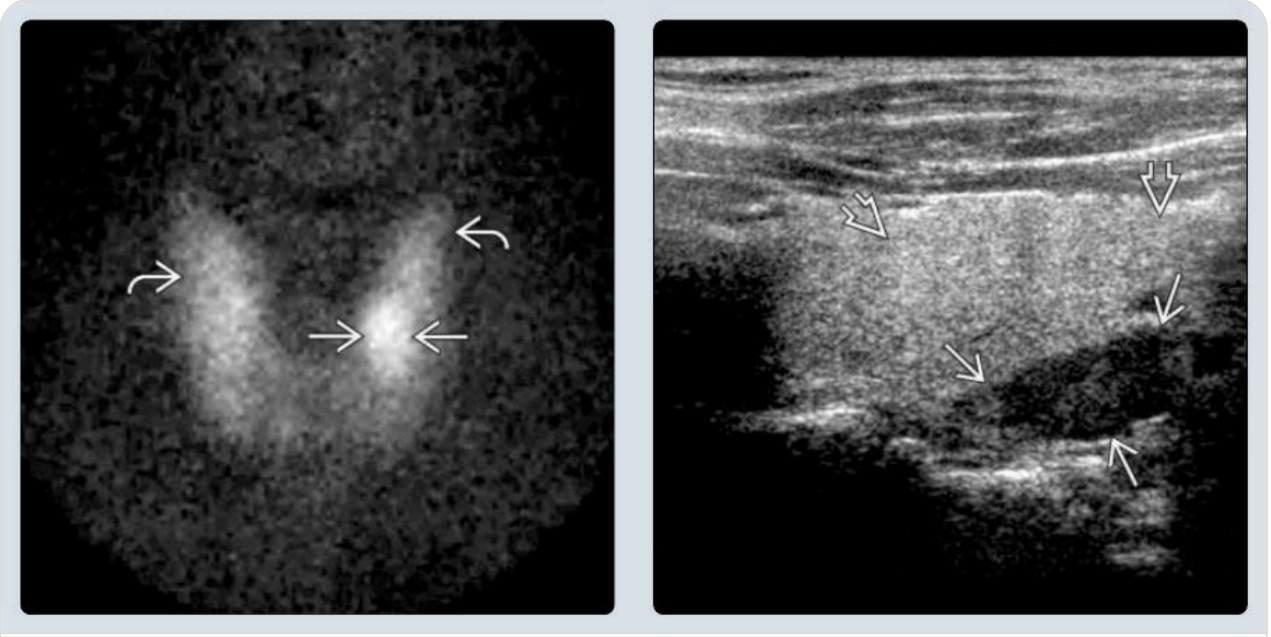


FIGURE 5 Parathyroid adenoma. Left: delayed sestamibi scan, persistent uptake below the left lobe. Right: ultrasound, a hypoechoic oval nodule behind the lower left lobe.
Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 234

2 CHAPTER TWO The parotid

Superficial, deep and the nerve.

2.1 The facial nerve plane

The facial nerve cannot be seen on routine imaging within the parotid, so its plane is inferred. The **retromandibular vein** runs just medial to the nerve, and the nerve lies lateral to it; a tumour lateral to the vein is in the superficial lobe, one medial to it in the deep lobe. A **deep lobe** tumour passes through the **stylomandibular gap** (between the mandibular ramus and the styloid process and its ligament) and pushes the parapharyngeal fat medially. Ultrasound shows the superficial lobe well but not the deep lobe, which is shadowed by the mandible.

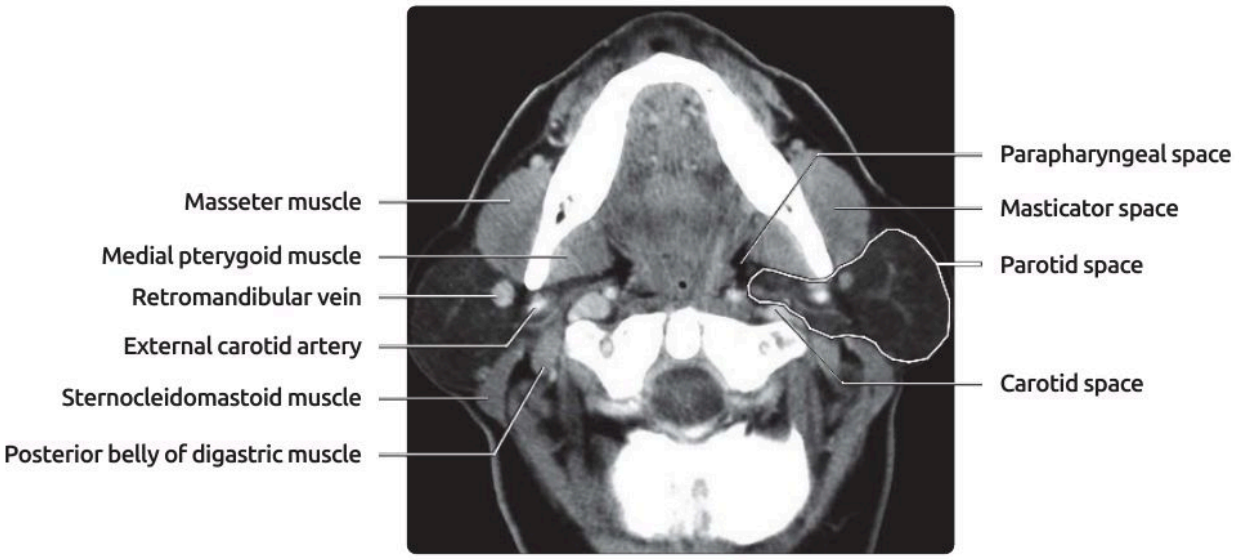


FIGURE 6 Axial contrast CT at the mid oropharynx: the parotid space outlined, with the retromandibular vein lateral to the external carotid artery. The facial nerve runs just lateral to the vein.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 326

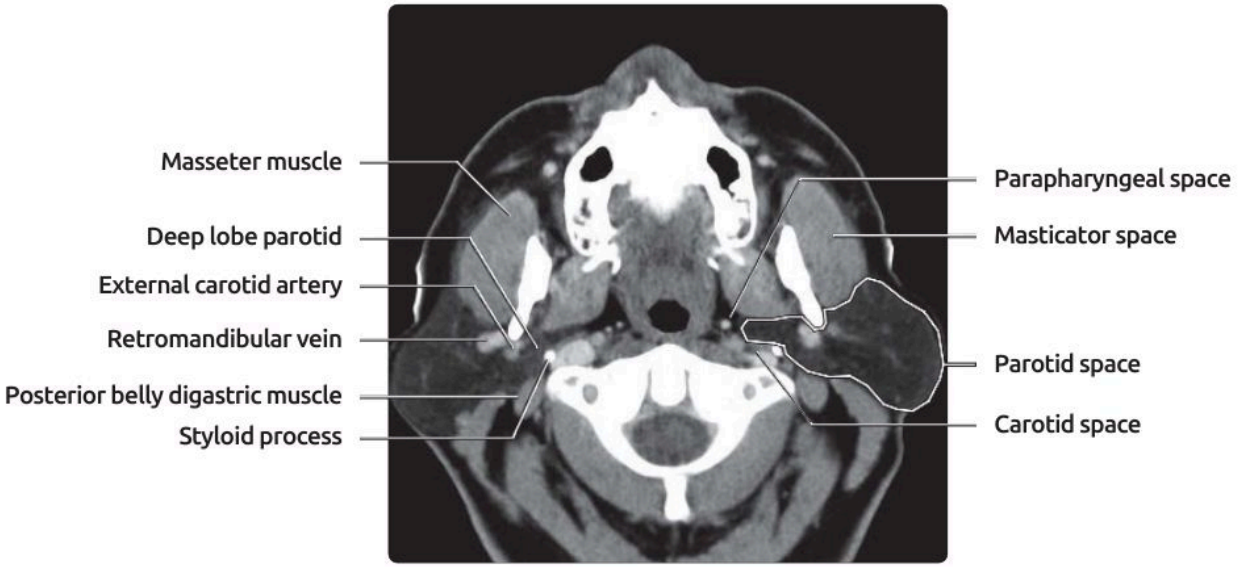


FIGURE 7 The deep lobe of the parotid projecting through the stylomandibular gap to touch the parapharyngeal space.

Chapman and Koontz, Imaging Anatomy: Head and Neck, p. 325

2.2 Benign tumours

A **pleomorphic adenoma** is well defined, lobulated and typically **very bright on T2**, with a capsule; it is the commonest salivary tumour and can arise anywhere in the

gland or its deep lobe. A **Warthin tumour** occurs in older men who smoke, in the **tail** of the parotid, is often **multiple or bilateral**, may be partly cystic, and is avid on PET.



FIGURE 8 Pleomorphic adenoma. Left: drawing of a small superficial lobe tumour. Right: fat-suppressed T2, a sharply marginated, very bright mass in the superficial lobe, close to the plane of the facial nerve.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 88

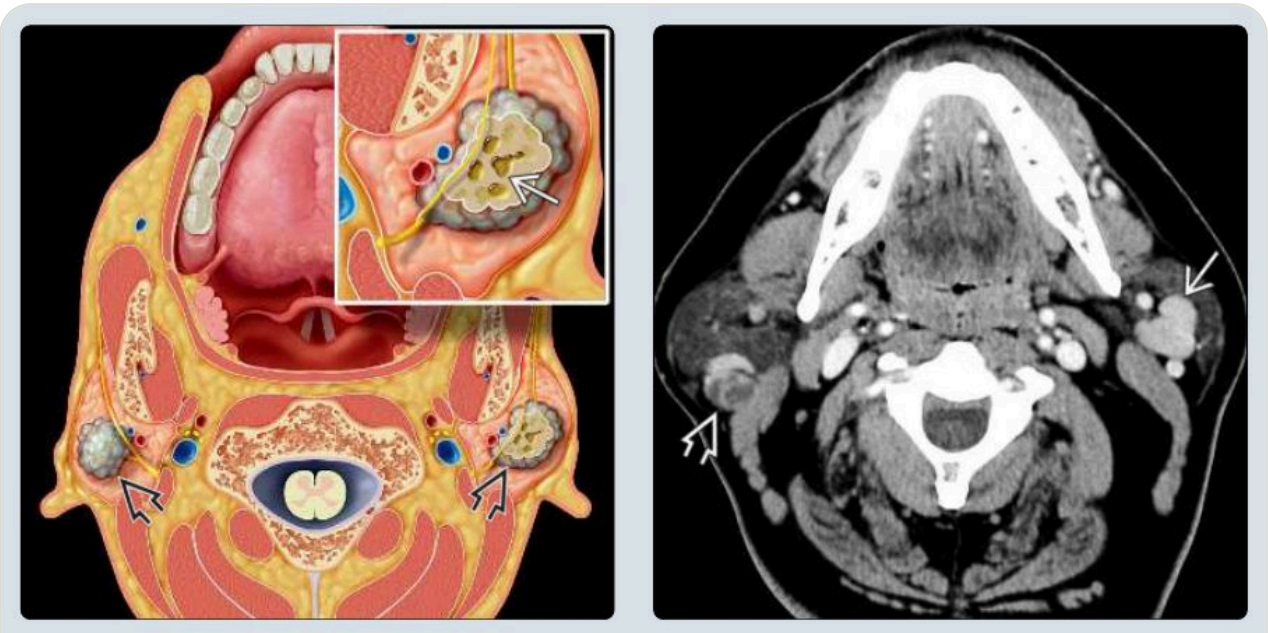


FIGURE 9 Warthin tumours. Left: drawing of bilateral tumours in the parotid tails. Right: contrast CT, a tumour in each parotid.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 92

2.3 Features of malignancy

Ill-defined or infiltrative margins, **low or intermediate T2 signal**, low ADC on diffusion, invasion of the masseter, skin or deep lobe, and abnormal nodes suggest malignancy, although a small low-grade carcinoma can look benign and FNA or core biopsy is still needed. Facial nerve weakness or pain with a parotid mass suggests **perineural spread**: follow the nerve up through the stylomastoid foramen and the mastoid segment of the facial canal on post-gadolinium MRI, and look for loss of the fat in the stylomastoid foramen.

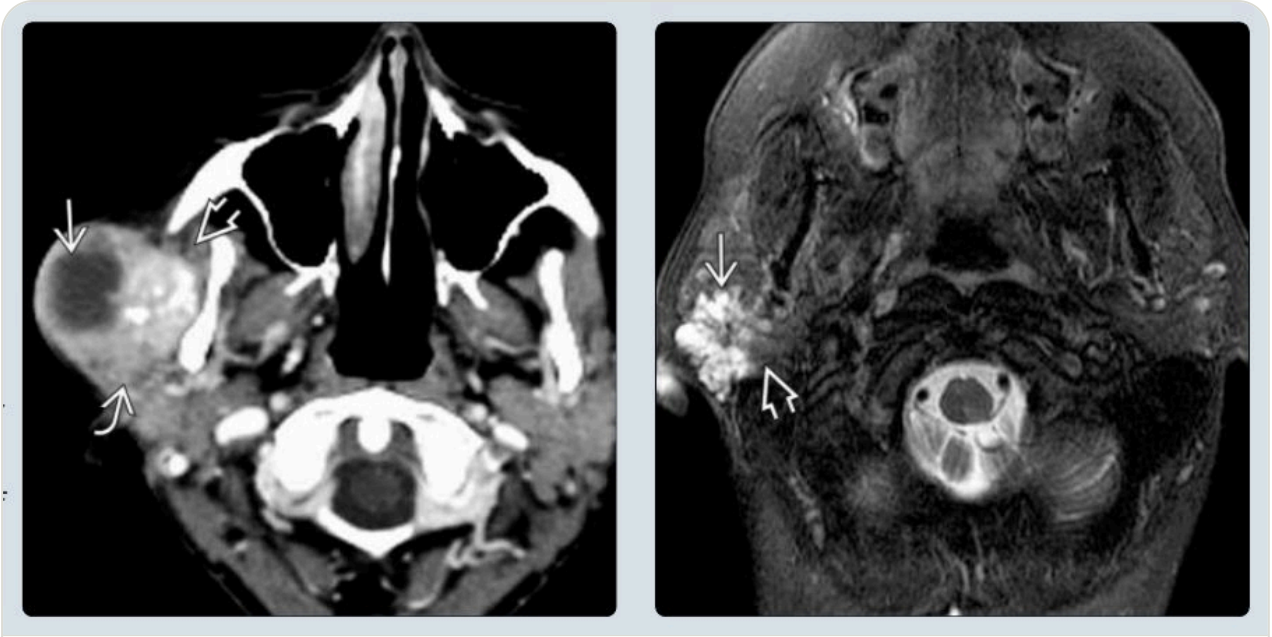


FIGURE 10 Mucoepidermoid carcinoma. Left: contrast CT, a large heterogeneous mass with cystic parts and ill-defined borders invading the masseter. Right: T2, irregular mixed signal with a low-signal solid part.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 98

2.4 Infection, stones and Sjögren's

Sialolithiasis is shown by non-contrast CT or ultrasound; most stones are in the submandibular duct. Acute **parotitis** enlarges and enhances the gland, and contrast CT shows an abscess. **Sjögren's syndrome** gives enlarged glands with many small cysts or a speckled "honeycomb" pattern on ultrasound and MRI; a new dominant solid mass in such a gland raises the question of lymphoma.

3 CHAPTER THREE
The submandibular gland

Gland or node?

3.1 Telling them apart

Masses in the submandibular space are either in the gland or in a level IB node next to it. A mass arising in the gland is surrounded by a "claw" of gland tissue; a node lies outside the gland, usually **in front of the facial vein**, whereas gland masses lie behind it. Submandibular gland tumours are more often malignant than parotid tumours, so a solid mass in the gland needs sampling.

READING A GLAND ULTRASOUND OR SCAN

1. Which gland and where
   - → thyroid lobe and pole
   - → parotid superficial or deep to the retromandibular vein
   - → submandibular gland or node
2. Lesion
   - → composition
   - → echogenicity or T2 signal
   - → shape (taller than wide)
   - → margins
   - → calcification
   - → vascularity

3 **Grade**

- → U or TI-RADS for thyroid
- → benign versus malignant features for salivary

4 **Beyond the gland**

- → extrathyroidal extension
- → trachea, larynx, oesophagus, recurrent nerve
- → masseter, skin, facial nerve and stylomastoid foramen

5 **Nodes**

- → levels III, IV, VI for thyroid
- → intraparotid and level II for parotid

6 **Next step**

- → FNA or core biopsy
- → CT or MRI for retrosternal, deep or invasive disease

4 CHAPTER FOUR For the examination

What the examiner wants to hear.

4.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Ultrasound first** for thyroid, salivary and superficial lumps, with FNA.
- **Suspicious thyroid features:** solid, hypoechoic, taller than wide, irregular, microcalcification.
- **U grading and TI-RADS**, and when to sample.

- **CT for retrosternal goitre** and invasive thyroid cancer; the contrast and radioactive iodine issue.
- **Parathyroid localisation:** ultrasound, sestamibi, 4D-CT.
- **Facial nerve plane** and the retromandibular vein; the deep lobe and the stylomandibular gap.
- **Pleomorphic adenoma versus Warthin tumour,** and the features of malignancy.

4.2 Common pitfalls

COMMON PITFALLS

- Ordering CT for a thyroid nodule instead of ultrasound.
- Giving iodinated contrast casually to a patient about to have radioactive iodine.
- Calling a deep lobe parotid tumour "parapharyngeal".
- Missing perineural spread to the stylomastoid foramen in a parotid cancer.
- Mistaking a level IB node for a submandibular gland tumour.

Written for this app from Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), visceral and parotid space chapters, Imaging Anatomy: Head and Neck (Chapman and Koontz), parotid and visceral spaces, ESR Modern Radiology, Head and Neck Imaging, Dunnebier, Imaging for Otolaryngologists. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 15

Reading a head and neck cancer scan

A staging scan for head and neck cancer is read to answer a fixed list of questions, and the multidisciplinary meeting expects those answers in order: what is the primary and how far has it gone, is there perineural spread, which nodes are involved and how, has it spread to the chest or bone, and is there a second primary. This lesson puts the regional lessons together into one routine you can use on any cancer scan, at any site.

AT A GLANCE

- **CT** for larynx and neck; **MRI** for nasopharynx, oral cavity, oropharynx and sinonasal; **PET-CT** for advanced disease and the unknown primary.
- Routine: **primary**, local extent and **T4 structures**, **perineural**, **vessels**, **nodes**, **chest and bones**, second primary.
- T4b usually means unresectable: carotid encasement, prevertebral fascia, skull base, mediastinum.
- **Extranodal extension** (ill-defined margins, fat, muscle, skin) changes stage and treatment.
- Unknown primary: image **before** panendoscopy; PET-CT; HPV and EBV status of the node.

1 CHAPTER ONE Choosing the scan

Match the modality to the site.

1.1 CT, MRI and PET-CT

Contrast CT of the neck and chest is quick, reproducible and tolerated by patients with secretions or pain; it is the usual first staging study for the larynx, hypopharynx and most of the neck. **MRI** is preferred for the **nasopharynx** (skull base and intracranial spread), the **oral cavity and oropharynx** (amalgam, depth of invasion, small tonsillar primaries), the **sinonasal** tract (tumour versus secretions, orbit and dura) and for **perineural spread**; in the larynx it is kept for equivocal cartilage invasion. **PET-CT** is used for advanced disease (to find distant metastases and second primaries), for the **unknown primary** before panendoscopy, and after treatment. A **CT of the chest** is part of every staging work-up for squamous carcinoma.

2 CHAPTER TWO The primary

Where it is, how big, what it touches.

2.1 The core questions

Describe the **subsite** of origin, the **size** in three planes (and, in the oral cavity, the depth of invasion), and then go through the **structures that change the stage or the operation** for that site. These are the T4 structures: for a T4a tumour the disease is still usually resectable; for a **T4b** tumour (encasing the carotid, invading the prevertebral fascia or muscles, the skull base or the mediastinum, or the masticator space and pterygoid plates for some sites) it usually is not.

KEY STRUCTURES TO REPORT BY SITE

SITE	REPORT SPECIFICALLY
Nasopharynx	Parapharyngeal space, skull base (clivus, foramen lacerum, foramen ovale), cavernous sinus, intracranial spread, retropharyngeal nodes
Oral cavity	Depth of invasion, lingual septum, neurovascular bundles, mandible cortex and marrow, inferior alveolar canal, skin
Oropharynx	Tongue base and vallecula, soft palate, parapharyngeal space, pterygoids, mandible, larynx, carotid
Larynx	Pre-epiglottic and paraglottic fat, commissures, subglottis, cartilage, extralaryngeal tissues, cord movement
Hypopharynx	Pyriform apex, postcricoid, oesophagus, thyroid and cricoid cartilage, prevertebral fascia
Sinonasal	Orbit (periorbita), anterior skull base and dura, pterygopalatine fossa, infratemporal fossa, skin

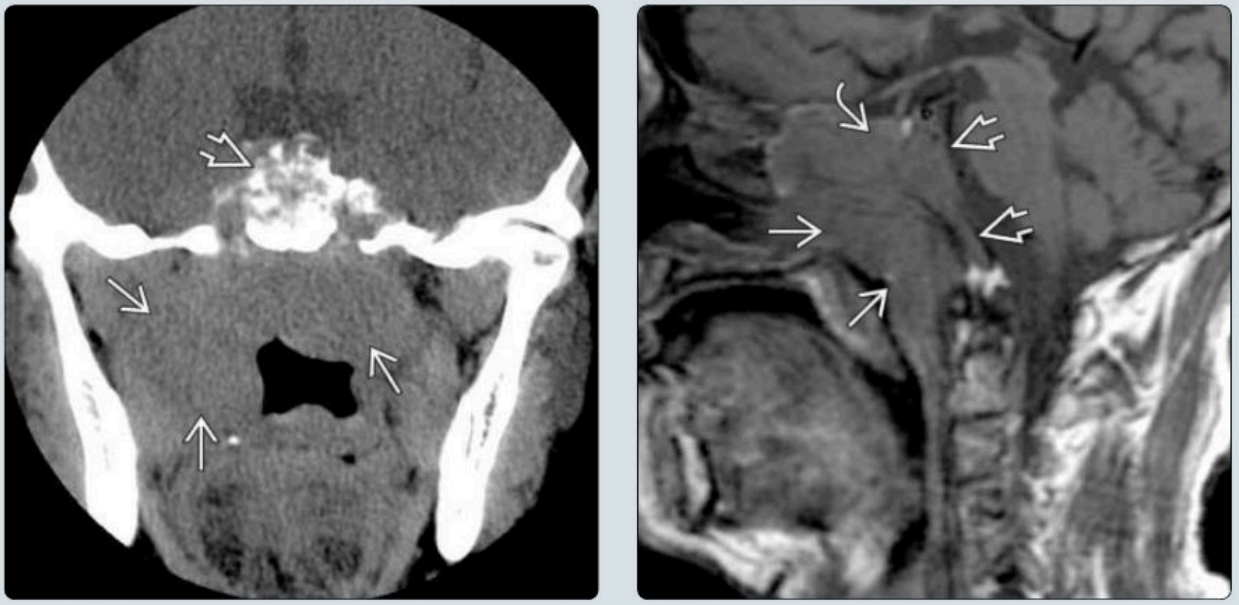


FIGURE 1 Nasopharyngeal carcinoma. Left: coronal CT, an asymmetric bulky nasopharyngeal mass. Right: sagittal T1, the tumour extends up into the sphenoid and sella and replaces the bright fatty marrow of the clivus, the sign of skull base invasion.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 484

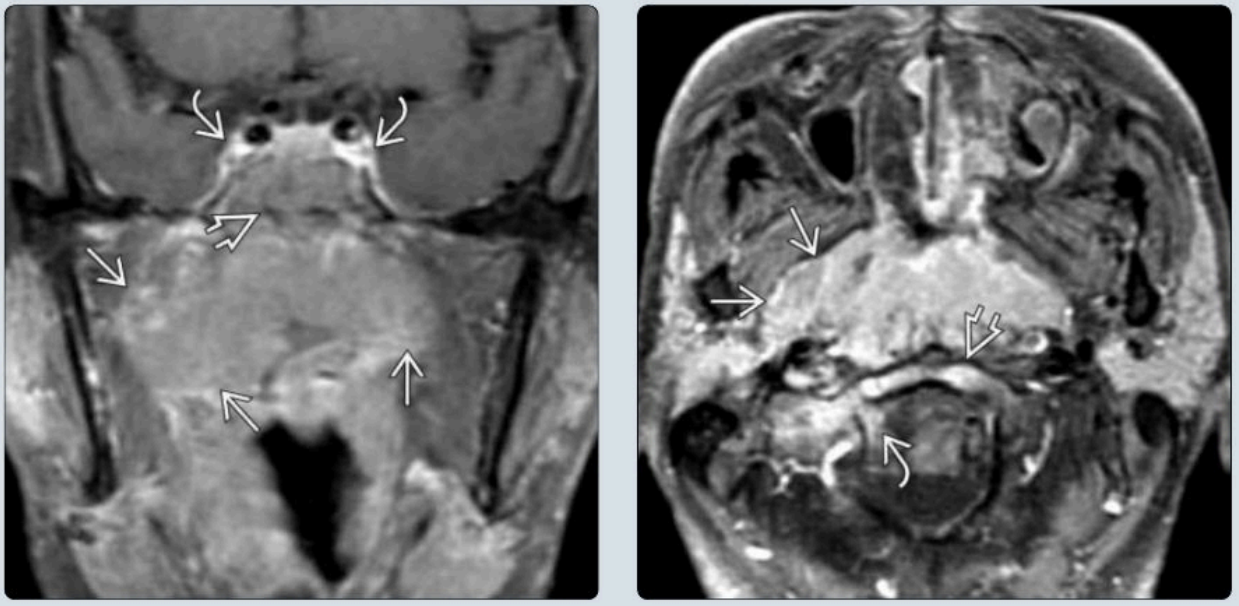


FIGURE 2 The same tumour after gadolinium. Left: coronal, enhancing tumour infiltrating the sphenoid. Right: axial, infiltration of the parapharyngeal fat and the basisphenoid marrow. MRI is the staging study for the nasopharynx.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 484

2.2 Perineural spread

Perineural tumour often causes no symptoms and is easily missed, but it changes the surgical margin and the radiotherapy field. Follow the nerves at risk for each site: **V2** (palate, maxilla, cheek) through the pterygopalatine fossa and foramen rotundum; **V3** (masticator space, retromolar trigone) through the foramen ovale; the **facial nerve** (parotid, skin of the face) through the stylomastoid foramen; the **hypoglossal** nerve (tongue base, skull base). Look for nerve enlargement and enhancement, lost fat at a foramen, widened foramina and denervation of muscle.



FIGURE 3 Perineural spread. Left: drawing of tumour following a nerve through the skull base. Right: coronal post-gadolinium MRI in maxillary carcinoma, a grossly enlarged, enhancing V2 in the foramen rotundum compared with the normal side.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 536



FIGURE 4 The same pattern on CT. Left: loss of the fat in the right pterygopalatine fossa compared with the left. Right: bone CT, smooth widening of the right foramen rotundum and pterygopalatine fossa.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 536

CHAPTER THREE

3 The nodes and beyond

N and M.

3.1 Nodes

Go through every level on both sides and the retropharyngeal nodes, as in the nodes lesson, and report the **number, size, side and level** of abnormal nodes, any **necrosis**, and any **extranodal extension**. Radiological extranodal extension (ill-defined margins, infiltration of fat, muscle or skin) matters: in the eighth edition of the TNM, clinically and radiologically overt extranodal extension makes HPV-negative nodal disease N3b. Report **carotid contact** in degrees: 270° or more suggests encasement. HPV-positive oropharyngeal cancer is staged with its own, more favourable system.

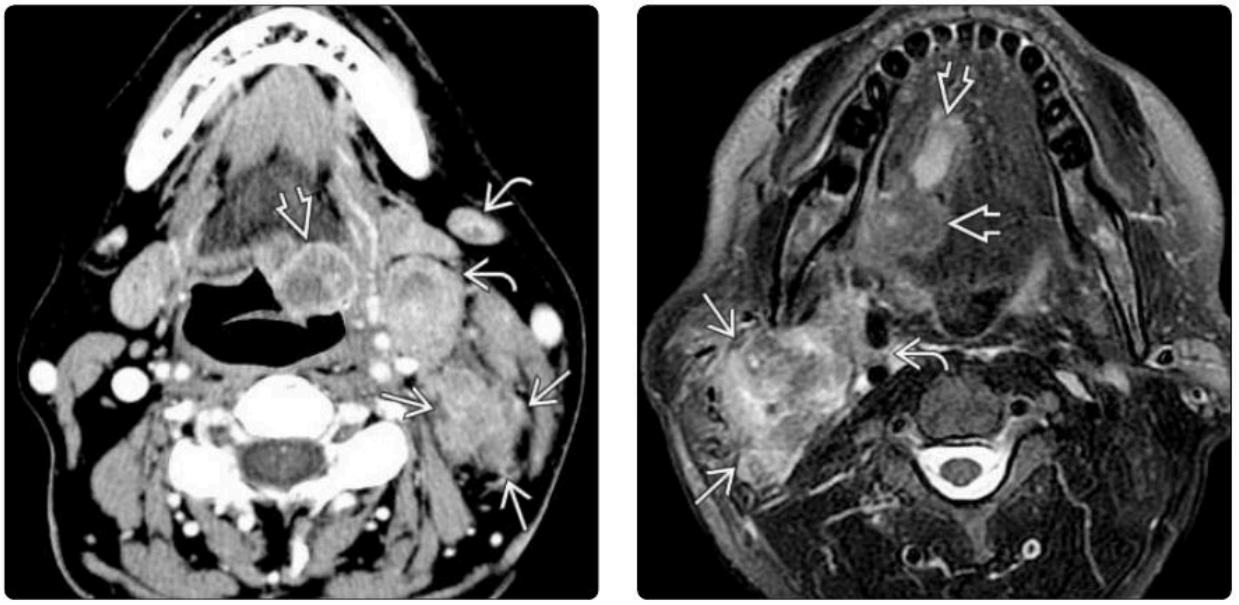


FIGURE 5 Extranodal extension. Left: pathological left neck nodes from a tongue base primary, the level IIB node with ill-defined contours. Right: a large level II nodal mass from a tongue cancer infiltrating the surrounding soft tissues.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 544

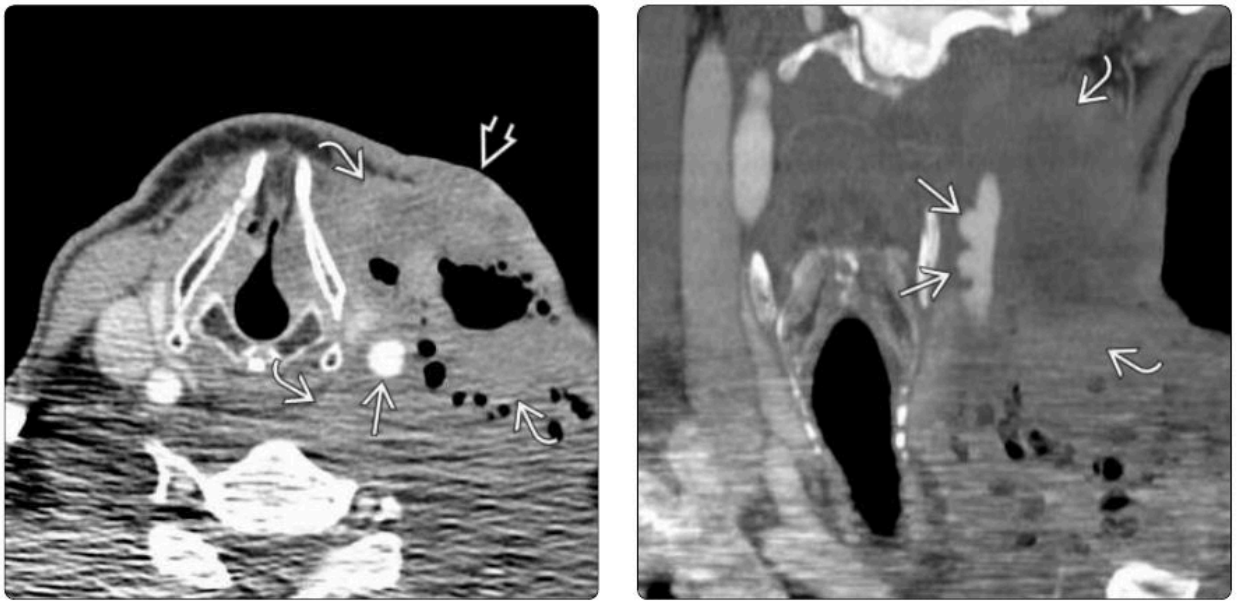


FIGURE 6 Advanced extranodal disease. Left: tumour surrounding the left common carotid and thickening the skin. Right: CT angiography showing outpouchings of a weakened carotid wall, a warning of carotid blowout.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 544

3.2 Distant disease and second primaries

At staging, look at the **lung apices**, the **bones** (including the mandible and spine) and the **mediastinal nodes** on every neck scan, and review the chest CT. Patients with smoking-related cancers carry a high risk of a **second primary** in the upper aerodigestive tract, lung or oesophagus, most often with hypopharyngeal cancer; PET-CT helps to find them.

CHAPTER FOUR

4 The unknown primary

When only the node is known.

4.1 A sequence that finds the primary

When squamous carcinoma is found in a neck node and examination is normal, imaging is done **before** panendoscopy and biopsy, so that biopsy changes do not confuse it. Contrast CT or MRI looks for asymmetry at the usual hiding places (the tonsils, the tongue base, the nasopharynx and the pyriform apex), and **PET-CT** finds a further proportion. HPV and EBV status of the node points to the oropharynx or the nasopharynx. The search ends with panendoscopy, directed biopsies and often tonsillectomy and tongue base mucosectomy.

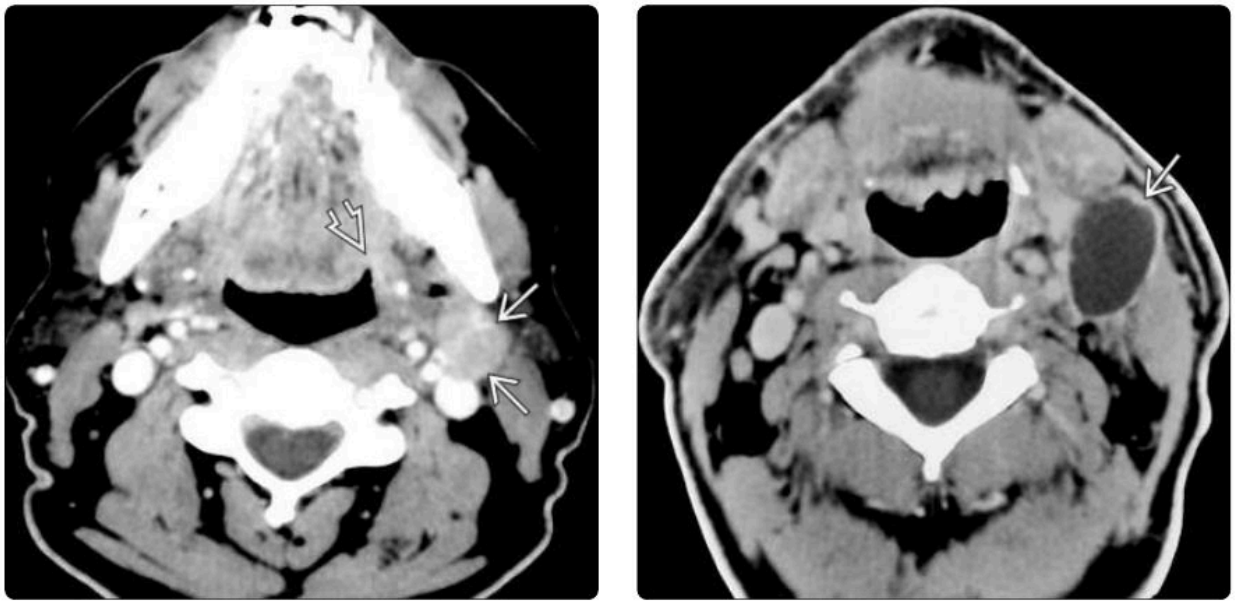


FIGURE 7 Unknown primary. Left: a unilateral necrotic level IIA node; panendoscopy was negative but directed biopsy of the glossotonsillar sulcus found the primary. Right: a cystic, rim-enhancing level II node from an occult lingual tonsil carcinoma.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 543

READING A HEAD AND NECK CANCER STAGING SCAN

1 Primary

- → subsite
- → size in three planes
- → depth (oral cavity)
- → crossing the midline

2 Local extent

- → site-specific T4 structures
- → cartilage, bone, skull base, orbit, dura
- → parapharyngeal space and pterygoids

3 Perineural

- → nerves at risk for the site

- → foramina and fat pads
- → denervation

4 **Vessels**

- → carotid contact in degrees
- → jugular vein

5 **Nodes**

- → every level both sides and retropharyngeal
- → size, necrosis, extranodal extension

6 **Beyond the neck**

- → lung apices and chest CT
- → bones
- → mediastinum
- → second primary

7 **Summary**

- → TNM stage
- → resectability
- → what would change the plan (biopsy, PET-CT, MRI)

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Which scan for which site:** CT for larynx and neck, MRI for nasopharynx, oral cavity, oropharynx and sinonasal, PET-CT for advanced and unknown primary.
- **A routine:** primary, local extent, perineural, vessels, nodes, chest and second primary.
- **T4a versus T4b:** what makes a tumour unresectable.
- **Perineural spread:** the nerves at risk for each site and the signs.
- **Extranodal extension** and carotid encasement.
- **The unknown primary:** imaging before panendoscopy, PET-CT, HPV and EBV.

5.2 Common pitfalls

COMMON PITFALLS

- Staging a nasopharyngeal cancer on CT alone.
- Forgetting the chest, the bones and the second primary.
- Missing perineural spread because the post-contrast images were not fat-suppressed.
- Imaging the unknown primary after biopsies have distorted the mucosa.

Written for this app from Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), squamous cell carcinoma overview and diagnoses, ESR Modern Radiology, Head and Neck Imaging, Dunnebier, Imaging for Otolaryngologists. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 16

The treated neck

After surgery, radiotherapy or chemoradiotherapy the neck no longer looks like the anatomy books. Fat planes vanish, tissues swell and then scar, muscles waste, flaps bring fat and muscle from elsewhere, and the radiologist's question changes from "what is this?" to "is anything new?". The key to reading the treated neck is comparison: a good post-treatment baseline scan, knowledge of exactly what was done, and a standard way of grading suspicion. This lesson covers all three, with the NI-RADS system that many centres now use.

AT A GLANCE

- Baseline **PET-CT at about 12 weeks** after chemoradiotherapy (PET-NECK); compare every later scan with it.
- Expected change is **symmetric, smooth and diffuse**: oedema early, fibrosis and atrophy late.
- Recurrence is **discrete, new or enlarging**, nodular at flap margins, restricts diffusion and is focally FDG-avid.
- **NI-RADS** 1 to 4 for primary and neck, each tied to an action.
- Mimics: hypertrophied levator scapulae, residual thyroid lobe, **osteoradionecrosis**.

1 CHAPTER ONE The baseline

When to scan, and with what.

1.1 Timing

Inflammation after radiotherapy makes early scans unreliable. The first **post-treatment baseline** is usually a **PET-CT at about 12 weeks** after chemoradiotherapy (the PET-NECK trial showed this is as safe as planned neck dissection while sparing most patients the operation), or a contrast CT or MRI at 8 to 12 weeks after surgery or radiotherapy. Scans earlier than this are more likely to show false-positive uptake and swelling. The baseline becomes the map against which every later scan is compared; surveillance then continues at intervals guided by the risk of recurrence, which is highest in the first two years.

1.2 Know the operation

Read the operation note before the scan. A **selective neck dissection** may leave only subtle changes: loss of the fat around the carotid sheath and under the sternocleidomastoid. A **radical neck dissection** removes the sternocleidomastoid, the jugular vein and the accessory nerve; the trapezius wastes and the **levator scapulae** enlarges in compensation, which can be mistaken for a mass. Resection of one thyroid lobe at laryngectomy leaves the other, which can mimic a node.

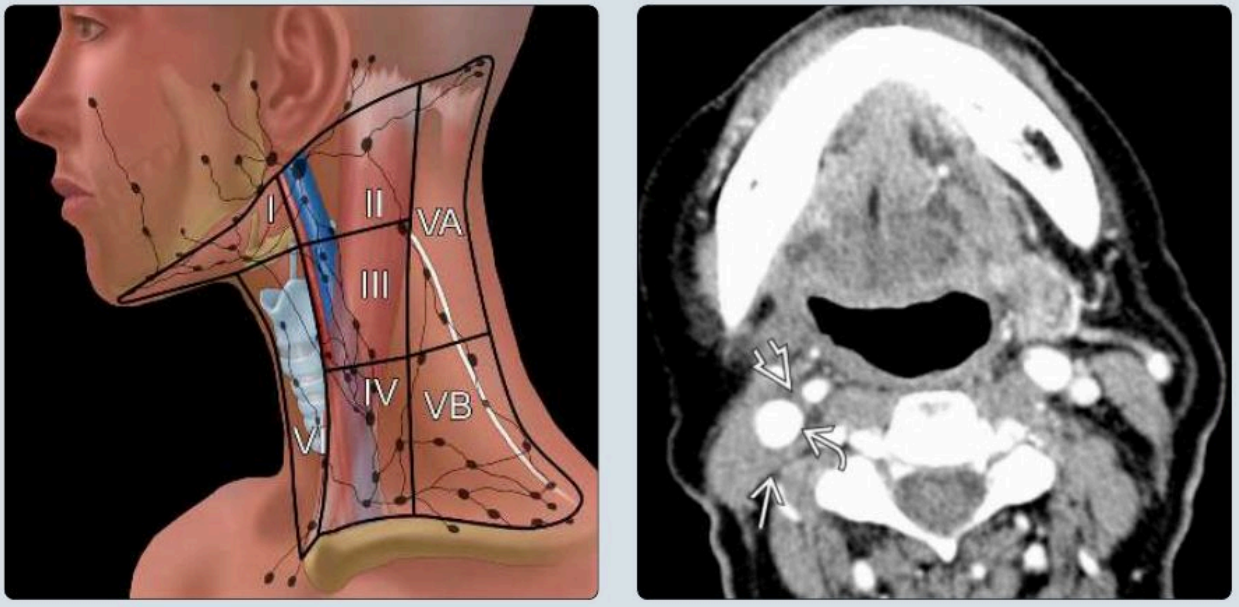


FIGURE 1 Neck dissection on imaging. Left: drawing of the nodal levels removed in different dissections. Right: contrast CT after selective neck dissection at level II on the right, with loss of the fat beneath the sternocleidomastoid and around the carotid space.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 548

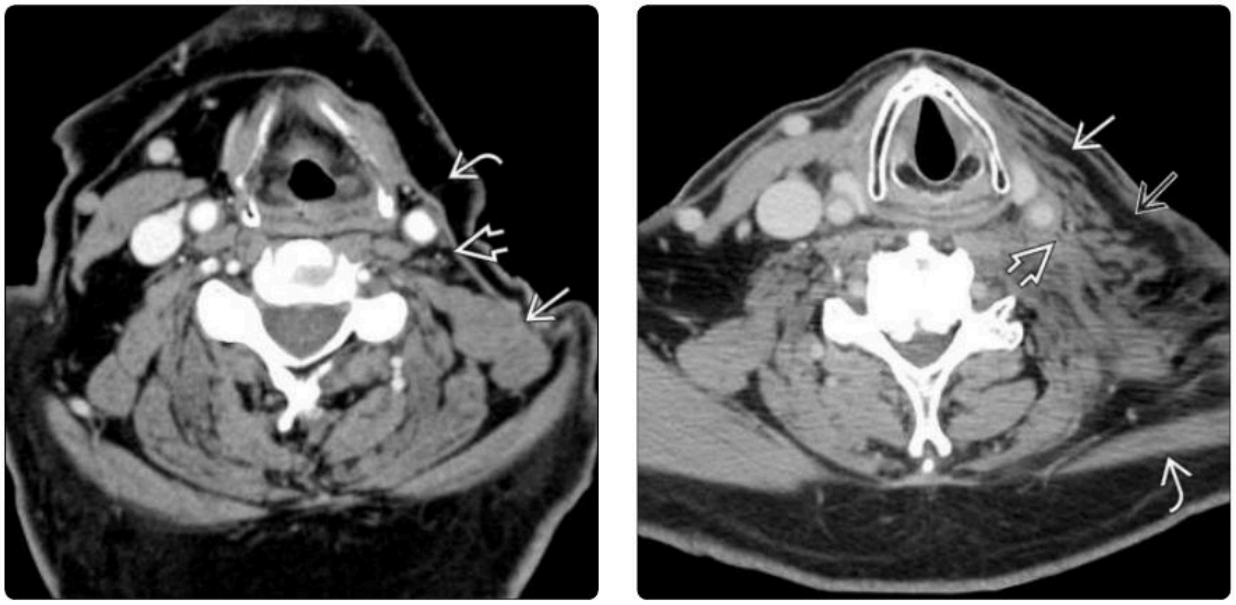


FIGURE 2 After radical neck dissection. Left: the left sternocleidomastoid and jugular vein are absent and the levator scapulae has enlarged, a mimic of tumour. Right: absent sternocleidomastoid and jugular vein, a wasted trapezius and a pectoralis flap.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 551

2 CHAPTER TWO Expected changes

What is normal after treatment.

2.1 After radiotherapy

In the first months after radiotherapy, everything in the field swells: the **skin and platysma thicken**, the **subcutaneous and deep fat becomes streaky**, the **mucosa and submucosa** of the pharynx and larynx are oedematous and enhance, the **retropharyngeal space** fills with fluid (oedema, not a mass), and the salivary glands enhance and later shrink. Over months to years, swelling turns into **fibrosis and atrophy**: the tissues become thin, dark on T2 and enhance little. Expected changes are **symmetric, smooth, diffuse and without a discrete mass**.

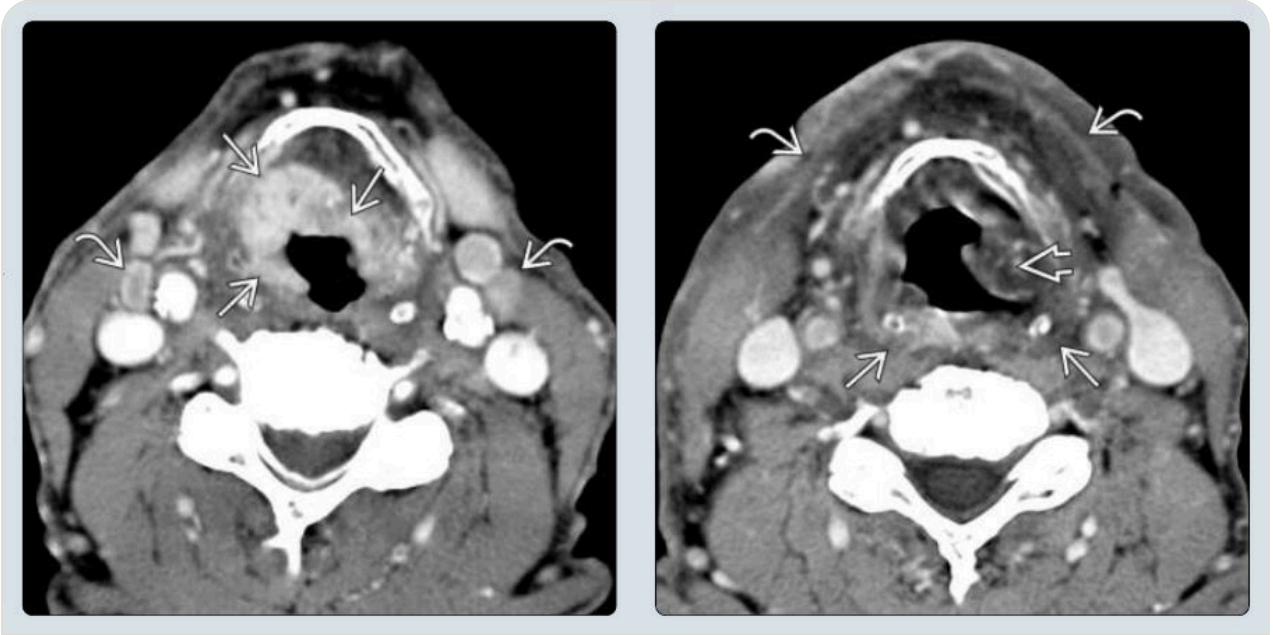


FIGURE 3 Expected changes after chemoradiotherapy. Left: a supraglottic carcinoma before treatment. Right: three months later, no residual tumour or nodes, but every fat plane and the aryepiglottic folds are swollen and hazy: expected oedema, not disease.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 556

2.2 After reconstruction

Free and pedicled **flaps** bring fat, muscle and sometimes bone. The fat component is easy to recognise; the muscle component is **denervated**, so it looks striated, may enhance at first and then wastes and becomes fatty. Bone flaps (fibula, iliac crest, scapula) show osteotomies and plates. The edges of a flap, where it meets native tissue, are where recurrence most often appears.

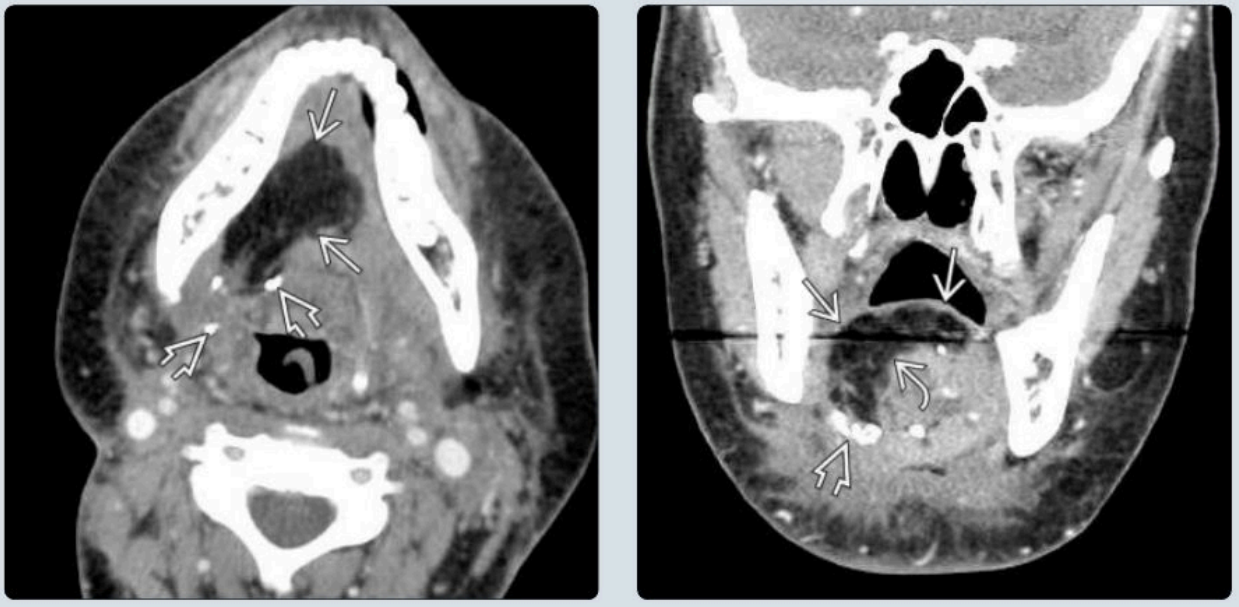


FIGURE 4 A radial forearm free flap. Left: axial contrast CT, a well-defined, rounded fatty flap filling the defect, with surgical clips. Right: coronal CT, the flap restoring the contour of the tongue.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 552

3 CHAPTER THREE Is it recurrence?

Patterns that worry.

3.1 The signs

Recurrence is suggested by a **discrete, mass-like** lesion rather than diffuse change, **new or enlarging** soft tissue compared with the baseline, **nodular enhancement** (especially at the margins of a flap or at an anastomosis), **restricted diffusion** (low ADC), **intense, focal FDG uptake** after the inflammatory period, **bone destruction**, new **necrotic nodes**, and new **perineural** enhancement. Most recurrences appear at the primary site or at the edge of the treated field, in the first two years.

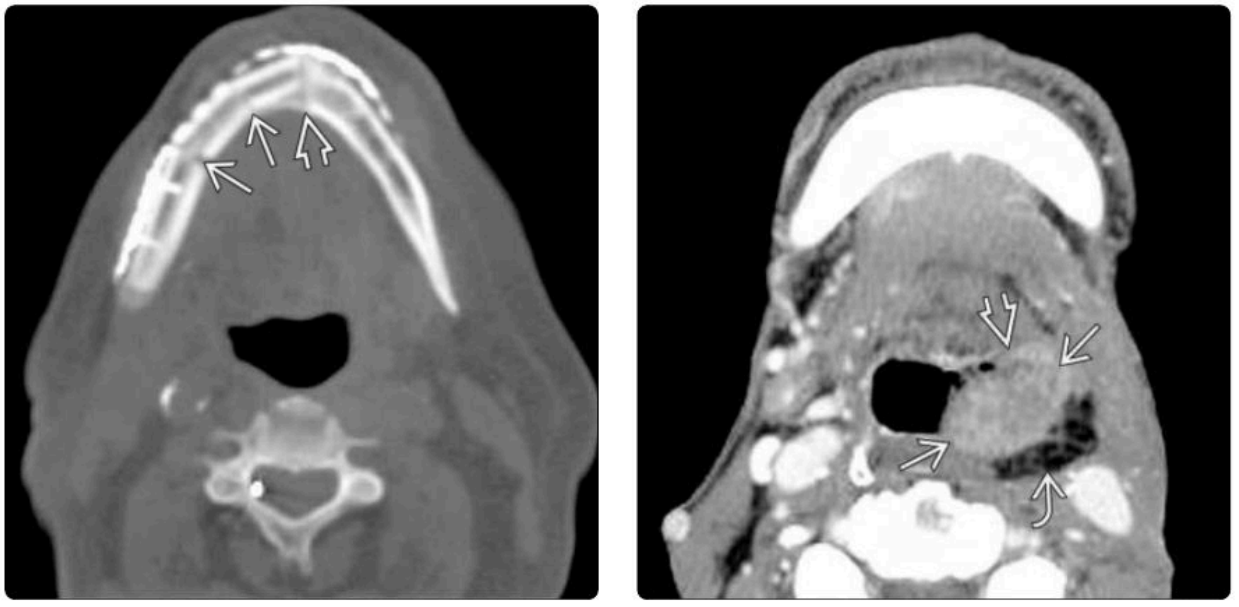


FIGURE 5 Left: a fibular free flap reconstructing the mandible, with osteotomies that allow the bone to be contoured. Right: recurrence at the junction of the normal tongue base and a myocutaneous flap, a discrete, uniformly enhancing nodule on the mucosal side.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 555

3.2 NI-RADS

The **Neck Imaging Reporting and Data System** (NI-RADS) gives a suspicion category for the primary site and for the neck on each surveillance scan, each tied to an action.

NI-RADS CATEGORIES

CATEGORY	PRIMARY SITE	NECK	USUAL ACTION
1, no evidence of recurrence	Expected changes only; non-mass-like distortion, diffuse mucosal enhancement	No residual or new abnormal nodes	Routine surveillance
2a, low suspicion	Focal **mucosal** enhancement, not mass-like	Residual or enlarging nodes without new necrosis or extranodal extension	Direct inspection
2b, low suspicion	**Deep**, ill-defined soft tissue with only mild differential enhancement	(as 2)	Short-interval follow-up or PET

CATEGORY	PRIMARY SITE	NECK	USUAL ACTION
3, high suspicion	**New or enlarging discrete** nodule or mass with intense enhancement or FDG uptake	New or enlarging necrosis or extranodal extension, or intensely FDG-avid node	Biopsy
4, definite recurrence	Proven by pathology, or definite progression	Proven or definite	Treat

4 CHAPTER FOUR Complications

Not every change is tumour.

4.1 Osteoradionecrosis and chondronecrosis

Osteoradionecrosis of the mandible appears on CT as mixed lysis and sclerosis, **cortical breaks**, **gas** in the bone, fragmentation and sequestra, and sometimes a pathological fracture, usually in the body of the mandible within the radiation field and often after a dental extraction. MRI shows marrow oedema and enhancement of the bone and surrounding soft tissue, which can look like tumour; the pattern of bone and gas on CT, and biopsy when doubtful, separate them. **Chondronecrosis** of the laryngeal cartilages shows fragmentation, collapse and gas.

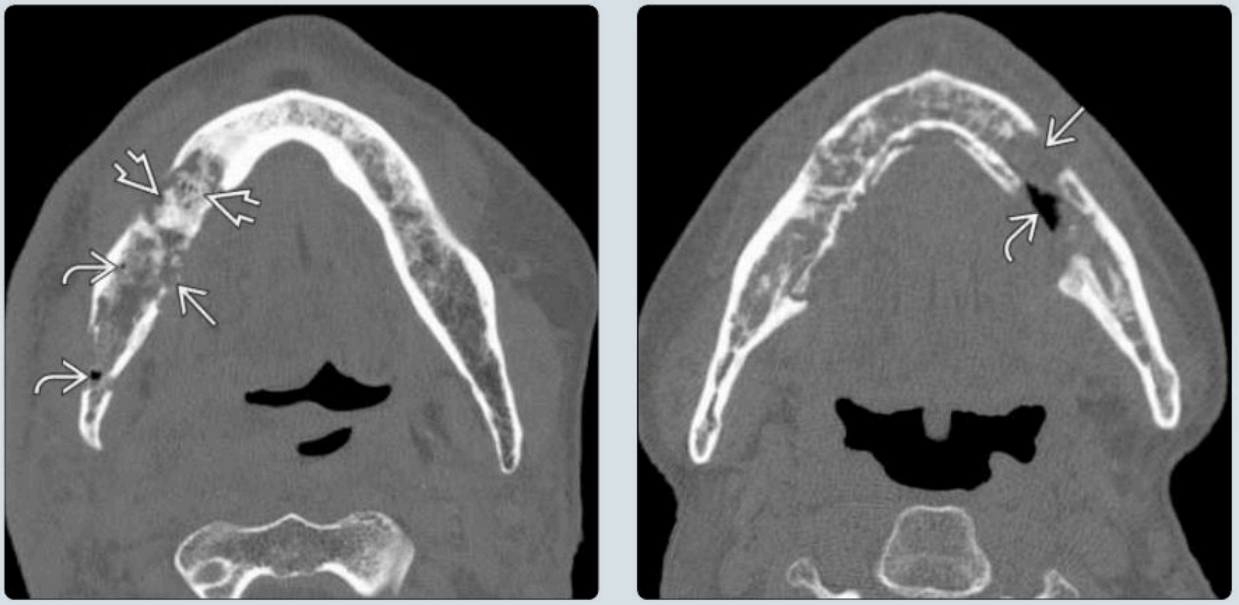


FIGURE 6 Osteoradionecrosis of the mandible. Left: bone CT, mixed lytic and sclerotic change with cortical interruption and gas bubbles in the bone. Right: a predominantly lytic pattern in another patient.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 564

4.2 Other late effects

Radiotherapy can also cause **temporal lobe necrosis** after nasopharyngeal treatment, **carotid stenosis**, hypothyroidism, fibrosis of the pharynx with dysphagia and aspiration, and, rarely, a **radiation-induced sarcoma** years later.

4.3 After laryngectomy

After total laryngectomy the **neopharynx** is a midline tube of muscle and mucosa from the tongue base to the oesophagus, in front of the spine. Recurrence appears as soft tissue thickening, often at the upper or lower anastomosis, the stoma or the paratracheal region.



FIGURE 7 Total laryngectomy. Left: contrast CT of a T4 laryngeal carcinoma with subglottic and prelaryngeal extension. Right: drawing of the expected anatomy after laryngectomy: the neopharynx joins the oral cavity to the oesophagus, and the trachea ends at the stoma.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 568

READING A POST-TREATMENT SCAN

1 Before you start

- → what was done and when
- → the pre-treatment scan
- → the baseline after treatment

2 Expected change

- → symmetric, smooth, diffuse
- → oedema early, fibrosis late
- → flaps and dissection changes

3 Primary site

- → discrete mass or nodular enhancement
- → flap margins and anastomoses

- → DWI and PET

4 **Neck**

- → new or enlarging nodes
- → new necrosis or extranodal extension
- → the other side

5 **Complications**

- → osteoradionecrosis
- → chondronecrosis
- → carotid
- → temporal lobes

6 **Grade**

- → NI-RADS for primary and neck
- → the action that goes with it

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Baseline scan timing:** PET-CT at about 12 weeks after chemoradiotherapy; PET-NECK.
- **Expected changes** after radiotherapy and their time course.

- **Neck dissection and flap** appearances, and the mimics (levator scapulae, residual thyroid lobe).
- **Signs of recurrence:** discrete, new, nodular, restricted diffusion, focal FDG.
- **NI-RADS** categories and actions.
- **Osteoradionecrosis** versus recurrence.

5.2 Common pitfalls

COMMON PITFALLS

- Scanning too early after radiotherapy and acting on inflammatory uptake.
- Reading a surveillance scan without the baseline for comparison.
- Calling a hypertrophied levator scapulae or a remaining thyroid lobe a recurrence.
- Calling osteoradionecrosis tumour, or tumour osteoradionecrosis, without biopsy.

Written for this app from Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), post-treatment neck chapters, ESR Modern Radiology, Head and Neck Imaging, Dunnebier, Imaging for Otolaryngologists. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 17

Trauma

In trauma, the scan has to answer quickly what the surgeon cannot see: which bones are broken and how they are displaced, whether the airway, the eye, the optic nerve, the facial nerve, the dura or the great vessels are at risk, and what needs fixing urgently. Facial and temporal bone fractures follow recognisable patterns, and recognising the pattern is what lets you look for the right complication. This lesson covers the face, the temporal bone, the larynx and penetrating injuries of the neck.

AT A GLANCE

- Thin **non-contrast CT** with 3D reformats; **CT angiography** for carotid canal fractures, penetrating injury or unexplained deficit.
- Le Fort fractures all break the **pterygoid plates**; unique fractures: nasal aperture (I), inferior orbital rim (II), zygomatic arch (III).
- NOE: the **medial canthal tendon** fragment. Frontal sinus: anterior table, **posterior table**, outflow.
- The mandible often breaks in **two places**; fractures through teeth are open.
- Temporal bone: **otic capsule sparing or violating**; check ossicles, facial canal, tegmen and carotid canal.

1 CHAPTER ONE The study

Thin CT, then angiography if needed.

1.1 Protocol

Non-contrast thin-section CT (about 1 mm or less) in a bone algorithm, with **multiplanar and 3D reformats**, is the standard for facial, temporal bone and laryngeal fractures; it is done in one pass with the head CT and, where indicated, the cervical spine. **CT angiography** is added when a vessel may be injured: fractures through the **carotid canal**, the **foramen transversarium** or the skull base, a penetrating neck wound, an expanding haematoma, a neurological deficit not explained by the brain scan, or severe midface fractures. **MRI** is kept for the optic nerve, brain, ligaments and the facial nerve when needed later.

2 CHAPTER TWO The face

Buttresses and patterns.

2.1 The buttresses

The midface is a frame of thick **vertical buttresses** (nasomaxillary, zygomaticomaxillary and pterygomaxillary) joined by **horizontal** ones (the frontal bar, the infraorbital rims and orbital floors, the zygomatic arches and the palate and alveolus). Fractures are described by which buttresses are broken, because repair restores them. The **pterygoid plates** link the midface to the skull base, and are broken in every Le Fort fracture.

2.2 Le Fort fractures

Le Fort fractures separate part of the midface from the skull base, and each type has a **unique fracture** that identifies it: **Le Fort I** (horizontal, separating the alveolus and palate) breaks the **lateral margin of the nasal aperture**; **Le Fort II** (pyramidal) breaks the **inferior orbital rim**; **Le Fort III** (craniofacial dissociation) breaks the **zygomatic arch** and the lateral orbital wall. All three break the pterygoid plates. Fractures are often mixed and different on the two sides.

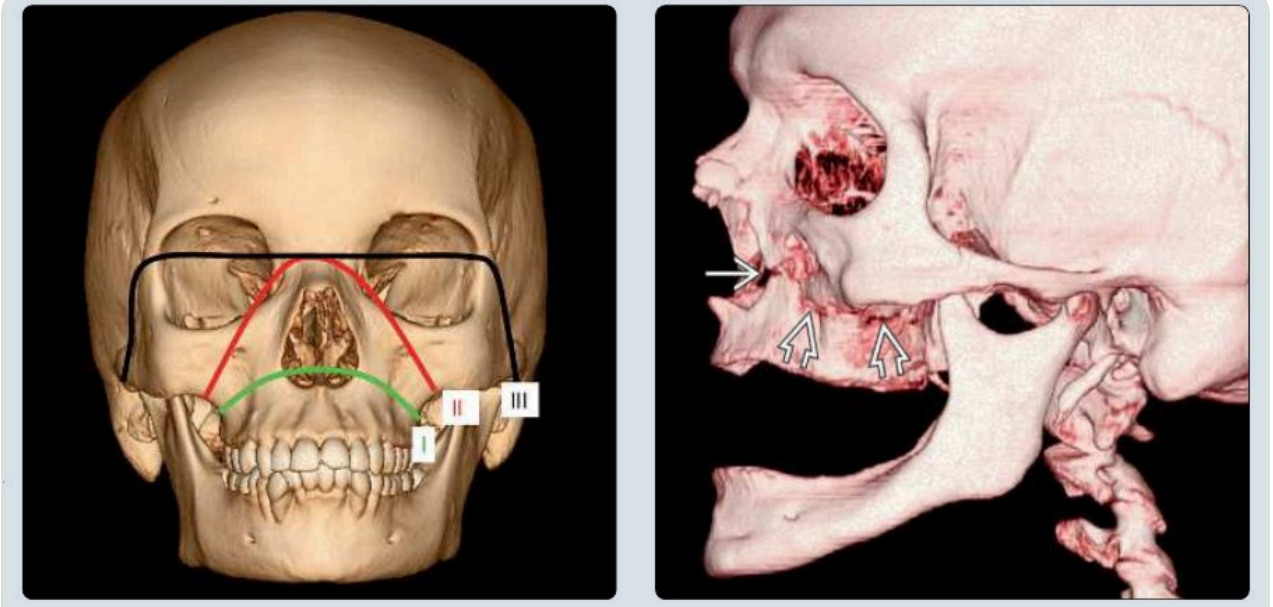


FIGURE 1 Le Fort fractures. Left: drawing of the three levels. Right: 3D CT of a Le Fort I fracture separating the maxillary alveolus from the midface, through the nasal aperture.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 982

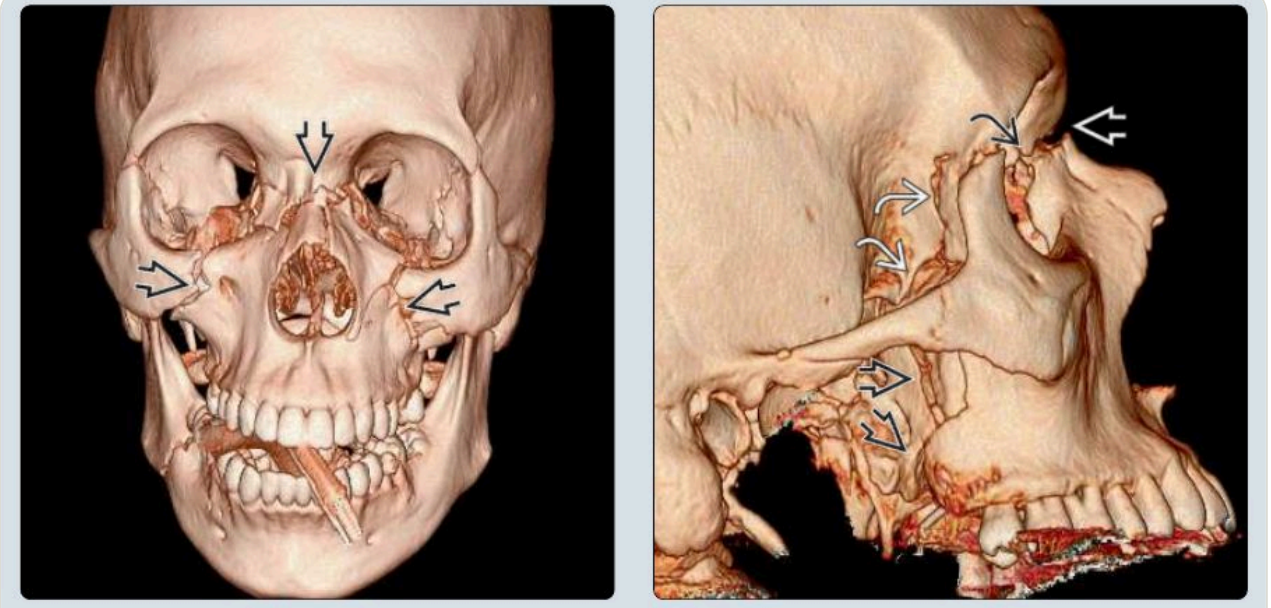


FIGURE 2 Left: 3D CT of a Le Fort II fracture through both inferior orbital rims, with rotation of the midface and malocclusion. Right: a Le Fort III fracture with nasofrontal separation, medial and lateral orbital wall and pterygoid plate fractures.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 982

2.3 Zygomaticomaxillary complex

The zygoma is attached at four points, and a **zygomaticomaxillary complex** (tetrapod) fracture separates it at all of them: the **zygomaticofrontal** suture

(lateral orbital rim), the **zygomatic arch**, the **inferior orbital rim and floor**, and the **anterior and posterolateral maxillary walls**, with the zygomaticosphenoid suture in the lateral orbital wall. Report the displacement and rotation of the zygoma, the orbital floor defect, and the infraorbital canal.

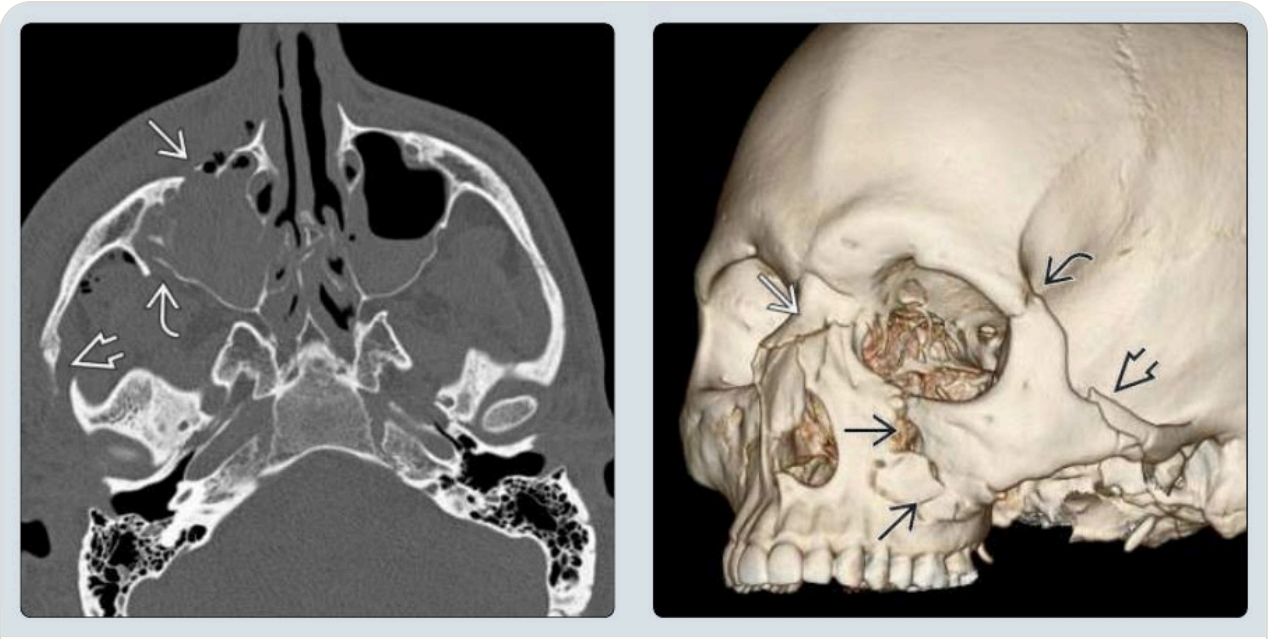


FIGURE 3 Zygomaticomaxillary complex fracture. Left: axial CT, fractures of the anterior and lateral walls of the maxillary sinus with swelling over the cheek. Right: 3D CT, the classic pattern through the maxillary sinus walls, zygomatic arch and lateral orbital wall.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 986

2.4 Nasoorbitoethmoid and frontal sinus fractures

Nasoorbitoethmoid fractures drive the nasal bones back between the orbits. The key question is whether the fragment carrying the **medial canthal tendon** is intact, displaced or comminuted, which decides the repair and the risk of telecanthus. Look also at the **nasolacrimal ducts**, the **frontal recess** and the anterior skull base. **Frontal sinus** fractures are described by the **anterior table**, the **posterior table** (dura and CSF leak) and the **frontal sinus outflow** through the frontal recess, which predicts later mucocele.

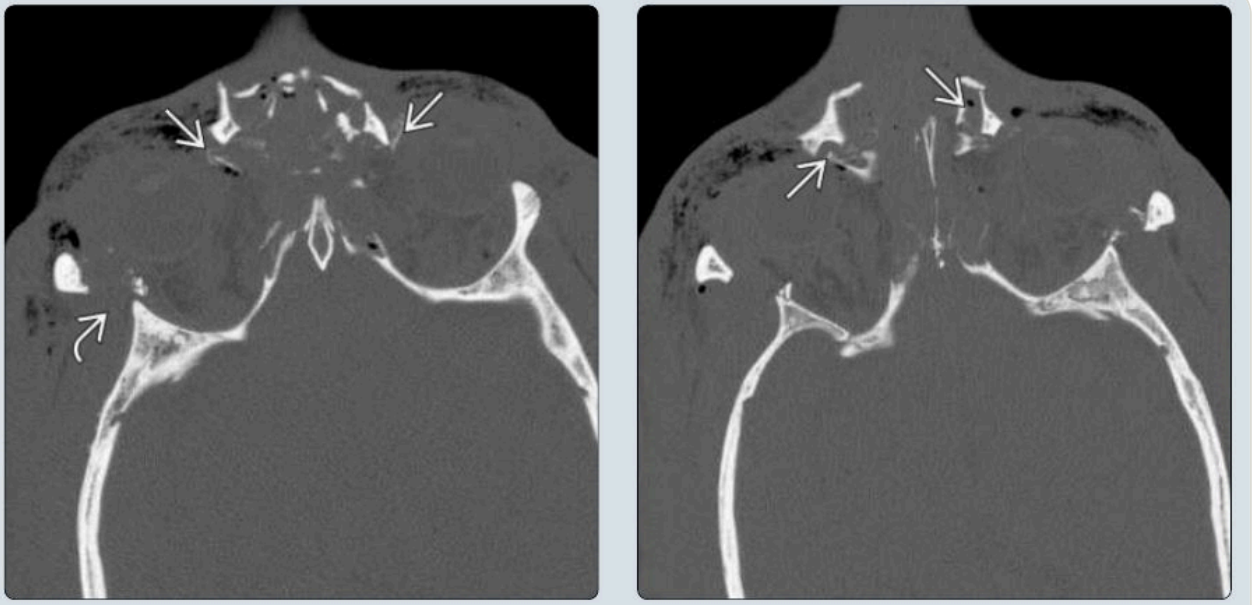


FIGURE 4 Nasoorbitoethmoid fracture. Left: axial CT, comminuted fractures of the nasoorbitoethmoid complex. Right: lower down, the fractures involve both nasolacrimal ducts; epiphora is expected.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 990

2.5 The orbit

Blowout fractures, muscle entrapment, retrobulbar haemorrhage and optic canal fractures are covered in the orbit lesson. In any midface fracture, look at the orbital floor and medial wall, the globe, the optic canal and the superior orbital fissure.

2.6 The mandible

The mandible is a ring with the skull base, so it often breaks in **two places** (for example an angle on one side and the parasymphysis on the other). Report each fracture by site (condyle, ramus, angle, body, parasymphysis, symphysis), displacement, involvement of **teeth** (a fracture through a tooth socket is open), the **inferior alveolar canal** and the **temporomandibular joint**. A panoramic radiograph complements CT.

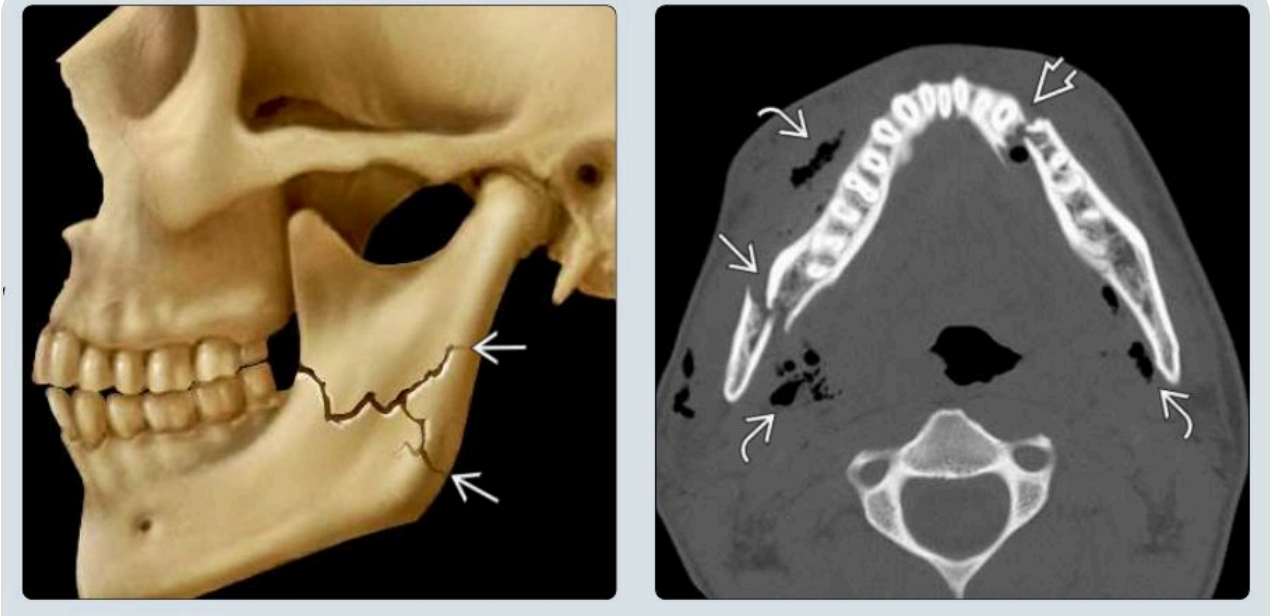


FIGURE 5 Mandible fractures. Left: drawing of a ramus fracture crossing the inferior alveolar canal. Right: axial CT, displaced fractures of the right angle and left parasymphysis, the typical two-site pattern.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 992

CHAPTER THREE

3 The temporal bone

Capsule sparing or violating.

3.1 The pattern and the structures

Temporal bone fractures are classed as **longitudinal**, **transverse** or **oblique** to the petrous ridge, but the classification that predicts complications is whether the **otic capsule is spared or violated**. Capsule-sparing fractures (the large majority) run through the squamous bone, external canal and middle ear, disrupt the **ossicles** (incudostapedial separation and incus dislocation are commonest) and cause conductive loss. Capsule-violating fractures cross the cochlea, vestibule or canals, cause sensorineural loss and vertigo, and are more often associated with **facial palsy** and **CSF leak**. In every fracture check the ossicles, the facial canal (the **geniculate** region is the commonest site of injury), the **tegmen**, the **carotid canal** (CT angiography if involved), the jugular foramen and the sigmoid sinus.

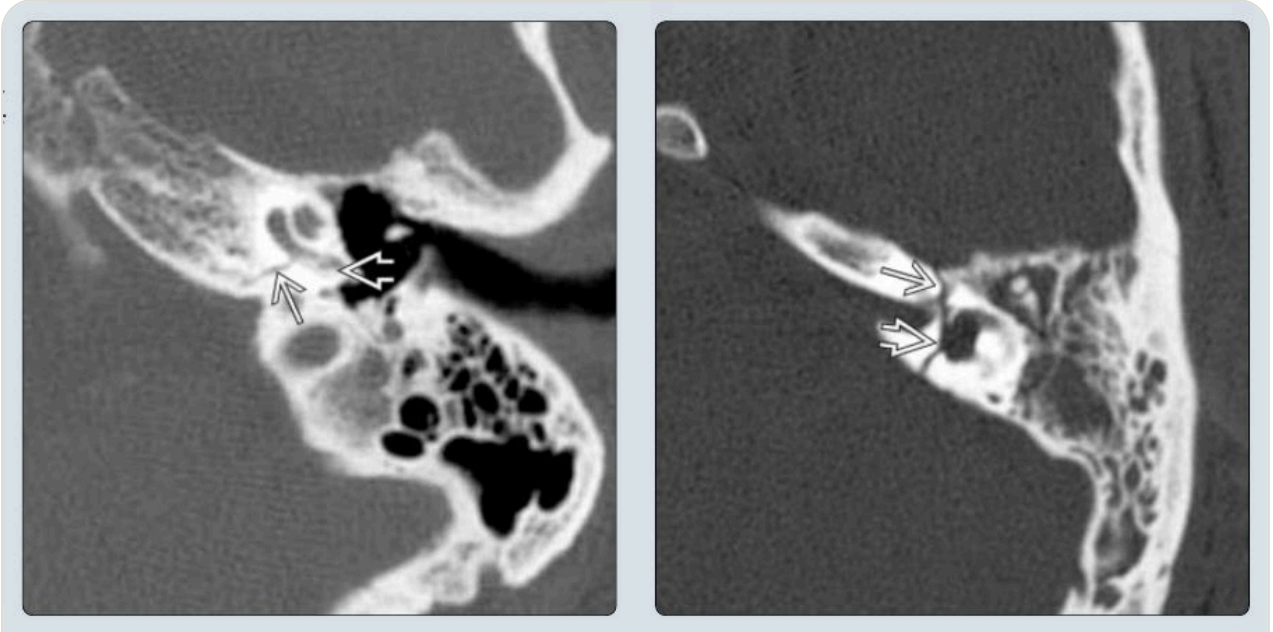


FIGURE 6 Otic capsule-violating fractures. Left: an old transverse fracture through the otic capsule to the basal turn, with new bone in the cochlea (labyrinthitis ossificans). Right: a fracture through the labyrinthine facial nerve canal and the vestibule, typically with permanent deafness and facial palsy.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 968

DO NOT CALL A SUTURE A FRACTURE

The temporal bone is full of sutures (petrotympanic, tympanosquamous, occipitomastoid), fissures and small canals (the cochlear and vestibular aqueducts, the subarcuate canal). Compare with the other side and follow the line across slices before calling it a fracture.

4 CHAPTER FOUR The larynx and the neck

Airway and vessels.

4.1 Laryngeal trauma

After blunt injury to the larynx, CT (when the airway is secure) shows fractures of the **thyroid** and **cricoid** cartilages, displacement of fragments, **arytenoid dislocation**, haematoma narrowing the airway, and air in the soft tissues. A fractured cricoid or a displaced thyroid fracture usually needs open repair; isolated

haematoma and minor oedema are usually managed conservatively. Endoscopy remains essential for the mucosa.

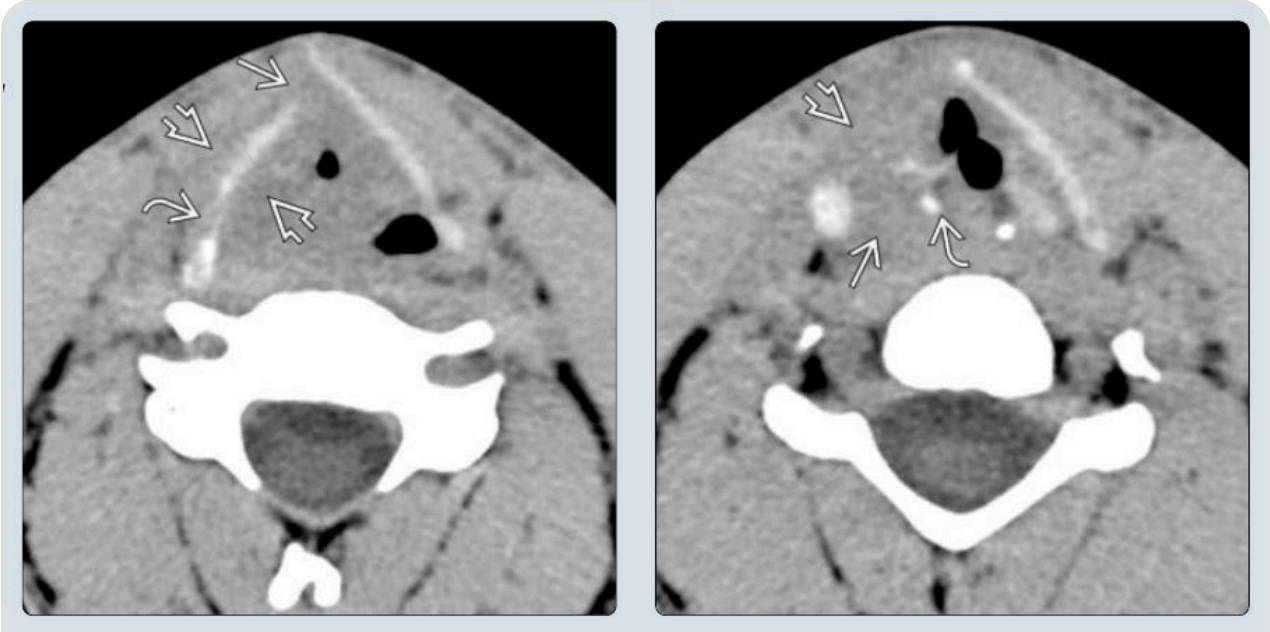


FIGURE 7 Laryngeal fracture. Left: a step at the thyroid notch and a subtle fracture of the right thyroid lamina with inward rotation of the fragment. Right: lower down, displacement of the right arytenoid and widening of the cricothyroid space.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 278

4.2 Penetrating neck injury

The neck is traditionally divided into **zones**: zone I from the clavicles to the cricoid, zone II from the cricoid to the angle of the mandible, zone III above the angle to the skull base. Unstable patients with hard signs (expanding haematoma, pulsatile bleeding, airway compromise, shock) go to theatre. In stable patients, **CT angiography** of the neck is now the first test whatever the zone; it shows the track, vascular injury (occlusion, dissection, pseudoaneurysm, extravasation), and injury to the airway, pharynx and oesophagus, which may need contrast swallow or endoscopy if the track passes close to them.



FIGURE 8 Penetrating laryngeal injury on CT angiography. Left: a pen through the supraglottis, its tip close to the left vertebral artery. Right: a gunshot wound with soft tissue air, shrapnel near the spine and comminuted fractures of the cricoid and arytenoid.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 278

READING A FACIAL OR NECK TRAUMA CT

1 Airway

- → narrowing, haematoma, displaced fragments
- → laryngeal cartilages

2 Midface

- → buttresses
- → pterygoid plates
- → Le Fort unique fractures
- → zygoma
- → NOE and canthal fragment

3 Orbit

- → floor and medial wall

- → muscle
- → globe
- → retrobulbar blood or air
- → optic canal

4 Frontal sinus and skull base

- → anterior and posterior tables
- → frontal recess
- → cribriform plate
- → pneumocephalus

5 Mandible

- → each fracture site
- → teeth
- → inferior alveolar canal
- → condyles and joints

6 Temporal bone

- → otic capsule
- → ossicles
- → facial canal
- → tegmen
- → carotid canal

7 Vessels

- → indications for CT angiography

→ carotid and vertebral arteries

→ venous sinuses

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Thin CT with 3D reformats;** indications for CT angiography.
- **Facial buttresses** and the pterygoid plates.
- **Le Fort unique fractures:** nasal aperture (I), inferior orbital rim (II), zygomatic arch (III).
- **Zygomaticomaxillary complex** fracture components.
- **NOE fractures:** the medial canthal tendon fragment.
- **Frontal sinus:** anterior table, posterior table, outflow.
- **Mandible:** two fractures, open fractures through teeth.
- **Temporal bone:** capsule sparing versus violating; structures to check.
- **Penetrating neck injury:** zones, hard signs, CT angiography.

5.2 Common pitfalls

COMMON PITFALLS

- Missing the second mandibular fracture.
- Calling a Le Fort fracture without broken pterygoid plates.
- Calling a temporal bone suture a fracture.
- Missing a carotid canal fracture and the need for CT angiography.

- Missing a posterior table frontal sinus fracture and a CSF leak.

Written for this app from Diagnostic Imaging: Head and Neck (Koch et al., 3rd ed.), skull base and facial trauma, ESR Modern Radiology, Head and Neck Imaging, Dunnebier, Imaging for Otolaryngologists. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING COURSE · LESSON 18

The child's head and neck

Imaging a child is different in three ways. The diseases are different: most neck lumps are reactive nodes or congenital cysts, nasal obstruction in a newborn may be bony, and deafness is often due to a malformation. The tests are different: ultrasound and MRI come first because of radiation, and MRI may need sedation. And the anatomy is still developing, so normal growth can mimic disease. This lesson gathers the paediatric imaging an ENT surgeon meets most often.

AT A GLANCE

- **Ultrasound and MRI first**; low-dose CT for bone and emergencies.
- Midline: **thyroglossal cyst** (hyoid, normal thyroid). Lateral, in front of SCM: **second branchial cleft cyst**. Left-sided recurrent abscess: third or fourth arch.
- **Haemangioma**: solid, high-flow, involutes. **Lymphatic malformation**: multicystic, transspatial, fluid levels.
- Newborn nose: **choanal atresia** (CHARGE) at the back, **pyriform aperture stenosis** (central incisor) at the front.
- Deaf child: **MRI** for the cochlear nerve and labyrinth, **CT** for bony anatomy; classify the malformation.

1 CHAPTER ONE Principles

The least harmful test that answers the question.

1.1 Choosing the test

Children are more sensitive to radiation than adults and have more years ahead in which harm could appear, so **ultrasound** is the first test for neck lumps, salivary and thyroid lesions, and **MRI** for soft tissue, the inner ear and nerves and intracranial connections. MRI under six months can often be done with a **feed and wrap** technique; older infants and young children may need sedation or anaesthesia, which can be combined with another procedure. **CT** is used when the question is about bone (choanal atresia, the temporal bone, fractures) or in an emergency (deep neck infection with airway concern), with a low-dose paediatric protocol.

2 CHAPTER TWO The neck lump

Position is the key.

2.1 Midline lumps

A **thyroglossal duct cyst** is the commonest congenital neck lump. It lies in the midline or just off it, anywhere from the foramen caecum of the tongue to the thyroid bed, most often at or just below the **hyoid**, and infrahyoid cysts are embedded in the **strap muscles**. Ultrasound confirms a cyst and, importantly, a **normal thyroid gland** in the lower neck. A solid component in an adult raises the question of carcinoma. A **dermoid cyst** of the floor of mouth or neck may contain fat globules ("a sack of marbles"); position relative to the mylohyoid decides the surgical approach.

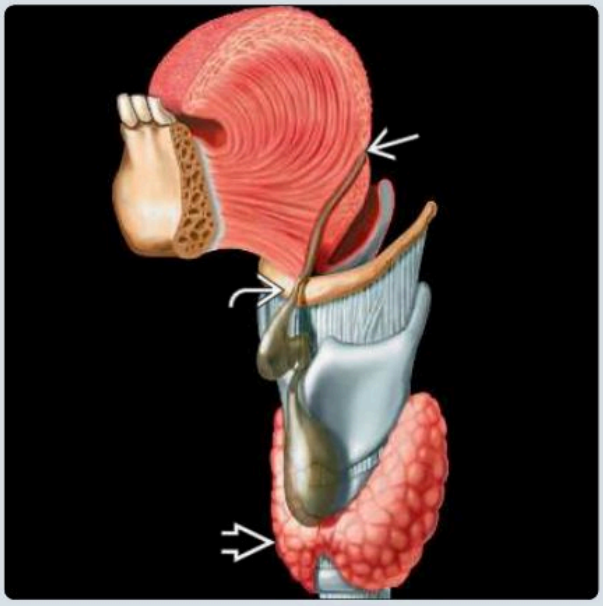




FIGURE 1 Thyroglossal duct cyst. Left: drawing of the course of the thyroglossal duct from the foramen caecum to the thyroid. Right: sagittal contrast CT, a well-defined cyst at the midline base of the tongue.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 590

2.2 Lateral lumps

The **second branchial cleft cyst** is the commonest lateral cyst: it sits at the angle of the mandible, **in front of the sternocleidomastoid**, lateral to the carotid sheath and behind the submandibular gland, displacing all three. Infection thickens its wall. A cystic lump here in an **adult** must be treated as a cystic nodal metastasis until proven otherwise. A **first branchial cleft** anomaly lies near the external auditory canal, parotid or angle of the mandible, close to the facial nerve. **Third and fourth** branchial anomalies are usually on the **left**, run from the pyriform sinus, and present as recurrent abscesses or **suppurative thyroiditis**; a barium swallow or CT after contrast may show the sinus tract.

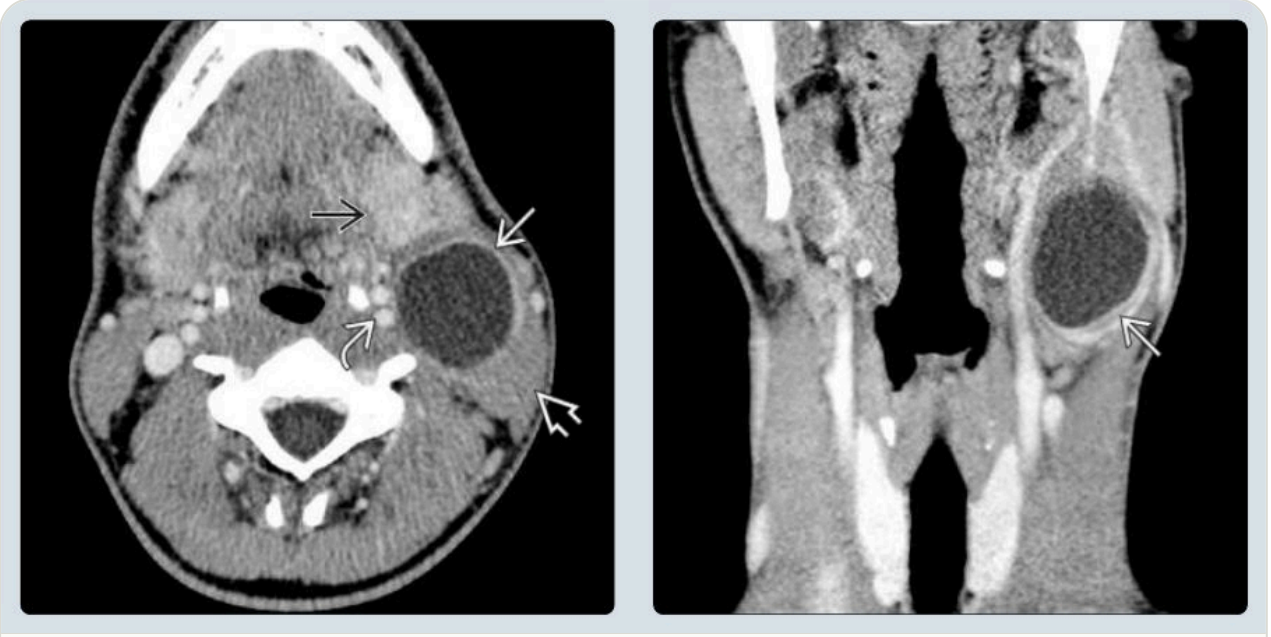


FIGURE 2 Second branchial cleft cyst. Left: contrast CT in a teenager, a well-defined low-density mass in front of the sternocleidomastoid and lateral to the carotid space. Right: coronal CT, a thicker wall inferiorly.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 602

2.3 Vascular anomalies

Vascular anomalies are divided into **tumours** and **malformations**. **Infantile haemangioma** appears in the first weeks, grows rapidly, then involutes; on imaging it is a lobulated, **solid, intensely enhancing** mass with **high-flow vessels**, often in the parotid. **Lymphatic malformations** are present from birth, are multicystic with thin septa, insinuate between structures across several spaces (**transspatial**), and

show **fluid–fluid levels** after bleeding. **Venous malformations** are bright on T2, enhance slowly and contain **phleboliths**, best seen on CT or ultrasound.

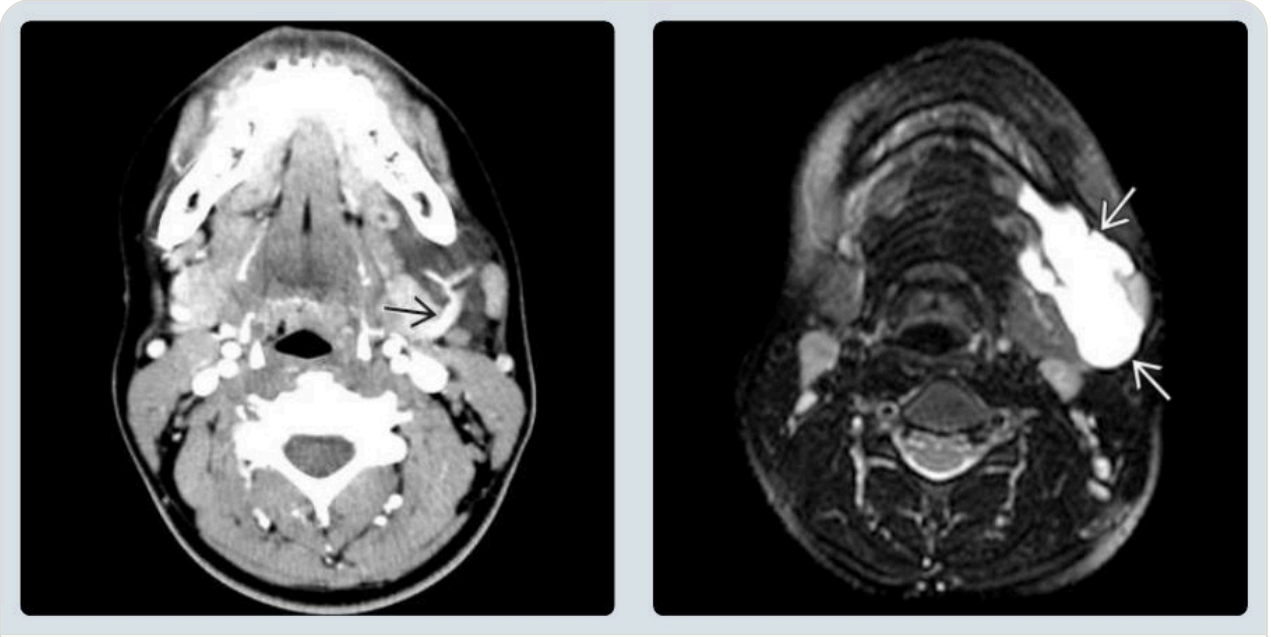


FIGURE 3 Lymphatic malformation. Left: contrast CT, a low-density lobulated mass in the submandibular space. Right: fat-suppressed T2, high fluid signal with lobular margins.
Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 580

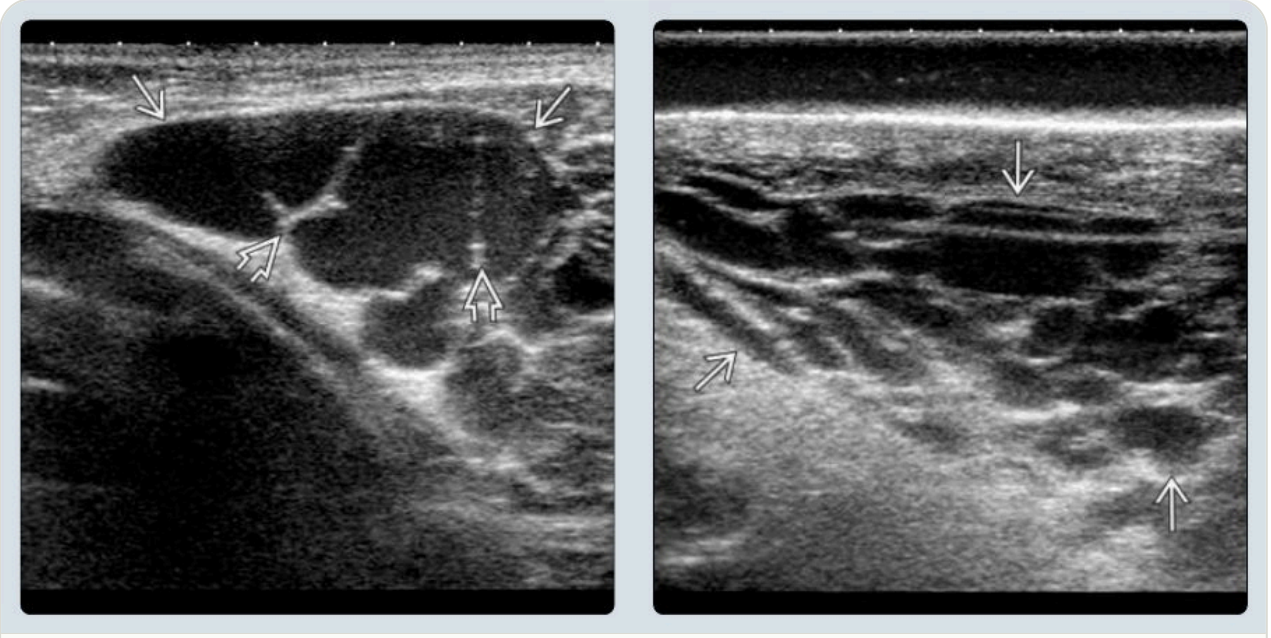


FIGURE 4 Lymphatic malformation on ultrasound. Left: a well-defined multicystic mass with thin septa (macrocystic). Right: a largely anechoic, septated mass in the subcutaneous tissues of a newborn.
Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 580

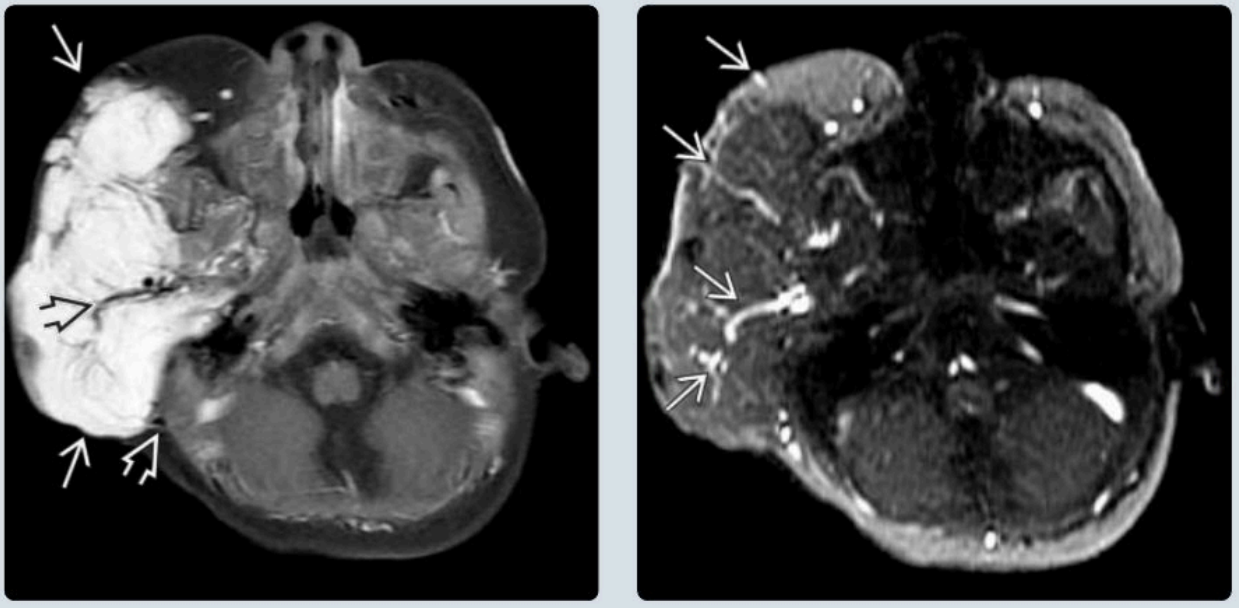


FIGURE 5 Infantile haemangioma. Left: post-gadolinium MRI, a large lobulated, intensely enhancing mass replacing the parotid gland of a five-month-old, with prominent vessels. Right: a flow-sensitive sequence showing the high-flow vessels.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 620

2.4 Malignancy and the rest

Most solid neck lumps in children are **reactive or suppurative nodes**, assessed by ultrasound. A firm, growing mass, a mass in the first months of life outside the usual sites, or nodes with systemic symptoms raise the possibility of **lymphoma**, **rhabdomyosarcoma** (orbit, nasopharynx, ear, parameningeal sites) or **neuroblastoma** (calcification). **Fibromatosis colli** is a fusiform enlargement of the sternocleidomastoid in the first weeks of life with torticollis, diagnosed on ultrasound and not to be biopsied.

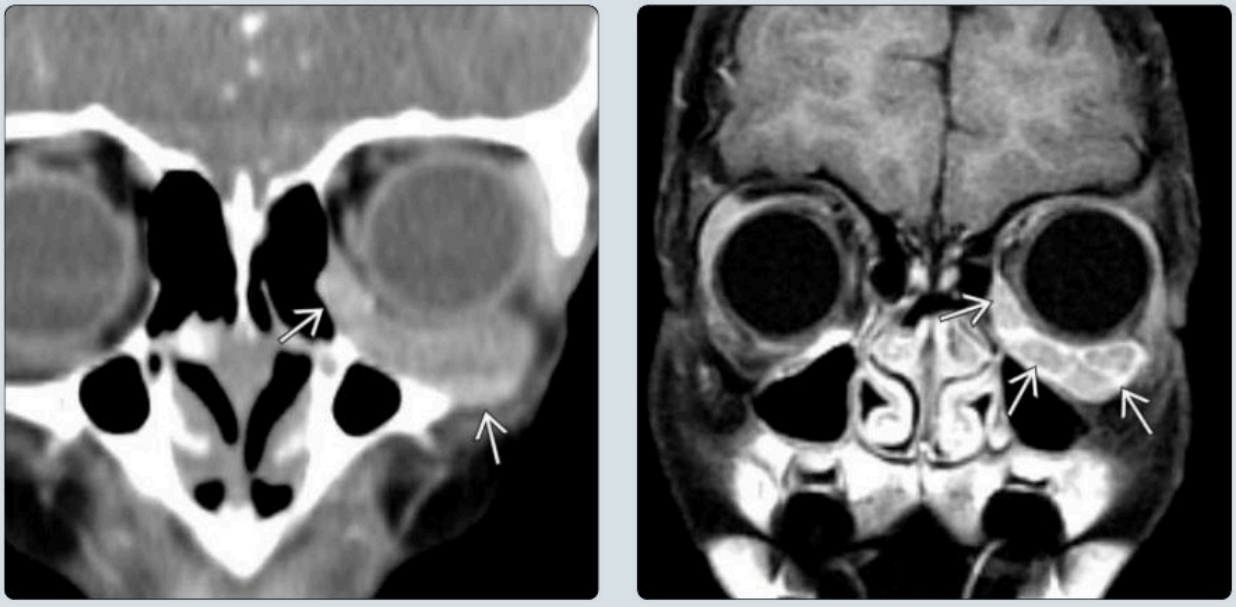


FIGURE 6 Orbital rhabdomyosarcoma in an eight-year-old. Left: coronal CT, a non-specific extraconal mass in the lower orbit without bone destruction. Right: post-gadolinium MRI, a heterogeneously enhancing solid mass.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 624

3 CHAPTER THREE The newborn nose

Bone, membrane or mass?

3.1 Choanal atresia and pyriform aperture stenosis

A newborn who cannot breathe through the nose needs **CT** after suction and topical decongestion. **Choanal atresia** shows narrowing of the posterior choanae with a **thickened vomer** and **medially bowed posterior maxilla**, closed by bone, membrane or (most often) both; bilateral atresia presents with cyclical cyanosis and is associated with **CHARGE** syndrome, so look at the inner ears too. **Congenital pyriform aperture stenosis** narrows the **front** of the nose through overgrowth of the maxillae, and is associated with a **single central incisor**, which calls for assessment of the pituitary and brain.

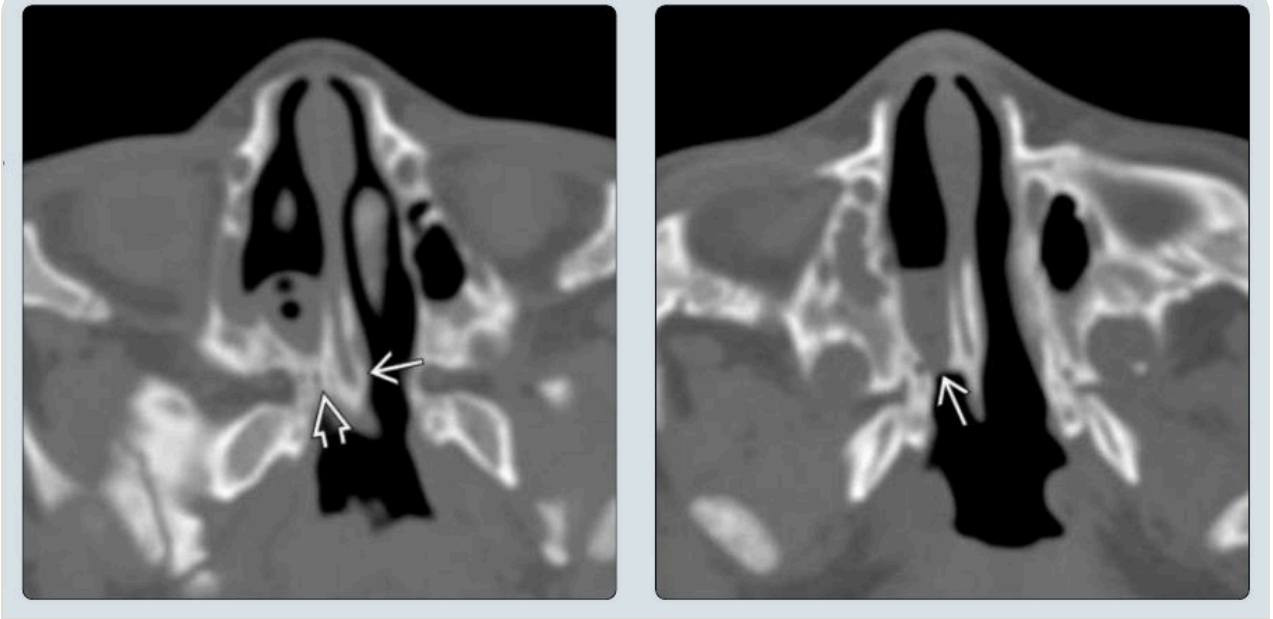


FIGURE 7 Choanal atresia. Left: axial CT, bony obstruction of the upper choana by an enlarged vomer and a thickened, medially placed posterior maxilla. Right: lower down, membranous atresia bridging the narrowed choana, with retained secretions.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 670



FIGURE 8 Congenital pyriform aperture stenosis. Left: axial CT in a newborn, overgrowth of the anterior maxillae narrowing the front of the nose. Right: a single median maxillary central incisor at the palate.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 686

3.2 Midline nasal masses

A midline nasal lump or pit in a child may be a **dermoid** (with a dermal sinus), a **glioma** (heterotopic brain) or an **encephalocele**, and any of them may connect with the cranial cavity through the **foramen caecum**. **MRI** (with thin CT of the anterior skull base) is mandatory **before any biopsy or excision**: an enlarged foramen caecum, a bifid crista galli or a tract to the dura means a combined approach.

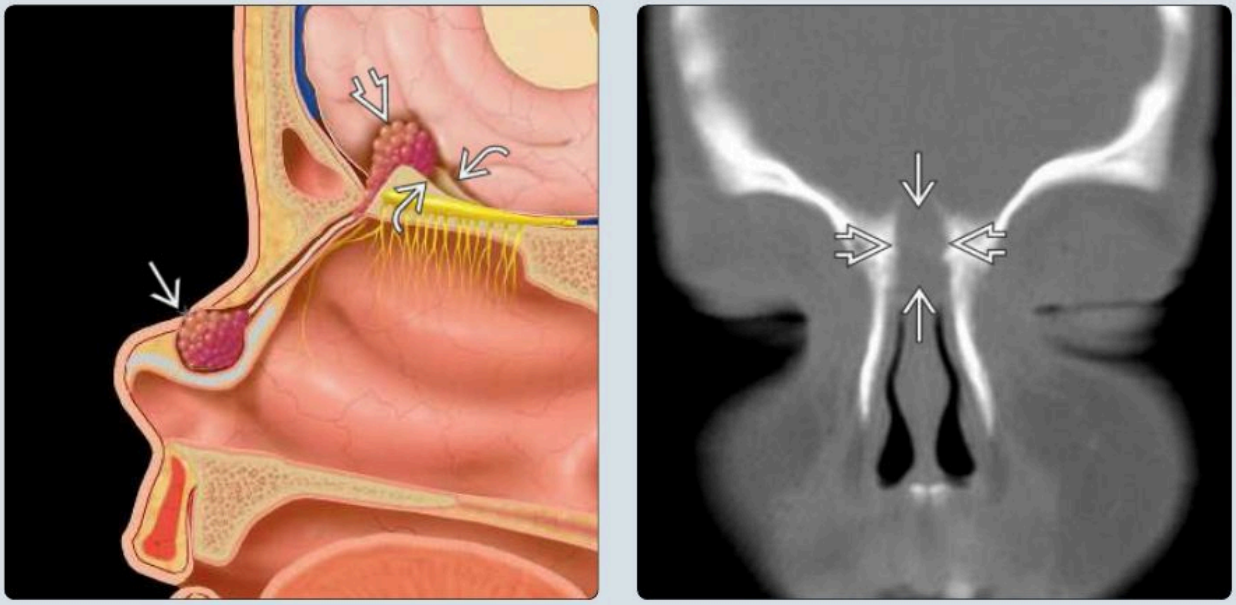


FIGURE 9 Nasal dermal sinus and dermoid. Left: drawing of a sinus tract from a nasal pit with dermoids outside and inside the skull. Right: coronal CT, a midline dermoid remodelling the margins of the foramen caecum.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 678

4 CHAPTER FOUR The deaf child

Malformations and the nerve.

4.1 Which test

In a child with **sensorineural** loss, **MRI** with high-resolution T2 shows the inner ear fluid, the IAC and, crucially, whether the **cochlear nerve** is present; **CT** shows the bony labyrinth, the cochlear nerve canal, the vestibular aqueduct and the middle ear, and is often added when a malformation is found or before cochlear

implantation. In a child with **conductive** loss and a normal drum, CT looks for ossicular malformation, oval window atresia, and congenital cholesteatoma.

4.2 The malformations

Inner ear malformations are described by how far development went, in the Sennaroglu classification: complete labyrinthine aplasia (Michel), cochlear aplasia, **common cavity**, **cochlear hypoplasia**, and the **incomplete partitions**. **IP-I** is a cystic cochlea without a modiolus; **IP-II** (the Mondini deformity) has a normal basal turn but a cystic apex where the middle and apical turns fuse, with a dilated vestibule and a **large vestibular aqueduct**; **IP-III** (X-linked) lacks the modiolus and carries a risk of a CSF **gusher** at surgery. A **narrow or absent cochlear nerve canal** or an absent cochlear nerve may make a cochlear implant pointless.

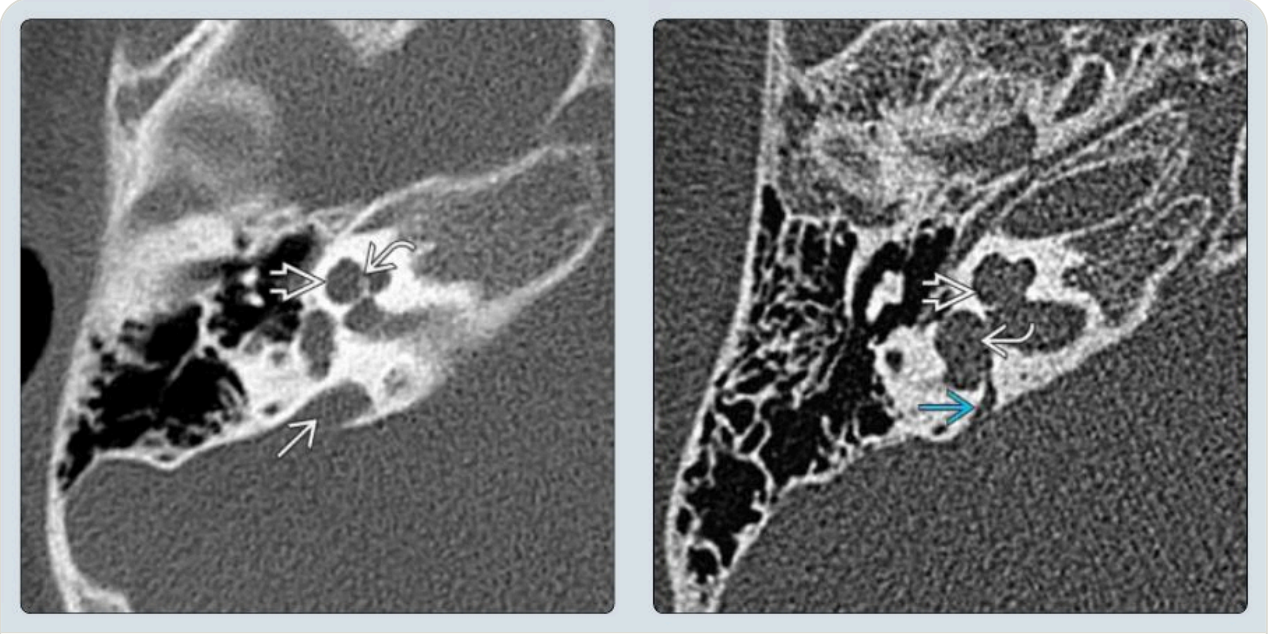


FIGURE 10 IP-II (Mondini) malformation. Left: axial CT, a large vestibular aqueduct. Right: a mild large vestibular aqueduct with plump middle and apical turns that lack the interscalar septum and modiolus.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1102

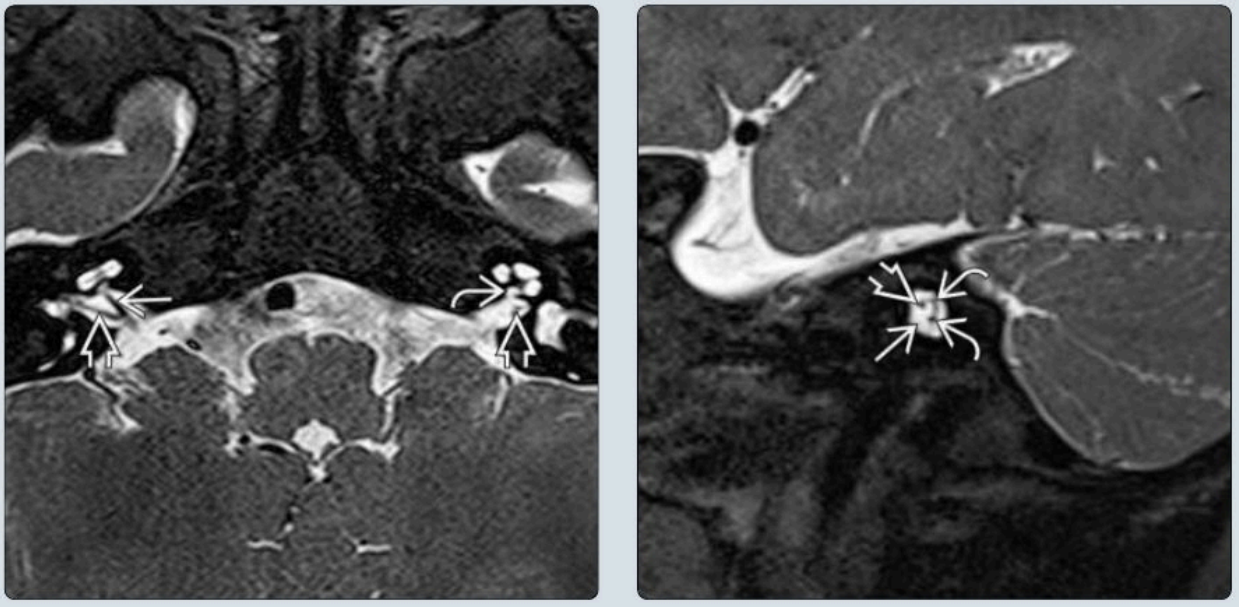


FIGURE 11 Cochlear nerve deficiency. Left: axial T2 in a one-year-old with profound left loss: normal right cochlear nerve, absent left. Right: oblique sagittal T2, no cochlear nerve in the left IAC, with normal facial and vestibular nerves.

Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 1108

IMAGING A CHILD

1 Choose the test

- → ultrasound for lumps and glands
- → MRI for soft tissue, inner ear and intracranial
- → low-dose CT for bone and emergencies

2 Neck lump

- → **midline**
  thyroglossal or dermoid
- → **lateral**
  branchial cleft (which one)
- → **transspatial and cystic**
  lymphatic malformation
- → **solid and high-flow in infancy**
  haemangioma
- → **solid and growing**

exclude malignancy

3 **Newborn nose**

- → front (pyriform aperture) or back (choanae)
- → bone or membrane
- → CHARGE
- → central incisor

4 **Midline nasal mass**

- → foramen caecum, crista galli, tract to dura before any biopsy

5 **Deaf child**

- → MRI for cochlear nerve and labyrinth
- → CT for bony anatomy
- → classify any malformation

5 CHAPTER FIVE For the examination

What the examiner wants to hear.

5.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Ultrasound and MRI first** in children; when CT is right.
- **Thyroglossal duct cyst:** relation to the hyoid and a normal thyroid on ultrasound.
- **Second branchial cleft cyst** position; third and fourth arch anomalies and suppurative thyroiditis.
- **Haemangioma versus vascular malformations.**
- **Choanal atresia versus pyriform aperture stenosis**, and their associations.

- **Midline nasal mass:** image before biopsy.
- **Inner ear malformations** (common cavity, IP-I, IP-II, IP-III) and the cochlear nerve.

5.2 Common pitfalls

COMMON PITFALLS

- CT for a child's neck lump when ultrasound would do.
- Removing a thyroglossal cyst without confirming a normal thyroid.
- Biopsying a midline nasal mass with an intracranial connection.
- Implanting an ear with no cochlear nerve.
- Calling an adult's cystic level II mass a branchial cyst.

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